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Purpose of Document				Type of Document
Information		for Approval		As Built

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Place for digital signature of Revisor

AS BUILT

Rev.	Revision Description	Drawn by	Checked by	Approved by Designer	Approved by Contractor	Date
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Employer: epcg Elektroprivreda Crne Gore AD

Consultant: ENERGOPROJEKT HIDROINŽENJERING a.d.

Contractor: VOITH HYDRO GmbH&CoKG St. Pölten

Project:

Reconstruction and Modernisation of HPP Perucica - Phase II

Title:

Unit 7 process control circuit diagram

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska
Approved by Designer	Grga Kolić	01.04.2026.			
Checked by	Antonio Aleksić	01.04.2026.			
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER

Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
TITLE PAGE	Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 1 PAGE 1 / 204

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Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00	+ U7CJA10
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Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska		PERU-01-17-170-31-001	00 + U7CJA10
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SYSTEM/EQUIPMENT

=U7AEA10GH100+U7AEA10	Marsalling kiosk
=U7BJA10(1,2)GH100+U7BJA10(1,2)	400 VAC sub-distribution
=U7BUA10GH100+U7BUA10	220VDC & 230VAC UPS sub-distribuition
=U7BAT10GH100+U7BAT10	Step-up transformer cooling control cubicle
=U7CJA10GH100+U7CJA10	Process control
=U7CBC10GH100+U7CBC10	Turbine governor
=U7CHA10GH100+U7CHA10	Electrical protection 1
=U7CHB10GH100+U7CHB10	Electrical protection 2
=U7CMB10GH100+U7CMB10	Machine condition monitoring
=U7MKC10GH100+U7MKC11	Excitation

BASIC:	RITTAL VX 8826.000 x 2
MEASUREMENT:	2400x1600X600 mm (HxWxD)
COLOR:	RAL 7035
DOORS:	FRONTSIDE
TYPE OF PROTECTION:	IP55
VOLTAGE LEVEL:	230VAC / 220VDC / 24VDC
CROSS SWCTION IF NOT DESIGNED DIFFERENT:	0,75QMM

DEVICE

A -	Assemblies, subassemblies
B -	Transducers from non-electric to electric quantities or vice versa
C -	Capacitors
D -	Binary elements
E -	Miscellaneous
F -	Protective devices
G -	Generators, power supplies
H -	Signaling devices
J -	
K -	Relays, contactors
L -	Inductors, reactors
M -	Motors
N -	Amplifiers, controllers
P -	Measuring instruments, test equipment
Q -	Switching devices for power circuits
R -	Resistors
S -	Switches for control circuits, selectors
T -	Transformers
U -	Modulators
V -	Tubes, semiconductors
W -	Transmission paths, cables, busbars, antennas
X -	Terminals, plugs, sockets
Y -	Electricaly operated mechanical devices
Z -	Terminations, hybrid transformers, filters, equilizers, limiters

100..199 -	Station services
200..299 -	Control
300..399 -	Protection
400..499 -	Measuring
500..599 -	Freely disposable

DOCUMENT

001...099 -	Text documents, chapters
100...999 -	Drawings: Block-diagrams, Single line diagrams, General arrangement drawings, Circuit diagrams, Cubicle layouts, Cubicle assigned drawings, Cable lists, cable core lists, Grounding and lightning, Unassigned drawings,

CIRCUIT DIAGRAM INDEX

A -	Indexes, title sheet, document list
B -	Function diagrams
C -	Designation code
D -	AC power supply (heating, lighting voltage distribution, fuses...)
E -	
F -	
G -	DC power supply (voltage distribution...)
H -	Signaling devices
J -	
K -	
L -	Operating mechanisms for switching devices (circuit breakers, disconnectors...)
M -	Control (local, remote, station...)
N -	Protection
P -	Interlocking
Q -	Position indication (local, remote, station...)
R -	Alarm (signaling, monitoring...)
S -	Instruments transformers circuits (measuring, protection...)
T -	Measuring (local, remote station... unless under "S")
U -	Feedback control (voltage regulator...)
V -	Communication
W -	Wiring diagrams, terminal matrix
X -	Cable list, cable core list
Y -	
Z -	Equipment documents for "C" to "W"

WIRE AND CABLE COLOURS

230V AC PHASE	Black
230V AC ZERO	Blue
230V AC NEUTRAL	Yellow/Green
220 DC	Red
CT 100-110V AC	Orange
CT 1/5A	Brown
DI/DO 24V DC	Grey
AI/AO 4-20mA/PT100	White

BK -	Black
BN -	Brown
RD -	Red
OG -	Orange
YE -	Yellow
GN -	Green
BU -	Blue
VT -	Violet
GY -	Grey
WH -	White
PK -	Pink
TO -	Turquaise
GNYE -	Green yellow
LB -	Light Blue

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Approved by Designer	Grga Kolić	01.04.2026.							+ U7CJA10
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SERVICE POWER SUPPLY

CUBICLE LIGHTING

CUBICLE HEATER

=U7BJA10GH100
+U7BJA11

3x400 VAC

L1 L2 L3

-1F08
16A/C

PE

-X11 22

400VAC SUB-DISTRIBUTION

+U7BJA11-W1008
NYY-J 3x2.5 mm²

BN BU GNYE

-X0

-F01
/146.4:B
16 A
30 mA

7.1:E / X1PE

-X1

-S101
/168.4:D

-S102
/168.4:D

PLUG-IN CABLE
1.5 m

-X101
/180.4:F
16A

-E101
/145.4:E
LED lamp

-E102
/145.4:E
LED lamp

5SU1354-1KK13
5ST3016

-B101
THR02
/145.4:B

-B102
10...90%
/145.4:C

-R101
/168.4:C
100W

-R102
/168.4:C
100W

-B103
THV02
/145.4:D

-M101
/167.4:D
37 W

-M102
/167.4:E
37 W

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Approved by Designer	Grga Kolić	01.04.2026.				AC DISTRIBUTION 230VAC	+ U7CJA10
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DC SYSTEM 1 INCOMING | DC SYSTEM 2 INCOMING | 220/24 VDC CONVERTER 1 | 220/24 VDC CONVERTER 2 | PANEL PC 220VDC | SPARE

-U1
/179.4:E
220VDC+
1P DISTRIBUTION BLOCK

220VdcL+ ... 220VdcL+

220VdcL+ ... 220VdcL-

220VDC-
1P DISTRIBUTION BLOCK
-U2
/179.4:E

-V1
100A
800V
/180.4:D

-V2
100A
800V
/180.4:D

-SV1
/179.4:D
20A

-SV2
/179.4:D
20A

-F11
/146.4:D
10A/B

-F12
/146.4:D
10A/B

-F13
/146.4:E
6A/B

-F14
/146.4:E
6A/B

-A100
/107.2:C
L N PE

CONV1 / 8.2:B
CONV1-
220VDC / 8.2:B

CONV2 / 8.5:B
CONV2-
220VDC / 8.5:B

6.4:C / X1PE
GNYE 2.5
PE
Cu 25 x 5
PE / 8.1:A

PE_SH
C30 29x15x2 mm
PE-SH
GNYE 6

-X10

*FOR GROUNDING
CUBICLE COMPONENTS
GNYE 6mm2

Cu 1x25 mm²

+U7BUA10-W2104
NYY-O 2x4 mm²

+U7BUA10-W2204
NYY-O 2x4 mm²

=U7BUA10GH100
+U7BUA10
DC SYSTEM 1
220Vdc, In=125A
-1QF4
10A/C
-X220.1

DC SYSTEM 2
220Vdc, In=125A
-2QF4
10A/C
-X220.2

220VDC & 230VAC UPS
SUB-DISTRIBUTION

PLANT GROUNDING SYSTEM

-X13
-X14

/21.3:D
/17.2:B
/17.3:B

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Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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220/24 VDC
CONVERTER 1

220/24 VDC
CONVERTER 2

7.8:E / PE

PE

7.4:E / CONV1

7.4:E / CONV1-
220VDC

7.6:E / CONV2

7.6:E / CONV2-
220VDC

BK 2.5
BU 2.5
GNYE 2.5

-U101
/180.4:B
In: AC 120-230 V
DC 110-220 V
Out: DC 24 V/20 A

-U102
/180.4:C
In: AC 120-230 V
DC 110-220 V
Out: DC 24 V/20 A

L1 N PE

SIEMENS
SITOP PSU8200
6EP1336-3BA10

ADJUST
24...28.8V
DEFAULT VALUE
24V

PARALLEL OPERATION
CONSTANT CURRENT
ON
OFF
A B

24V O.K.
13 14

1+ 2+ 1- 2-

RD 1
DC1_OK / 17.2:C

RD 1
DC1_OK / 17.2:C

GNYE 2.5

RD 1
DC2_OK / 17.3:C

RD 1
DC2_OK / 17.3:C

GNYE 2.5

GNYE 16

RD 4
BU 4

-V101
In/Out: DC 24-48V/40A
/180.4:E

DC IN1 DC IN2

DC OUT1

RD 10

24VdcL- 24VDC-BUSBAR

24VdcL- / 9.1:C

24VdcL+

24VdcL+ / 9.1:A

24VDC-
1P DISTRIBUTION BLOCK
-U10
/179.4:E

7 9

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Approved by Designer	Grga Kolić	01.04.2026.							+ U7CJA10
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Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 8 / 204

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PLC, MULTIMETER | DIGITAL INPUTS | DIGITAL OUTPUTS | ANALOG I/O | COMUNICATION POWER SUPPLY 1 | COMUNICATION POWER SUPPLY 2 | CONTROL 1

-U10
24VDC+
1P DISTRIBUTION BLOCK

8.8:F / 24VdcL+ → 24VdcL+ / 10.1:A

RD 4

5ST3733 - Pin busbar

	-F31	-F32	-F33	-F34	-F35	-F36	-F37
	/146.4:F	/146.4:F	/147.4:B	/147.4:C	/147.4:D	/147.4:D	/147.4:D
	6 A/B	6 A/B	6 A/B	2 A/B	6 A/B	6 A/B	6 A/B
	/21.4:B	/21.4:C	/21.4:B	/21.4:C	/21.4:C	/21.4:C	/21.4:C

8.8:F / 24VdcL- → 24VdcL- / 10.1:C

RD 2.5 | BU 2.5

	-X31	-X32	-X33	-X34	-X35	-X36	-X37
B	31L+ 24VDC / 13.2:E	32L+ 24VDC / 14.3:E	33L+ 24VDC / 15.4:E	34L+ 24VDC / 16.1:E	35L+ 24VDC / 104.4:E	36L+ 24VDC / 104.4:F	37L+ 24VDC / 65.3:B
A	1	1	1	1	1	1	1
B	31L- 24VDC / 13.2:E	32L- 24VDC / 14.3:F	33L- 24VDC / 15.4:E	34L- 24VDC / 16.1:E	35L- 24VDC / 104.4:E	36L- 24VDC / 104.4:F	37L- 24VDC / 65.3:F
A	6	23	3	5	4	4	6

RD 2.5 | BU 2.5

-P101
SENTRON
/139.2:B
+ L | - N

8 | 10

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				DC DISTRIBUTION 24VDC	Designer No.:	SALES No.: PAGE 9
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 9 / 204

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CWS | SYNCHRONIZATION CONTROL | AUTOMATIC SYNCHRONIZATION | ALARM ANNUNCIATOR | FEEDBACK SIGNAL LAMPS | MCR DIGITAL DISPLAY AND TEMPERATURE SIGNAL CONVERTERS | EXTERNAL SIGNALS - LV SWITCHGEAR

-U10
24VDC+
1P DISTRIBUTION BLOCK

9.8:A / 24VdcL+ → 24VdcL+ / 11.1:A

RD 4

5ST3733 - Pin busbar

	-F38	-F39	-F40	-F41	-F42	-F43	-F44
	/147.4:D 6 A/B	/147.4:D 6 A/B	/147.4:D 6 A/B	/147.4:E 6 A/B	/147.4:F 2 A/B	/147.4:F 2 A/B	/147.4:F 2 A/B
	/21.4:D	/21.4:D	/21.4:D	/21.4:B	/21.4:C	/21.4:C	/21.4:C

9.8:C / 24VdcL- → 24VdcL- / 11.1:C

RD 2.5, BU 2.5

-X38 B A1 → B A2; -X38 B A3 → B A4

+U7CJA10-W5003
NYY-J 3x2.5 mm²
BN, BU

PS_24- / 106.3:F
PS_24+ / 106.3:F

	-X39	-X40	-X41	-X42	-X43	-X44
	39L+ 24VDC / 70.1:B	40L+ 24VDC / 77.1:B	41L+ 24VDC / 86.1:B	42L+ 24VDC / 80.1:B	43L+ 24VDC / 58.1:D	44L+ 24VDC / 22.1:B
	39L- 24VDC / 70.1:F	40L- 24VDC / 77.1:F	41L- 24VDC / 88.1:F	42L- 24VDC / 80.1:F	43L- 24VDC / 58.1:E	44L- 24VDC / 22.1:D

-F25G
SPM-D10
/143.2:B
1 (+), 2 (-), GND

-H01
/140.2:B
X1:1 L1, X1:2 N, X1:3 PE

BU 1 41L1- 24VDC / 86.8:D

-PT10
/144.2:B
1 (+), 2 (-), GND

GNYE 2.5

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.							+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				DC DISTRIBUTION 24VDC	Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 10
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 10 / 204

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EXTERNAL SIGNALS - MARSHALLING KIOSK

EXTERNAL SIGNALS - PROTECTION, GOVERNOR, MONITORING

EXTERNAL SIGNALS - EXCITATION

EXTERNAL SIGNALS - GENERATOR, BUSBAR DUCT FAN

EXTERNAL SIGNALS - TRANSFORMATOR

EXTERNAL SIGNALS - MONITORING

SPARE

-U10
24VDC+
1P DISTRIBUTION BLOCK

10.8:A / 24VdcL+ → 24VdcL+ / 12.1:A

RD 4

5ST3733 - Pin busbar

	-F45	-F46	-F47	-F48	-F49	-F50	-F51
	/148.4:B	/148.4:B	/148.4:B	/148.4:C	/148.4:C	/148.4:C	/148.4:C
	2 A/B	2 A/B	2 A/B	2 A/B	2 A/B	2 A/B	2 A/B
	1 / 12 14 / 2 / 11	1 / 12 14 / 2 / 11	1 / 12 14 / 2 / 11	1 / 12 14 / 2 / 11	1 / 12 14 / 2 / 11	1 / 12 14 / 2 / 11	1 / 12 14 / 2 / 11
	/21.4:D	/21.4:D	/21.4:D	/21.5:B	/21.5:C	/21.4:C	

10.8:C / 24VdcL- → 24VdcL- / 12.1:C

RD 2.5 / BU 2.5

	-X45	-X46	-X47	-X48	-X49	-X50	-X51
B A1	45L+ 24VDC / 29.3:B	46L+ 24VDC / 35.1:B	47L+ 24VDC / 35.7:B	48L+ 24VDC / 40.2:B	49L+ 24VDC / 41.2:B	50L+ 24VDC / 38.7:B	51L+ 24VDC / 45.5:B
B	45L- 24VDC / 29.3:D	46L- 24VDC / 35.1:D	47L- 24VDC / 35.7:D	48L- 24VDC / 40.2:D	49L- 24VDC / 41.2:D	50L- 24VDC / 38.7:D	51L- 24VDC / 45.5:D
	A14	A9	A4	A5	A10	A4	A12

Format ISO/DIN A3

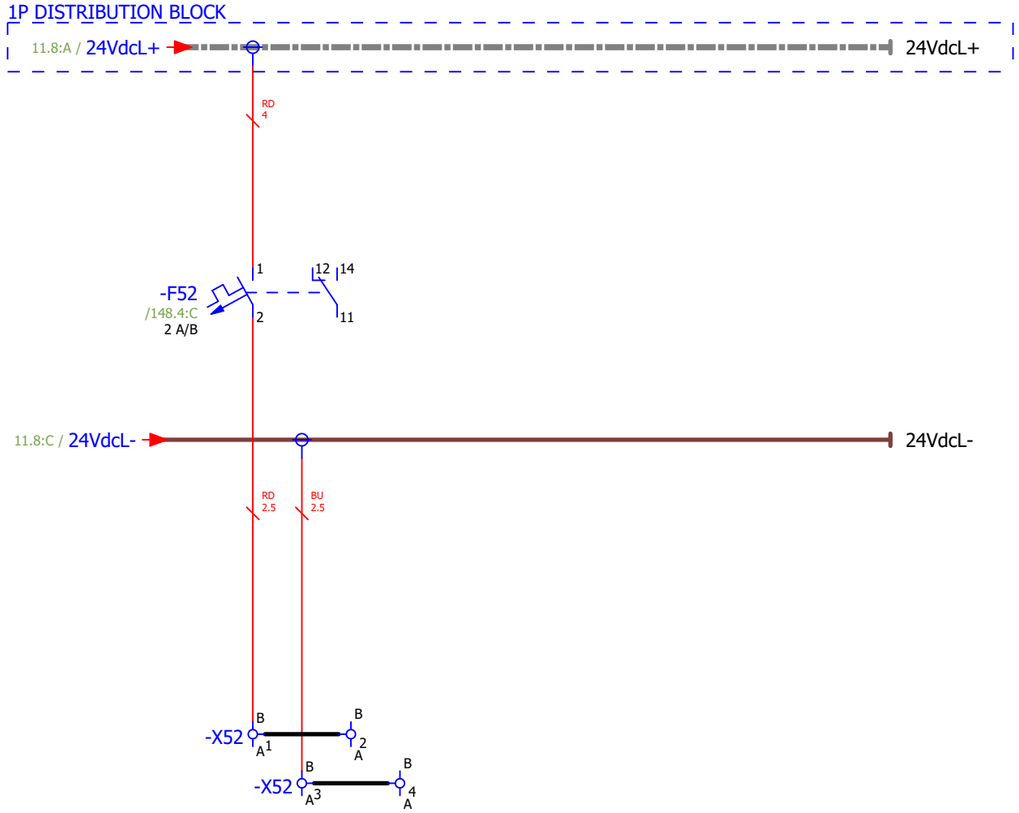
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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II		SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DC DISTRIBUTION 24VDC	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00
Approved by Designer	Grga Kolić	01.04.2026.						= U7CJA10GH100 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			PAGE 11 PAGE 11 / 204

SPARE

-U10
24VDC+
1P DISTRIBUTION BLOCK



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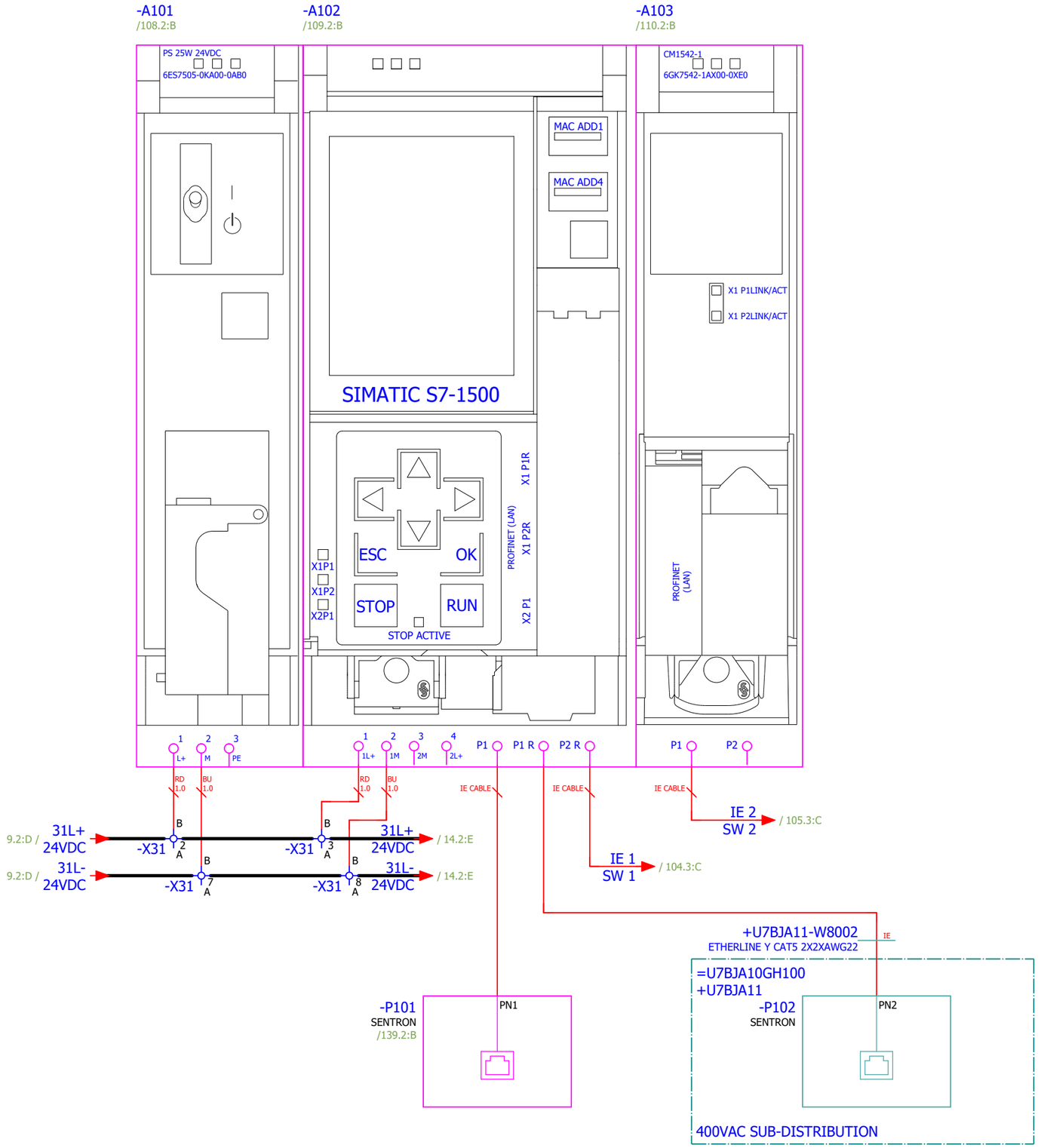
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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DC DISTRIBUTION 24VDC	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 12 PAGE 12 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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12 14

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	PLC CONFIGURATION	Designer No.: 22044-DD-EE-08-07-100	SALES No.:
Checked by	Antonio Aleksić	01.04.2026.						
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			

= U7CJA10GH100
+ U7CJA10
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-P101
SENTRON
/139.2:B

PN2

IE CABLE

106.5:B / CWS_COMM

-A201
/111.2:B

BA 2xRJ45

P1 R

P2 R

BA

P1R
P2R
X1
PROFINET
(LAN)

6ES7193-6AR00-0AA0

LK1
LK2

SIEMENS

RN ER MT

SIMATIC
ET 200SP

IM155-6PN/2 HF

PWR

6ES7155-6AU01-0CN0

MAC ADDRESS

X80
24VDC

1L+ 1M 2L+ 2M

-X80
1 2 3 4
1L+ 1M 2M 2L+

-A211 /112.2:B
-A212 /113.2:B
-A213 /114.2:B
-A214 /115.2:B
-A215 /116.2:B
-A216 /117.2:B
-A217 /118.2:B
-A218 /119.2:B
-A219 /120.2:B

DI 16x24VDC ST
DIAG
PWR CC00
6ES7131-6BH01-0BA0

P1 P2 AUX L+ M

RD 1.0
BU 1.0

13.3:E / 31L+ 24VDC
13.3:E / 31L- 24VDC
-X31

9.3:D / 32L+ 24VDC
9.3:D / 32L- 24VDC
-X32

32L+ 24VDC / 17.1:B
32L- 24VDC / 17.1:F

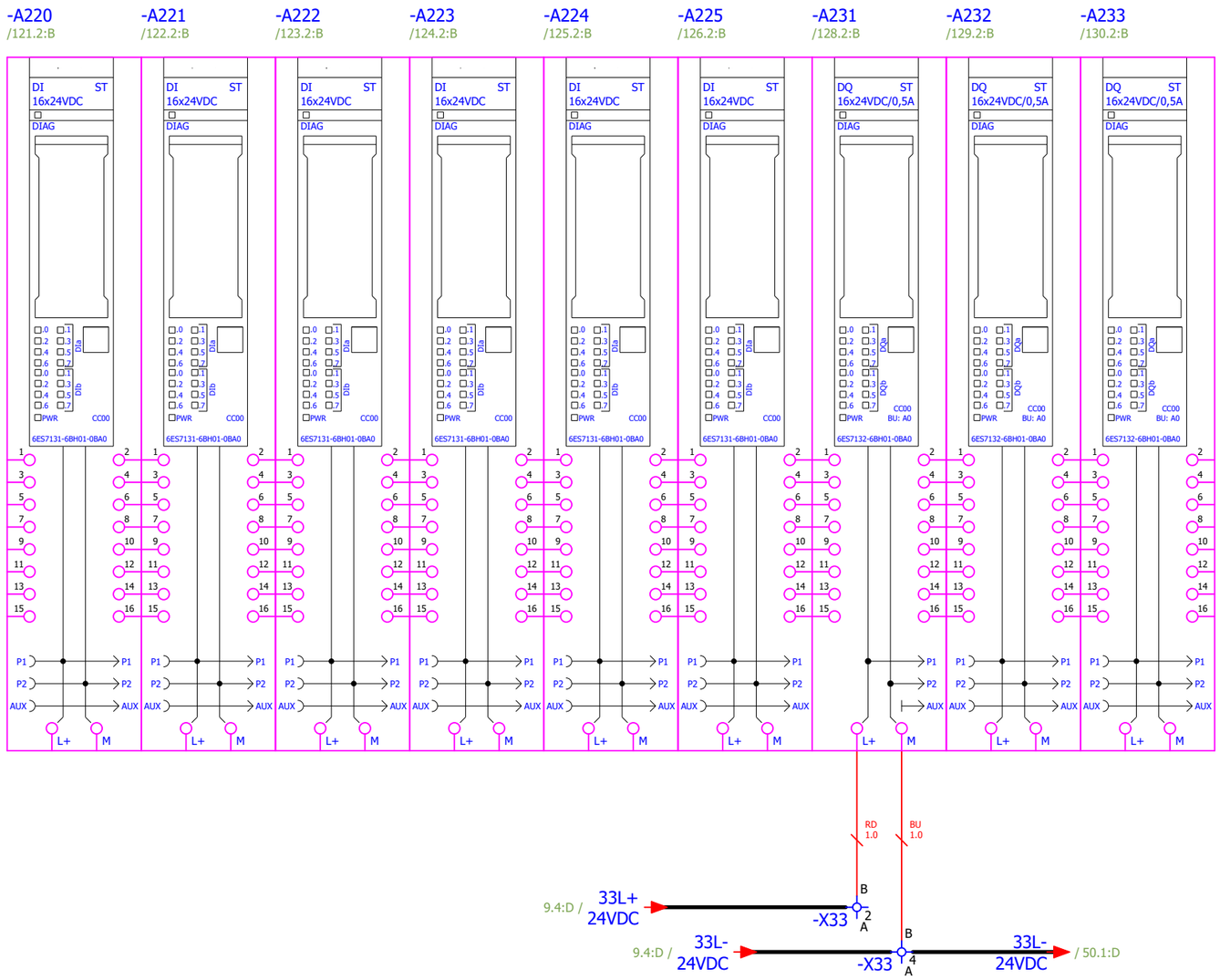
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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				PLC CONFIGURATION	Designer No.:	SALES No.:	PAGE 14
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 14 / 204

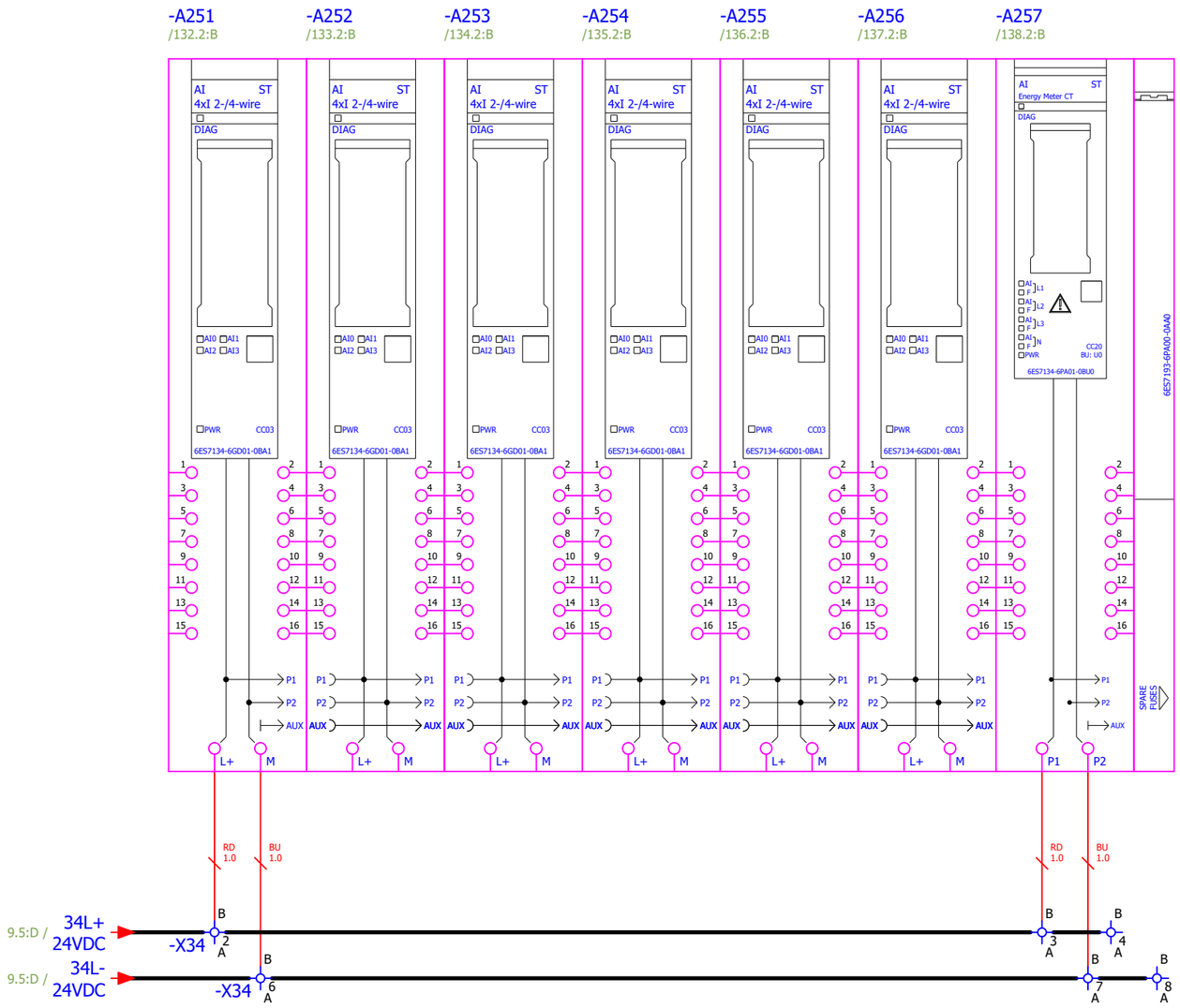
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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	PLC CONFIGURATION			+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 15
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 15 / 204

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	PLC CONFIGURATION			+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 16
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 16 / 204

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24V DC POWER
SUPPLY DEVICE I
FAULT - ALARM

24V DC POWER
SUPPLY DEVICE II
FAULT - ALARM

14.4:E / 32L+ 24VDC → -X32 A B3 — A B4 → 32L+ 24VDC / 18.1:C

RD 1.0

-F11 /7.4 — 11 — 14 12

-F12 /7.5 — 11 — 14 12

8.4:D / DC1_OK

8.7:D / DC2_OK

-K201 /165.4:C
24VDC
24V DC POWER SUPPLY
DEVICE I FAULT - ALARM
A1 / A2

-K202 /165.4:C
24VDC
24V DC POWER SUPPLY
DEVICE II FAULT - ALARM
A1 / A2

BU 1.0

14.4:F / 32L- 24VDC → -X32 B 25 A — B 26 A

RXG22BD
SE.RZM040W
14 — 11 /20.4:D
12
24 — 21 /86.4:C
22

RXG22BD
SE.RZM040W
14 — 11 /20.5:D
12
24 — 21 /86.4:C
22

16 18

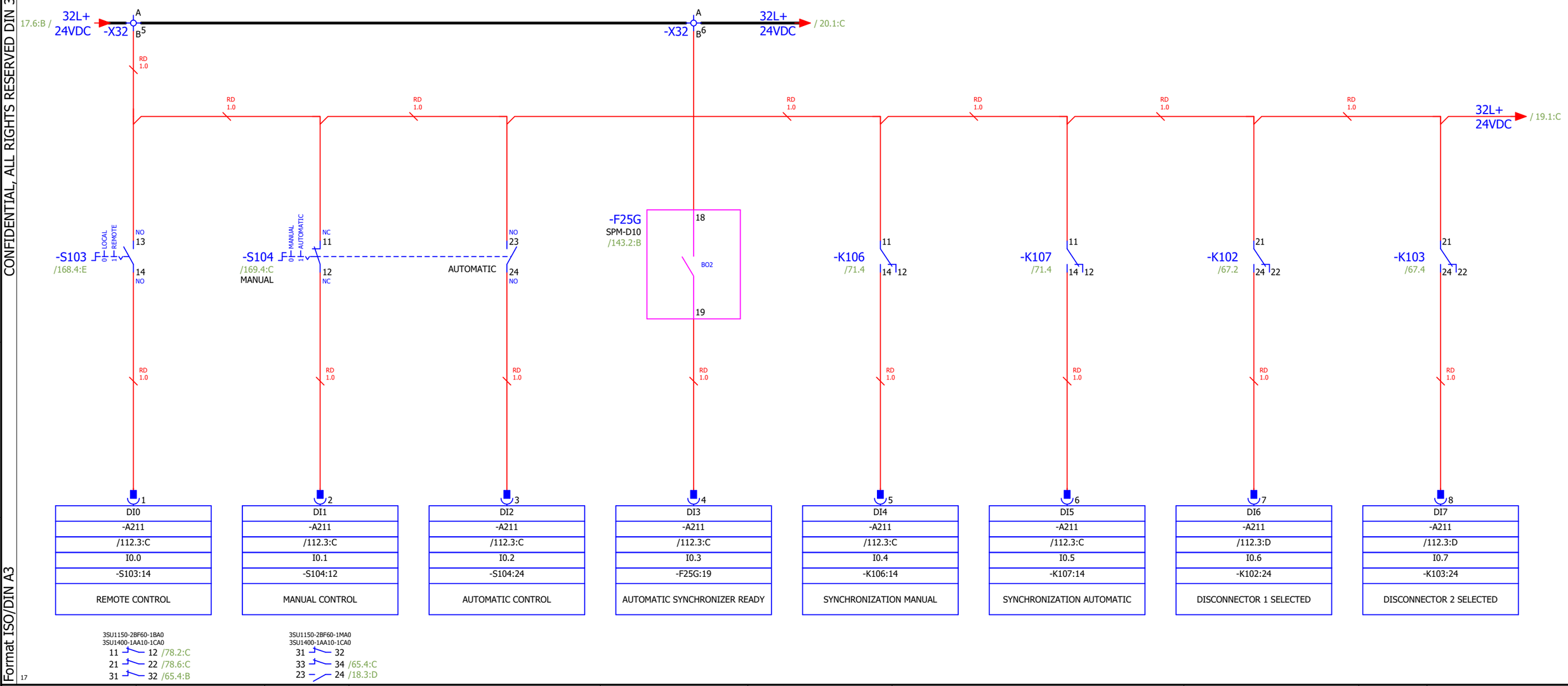
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 0.0 - 0.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 17 PAGE 17 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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REMOTE CONTROL	MANUAL CONTROL	AUTOMATIC CONTROL	AUTOMATIC SYNCHRONIZER READY	SYNCHRONIZATION MANUAL	SYNCHRONIZATION AUTOMATIC	DISCONNECTOR 1 SELECTED	DISCONNECTOR 2 SELECTED



Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.							+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	DIGITAL INPUTS: 0.0 - 0.7	Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 18
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 18 / 204

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GENERATOR CIRCUIT BREAKER SELECTED	OFF COMMAND	ON COMMAND	TRANSFORMER OIL PUMP 1 SELECTED	TRANSFORMER OIL PUMPS 1 & 2 SELECTED	ESD PUSHBUTTON OK	FIRE EXTINGUISHING SYSTEM PUSHBUTTON OK	SPEED LOWER

18.8:C / 32L+ 24VDC

RD 1.0

-K104 /67.6 — 21, 24, 22

-S114 /170.4:C — NO 13, 14 NO — GREEN

-S115 /170.4:E — NC 11, 12 NC — RED

-S141 /172.4:F — PUMPA I, PUMPA II, PUMPA I&II — 13, 14 — LEFT; 23, 24 — RIGHT

-S301 /178.4:F — 21, 22 — EMERGENCY SHUTDOWN

-S302 /179.4:C — 21, 22 — FIRE EXTINGUISHING SYSTEM

-S221 /176.4:D — LOWER, HIGHER — NO 13, 14 NO

	9	10	11	12	13	14	15	16
	DI8	DI9	DI10	DI11	DI12	DI13	DI14	DI15
	-A211	-A211	-A211	-A211	-A211	-A211	-A211	-A211
	/112.3:D	/112.3:D	/112.3:D	/112.3:E	/112.3:E	/112.3:E	/112.3:E	/112.3:E
	I1.0	I1.1	I1.2	I1.3	I1.4	I1.5	I1.6	I1.7
	-K104:24	-S114:14	-S115:12	-S141:14	-S141:24	-S301:22	-S302:22	-S221:14
	GENERATOR CIRCUIT BREAKER SELECTED	OFF COMMAND	ON COMMAND	TRANSFORMER OIL PUMP 1 SELECTED	TRANSFORMER OIL PUMPS 1 & 2 SELECTED	ESD PUSHBUTTON OK	FIRE EXTINGUISHING SYSTEM PUSHBUTTON OK	SPEED LOWER
		3SU1152-0AB40-1BA0 3SU1900-0AP10-0AA0	3SU1152-0AB20-1CA0 3SU1900-0AP10-0AA0	3SU1052-2BL60-0AA0 3SU1550-0AA10-0AA0		3SU1051-1BA20-0AA0 3SU1550-0AA10-0AA0	3SU1051-1BA20-0AA0 3SU1550-0AA10-0AA0	3SU1052-2BM20-0AA0 3SU1400-1AA10-1BA0
		43 — 44 /68.6:C	43 — 44 /72.2:C	13 — 14 /19.4:D		13 — 14 /82.2:C	13 — 14 /82.3:C	23 — 24 /78.6:D
		23 — 24 /68.2:C	23 — 24 /68.3:C	23 — 24 /19.5:D		11 — 12 /76.3:B	11 — 12 /76.3:C	33 — 34 /78.7:D
		33 — 34 /68.4:C	33 — 34 /68.5:C	33 — 34 /69.2:B		21 — 22 /19.6:D	21 — 22 /19.7:D	43 — 44 /20.1:D
				43 — 44 /69.4:B			31 — 32	23 — 24 /76.7:C

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001 / REV No.: 00 / = U7CJA10GH100 / + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.				DIGITAL INPUTS: 1.0 - 1.7	Designer No.: 22044-DD-EE-08-07-100 / SALES No.: / PAGE 19 / PAGE 19 / 204
Checked by	Antonio Aleksić	01.04.2026.					
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		

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1	2	3	4	5	6	7	8
SPEED HIGHER	VOLTAGE LOWER	VOLTAGE HIGHER	DC CONVERTER 1 OK	DC CONVERTER 2 OK	SPARE	SWITCH 1 OK	SWITCH 2 OK
DI0	DI1	DI2	DI3	DI4	DI5	DI6	DI7
-A212	-A212	-A212	-A212	-A212	-A212	-A212	-A212
/113.3:C	/113.3:C	/113.3:C	/113.3:C	/113.3:C	/113.3:C	/113.3:D	/113.3:D
I2.0	I2.1	I2.2	I2.3	I2.4	I2.5	I2.6	I2.7
-S221:44	-S222:14	-S222:44	-K201:14	-K202:14		-U111:-X10:F2	-U112:-X10:F2
SPEED HIGHER	VOLTAGE LOWER	VOLTAGE HIGHER	DC CONVERTER 1 OK	DC CONVERTER 2 OK	SPARE	SWITCH 1 OK	SWITCH 2 OK

18.5:C / 32L+ 24VDC -X32 A B7 32L+ 24VDC / 22.1:D

RD 1.0

-S221 43 44

-S222 /177.4:C LOWER HIGHER NO 13 14 NO

-K201 /17.2 11 14 12

-K202 /17.3 11 14 12

-U111 /141.2:B F1 FAULT FAULT F2

-U112 /142.2:B F1 FAULT FAULT F2

32L+ 24VDC / 21.1:C

3SU1052-2BM50-0AA0
3SU1400-1AA10-1BA0
23 24 /78.2:D
33 34 /78.3:D
43 44 /20.3:D

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 2.0 - 2.7	DRAWING No.: PERU-01-17-170-31-001
Approved by Designer	Grga Kolić	01.04.2026.					REV No.: 00
Checked by	Antonio Aleksić	01.04.2026.					= U7CJA10GH100 + U7CJA10
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER	Designer No.: 22044-DD-EE-08-07-100	SALES No.:
							PAGE 20 PAGE 20 / 204

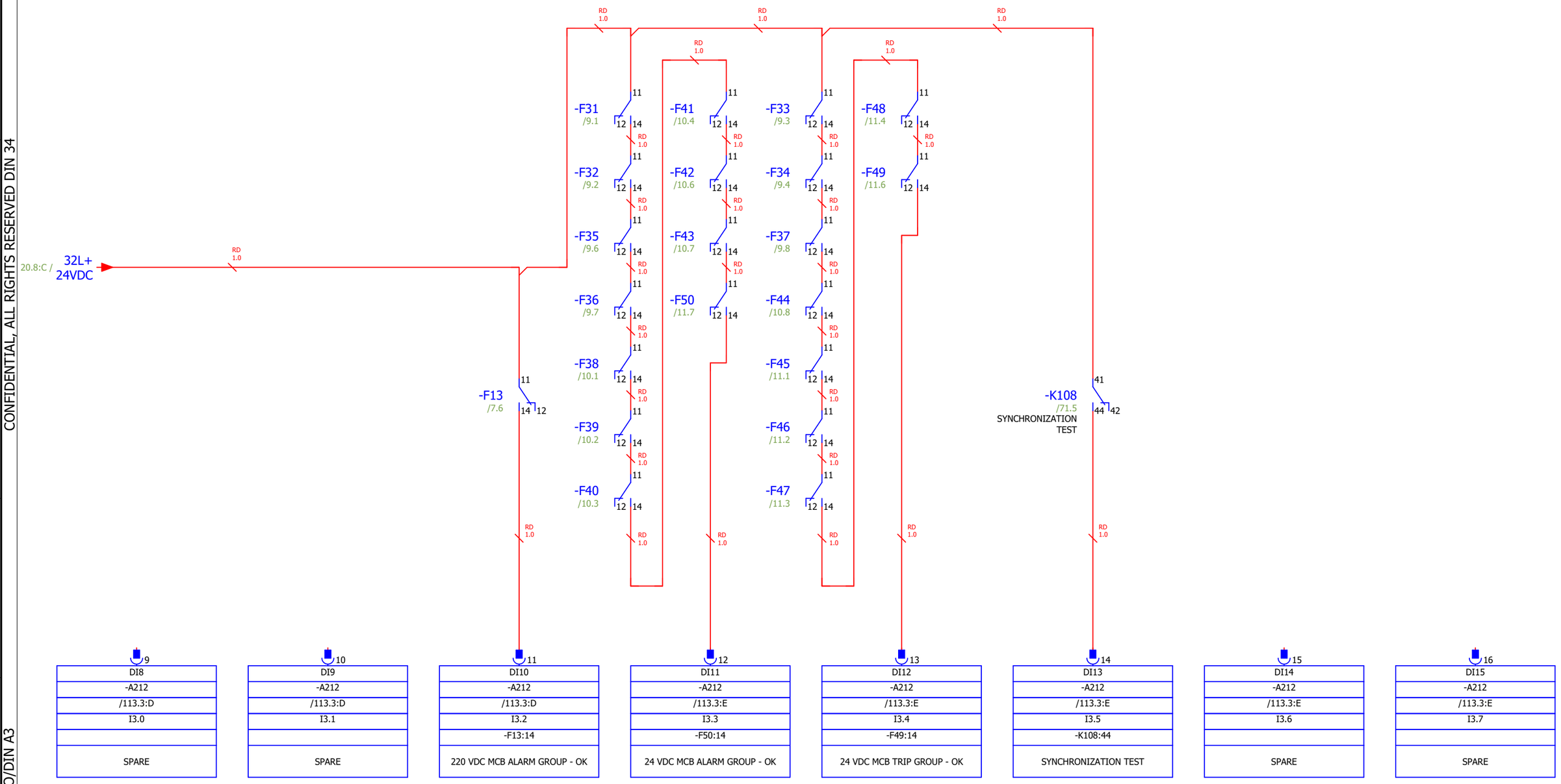
Format ISO/DIN A3

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9	10	11	12	13	14	15	16
SPARE	SPARE	220 VDC MCB ALARM GROUP - OK	24 VDC MCB ALARM GROUP - OK	24 VDC MCB TRIP GROUP - OK	SYNCHRONIZATION TEST	SPARE	SPARE



9	10	11	12	13	14	15	16
DI8	DI9	DI10	DI11	DI12	DI13	DI14	DI15
-A212	-A212	-A212	-A212	-A212	-A212	-A212	-A212
/113.3:D	/113.3:D	/113.3:D	/113.3:E	/113.3:E	/113.3:E	/113.3:E	/113.3:E
I3.0	I3.1	I3.2	I3.3	I3.4	I3.5	I3.6	I3.7
		-F13:14	-F50:14	-F49:14	-K108:44		
SPARE	SPARE	220 VDC MCB ALARM GROUP - OK	24 VDC MCB ALARM GROUP - OK	24 VDC MCB TRIP GROUP - OK	SYNCHRONIZATION TEST	SPARE	SPARE

Format ISO/DIN A3

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 3.0 - 3.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 21
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 21 / 204

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AC MAIN SUPPLY REMOTE | AC MAIN SUPPLY CB OFF | AC MAIN SUPPLY CB INSERTED | AC MAIN SUPPLY CB READY | AC AUXILIARY SUPPLY CB OFF | AC AUXILIARY SUPPLY CB INSERTED | AC VOLTAGE OK | AC VOLTAGE OK MCB OK

10.8:D / 44L+ 24VDC -X44 B 2 A

+U7BJA11-W5021 NYCY 10x1.5 mm² 1

=U7BJA10GH100 +U7BJA11 -X21 1

+U7BJA11-W5031 NYCY 7x1.5 mm² 1

=U7BJA10GH100 +U7BJA11 -X31 1

44L+ 24VDC / 23.1:B

-X21 6 2 8 7

400VAC SUB-DISTRIBUTION

+U7BJA11-W5021 NYCY 10x1.5 mm² 6 2 8 7

-X31 2 4

400VAC SUB-DISTRIBUTION

+U7BJA11-W5031 NYCY 7x1.5 mm² 4 2

-X21 3 4

+U7BJA11-W5021 NYCY 10x1.5 mm² 3 4

-1K11 /148.4:F | -1K12 /149.4:B | -1K13 /149.4:B | -1K14 /149.4:B | -1K15 /149.4:B | -1K16 /149.4:B | -1K17 /149.4:B | -1K18 /149.4:C

A1 A2

BU 1.0

44L- 24VDC / 23.1:D

10.8:D / 44L- 24VDC -X44 B 20 A 44L- 24VDC / 24.1:D

20.2:C / 32L+ 24VDC -X32 A 8 B 32L+ 24VDC / 24.1:D

RD 1.0

32L+ 24VDC / 23.1:E

-1K11 /22.1 | -1K12 /22.2 | -1K13 /22.3 | -1K14 /22.4 | -1K15 /22.5 | -1K16 /22.6 | -1K17 /22.7 | -1K18 /22.8

11 14 12

	DI0	DI1	DI2	DI3	DI4	DI5	DI6	DI7
	-A213	-A213	-A213	-A213	-A213	-A213	-A213	-A213
	/114.3:C	/114.3:C	/114.3:C	/114.3:C	/114.3:C	/114.3:C	/114.3:D	/114.3:D
	I4.0	I4.1	I4.2	I4.3	I4.4	I4.5	I4.6	I4.7
	-1K11:14	-1K12:14	-1K13:14	-1K14:14	-1K15:14	-1K16:14	-1K17:14	-1K18:14
	AC MAIN SUPPLY REMOTE	AC MAIN SUPPLY CB OFF	AC MAIN SUPPLY CB INSERTED	AC MAIN SUPPLY CB READY	AC AUXILIARY SUPPLY CB OFF	AC AUXILIARY SUPPLY CB INSERTED	AC VOLTAGE OK	AC VOLTAGE OK MCB OK

PLC-RSC-24DC/21 14 12 11 /22.1:E | PLC-RSC-24DC/21 14 12 11 /22.2:E | PLC-RSC-24DC/21 14 12 11 /22.3:E | PLC-RSC-24DC/21 14 12 11 /22.4:E | PLC-RSC-24DC/21 14 12 11 /22.5:E | PLC-RSC-24DC/21 14 12 11 /22.6:E | PLC-RSC-24DC/21 14 12 11 /22.7:E | PLC-RSC-24DC/21 14 12 11 /22.8:E

21 23

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 4.0 - 4.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 22
Checked by	Antonio Aleksić	01.04.2026.							PAGE 22 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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AC SURGE ARRESTER OK

230 VAC MARSHALLING KIOSK MCB ALARM GROUP - OK

GENERATOR COOLING WATER VALVE FEEDER INSERTED

GENERATOR COOLING WATER VALVE READY

TRANSFORMER COOLING WATER VALVE FEEDER INSERTED

TRANSFORMER COOLING WATER VALVE READY

EXCITATION COOLING FANS & AC FIELD FLASHING FEEDER INSERTED

EXCITATION COOLING FANS & AC FIELD FLASHING READY

22.8:B / 44L+ 24VDC

-X44 4 A B

+U7BJA11-W5001 NYCY 7x1.5 mm² 1

+U7BJA13-W5002 NYCY 12x1.5 mm² 1

+U7BJA13-W5071 NYCY 2x1.5/1.5 mm² BN

+U7BJA13-W5081 NYCY 2x1.5/1.5 mm² BN

+U7BJA13-W5111 NYCY 2x1.5/1.5 mm² BN

44L+ 24VDC / 24.1:B

=U7BJA10GH100
+U7BJA11

-X12 1

-X21 5

-X12 2

400VAC SUB-DISTRIBUTION

=U7BJA10GH100
+U7BJA13.LA001

-X20 1 3 24VDC

20L+ / 23.5:B

RJ45

2 connected

-X20 5

-3Q7 15 21 24 22 16

400VAC SUB-DISTRIBUTION
GENERATOR COOLING WATER
VALVE CONTROL CUBICLE

=U7BJA10GH100
+U7BJA13.MA001

20L+ / 23.3:B

-X20 2 3 24VDC

20L+ / 25.5:B

RJ45

2 connected

-X20 6

-3Q8 15 21 24 22 16

400VAC SUB-DISTRIBUTION
TRANSFORMER COOLING WATER
VALVE CONTROL CUBICLE

=U7BJA10GH100
+U7BJA13.CA002

-20L+ / 25.5:B

-X20 1 3 24VDC

20L+ / 24.1:B

RJ45

2 connected

-X20 10

-3Q11 15 21 24 22 16

400VAC SUB-DISTRIBUTION
EXCITATION COOLING FANS
& AC FIELD FLASHING

+U7BJA11-W5021 NYCY 10x1.5 mm² 5

+U7BJA11-W5001 NYCY 7x1.5 mm² 2

+U7BJA13-W5004 NYCY 12x1.5 mm² 2

+U7BJA13-W5071 NYCY 2x1.5/1.5 mm² BU

+U7BJA13-W5004 NYCY 12x1.5 mm² 3

+U7BJA13-W5081 NYCY 2x1.5/1.5 mm² BU

+U7BJA13-W5002 NYCY 12x1.5 mm² 2

+U7BJA13-W5111 NYCY 2x1.5/1.5 mm² BU

-1K21 /149.4:C A1 A2

-1K22 /149.4:C A1 A2

-1K23 /149.4:C A1 A2

-1K24 /149.4:C A1 A2

-1K25 /149.4:C A1 A2

-1K26 /149.4:C A1 A2

-1K27 /149.4:C A1 A2

-1K28 /149.4:D A1 A2

BU 1.0

22.8:D / 44L- 24VDC

22.8:E / 32L+ 24VDC RD 1.0

-1K21 /23.1 11 14 12

-1K22 /23.2 11 14 12

-1K23 /23.3 11 14 12

-1K24 /23.4 11 14 12

-1K25 /23.5 11 14 12

-1K26 /23.6 11 14 12

-1K27 /23.7 11 14 12

-1K28 /23.8 11 14 12

RD 1.0

9	10	11	12	13	14	15	16
DI8	DI9	DI10	DI11	DI12	DI13	DI14	DI15
-A213	-A213	-A213	-A213	-A213	-A213	-A213	-A213
/114.3:D	/114.3:D	/114.3:D	/114.3:E	/114.3:E	/114.3:E	/114.3:E	/114.3:E
I5.0	I5.1	I5.2	I5.3	I5.4	I5.5	I5.6	I5.7
-1K21:14	-1K22:14	-1K23:14	-1K24:14	-1K25:14	-1K26:14	-1K27:14	-1K28:14
AC SURGE ARRESTER OK	230 VAC MARSHALLING KIOSK MCB ALARM GROUP - OK	GENERATOR COOLING WATER VALVE FEEDER INSERTED	GENERATOR COOLING WATER VALVE READY	TRANSFORMER COOLING WATER VALVE FEEDER INSERTED	TRANSFORMER COOLING WATER VALVE READY	EXCITATION COOLING FANS & AC FIELD FLASHING FEEDER INSERTED	EXCITATION COOLING FANS & AC FIELD FLASHING READY

PLC-RSC-24DC/21 14 11 12 /23.1:E

PLC-RSC-24DC/21 14 11 12 /23.2:E

PLC-RSC-24DC/21 14 11 12 /23.3:E

PLC-RSC-24DC/21 14 11 12 /23.4:E

PLC-RSC-24DC/21 14 11 12 /23.5:E

PLC-RSC-24DC/21 14 11 12 /23.6:E

PLC-RSC-24DC/21 14 11 12 /23.7:E

PLC-RSC-24DC/21 14 11 12 /23.8:E

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 5.0 - 5.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 23 PAGE 23 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

BUS DUCT FAN FEEDER INSERTED | BUS DUCT FAN REMOTE | BUS DUCT FAN AUTO | BUS DUCT FAN ON | BUS DUCT FAN READY 1 | BUS DUCT FAN READY 2 | GRAPHITE DUST SUCTION SYSTEM FEEDER INSERTED | GRAPHITE DUST SUCTION SYSTEM REMOTE

23.8:B / 44L+ 24VDC

-X44 B 8 A

44L+ 24VDC / 25.1:B

+U7CJA10-W5004 NYCY 10x1.5 mm² 1

+U7BJA13-W5121 NYCY 2x1.5/1.5 mm² BN

=U7BJA10GH100 +U7BJA13.DB002

20L+ / 23.7:B → -X20 2 → 20L+ / 24.7:B

3 24VDC

RJ45

2 connected

400VAC SUB-DISTRIBUTION BUS DUCT TUNNEL FAN CONTROL CUBICLE

-X20 11

=U7SAA10GH100 +U7SAA10

-X71 1

-K10 31 34 32

-K11 21 24 22

-K1 24 23

BDF / 24.6:B

-X71 2 | 3 | 4

BUSBAR DUCT FAN CONTROL CUBICLE

=U7BJA10GH100 +U7BJA13.DB002

15

-3Q12 21 24 22

16

400VAC SUB-DISTRIBUTION BUS DUCT TUNNEL FAN CONTROL CUBICLE

=U7SAA10GH100 +U7SAA10

BDF / 24.4:B

-Q2 14 13

-X71 5

BUSBAR DUCT FAN CONTROL CUBICLE

=U7BJA10GH100 +U7BJA13.FA002

20L+ / 24.1:B → -X20 3 → 20L+ / 26.3:B

3 24VDC

RJ45

2 connected

-X20 12

400VAC SUB-DISTRIBUTION GRAPHITE DUST SUCTION SYSTEM

-3S13 24 23 22

GDS / 25.2:B

+U7BJA13-W5002 NYCY 12x1.5 mm² 3

+U7CJA10-W5004 NYCY 10x1.5 mm² 2 3 4

+U7BJA13-W5121 NYCY 2x1.5/1.5 mm² BU

+U7CJA10-W5004 NYCY 10x1.5 mm² 5

+U7BJA13-W5002 NYCY 12x1.5 mm² 4

+U7BJA13-W5131 NYCY 7x1.5 mm² 2

-1K31 /149.4:D | -1K32 /149.4:D | -1K33 /149.4:D | -1K34 /149.4:D | -1K35 /149.4:D | -1K36 /149.4:D | -1K37 /149.4:E | -1K38 /149.4:E

A1 A2

BU 1.0

44L- 24VDC / 25.1:D

BU 1.0

22.2:D / 44L- 24VDC -X44 B 21 A 44L- 24VDC / 26.1:D

22.2:D / 32L+ 24VDC -X32 A 9 B 32L+ 24VDC / 26.1:D

RD 1.0

32L+ 24VDC / 25.1:E

-1K31 /24.1 | -1K32 /24.2 | -1K33 /24.3 | -1K34 /24.4 | -1K35 /24.5 | -1K36 /24.6 | -1K37 /24.7 | -1K38 /24.8

11 14 12

1 | 2 | 3 | 4 | 5 | 6 | 7 | 8

DI0	DI1	DI2	DI3	DI4	DI5	DI6	DI7
-A214	-A214	-A214	-A214	-A214	-A214	-A214	-A214
/115.3:C	/115.3:C	/115.3:C	/115.3:C	/115.3:C	/115.3:C	/115.3:D	/115.3:D
I6.0	I6.1	I6.2	I6.3	I6.4	I6.5	I6.6	I6.7
-1K31:14	-1K32:14	-1K33:14	-1K34:14	-1K35:14	-1K36:14	-1K37:14	-1K38:14
BUS DUCT FAN FEEDER INSERTED	BUS DUCT FAN REMOTE	BUS DUCT FAN AUTO	BUS DUCT FAN ON	BUS DUCT FAN READY 1	BUS DUCT FAN READY 2	GRAPHITE DUST SUCTION SYSTEM FEEDER INSERTED	GRAPHITE DUST SUCTION SYSTEM REMOTE

PLC-RSC-24DC/21 14 12 11 /24.1:E | /24.2:E | /24.3:E | /24.4:E | /24.5:E | /24.6:E | /24.7:E | /24.8:E

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 6.0 - 6.7	DRAWING No.: PERU-01-17-170-31-001 / REV No.: 00 / = U7CJA10GH100 / + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100 / SALES No.: / PAGE 24
Checked by	Antonio Aleksić	01.04.2026.					PAGE 24 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		

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BUSBAR DUCT VOLTAGE OK | GRAPHITE DUST SUCTION SYSTEM ON | GRAPHITE DUST SUCTION SYSTEM READY | GRAPHITE DUST SUCTION SYSTEM SPARE | GENERATOR HEATERS FEEDER INSERTED | GENERATOR HEATERS REMOTE | GENERATOR HEATERS AUTO | GENERATOR HEATERS ON

24.8:B / 44L+ 24VDC — -X44 B 9 A — 44L+ 24VDC / 26.1:B

+U7BJA13-W5131 NYCY 7x1.5 mm² 1

=U7SAA10GH100 +U7SAA10

BUSBAR DUCT FAN CONTROL CUBICLE

-X71 6

+U7CJA10-W5004 NYCY 10x1.5 mm² 8

=U7BJA10GH100 +U7BJA13.FA002

24.8:B / GDS

-3KM13 74 73

-3Q13 19 24 23

-3F13 11 14 12

400VAC SUB-DISTRIBUTION GRAPHITE DUST SUCTION SYSTEM

21 20

+U7BJA13-W5131 NYCY 7x1.5 mm² 4 3

=U7BJA10GH100 +U7BJA13.NB001

20L+ / 23.5:B — -X20 3 — 20L+ / 23.7:B

3 24VDC — RJ45 — 2 connected — -X20 7

400VAC SUB-DISTRIBUTION GENERATOR HEATER CONTROL CUBICLE

+U7BJA13-W5004 NYCY 12x1.5 mm² 4

-3S91 24 23 — -3S92 24 23 — -3KM9 74 73 — GH / 26.1:B

23 24 22

+U7BJA13-W5091 NYCY 7x1.5 mm² 2 3 4

-1K41 /149.4:E	-1K42 /149.4:E	-1K43 /149.4:E	-1K44 /149.4:E	-1K45 /149.4:E	-1K46 /149.4:E	-1K47 /149.4:F 24VDC	-1K48 /149.4:F 24VDC

A1 / A2 — BU 1.0

24.8:D / 44L- 24VDC

24.8:E / 32L+ 24VDC — RD 1.0

-1K41 /25.1	-1K42 /25.2	-1K43 /25.3	-1K44 /25.4	-1K45 /25.5	-1K46 /25.6	-1K47 /25.7	-1K48 /25.8

11 — 14 12 — RD 1.0

9	10	11	12	13	14	15	16
DI8	DI9	DI10	DI11	DI12	DI13	DI14	DI15
-A214	-A214	-A214	-A214	-A214	-A214	-A214	-A214
/115.3:D	/115.3:D	/115.3:D	/115.3:E	/115.3:E	/115.3:E	/115.3:E	/115.3:E
I7.0	I7.1	I7.2	I7.3	I7.4	I7.5	I7.6	I7.7
-1K41:14	-1K42:14	-1K43:14	-1K44:14	-1K45:14	-1K46:14	-1K47:14	-1K48:14
BUSBAR DUCT VOLTAGE OK	GRAPHITE DUST SUCTION SYSTEM ON	GRAPHITE DUST SUCTION SYSTEM READY	GRAPHITE DUST SUCTION SYSTEM SPARE	GENERATOR HEATERS FEEDER INSERTED	GENERATOR HEATERS REMOTE	GENERATOR HEATERS AUTO	GENERATOR HEATERS ON

PLC-RSC-24DC/21 — 14 12 — 11 /25.1:E / 25.2:E / 25.3:E / 25.4:E / 25.5:E / 25.6:E / 25.7:E / 25.8:E

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 7.0 - 7.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 25 PAGE 25 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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GENERATOR HEATER READY | GENERATOR HEATERS SPARE | DRAINAGE UNDER GENERATOR FEEDER INSERTED | SPARE | DRAINAGE UNDER GENERATOR AUTO | DRAINAGE UNDER GENERATOR ON | DRAINAGE UNDER GENERATOR READY | DRAINAGE UNDER GENERATOR SPARE

25.8:B / 44L+ 24VDC -X44 B 10 A
B 11 A
B 12 A 44L+ 24VDC / 27.1:B

+U7BJA13-W5091 NYCY 7x1.5 mm² 1
+U7BJA13-W5141 NYCY 7x1.5 mm² 1

=U7BJA10GH100
+U7BJA13.NB001
20
25.8:B / GH
21
-3Q9
24 22
11
-3F9
14 12
21
400VAC SUB-DISTRIBUTION GENERATOR HEATER CONTROL CUBICLE

=U7BJA10GH100
+U7BJA13.GA002
20L+ / 24.7:B
20L+ / 27.1:B
-X20 4
3 24VDC
RJ45
2 connected
-X20 13
400VAC SUB-DISTRIBUTION DRAINAGE UNDER GENERATOR

-3S142 24 23
-3KM14 74 73
-3Q14 19 24 23
-3F14 11 14 12
23 21 20

+U7BJA13-W5091 NYCY 7x1.5 mm² 5
+U7BJA13-W5002 NYCY 12x1.5 mm² 5
+U7BJA13-W5141 NYCY 7x1.5 mm² 2 3 4

-1K51 /149.4:F | -1K52 /149.4:F | -1K53 /149.4:F | -1K54 /150.4:B | -1K55 /150.4:B | -1K56 /150.4:B | -1K57 /150.4:B | -1K58 /150.4:B
A1 / A2

BU 1.0

BU 1.0
24.2:D / 44L- 24VDC -X44 B 22 A 44L- 24VDC / 28.1:D

24.2:D / 32L+ 24VDC -X32 A 10 B 32L+ 24VDC / 28.1:D

RD 1.0

-1K51 /26.1 | -1K52 /26.2 | -1K53 /26.3 | -1K54 /26.4 | -1K55 /26.5 | -1K56 /26.6 | -1K57 /26.7 | -1K58 /26.8
11 14 12

32L+ 24VDC / 27.1:E

	DI0	DI1	DI2	DI3	DI4	DI5	DI6	DI7
	-A215	-A215	-A215	-A215	-A215	-A215	-A215	-A215
	/116.3:C	/116.3:C	/116.3:C	/116.3:C	/116.3:C	/116.3:C	/116.3:D	/116.3:D
	I8.0	I8.1	I8.2	I8.3	I8.4	I8.5	I8.6	I8.7
	-1K51:14	-1K52:14	-1K53:14	-1K54:14	-1K55:14	-1K56:14	-1K57:14	-1K58:14
	GENERATOR HEATER READY	GENERATOR HEATERS SPARE	DRAINAGE UNDER GENERATOR FEEDER INSERTED	SPARE	DRAINAGE UNDER GENERATOR AUTO	DRAINAGE UNDER GENERATOR ON	DRAINAGE UNDER GENERATOR READY	DRAINAGE UNDER GENERATOR SPARE

PLC-RSC-24DC/21 14 12 11 /26.1:E | /26.2:E | /26.3:E | /26.4:E | /26.5:E | /26.6:E | /26.7:E | /26.8:E

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 8.0 - 8.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 26
Checked by	Antonio Aleksić	01.04.2026.							PAGE 26 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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DRAINAGE UNDER TURBINE A FEEDER INSERTED | SPARE | DRAINAGE UNDER TURBINE A AUTO | DRAINAGE UNDER TURBINE A ON | DRAINAGE UNDER TURBINE A READY | DRAINAGE UNDER TURBINE A SPARE | DRAINAGE UNDER TURBINE B FEEDER INSERTED | SPARE

26.8:B / 44L+ 24VDC

-X44 B 13 A

44L+ 24VDC / 28.1:B

+U7BJA13-W5151 NYCY 7x1.5 mm² 1

=U7BJA10GH100
+U7BJA13.HA002

20L+ / 26.3:B

-X20 5 3 24VDC

RJ45

2 connected

-X20 14

400VAC SUB-DISTRIBUTION
DRAINAGE UNDER TURBINE A

+U7BJA13-W5002 NYCY 12x1.5 mm² 6

-3S152 24 23

-3KM15 74 73

-3Q15 19 24 23

-3F15 11 14 12

23 21 20

+U7BJA13-W5151 NYCY 7x1.5 mm² 2 3 4

20L+ / 27.7:B

=U7BJA10GH100
+U7BJA13.JB002

20L+ / 27.1:B

-X20 6 3 24VDC

RJ45

2 connected

-X20 15

400VAC SUB-DISTRIBUTION
DRAINAGE UNDER TURBINE B

+U7BJA13-W5002 NYCY 12x1.5 mm² 7

-1K61 /150.4:B
-1K62 /150.4:C
-1K63 /150.4:C
-1K64 /150.4:C
-1K65 /150.4:C
-1K66 /150.4:C
-1K67 /150.4:C 24VDC
-1K68 /150.4:C 24VDC

A1 A2

BU 1.0

26.8:D / 44L- 24VDC

26.8:E / 32L+ 24VDC

RD 1.0

-1K61 /27.1
-1K62 /27.2
-1K63 /27.3
-1K64 /27.4
-1K65 /27.5
-1K66 /27.6
-1K67 /27.7
-1K68 /27.8

11 14 12

9 10 11 12 13 14 15 16

DI8	DI9	DI10	DI11	DI12	DI13	DI14	DI15
-A215	-A215	-A215	-A215	-A215	-A215	-A215	-A215
/116.3:D	/116.3:D	/116.3:D	/116.3:E	/116.3:E	/116.3:E	/116.3:E	/116.3:E
I9.0	I9.1	I9.2	I9.3	I9.4	I9.5	I9.6	I9.7
-1K61:14	-1K62:14	-1K63:14	-1K64:14	-1K65:14	-1K66:14	-1K67:14	-1K68:14
DRAINAGE UNDER TURBINE A FEEDER INSERTED	SPARE	DRAINAGE UNDER TURBINE A AUTO	DRAINAGE UNDER TURBINE A ON	DRAINAGE UNDER TURBINE A READY	DRAINAGE UNDER TURBINE A SPARE	DRAINAGE UNDER TURBINE B FEEDER INSERTED	SPARE

PLC-RSC-24DC/21 14 12 11 /27.1:E
PLC-RSC-24DC/21 14 12 11 /27.2:E
PLC-RSC-24DC/21 14 12 11 /27.3:E
PLC-RSC-24DC/21 14 12 11 /27.4:E
PLC-RSC-24DC/21 14 12 11 /27.5:E
PLC-RSC-24DC/21 14 12 11 /27.6:E
PLC-RSC-24DC/21 14 12 11 /27.7:E
PLC-RSC-24DC/21 14 12 11 /27.8:E

26 28

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 9.0 - 9.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 27
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 27 / 204

DRAINAGE UNDER TURBINE B AUTO | DRAINAGE UNDER TURBINE B ON | DRAINAGE UNDER TURBINE B READY | DRAINAGE UNDER TURBINE B SPARE | 220 VDC SYSTEM 1 MCB TRIP GROUP - OK | 220 VDC SYSTEM 2 MCB TRIP GROUP - OK | AC AUXILIARY SUPPLY CB READY | 24 VDC MCB TRIP GROUP - OK

27.8:B / 44L+ 24VDC — -X44 B14 A — B15 A — +U7BUA10-W5001 BN NYCY 2x1.5/1.5 mm² B16 A — +U7BUA10-W5002 BN — +U7BJA13-W5161 NYCY 7x1.5 mm² 1 — +U7BUA10-W5003 NYCY 7x1.5 mm² B17 A 1 — 44L+ 24VDC / 29.1:B

=U7BJA10GH100 +U7BJA13.JB002

19 — 24 — -3Q16 — 23 — 11 — -3F16 — 14 12

-3S162 24 23 — -3KM16 74 73

23 — 21 — 20

400VAC SUB-DISTRIBUTION DRAINAGE UNDER TURBINE B

+U7BJA13-W5161 NYCY 7x1.5 mm² 2 — 3 — 4

A1 A2

-1K71 /150.4:C — -1K72 /150.4:D — -1K73 /150.4:D — -1K74 /150.4:D — -1K75 /150.4:E 24VDC — -1K76 /150.4:E 24VDC — -1K77 /150.4:F 24VDC — -1K78 /150.4:F 24VDC

BU 1.0

=U7BJA10GH100 +U7BUA10 — -X11 1 — -X21 1 — -X11 2 — -X21 2

220VDC & 230VAC UPS SUB-DISTRIBUTION

+U7BUA10-W5001 NYCY 2x1.5/1.5 mm² BU — +U7BUA10-W5002 NYCY 2x1.5/1.5 mm² BU

=U7BJA10GH100 +U7BJA11 — -X31 3

400VAC SUB-DISTRIBUTION

+U7BJA11-W5031 NYCY 7x1.5 mm² 3

=U7BUA10GH100 +U7BUA10 — -X41 1 — -X41 4

220VDC & 230VAC UPS SUB-DISTRIBUTION

+U7BUA10-W5003 NYCY 7x1.5 mm² 4

44L- 24VDC / 29.1:D

26.2:D / 44L- 24VDC -X44 B23 A — B24 A

26.2:D / 32L+ 24VDC -X32 A B11 — 32L+ 24VDC / 30.1:D

RD 1.0

32L+ 24VDC / 29.1:E

-1K71 /28.1 — -1K72 /28.2 — -1K73 /28.3 — -1K74 /28.4 — -1K75 /28.5 — -1K76 /28.6 — -1K77 /28.7 — -1K78 /28.8

11 — 14 12

	DI0	DI1	DI2	DI3	DI4	DI5	DI6	DI7
1	-A216	-A216	-A216	-A216	-A216	-A216	-A216	-A216
2	/117.3:C	/117.3:C	/117.3:C	/117.3:C	/117.3:C	/117.3:C	/117.3:D	/117.3:D
3	I10.0	I10.1	I10.2	I10.3	I10.4	I10.5	I10.6	I10.7
4	-1K71:14	-1K72:14	-1K73:14	-1K74:14	-1K75:14	-1K76:14	-1K77:14	-1K78:14
5	DRAINAGE UNDER TURBINE B AUTO	DRAINAGE UNDER TURBINE B ON	DRAINAGE UNDER TURBINE B READY	DRAINAGE UNDER TURBINE B SPARE	220 VDC SYSTEM 1 MCB TRIP GROUP - OK	220 VDC SYSTEM 2 MCB TRIP GROUP - OK	AC AUXILIARY SUPPLY CB READY	24 VDC MCB TRIP GROUP - OK

PLC-RSC-24DC/21 14 12 — 11 /28.1:E — 11 /28.2:E — 11 /28.3:E — 11 /28.4:E — 11 /28.5:E — 24 22 — 21 /86.7:C — 11 /28.6:E — 24 22 — 21 /86.8:C — 11 /28.7:E — 11 /28.8:E

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Format ISO/DIN A3

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 10.0 - 10.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 28
Checked by	Antonio Aleksić	01.04.2026.							PAGE 28 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

DC CONVERTER 1 OK | DC CONVERTER 2 OK | DISCONNECTOR 1 OFF | DISCONNECTOR 1 ON | DISCONNECTOR 2 OFF | DISCONNECTOR 2 ON | GENERATOR CIRCUIT BREAKER OFF | GENERATOR CIRCUIT BREAKER ON

28.8:B / 44L+ 24VDC -X44 B 18 A

11.2:D / 45L+ 24VDC -X45 B 2 A

B 3 A

B 4 A

45L+ 24VDC / 30.1:B

+U7AEA10-W3056 NYCY 16x2.5 mm² 1 4

+U7AEA10-W3053 NYCY 16x2.5 mm² 11

=U7BUA10GH100
+U7BUA10

=U7AEA10GH100
+U7AEA10

-X10 3

-X11 3

-X09 8

-X41 2 3

220VDC & 230VAC UPS SUB-DISTRIBUTION

+U7BUA10-W5003 NYCY 7x1.5 mm² 2 3

MARSALLING KIOSK

-X10 18 17

-X11 18 17

-X09 29 30

+U7AEA10-W3056 NYCY 16x2.5 mm² 3 2 6 5

+U7AEA10-W3053 NYCY 16x2.5 mm² 12 13

+ A1

-1K81 /150.4:F

-1K82 /150.4:F

-1K83 /151.4:B 24VDC

-1K84 /151.4:B 24VDC

-1K85 /151.4:B 24VDC

-1K86 /151.4:B 24VDC

-1K87 /151.4:B 24VDC

-1K88 /151.4:B 24VDC

A2 -

BU 1.0

28.8:D / 44L- 24VDC

45L- 24VDC / 30.1:D

BU 1.0

11.2:D / 45L- 24VDC -X45 B 15 A 45L- 24VDC / 31.1:D

28.8:E / 32L+ 24VDC RD 1.0

-1K81 /29.1

-1K82 /29.2

-1K83 /29.3

-1K84 /29.4

-1K85 /29.5

-1K86 /29.6

-1K87 /29.7

-1K88 /29.8

11 14 12

RD 1.0

9 10 11 12 13 14 15 16

DI8	DI9	DI10	DI11	DI12	DI13	DI14	DI15
-A216	-A216	-A216	-A216	-A216	-A216	-A216	-A216
/117.3:D	/117.3:D	/117.3:D	/117.3:E	/117.3:E	/117.3:E	/117.3:E	/117.3:E
I11.0	I11.1	I11.2	I11.3	I11.4	I11.5	I11.6	I11.7
-1K81:14	-1K82:14	-1K83:14	-1K84:14	-1K85:14	-1K86:14	-1K87:14	-1K88:14
DC CONVERTER 1 OK	DC CONVERTER 2 OK	DISCONNECTOR 1 OFF	DISCONNECTOR 1 ON	DISCONNECTOR 2 OFF	DISCONNECTOR 2 ON	GENERATOR CIRCUIT BREAKER OFF	GENERATOR CIRCUIT BREAKER ON

PLC-RSC-24DC/21 14 11 /29.1:E 12

PLC-RSC-24DC/21 14 11 /29.2:E 12

PLC-RSC-24DC/21-21 14 11 /29.3:E 12 24 21 /83.2:C 22

PLC-RSC-24DC/21-21 14 11 /29.4:E 12 24 21 /73.5:C 22

PLC-RSC-24DC/21-21 14 11 /29.5:E 12 24 21 /83.4:C 22

PLC-RSC-24DC/21-21 14 11 /29.6:E 12 24 21 /73.6:C 22

PLC-RSC-24DC/21-21 14 11 /29.7:E 12 24 21 /73.2:C 22

PLC-RSC-24DC/21-21 14 11 /29.8:E 12 24 21 /73.2:C 22

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 11.0 - 11.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 29 PAGE 29 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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BRAKING DISCONNECTOR OFF | BRAKING DISCONNECTOR ON | BRAKING EARTH SWITCH OFF | BRAKING EARTH SWITCH ON | TRANSFORMER NEUTRAL POINT DISCONNECTOR OFF | TRANSFORMER NEUTRAL POINT DISCONNECTOR ON | GENERATOR CIRCUIT BREAKER SF6 ALARM | GENERATOR CIRCUIT BREAKER SF6 TRIP

29.8:B / 45L+ 24VDC -X45 B 5 A

+U7AEA10-W3056 NYCY 16x2.5 mm² 7 -X12 10

+U7AEA10-W3056 NYCY 16x2.5 mm² 10 -X13 10

+U7AEA10-W3056 NYCY 16x2.5 mm² 13 -X14 6

+U7AEA10-W3053 NYCY 16x2.5 mm² 1 -X08 15

B 6 A B 7 A B 8 A 45L+ 24VDC / 31.1:B

=U7AEA10GH100
+U7AEA10

MARSALLING KIOSK

-X12 32 | -X13 32 | -X14 16 | -X08 16

+U7AEA10-W3056 NYCY 16x2.5 mm² 9

+U7AEA10-W3056 NYCY 16x2.5 mm² 12

+U7AEA10-W3056 NYCY 16x2.5 mm² 14

+U7AEA10-W3053 NYCY 16x2.5 mm² 2

-X12 31 | -X13 31 | -X14 17 | -X08 17

-1K91 /151.4:C 24VDC	-1K92 /151.4:C 24VDC	-1K93 /151.4:C 24VDC	-1K94 /151.4:C 24VDC	-1K95 /151.4:C 24VDC	-1K96 /151.4:C 24VDC	-1K97 /151.4:C 24VDC	-1K98 /151.4:C 24VDC

29.8:D / 45L- 24VDC — BU 1.0

28.2:D / 32L+ 24VDC -X32 A B 12 32L+ 24VDC / 32.1:D — RD 1.0

32L+ 24VDC / 31.1:E

-1K91 /30.1	-1K92 /30.2	-1K93 /30.3	-1K94 /30.4	-1K95 /30.5	-1K96 /30.6	-1K97 /30.7	-1K98 /30.8

DI0	DI1	DI2	DI3	DI4	DI5	DI6	DI7
-A217	-A217	-A217	-A217	-A217	-A217	-A217	-A217
/118.3:C	/118.3:C	/118.3:C	/118.3:C	/118.3:C	/118.3:C	/118.3:D	/118.3:D
I12.0	I12.1	I12.2	I12.3	I12.4	I12.5	I12.6	I12.7
-1K91:14	-1K92:14	-1K93:14	-1K94:14	-1K95:14	-1K96:14	-1K97:14	-1K98:14
BRAKING DISCONNECTOR OFF	BRAKING DISCONNECTOR ON	BRAKING EARTH SWITCH OFF	BRAKING EARTH SWITCH ON	TRANSFORMER NEUTRAL POINT DISCONNECTOR OFF	TRANSFORMER NEUTRAL POINT DISCONNECTOR ON	GENERATOR CIRCUIT BREAKER SF6 ALARM	GENERATOR CIRCUIT BREAKER SF6 TRIP

PLC-RSC-24DC/21-21: 14–11 /30.1:E; 12; 24–21 /84.2:C; 22

PLC-RSC-24DC/21-21: 14–11 /30.2:E; 12; 24–21 /84.3:C; 22

PLC-RSC-24DC/21-21: 14–11 /30.3:E; 12; 24–21 /84.4:C; 22

PLC-RSC-24DC/21-21: 14–11 /30.4:E; 12; 24–21 /84.5:C; 22

PLC-RSC-24DC/21-21: 14–11 /30.5:E; 12; 24–21 /84.6:C; 22

PLC-RSC-24DC/21-21: 14–11 /30.6:E; 12; 24–21 /84.7:C; 22

PLC-RSC-24DC/21-21: 14–11 /30.7:E; 12; 24–21 /86.3:C; 22

PLC-RSC-24DC/21-21: 14–11 /30.8:E; 12; 24–21 /87.3:C; 22

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	DIGITAL INPUTS: 12.0 - 12.7			+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 30
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 30 / 204

GENERATOR CIRCUIT BREAKER MOTOR NOT READY 1

GENERATOR CIRCUIT BREAKER MCB ALARM GROUP NOT OK

GENERATOR CIRCUIT BREAKER REMOTE

GENERATOR CIRCUIT BREAKER ON/OFF RELEASE

GENERATOR CIRCUIT BREAKER NOT READY ON (SPRING NOT CHARGED)

DISCONNECTOR 1 REMOTE

DISCONNECTOR 1 MOTOR READY

DISCONNECTOR 1 DOOR OPEN

30.8:B / 45L+ 24VDC

B -X45 9 A

45L+ 24VDC / 32.1:B

+U7AEA10-W3058 NYCY 16x2.5 mm² 1

-X16 1

=U7AEA10GH100
+U7AEA10

MARSALLING KIOSK

-X08 22 | 23 | 25 | 18 | 19

-X16 8 | 10 | 12

+U7AEA10-W3053 NYCY 16x2.5 mm² 8 | 9 | 10 | 4 | 5

+U7AEA10-W3058 NYCY 16x2.5 mm² 2 | 3 | 4

	-1K101	-1K102	-1K103	-1K104	-1K105	-1K106	-1K107	-1K108
	/151.4:D	/151.4:D	/151.4:E	/151.4:E	/151.4:E	/151.4:E	/151.4:E	/151.4:E
	24VDC	24VDC	24VDC	24VDC	24VDC	24VDC	24VDC	24VDC
	A1	A1	A1	A1	A1	A1	A1	A1
	A2	A2	A2	A2	A2	A2	A2	A2

BU 1.0

45L- 24VDC / 32.1:D

BU 1.0

29.3:D / 45L- 24VDC -X45 B 16 A 45L- 24VDC / 33.1:D

30.8:E / 32L+ 24VDC

RD 1.0

	-1K101	-1K102	-1K103	-1K104	-1K105	-1K106	-1K107	-1K108
	/31.1	/31.2	/31.3	/31.4	/31.5	/31.6	/31.7	/31.8
	11	11	11	11	11	11	11	11
	14 12	14 12	14 12	14 12	14 12	14 12	14 12	14 12
	RD 1.0	RD 1.0	RD 1.0	RD 1.0	RD 1.0	RD 1.0	RD 1.0	RD 1.0
	9	10	11	12	13	14	15	16
	DI8	DI9	DI10	DI11	DI12	DI13	DI14	DI15
	-A217	-A217	-A217	-A217	-A217	-A217	-A217	-A217
	/118.3:D	/118.3:D	/118.3:D	/118.3:E	/118.3:E	/118.3:E	/118.3:E	/118.3:E
	I13.0	I13.1	I13.2	I13.3	I13.4	I13.5	I13.6	I13.7
	-1K101:14	-1K102:14	-1K103:14	-1K104:14	-1K105:14	-1K106:14	-1K107:14	-1K108:14
	GENERATOR CIRCUIT BREAKER MOTOR NOT READY 1	GENERATOR CIRCUIT BREAKER MCB ALARM GROUP NOT OK	GENERATOR CIRCUIT BREAKER REMOTE	GENERATOR CIRCUIT BREAKER ON/OFF RELEASE	GENERATOR CIRCUIT BREAKER NOT READY ON (SPRING NOT CHARGED)	DISCONNECTOR 1 REMOTE	DISCONNECTOR 1 MOTOR READY	DISCONNECTOR 1 DOOR OPEN
	PLC-RSC-24DC/21	PLC-RSC-24DC/21	PLC-RSC-24DC/21	PLC-RSC-24DC/21	PLC-RSC-24DC/21	PLC-RSC-24DC/21	PLC-RSC-24DC/21	PLC-RSC-24DC/21
	14 11 /31.1:E 12	14 11 /31.2:E 12	14 11 /31.3:E 12	14 11 /31.4:E 12	14 11 /31.5:E 12	14 11 /31.6:E 12	14 11 /31.7:E 12	14 11 /31.8:E 12

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 13.0 - 13.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 31
Checked by	Antonio Aleksić	01.04.2026.							PAGE 31 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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DISCONNECTOR 1 HEATER MCB OK | DISCONNECTOR 2 REMOTE | DISCONNECTOR 2 MOTOR READY | DISCONNECTOR 2 DOOR OPEN | DISCONNECTOR 2 HEATER MCB OK | BRAKING EARTHING SWITCH REMOTE | BRAKING EARTHING SWITCH MOTOR READY | BRAKING EARTHING SWITCH DOOR OPEN

31.8:B / 45L+ 24VDC → 45L+ 24VDC / 33.1:B

=U7AEA10GH100
+U7AEA10

-X16 13 15 17 19 20 29 31 33

MARSALLING KIOSK
+U7AEA10-W3058
NYCY 16x2.5 mm²
5 6 7 8 9 10 11 12

-1K111	-1K112	-1K113	-1K114	-1K115	-1K116	-1K117	-1K118
/151.4:E	/151.4:F	/151.4:F	/151.4:F	/151.4:F	/151.4:F	/151.4:F	/152.4:B
24VDC	24VDC	24VDC	24VDC	24VDC	24VDC	24VDC	24VDC
A1	A1	A1	A1	A1	A1	A1	A1
A2	A2	A2	A2	A2	A2	A2	A2

BU 1.0

31.8:D / 45L- 24VDC

30.2:D / 32L+ 24VDC -X32 A 13 B 32L+ 24VDC / 34.1:D

RD 1.0

32L+ 24VDC / 33.1:E

-1K111 /32.1	-1K112 /32.2	-1K113 /32.3	-1K114 /32.4	-1K115 /32.5	-1K116 /32.6	-1K117 /32.7	-1K118 /32.8
11 14 12	11 14 12	11 14 12	11 14 12	11 14 12	11 14 12	11 14 12	11 14 12
1	2	3	4	5	6	7	8
DI0	DI1	DI2	DI3	DI4	DI5	DI6	DI7
-A218	-A218	-A218	-A218	-A218	-A218	-A218	-A218
/119.3:C	/119.3:C	/119.3:C	/119.3:C	/119.3:C	/119.3:C	/119.3:D	/119.3:D
I14.0	I14.1	I14.2	I14.3	I14.4	I14.5	I14.6	I14.7
-1K111:14	-1K112:14	-1K113:14	-1K114:14	-1K115:14	-1K116:14	-1K117:14	-1K118:14
DISCONNECTOR 1 HEATER MCB OK	DISCONNECTOR 2 REMOTE	DISCONNECTOR 2 MOTOR READY	DISCONNECTOR 2 DOOR OPEN	DISCONNECTOR 2 HEATER MCB OK	BRAKING EARTHING SWITCH REMOTE	BRAKING EARTHING SWITCH MOTOR READY	BRAKING EARTHING SWITCH DOOR OPEN

PLC-RSC-24DC/21
14 — 11 /32.1:E / 12
PLC-RSC-24DC/21
14 — 11 /32.2:E / 12
PLC-RSC-24DC/21
14 — 11 /32.3:E / 12
PLC-RSC-24DC/21
14 — 11 /32.4:E / 12
PLC-RSC-24DC/21
14 — 11 /32.5:E / 12
PLC-RSC-24DC/21
14 — 11 /32.6:E / 12
PLC-RSC-24DC/21
14 — 11 /32.7:E / 12
PLC-RSC-24DC/21
14 — 11 /32.8:E / 12

Format ISO/DIN A3

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 14.0 - 14.7	DRAWING No.: PERU-01-17-170-31-001 / REV No.: 00 / = U7CJA10GH100 / + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100 / SALES No.: / PAGE 32
Checked by	Antonio Aleksić	01.04.2026.					PAGE 32 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		

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BRAKING DISCONNECTOR EARTHING SWITCH HEATER READY

VT MCB OK - MEASUREMENT

VT MCB FOR CONTROL SYSTEM (SYNCH) OK

VT MCB OK - 110kV MEASUREMENT DEVICES

GENERATOR CIRCUIT BREAKER MOTOR READY

DISCONNECTOR MOTORS READY

MARSALLING KIOSK MCB ALARM GROUP - OK

MARSALLING KIOSK LOCAL

32.8:B / 45L+ 24VDC

-X45 B 10 A

45L+ 24VDC / 34.1:B

=U7AEA10GH100
+U7AEA10

-X15 1

=U7AEA10GH100
+U7AEA10

-X16 34

-X15 11

12

-X15 13

14

15

17

19

MARSALLING KIOSK

MARSALLING KIOSK

+U7AEA10-W3058
NYCY 16x2.5 mm²
13

+U7AEA10-W3057
NYCY 16x2.5 mm²
2

3

+U7AEA10-W3057
NYCY 16x2.5 mm²
4

5

6

8

10

-1K121 /152.4:B 24VDC

-1K122 /152.4:B 24VDC

-1K123 /152.4:C 24VDC

-1K124 /152.4:D 24VDC

-1K125 /152.4:D 24VDC

-1K126 /152.4:D 24VDC

-1K127 /152.4:D 24VDC

-1K128 /152.4:D 24VDC

A1 A2

BU 1.0

45L- 24VDC / 34.1:D

BU 1.0

31.2:D / 45L- 24VDC -X45 B 17 A 45L- 24VDC / 73.1:F

32L+ 24VDC 32.8:E /

RD 1.0

-1K121 /33.1

-1K122 /33.2

-1K123 /33.3

-1K124 /33.4

-1K125 /33.5

-1K126 /33.6

-1K127 /33.7

-1K128 /33.8

11 14 12

9 10 11 12 13 14 15 16

DI8	DI9	DI10	DI11	DI12	DI13	DI14	DI15
-A218	-A218	-A218	-A218	-A218	-A218	-A218	-A218
/119.3:D	/119.3:D	/119.3:D	/119.3:E	/119.3:E	/119.3:E	/119.3:E	/119.3:E
I15.0	I15.1	I15.2	I15.3	I15.4	I15.5	I15.6	I15.7
-1K121:14	-1K122:14	-1K123:14	-1K124:14	-1K125:14	-1K126:14	-1K127:14	-1K128:14
BRAKING DISCONNECTOR EARTHING SWITCH HEATER READY	VT MCB OK - MEASUREMENT	VT MCB FOR CONTROL SYSTEM (SYNCH) OK	VT MCB OK - 110kV MEASUREMENT DEVICES	GENERATOR CIRCUIT BREAKER MOTOR READY	DISCONNECTOR MOTORS READY	MARSALLING KIOSK MCB ALARM GROUP - OK	MARSALLING KIOSK LOCAL

PLC-RSC-24DC/21

14 — 11 /33.1:E
12

14 — 11 /33.2:E
12

PLC-RSC-24DC/21-21

14 — 11 /33.3:E
12
24 — 21 /93.6:B
22

14 — 11 /33.4:E
12

14 — 11 /33.5:E
12

14 — 11 /33.6:E
12

14 — 11 /33.7:E
12

14 — 11 /33.8:E
12

32

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 15.0 - 15.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 33
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 33 / 204

MARSALLING KIOSK REMOTE | SIGNALISATION MCB OK | SYNHRONSIATION VOLTAGE MCB of VTs ON BUSBAR SYTEM OK | GRID VT MCB OK - SYNCHRONIZATION | SYNHRONSIATION VOLTAGE MCB of VTs ON BUSBAR SYTEM OK | MARSHALLING KIOSK MOTORS POWER SUPPLY ON | SPARE | SPARE

33.8:B / 45L+ 24VDC → -X45 B 11 A → 45L+ 24VDC / 73.1:B

=U7AEA10GH100
+U7AEA10
MARSALLING KIOSK

=U7BJA10GH100
+U7BJA11
400VAC SUB-DISTRIBUTION

-X15 20 | 16 | -X20 9 | -X15 18 | -X20 10 | -X12 3

+U7AEA10-W3057 NYCY 16x2.5 mm² (11, 7, 12, 9, 13)
+U7BJA11-W5001 NYCY 7x1.5 mm² (3)

-1K131 /152.4:D 24VDC	-1K132 /152.4:D 24VDC	-1K133 /152.4:E 24VDC	-1K134 /152.4:E 24VDC	-1K135 /153.4:B 24VDC	-1K136 /153.4:B 24VDC	-1K137 /153.4:B 24VDC	-1K138 /153.4:B 24VDC

33.8:D / 45L- 24VDC (BU 1.0)

32.1:D / 32L+ 24VDC -X32 A B 14 → 32L+ 24VDC / 36.1:D

RD 1.0 → 32L+ 24VDC / 35.1:E

-1K131 /34.1	-1K132 /34.2	-1K133 /34.3	-1K134 /34.4	-1K135 /34.5	-1K136 /34.6	-1K137 /34.7	-1K138 /34.8

DI0	DI1	DI2	DI3	DI4	DI5	DI6	DI7
-A219	-A219	-A219	-A219	-A219	-A219	-A219	-A219
/120.3:C	/120.3:C	/120.3:C	/120.3:C	/120.3:C	/120.3:C	/120.3:D	/120.3:D
I16.0	I16.1	I16.2	I16.3	I16.4	I16.5	I16.6	I16.7
-1K131:14	-1K132:14	-1K133:14	-1K134:14	-1K135:14	-1K136:14	-1K137:14	-1K138:14
MARSALLING KIOSK REMOTE	SIGNALISATION MCB OK	SYNHRONSIATION VOLTAGE MCB of VTs ON BUSBAR SYTEM OK	GRID VT MCB OK - SYNCHRONIZATION	SYNHRONSIATION VOLTAGE MCB of VTs ON BUSBAR SYTEM OK	MARSHALLING KIOSK MOTORS POWER SUPPLY ON	SPARE	SPARE

PLC-RSC-24DC/21 14 — 11 /34.1:E; 12
PLC-RSC-24DC/21 14 — 11 /34.2:E; 12
PLC-RSC-24DC/21 14 — 11 /34.3:E; 12
PLC-RSC-24DC/21-21 14 — 11 /34.4:E; 12; 24 — 21 /93.5:B; 22
PLC-RSC-24DC/21 14 — 11 /34.5:E; 12
PLC-RSC-24DC/21 14 — 11 /34.6:E; 12
PLC-RSC-24DC/21 14 — 11 /34.7:E; 12
PLC-RSC-24DC/21 14 — 11 /34.8:E; 12

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 16.0 - 16.7	DRAWING No.: PERU-01-17-170-31-001 / REV No.: 00 / = U7CJA10GH100 / + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100 / SALES No.: / PAGE 34 / PAGE 34 / 204
Checked by	Antonio Aleksić	01.04.2026.					
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		

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GENERATOR PROTECTION 1 RELAY TRIP | GENERATOR PROTECTION 1 INTERNAL FAULT | SPARE | GENERATOR PROTECTION 2 RELAY TRIP OK | GENERATOR PROTECTION 2 INTERNAL FAULT | SPARE | EXCITATION LOCAL | EXCITATION ON RELEASE

11.3:D / 46L+ 24VDC -X46 B 2 A
+U7CHA10-W3015 NYCY 4x2.5 mm² BN
=U7CHA10GH100
+U7CHA10
-X171 1A
-X171 1B 9B
ELECTRICAL PROTECTION A
+U7CHA10-W3015 NYCY 4x2.5 mm² BK GY

+U7CHB10-W3001 NYCY 4x2.5 mm² BN
=U7CHB10GH100
+U7CHB10
-X271 1A
-X271 1B 9B
ELECTRICAL PROTECTION B
+U7CHB10-W3001 NYCY 4x2.5 mm² BK GY

46L+ 24VDC / 37.4:B

11.4:D / 47L+ 24VDC -X47 B 2 A
+U7CJA10-W5201 NYCY 10x1.5 mm² 1
=U7MKC10GH100
+U7MKC11
-X050 1
-X050 3 4
EXCITATION
+U7CJA10-W5201 NYCY 10x1.5 mm² 2 3

47L+ 24VDC / 36.1:B

-1K141 /153.4:C 24VDC
-1K142 /153.4:C 24VDC
-1K143 /153.4:D 24VDC
-1K144 /153.4:E 24VDC
-1K145 /153.4:E 24VDC
-1K146 /153.4:F 24VDC
-1K147 /154.4:B 24VDC
-1K148 /154.4:B 24VDC
BU 1.0

11.3:D / 46L- 24VDC -X46 B 10 A
46L- 24VDC / 37.4:D

11.4:D / 47L- 24VDC -X47 B 5 A B 6 A
47L- 24VDC / 36.1:D

34.8:E / 32L+ 24VDC RD 1.0

-1K141 /35.1
-1K142 /35.2
-1K143 /35.3
-1K144 /35.4
-1K145 /35.5
-1K146 /35.6
-1K147 /35.7
-1K148 /35.8
11 14 12

9	10	11	12	13	14	15	16
DI8	DI9	DI10	DI11	DI12	DI13	DI14	DI15
-A219	-A219	-A219	-A219	-A219	-A219	-A219	-A219
/120.3:D	/120.3:D	/120.3:D	/120.3:E	/120.3:E	/120.3:E	/120.3:E	/120.3:E
I17.0	I17.1	I17.2	I17.3	I17.4	I17.5	I17.6	I17.7
-1K141:14	-1K142:14	-1K143:14	-1K144:14	-1K145:14	-1K146:14	-1K147:14	-1K148:14
GENERATOR PROTECTION 1 RELAY TRIP	GENERATOR PROTECTION 1 INTERNAL FAULT	SPARE	GENERATOR PROTECTION 2 RELAY TRIP OK	GENERATOR PROTECTION 2 INTERNAL FAULT	SPARE	EXCITATION LOCAL	EXCITATION ON RELEASE

PLC-RSC-24DC/21-21: 14 11 /35.1:E; 12; 24 21 /87.4:C; 22
PLC-RSC-24DC/21-21: 14 11 /35.2:E; 12; 24 21 /87.7:C; 22
PLC-RSC-24DC/21: 14 11 /35.3:E; 12
PLC-RSC-24DC/21-21: 14 11 /35.4:E; 12; 24 21 /87.4:C; 22
PLC-RSC-24DC/21-21: 14 11 /35.5:E; 12; 24 21 /87.7:C; 22
PLC-RSC-24DC/21: 14 11 /35.6:E; 12
PLC-RSC-24DC/21-21: 14 11 /35.7:E; 12; 24 21; 22
PLC-RSC-24DC/21-21: 14 11 /35.8:E; 12; 24 21; 22

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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 17.0 - 17.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.							+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 35
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 35 / 204

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EXCITATION ALARM OK | EXCITATION ON | EXCITATION OFF | EXCITATION EXTERNAL TRIP | EXCITATION THYRICON TRIP | EXCITATION FCR MODE SELECTED | EXCITATION AVR MODE SELECTED | EXCITATION VAR MODE SELECTED

35.8:B / 47L+ 24VDC → -X47 B 3 A

=U7MKC10GH100
+U7MKC11

-X050 5 6 7 9 10 11 12 13

EXCITATION

+U7CJA10-W5201
NYCY 10x1.5 mm²

+U7CJA10-W5202
NYCY 10x1.5 mm²

	-1K151	-1K152	-1K153	-1K154	-1K155	-1K156	-1K157	-1K158
	/154.4:B 24VDC	/154.4:B 24VDC	/154.4:B 24VDC	/154.4:C 24VDC	/154.4:D 24VDC	/154.4:E 24VDC	/154.4:E 24VDC	/154.4:E 24VDC

35.8:D / 47L- 24VDC → / 37.1:D

34.2:D / 32L+ 24VDC -X32 A B 15 32L+ 24VDC / 38.1:D → 32L+ 24VDC / 37.1:E

	-1K151	-1K152	-1K153	-1K154	-1K155	-1K156	-1K157	-1K158
	/36.1	/36.2	/36.3	/36.4	/36.5	/36.6	/36.7	/36.8
	1	2	3	4	5	6	7	8
	DI0	DI1	DI2	DI3	DI4	DI5	DI6	DI7
	-A220	-A220	-A220	-A220	-A220	-A220	-A220	-A220
	/121.3:C	/121.3:C	/121.3:C	/121.3:C	/121.3:C	/121.3:C	/121.3:D	/121.3:D
	I18.0	I18.1	I18.2	I18.3	I18.4	I18.5	I18.6	I18.7
	-1K151:14	-1K152:14	-1K153:14	-1K154:14	-1K155:14	-1K156:14	-1K157:14	-1K158:14
	EXCITATION ALARM OK	EXCITATION ON	EXCITATION OFF	EXCITATION EXTERNAL TRIP	EXCITATION THYRICON TRIP	EXCITATION FCR MODE SELECTED	EXCITATION AVR MODE SELECTED	EXCITATION VAR MODE SELECTED
	PLC-RSC-24DC/21-21	PLC-RSC-24DC/21-21	PLC-RSC-24DC/21-21	PLC-RSC-24DC/21	PLC-RSC-24DC/21-21	PLC-RSC-24DC/21	PLC-RSC-24DC/21	PLC-RSC-24DC/21
	14 — 11 /36.1:E	14 — 11 /36.2:E	14 — 11 /36.3:E	14 — 11 /36.4:E	14 — 11 /36.5:E	14 — 11 /36.6:E	14 — 11 /36.7:E	14 — 11 /36.8:E
	12	12	12	12	12	12	12	12
	24 — 21 /86.2:C	24 — 21 /80.3:C	24 — 21 /80.4:C		24 — 21 /87.2:C			
	22	22	22		22			

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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 18.0 - 18.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 36
Checked by	Antonio Aleksić	01.04.2026.							PAGE 36 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

EXCITATION PF MODE SELECTED | EXCITATION ELECTRICAL BRAKING READY | EXCITATION ELECTRICAL BRAKING ON | SPARE | GENERATOR POWER LOWER THEN MINIMAL | SPARE (TURBINE SPEED < ELECTRICAL BRAKING STOP LIMIT) | TURBINE SPEED > EXCITATION START LIMIT | SPARE

35.6:B / 46L+ 24VDC — -X46 — / 38.1:B 46L+ 24VDC

+U7CBC10-W5041 NYCY 10x1.5 mm²

=U7MKC10GH100
+U7MKC11

=U7CBC10GH100
+U7CBC10

-X050 14 17 18 — -X70 4 — 11 5

EXCITATION

TURBINE GOVERNOR

+U7CJA10-W5202 NYCY 10x1.5 mm² — 2 3 4

+U7CBC10-W5041 NYCY 10x1.5 mm² — 7 2

-1K161 /154.4:E 24VDC	-1K162 /154.4:E 24VDC	-1K163 /154.4:E 24VDC	-1K164 /154.4:F 24VDC	-1K165 /154.4:F 24VDC	-1K166 /154.4:F 24VDC	-1K167 /154.4:F 24VDC	-1K168 /154.4:F 24VDC

36.8:D / 47L- 24VDC — BU 1.0 — 46L- 24VDC / 38.1:D

35.2:D / 46L- 24VDC -X46 11 12 13

36.8:E / 32L+ 24VDC — RD 1.0

-1K161 /37.1	-1K162 /37.2	-1K163 /37.3	-1K164 /37.4	-1K165 /37.5	-1K166 /37.6	-1K167 /37.7	-1K168 /37.8
9	10	11	12	13	14	15	16
DI8	DI9	DI10	DI11	DI12	DI13	DI14	DI15
-A220	-A220	-A220	-A220	-A220	-A220	-A220	-A220
/121.3:D	/121.3:D	/121.3:D	/121.3:E	/121.3:E	/121.3:E	/121.3:E	/121.3:E
I19.0	I19.1	I19.2	I19.3	I19.4	I19.5	I19.6	I19.7
-1K161:14	-1K162:14	-1K163:14	-1K164:14	-1K165:14	-1K166:14	-1K167:14	-1K168:14
EXCITATION PF MODE SELECTED	EXCITATION ELECTRICAL BRAKING READY	EXCITATION ELECTRICAL BRAKING ON	SPARE	GENERATOR POWER LOWER THEN MINIMAL	SPARE (TURBINE SPEED < ELECTRICAL BRAKING STOP LIMIT)	TURBINE SPEED > EXCITATION START LIMIT	SPARE
PLC-RSC-24DC/21 14 12 11 /37.1:E	PLC-RSC-24DC/21 14 12 11 /37.2:E	PLC-RSC-24DC/21 14 12 11 /37.3:E	PLC-RSC-24DC/21 14 12 11 /37.4:E	PLC-RSC-24DC/21 14 12 11 /37.5:E	PLC-RSC-24DC/21 14 12 11 /37.6:E	PLC-RSC-24DC/21 14 12 11 /37.7:E	PLC-RSC-24DC/21 14 12 11 /37.8:E

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 19.0 - 19.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 37 PAGE 37 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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TURBINE SPEED > SYNCHRONIZATION START LIMIT

TURBINE SPEED > OVERSPEED LIMIT

TURBINE GOVERNOR SPARE

TURBINE GOVERNOR ALARM

TURBINE GOVERNOR TRIP

TURBINE GOVERNOR EMERGENCY SHUTDOWN OK

VIBRATION MONITORING TRIP

MONITORING SYSTEM ALARM 2

37.8:B / 46L+ 24VDC

-X46 B 5 A, B 6 A, B 7 A, B 8 A

11.7:D / 50L+ 24VDC -X50 B 2 A 50L+ 24VDC / 39.1:B

+U7CJA10-W5001 NYCY 10x1.5 mm² 1

=U7CBC10GH100
+U7CBC10

=U7CMB10GH100
+U7CMB10

-X13 1

-K30.1 /76.a.4 NO 13, 14 NO

-X70 6, 7, 8, 9

-X13 2, 4

TURBINE GOVERNOR

MACHINE CONDITION MONITORING

+U7CBC10-W5041 NYCY 10x1.5 mm² 3, 4, 5, 6

+U7CJA10-W5001 NYCY 10x1.5 mm² 2, 3

-1K171 /154.4:F 24VDC
-1K172 /155.4:B 24VDC
-1K173 /155.4:B 24VDC
-1K174 /155.4:C 24VDC
-1K175 /155.4:C 24VDC
-1K176 /155.4:D 24VDC
-1K177 /155.4:D 24VDC
-1K178 /155.4:E 24VDC

A1, A2, BU 1.0

37.8:D / 46L- 24VDC

50L- 24VDC / 39.1:D

BU 1.0

11.7:D / 50L- 24VDC -X50 B 5 A, B 6 A

36.2:D / 32L+ 24VDC -X32 A 16 B 32L+ 24VDC / 40.1:D

RD 1.0

32L+ 24VDC / 39.1:E

-1K171 /38.1
-1K172 /38.2
-1K173 /38.3
-1K174 /38.4
-1K175 /38.5
-1K176 /38.6
-1K177 /38.7
-1K178 /38.8

11, 12, 14

	DI0	DI1	DI2	DI3	DI4	DI5	DI6	DI7
	-A221	-A221	-A221	-A221	-A221	-A221	-A221	-A221
	/122.3:C	/122.3:C	/122.3:C	/122.3:C	/122.3:C	/122.3:C	/122.3:D	/122.3:D
	I20.0	I20.1	I20.2	I20.3	I20.4	I20.5	I20.6	I20.7
	-1K171:14	-1K172:14	-1K173:14	-1K174:14	-1K175:14	-1K176:14	-1K177:14	-1K178:14
	TURBINE SPEED > SYNCHRONIZATION START LIMIT	TURBINE SPEED > OVERSPEED LIMIT	TURBINE GOVERNOR SPARE	TURBINE GOVERNOR ALARM	TURBINE GOVERNOR TRIP	TURBINE GOVERNOR EMERGENCY SHUTDOWN OK	VIBRATION MONITORING TRIP	MONITORING SYSTEM ALARM 2

PLC-RSC-24DC/21 14, 12 — 11 /38.1:E
PLC-RSC-24DC/21 14, 12 — 11 /38.2:E
PLC-RSC-24DC/21 14, 12 — 11 /38.3:E
PLC-RSC-24DC/21-21 14, 12 — 11 /38.4:E; 24, 22 — 21 /86.2:C
PLC-RSC-24DC/21-21 14, 12 — 11 /38.5:E; 24, 22 — 21 /87.1:C
PLC-RSC-24DC/21 14, 12 — 11 /38.6:E
PLC-RSC-24DC/21-21 14, 12 — 11 /38.7:E; 24, 22 — 21 /95.5:B
PLC-RSC-24DC/21 14, 12 — 11 /38.8:E

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 20.0 - 20.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 38
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 38 / 204

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MONITORING SYSTEM ALARM 3 | MONITORING SYSTEM ALARM 4 | MONITORING SYSTEM ALARM 5 | MONITORING SYSTEM ALARM 6 | MONITORING SYSTEM ALARM 7 | MONITORING SYSTEM ALARM 8 | SHAFT CURRENT ALARM | SHAFT CURRENT TRIP

38.7:B / 50L+ 24VDC

-X50 B 3 A

+U7CJA10-W5001 NYCY 10x1.5 mm² 6

+U7CJA10-W5002 NYCY 10x1.5 mm² 1

=U7CMB10GH100
+U7CMB10

-X13 9

-X16 1 3 5

-X13 6 8 10 12 14 16

-X16 2 4

MACHINE CONDITION MONITORING

+U7CJA10-W5001 NYCY 10x1.5 mm² 4 5 7 8 9 10

+U7CJA10-W5002 NYCY 10x1.5 mm² 2 3

-1K181 /155.4:E 24VDC	-1K182 /155.4:F 24VDC	-1K183 /155.4:F 24VDC	-1K184 /155.4:F 24VDC	-1K185 /155.4:F 24VDC	-1K186 /155.4:F 24VDC	-1K187 /156.4:B 24VDC	-1K188 /156.4:B 24VDC

38.8:D / 50L- 24VDC / 40.1:D

38.8:E / 32L+ 24VDC

-1K181 /39.1	-1K182 /39.2	-1K183 /39.3	-1K184 /39.4	-1K185 /39.5	-1K186 /39.6	-1K187 /39.7	-1K188 /39.8
DI8	DI9	DI10	DI11	DI12	DI13	DI14	DI15
-A221	-A221	-A221	-A221	-A221	-A221	-A221	-A221
/122.3:D	/122.3:D	/122.3:D	/122.3:E	/122.3:E	/122.3:E	/122.3:E	/122.3:E
I21.0	I21.1	I21.2	I21.3	I21.4	I21.5	I21.6	I21.7
-1K181:14	-1K182:14	-1K183:14	-1K184:14	-1K185:14	-1K186:14	-1K187:14	-1K188:14
MONITORING SYSTEM ALARM 3	MONITORING SYSTEM ALARM 4	MONITORING SYSTEM ALARM 5	MONITORING SYSTEM ALARM 6	MONITORING SYSTEM ALARM 7	MONITORING SYSTEM ALARM 8	SHAFT CURRENT ALARM	SHAFT CURRENT TRIP
PLC-RSC-24DC/21	PLC-RSC-24DC/21	PLC-RSC-24DC/21	PLC-RSC-24DC/21	PLC-RSC-24DC/21	PLC-RSC-24DC/21	PLC-RSC-24DC/21	PLC-RSC-24DC/21
14 11 /39.1:E	14 11 /39.2:E	14 11 /39.3:E	14 11 /39.4:E	14 11 /39.5:E	14 11 /39.6:E	14 11 /39.7:E	14 11 /39.8:E
12	12	12	12	12	12	12	12

Format ISO/DIN A3

38 40

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 21.0 - 21.7	DRAWING No.: PERU-01-17-170-31-001 / REV No.: 00 / = U7CJA10GH100 / + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100 / SALES No.: / PAGE 39
Checked by	Antonio Aleksić	01.04.2026.					PAGE 39 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		

CONFIDENTIAL, ALL RIGHTS RESERVED DIN 34

SHAFT CURRENT SYSTEM OK | GENERATOR COOLING WATER VALVE OPEN | GENERATOR COOLING WATER VALVE CLOSED | TURBINE A LEAKAGE TANK LEVEL HIGH | TURBINE B LEAKAGE TANK LEVEL HIGH | GENERAL LOCKOUT LOSS OF SF6 | GENERATOR COOLING WATER VALVE OVERLOAD/BIMETAL WARNING | GENERATOR FIRE EXTINGUISHING SYSTEM TRIP

11.5:D / 48L+ 24VDC -X48 B 2 A

+U7CJA10-W5101 NYCY 12x1.5 mm² 1

+U7CJA10-W5102 LiYCY 4x1.5 mm² BN

+U7CJA10-W5103 LiYCY 4x1.5 mm² BN

B 3 A -X48 B 4 A

+U7CHA10-W3018 NYCY 7x2.5 mm² 2

=U7CMB10GH100 +U7CMB10

=7NU01 +SC44

-X3 1

=? +?

? ? 13 14 21 22

=? +?

? ? 13 14 21 22

=U7AEA10GH100 +U7AEA10

=7NU01 +SC44

=U7CHA10GH100 +U7CHA10

-X171 13A

-K327 21 24 22

-X171 14A

-X16 6

MACHINE CONDITION MONITORING

-X3 6

GENERATOR COOLING

-X3 5

-X08 20

MARSALLING KIOSK

-X3 8

GENERATOR COOLING

ELECTRICAL PROTECTION A

+U7CJA10-W5002 NYCY 10x1.5 mm² 4

+U7CJA10-W5101 NYCY 12x1.5 mm² 2

+U7CJA10-W5101 NYCY 12x1.5 mm² 3

+U7CJA10-W5102 LiYCY 4x1.5 mm² WH

+U7CJA10-W5103 LiYCY 4x1.5 mm² WH

+U7AEA10-W3053 NYCY 16x2.5 mm² 6

+U7CJA10-W5101 NYCY 12x1.5 mm² 5

+U7CHA10-W3018 NYCY 7x2.5 mm² 5

-1K191 /156.4:B 24VDC

-1K192 /156.4:C 24VDC

-1K193 /156.4:C 24VDC

-1K194 /156.4:D 24VDC

-1K195 /156.4:D 24VDC

-1K196 /156.4:D 24VDC

-1K197 /156.4:E 24VDC

-1K198 /156.4:E 24VDC

39.8:D / 50L- 24VDC

BU 1.0

48L1- 24VDC / 41.1:D

11.5:D / 48L- 24VDC -X48 B 6 A 48L- 24VDC / 45.1:D

38.2:D / 32L+ 24VDC -X32 A B 17 32L+ 24VDC / 42.1:D

RD 1.0

32L+ 24VDC / 41.1:E

-1K191 /40.1

-1K192 /40.2

-1K193 /40.3

-1K194 /40.4

-1K195 /40.5

-1K196 /40.6

-1K197 /40.7

-1K198 /40.8

11 12 14

DI0	DI1	DI2	DI3	DI4	DI5	DI6	DI7
-A222	-A222	-A222	-A222	-A222	-A222	-A222	-A222
/123.3:C	/123.3:C	/123.3:C	/123.3:C	/123.3:C	/123.3:C	/123.3:D	/123.3:D
I22.0	I22.1	I22.2	I22.3	I22.4	I22.5	I22.6	I22.7
-1K191:14	-1K192:14	-1K193:14	-1K194:14	-1K195:14	-1K196:14	-1K197:14	-1K198:14
SHAFT CURRENT SYSTEM OK	GENERATOR COOLING WATER VALVE OPEN	GENERATOR COOLING WATER VALVE CLOSED	TURBINE A LEAKAGE TANK LEVEL HIGH	TURBINE B LEAKAGE TANK LEVEL HIGH	GENERAL LOCKOUT LOSS OF SF6	GENERATOR COOLING WATER VALVE OVERLOAD/BIMETAL WARNING	GENERATOR FIRE EXTINGUISHING SYSTEM TRIP

PLC-RSC-24DC/21 14 12 11 /40.1:E

PLC-RSC-24DC/21-21 14 12 11 /40.2:E 24 22 21 /80.5:C

PLC-RSC-24DC/21-21 14 12 11 /40.3:E 24 22 21 /80.6:C

PLC-RSC-24DC/21 14 12 11 /40.4:E

PLC-RSC-24DC/21 14 12 11 /40.5:E

PLC-RSC-24DC/21 14 12 11 /40.6:E

PLC-RSC-24DC/21-21 14 12 11 /40.7:E 24 22 21 /86.5:C

PLC-RSC-24DC/21-21 14 12 11 /40.8:E 24 22 21 /87.4:C

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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 22.0 - 22.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.							+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 40
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 40 / 204

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CLOSING LOCKOUT OPERATING MECHANISAM | TRANSFORMER OIL PUMP 1 ON | TRANSFORMER OIL PUMP 2 ON | TRANSFORMER COOLING WATER VALVE CLOSED | TRANSFORMER COOLING WATER VALVE OPEN | TRANSFORMER BUCHHOLZ ALARM | TRANSFORMER BUCHHOLZ TRIP | TRANSFORMER TEMPERATURE ALARM

11.6:D / 49L+ 24VDC -X49 B 2 A

+U7BAT10-W5001 NYCY 10x1.5 mm² 1

+U7BAT10-W5002 NYCY 10x1.5 mm² 6

+U7CHA10-W3018 NYCY 7x1.5 mm² 1

49L+ 24VDC / 42.1:B

=U7AEA10GH100 +U7AEA10

=U7BAT10GH100 +U7BAT10

-X71 1

-X71 16

=U7CHA10GH100 +U7CHA10

-X171 11A

L+ 24VDC / 42.1:B

21 -K317 24 22

=U7BAT10GH100 +U7BAT10

-X08 21

-X71 6 7 11 10 32

-X171 13B

33

MARSALLING KIOSK

TRANSFORMER

ELECTRICAL PROTECTION A

TRANSFORMER

+U7AEA10-W3053 NYCY 16x2.5 mm² 7

+U7BAT10-W5001 NYCY 10x1.5 mm² 6

+U7BAT10-W5002 NYCY 10x1.5 mm² 7 1

+U7BAT10-W5001 NYCY 10x1.5 mm² 10

+U7BAT10-W5003 NYCY 10x1.5 mm² 1

+U7CHA10-W3018 NYCY 7x2.5 mm² 3

+U7BAT10-W5003 NYCY 10x1.5 mm² 2

-1K201 /156.4:F 24VDC

-1K202 /157.4:B 24VDC

-1K203 /157.4:B 24VDC

-1K204 /157.4:B 24VDC

-1K205 /157.4:B 24VDC

-1K206 /157.4:B 24VDC

-1K207 /157.4:B 24VDC

-1K208 /157.4:C 24VDC

A1 A2

BU 1.0

40.8:D / 48L1- 24VDC

49L- 24VDC / 42.1:D

BU 1.0

11.6:D / 49L- 24VDC -X49 B 11 A 49L- 24VDC / 43.1:D

40.8:E / 32L+ 24VDC

RD 1.0

-1K201 /41.1

-1K202 /41.2

-1K203 /41.3

-1K204 /41.4

-1K205 /41.5

-1K206 /41.6

-1K207 /41.7

-1K208 /41.8

11 14 12

9 10 11 12 13 14 15 16

DI8	DI9	DI10	DI11	DI12	DI13	DI14	DI15
-A222	-A222	-A222	-A222	-A222	-A222	-A222	-A222
/123.3:D	/123.3:D	/123.3:D	/123.3:E	/123.3:E	/123.3:E	/123.3:E	/123.3:E
I23.0	I23.1	I23.2	I23.3	I23.4	I23.5	I23.6	I23.7
-1K201:14	-1K202:14	-1K203:14	-1K204:14	-1K205:14	-1K206:14	-1K207:14	-1K208:14
CLOSING LOCKOUT OPERATING MECHANISAM	TRANSFORMER OIL PUMP 1 ON	TRANSFORMER OIL PUMP 2 ON	TRANSFORMER COOLING WATER VALVE CLOSED	TRANSFORMER COOLING WATER VALVE OPEN	TRANSFORMER BUCHHOLZ ALARM	TRANSFORMER BUCHHOLZ TRIP	TRANSFORMER TEMPERATURE ALARM

PLC-RSC-24DC/21

14 — 11 /41.1:E
12

14 — 11 /41.2:E
12
24 — 21 /80.7:C
22

14 — 11 /41.3:E
12
24 — 21 /80.8:C
22

14 — 11 /41.4:E
12
24 — 21 /81.3:C
22

14 — 11 /41.5:E
12
24 — 21 /81.2:C
22

14 — 11 /41.6:E
12
24 — 21
22

14 — 11 /41.7:E
12
24 — 21 /87.5:C
22

14 — 11 /41.8:E
12
24 — 21 /86.6:C
22

40 42

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 23.0 - 23.7	DRAWING No.: PERU-01-17-170-31-001 / REV No.: 00 / = U7CJA10GH100 / + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100 / SALES No.: / PAGE 41
Checked by	Antonio Aleksić	01.04.2026.					PAGE 41 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		

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TRANSFORMER TEMPERATURE TRIP | TRANSFORMER OIL LEVEL LOW | TRANSFORMER OIL LEVEL HIGH | TRANSFORMER COOLER 1 OIL FLOW OK | TRANSFORMER COOLER 1 PRESSURE LOW | TRANSFORMER COOLER 1 WATER FLOW OK | TRANSFORMER COOLER 1 TEMPARTURE ALARM | TRANSFORMER COOLER 2 TEMPARTURE ALARM

41.8:B / 49L+ 24VDC → -X49 B 5 A → -X49 B 6 A → 49L+ 24VDC / 43.1:B

=U7CHA10GH100
+U7CHA10
ELECTRICAL PROTECTION A

L+ 24VDC / 41.7:B
-K316 21 24 22
-X171 14B

=U7BAT10GH100
+U7BAT10
TRANSFORMER

-X71 44 37 28 38 30 40 41

+U7CHA10-W3018 NYCY 7x2.5 mm²	4						
+U7BAT10-W5003 NYCY 10x1.5 mm²		8	3				
+U7BAT10-W5002 NYCY 10x1.5 mm²				7			
+U7BAT10-W5003 NYCY 10x1.5 mm²					4		
+U7BAT10-W5002 NYCY 10x1.5 mm²						9	
+U7BAT10-W5003 NYCY 10x1.5 mm²							6 7

-1K211 /157.4:C 24VDC	-1K212 /157.4:D 24VDC	-1K213 /157.4:D 24VDC	-1K214 /157.4:D 24VDC	-1K215 /157.4:D 24VDC	-1K216 /157.4:D 24VDC	-1K217 /157.4:D 24VDC	-1K218 /157.4:D 24VDC

41.8:D / 49L- 24VDC → BU 1.0 → A2

40.2:D / 32L+ 24VDC → -X32 A B 18 → 32L+ 24VDC / 44.1:D

RD 1.0 → 32L+ 24VDC / 43.1:E

-1K211 /42.1	-1K212 /42.2	-1K213 /42.3	-1K214 /42.4	-1K215 /42.5	-1K216 /42.6	-1K217 /42.7	-1K218 /42.8

11 → 14 12 → RD 1.0 → 1 2 3 4 5 6 7 8

DI0	DI1	DI2	DI3	DI4	DI5	DI6	DI7
-A223	-A223	-A223	-A223	-A223	-A223	-A223	-A223
/124.3:C	/124.3:C	/124.3:C	/124.3:C	/124.3:C	/124.3:C	/124.3:D	/124.3:D
I24.0	I24.1	I24.2	I24.3	I24.4	I24.5	I24.6	I24.7
-1K211:14	-1K212:14	-1K213:14	-1K214:14	-1K215:14	-1K216:14	-1K217:14	-1K218:14
TRANSFORMER TEMPERATURE TRIP	TRANSFORMER OIL LEVEL LOW	TRANSFORMER OIL LEVEL HIGH	TRANSFORMER COOLER 1 OIL FLOW OK	TRANSFORMER COOLER 1 PRESSURE LOW	TRANSFORMER COOLER 1 WATER FLOW OK	TRANSFORMER COOLER 1 TEMPARTURE ALARM	TRANSFORMER COOLER 2 TEMPARTURE ALARM

PLC-RSC-24DC/21-21: 14 — 11 /42.1:E; 12; 24 — 21 /87.6:C; 22

PLC-RSC-24DC/21 (-1K212): 14 — 11 /42.2:E; 12

PLC-RSC-24DC/21 (-1K213): 14 — 11 /42.3:E; 12

PLC-RSC-24DC/21 (-1K214): 14 — 11 /42.4:E; 12

PLC-RSC-24DC/21 (-1K215): 14 — 11 /42.5:E; 12

PLC-RSC-24DC/21 (-1K216): 14 — 11 /42.6:E; 12

PLC-RSC-24DC/21 (-1K217): 14 — 11 /42.7:E; 12

PLC-RSC-24DC/21 (-1K218): 14 — 11 /42.8:E; 12

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001 / REV No.: 00 / = U7CJA10GH100 / + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.				DIGITAL INPUTS: 24.0 - 24.7	Designer No.: 22044-DD-EE-08-07-100 / SALES No.: / PAGE 42 / PAGE 42 / 204
Checked by	Antonio Aleksić	01.04.2026.					
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		

Sintaksa - A Voith Company
Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska

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TRANSFORMER COOLER 2 WATER FLOW OK

TRANSFORMER COOLER 2 PRESSURE LOW

TRANSFORMER COOLER 2 OIL FLOW OK

TRANSFORMER COOLING CONTROL CUBICLE AC VOLTAGE OK

TRANSFORMER OIL PUMP 1 READY

TRANSFORMER OIL PUMP 2 READY

TRANSFORMER COOLING SYSTEM REMOTE

TRANSFORMER COOLING VALVE READY

42.8:B / 49L+ 24VDC — -X49 B 7 A — B 8 A — 49L+ 24VDC / 44.1:B

=U7BAT10GH100
+U7BAT10

TRANSFORMER

-X71 31 39 29 2 4 5 8 9

+U7BAT10-W5002 NYCY 10x1.5 mm² 10
+U7BAT10-W5003 NYCY 10x1.5 mm² 5
+U7BAT10-W5002 NYCY 10x1.5 mm² 8
+U7BAT10-W5001 NYCY 10x1.5 mm² 2 4 5 8 9

-1K221 /157.4:D 24VDC	-1K222 /157.4:E 24VDC	-1K223 /157.4:E 24VDC	-1K224 /157.4:E 24VDC	-1K225 /157.4:E 24VDC	-1K226 /157.4:E 24VDC	-1K227 /157.4:E 24VDC	-1K228 /157.4:F 24VDC

BU 1.0

49L- 24VDC / 44.1:D

41.2:D / 49L- 24VDC — -X49 B 12 A — B 13 A

42.8:E / 32L+ 24VDC — RD 1.0

-1K221 /43.1	-1K222 /43.2	-1K223 /43.3	-1K224 /43.4	-1K225 /43.5	-1K226 /43.6	-1K227 /43.7	-1K228 /43.8

DI8	DI9	DI10	DI11	DI12	DI13	DI14	DI15
-A223	-A223	-A223	-A223	-A223	-A223	-A223	-A223
/124.3:D	/124.3:D	/124.3:D	/124.3:E	/124.3:E	/124.3:E	/124.3:E	/124.3:E
I25.0	I25.1	I25.2	I25.3	I25.4	I25.5	I25.6	I25.7
-1K221:14	-1K222:14	-1K223:14	-1K224:14	-1K225:14	-1K226:14	-1K227:14	-1K228:14
TRANSFORMER COOLER 2 WATER FLOW OK	TRANSFORMER COOLER 2 PRESSURE LOW	TRANSFORMER COOLER 2 OIL FLOW OK	TRANSFORMER COOLING CONTROL CUBICLE AC VOLTAGE OK	TRANSFORMER OIL PUMP 1 READY	TRANSFORMER OIL PUMP 2 READY	TRANSFORMER COOLING SYSTEM REMOTE	TRANSFORMER COOLING VALVE READY

PLC-RSC-24DC/21: 14–11 /43.1:E; 12

PLC-RSC-24DC/21: 14–11 /43.2:E; 12

PLC-RSC-24DC/21: 14–11 /43.3:E; 12

PLC-RSC-24DC/21: 14–11 /43.4:E; 12

PLC-RSC-24DC/21: 14–11 /43.5:E; 12

PLC-RSC-24DC/21: 14–11 /43.6:E; 12

PLC-RSC-24DC/21: 14–11 /43.7:E; 12

PLC-RSC-24DC/21-21: 14–11 /43.8:E; 12; 24–21 /86.6:C; 22

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 25.0 - 25.7	DRAWING No.: PERU-01-17-170-31-001 / REV No.: 00 / = U7CJA10GH100 / + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100 / SALES No.: / PAGE 43
Checked by	Antonio Aleksić	01.04.2026.					PAGE 43 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		

Format ISO/DIN A3

TRANSFORMER COOLING VALVE FAULT

TRANSFORMER COOLING WATER VALVE TORQUE SWITCH OPEN

TRANSFORMER COOLING WATER VALVE TORQUE SWITCH CLOSED

TRANSFORMER 230 VAC MCB GROUP OK - ALARM

TRANSFORMER COOLING SYSTEM 220 VDC MCBs OK - TRIP

SPARE

SPARE

SPARE

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43.8:B / 49L+ 24VDC -X49 B 9 A

=U7BAT10GH100
+U7BAT10

-X71 12 13 14 15 3

TRANSFORMER

+U7BAT10-W5002 2 3 4 5
NYCY 10x1.5 mm²

+U7BAT10-W5001 3
NYCY 10x1.5 mm²

-1K231 /158.4:B 24VDC	-1K232 /158.4:B 24VDC	-1K233 /158.4:B 24VDC	-1K234 /158.4:B 24VDC	-1K235 /158.4:B 24VDC	-1K236 /158.4:B 24VDC	-1K237 /158.4:C 24VDC	-1K238 /158.4:C 24VDC
A1	A1	A1	A1	A1	A1	A1	A1
A2	A2	A2	A2	A2	A2	A2	A2

BU 1.0

43.8:D / 49L- 24VDC

42.2:D / 32L+ 24VDC -X32 A B 19 32L+ 24VDC / 46.1:D

RD 1.0

32L+ 24VDC / 45.1:E

-1K231 /44.1	-1K232 /44.2	-1K233 /44.3	-1K234 /44.4	-1K235 /44.5	-1K236 /44.6	-1K237 /44.7	-1K238 /44.8
11 14 12	11 14 12	11 14 12	11 14 12	11 14 12	11 14 12	11 14 12	11 14 12
1	2	3	4	5	6	7	8
DI0	DI1	DI2	DI3	DI4	DI5	DI6	DI7
-A224	-A224	-A224	-A224	-A224	-A224	-A224	-A224
/125.3:C	/125.3:C	/125.3:C	/125.3:C	/125.3:C	/125.3:C	/125.3:D	/125.3:D
I26.0	I26.1	I26.2	I26.3	I26.4	I26.5	I26.6	I26.7
-1K231:14	-1K232:14	-1K233:14	-1K234:14	-1K235:14	-1K236:14	-1K237:14	-1K238:14
TRANSFORMER COOLING VALVE FAULT	TRANSFORMER COOLING WATER VALVE TORQUE SWITCH OPEN	TRANSFORMER COOLING WATER VALVE TORQUE SWITCH CLOSED	TRANSFORMER 230 VAC MCB GROUP OK - ALARM	TRANSFORMER COOLING SYSTEM 220 VDC MCBs OK - TRIP	SPARE	SPARE	SPARE
PLC-RSC-24DC/21 14 12 11 /44.1:E	PLC-RSC-24DC/21 14 12 11 /44.2:E	PLC-RSC-24DC/21 14 12 11 /44.3:E	PLC-RSC-24DC/21 14 12 11 /44.4:E	PLC-RSC-24DC/21 14 12 11 /44.5:E	PLC-RSC-24DC/21 14 12 11 /44.6:E	PLC-RSC-24DC/21 14 12 11 /44.7:E	PLC-RSC-24DC/21 14 12 11 /44.8:E

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 26.0 - 26.7	DRAWING No.: PERU-01-17-170-31-001 / REV No.: 00 / = U7CJA10GH100 / + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100 / SALES No.: / PAGE 44
Checked by	Antonio Aleksić	01.04.2026.					PAGE 44 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		

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GENERATOR COOLING WATER VALVE REMOTE

SPARE

SPARE

SPARE

SPARE

SPARE

SPARE

SPARE

11.8:D / 51L+ 24VDC -X51 B 2 A

B 3 A

51L+ 24VDC / 46.1:B

=7N601
+SC44

-X3 7

+U7CJA10-W5101 6
NYCY 12x1.5 mm²

-1K241 /158.4:C 24VDC
-1K242 /158.4:C 24VDC
-1K243 /158.4:C 24VDC
-1K244 /158.4:C 24VDC
-1K245 /158.4:C 24VDC
-1K246 /158.4:C 24VDC
-1K247 /158.4:D 24VDC
-1K248 /158.4:D 24VDC

BU 1.0

51L1- 24VDC / 46.1:D

40.2:D / 48L- 24VDC -X48 B 7 A

11.8:D / 51L- 24VDC -X51 B 13 A

51L- 24VDC / 47.1:D

44.8:E / 32L+ 24VDC

RD 1.0

-1K241 /45.1
-1K242 /45.2
-1K243 /45.3
-1K244 /45.4
-1K245 /45.5
-1K246 /45.6
-1K247 /45.7
-1K248 /45.8

DI8	DI9	DI10	DI11	DI12	DI13	DI14	DI15
-A224	-A224	-A224	-A224	-A224	-A224	-A224	-A224
/125.3:D	/125.3:D	/125.3:D	/125.3:E	/125.3:E	/125.3:E	/125.3:E	/125.3:E
I27.0	I27.1	I27.2	I27.3	I27.4	I27.5	I27.6	I27.7
-1K241:14	-1K242:14	-1K243:14	-1K244:14	-1K245:14	-1K246:14	-1K247:14	-1K248:14
GENERATOR COOLING WATER VALVE REMOTE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE

PLC-RSC-24DC/21
14 11 /45.1:E 12
14 11 /45.2:E 12
14 11 /45.3:E 12
14 11 /45.4:E 12
14 11 /45.5:E 12
14 11 /45.6:E 12
14 11 /45.7:E 12
14 11 /45.8:E 12

44

46

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 27.0 - 27.7	DRAWING No.: PERU-01-17-170-31-001
Approved by Designer	Grga Kolić	01.04.2026.					REV No.: 00
Checked by	Antonio Aleksić	01.04.2026.					= U7CJA10GH100 + U7CJA10
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER	Designer No.: 22044-DD-EE-08-07-100	SALES No.:
							PAGE 45 PAGE 45 / 204

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SPARE SPARE SPARE SPARE SPARE SPARE SPARE SPARE

45.8:B / 51L+ 24VDC -X51 B 4 A — B 5 A — 51L+ 24VDC / 47.1:B

45.8:D / 51L1- 24VDC

	-1K251	-1K252	-1K253	-1K254	-1K255	-1K256	-1K257	-1K258
	/158.4:E	/158.4:E	/158.4:E	/158.4:E	/158.4:E	/158.4:E	/158.4:E	/158.4:F
	24VDC	24VDC	24VDC	24VDC	24VDC	24VDC	24VDC	24VDC
	A1	A1	A1	A1	A1	A1	A1	A1
	A2	A2	A2	A2	A2	A2	A2	A2
BU	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0

44.2:D / 32L+ 24VDC -X32 A B20 A B21 A B22

RD 1.0

32L+ 24VDC / 47.1:E

	-1K251	-1K252	-1K253	-1K254	-1K255	-1K256	-1K257	-1K258
	/46.1	/46.2	/46.3	/46.4	/46.5	/46.6	/46.7	/46.8
	11	11	11	11	11	11	11	11
	14 12	14 12	14 12	14 12	14 12	14 12	14 12	14 12
RD	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	1	2	3	4	5	6	7	8
	DI0	DI1	DI2	DI3	DI4	DI5	DI6	DI7
	-A225	-A225	-A225	-A225	-A225	-A225	-A225	-A225
	/126.3:C	/126.3:C	/126.3:C	/126.3:C	/126.3:C	/126.3:C	/126.3:D	/126.3:D
	I28.0	I28.1	I28.2	I28.3	I28.4	I28.5	I28.6	I28.7
	-1K251:14	-1K252:14	-1K253:14	-1K254:14	-1K255:14	-1K256:14	-1K257:14	-1K258:14
	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE

PLC-RSC-24DC/21-21

	/46.1:E	/46.2:E	/46.3:E	/46.4:E	/46.5:E	/46.6:E	/46.7:E	/46.8:E
14	11	11	11	11	11	11	11	11
12								
24	21	21	21	21	21	21	21	21
22								

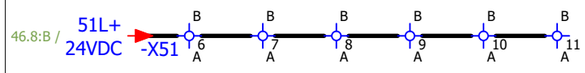
45 47

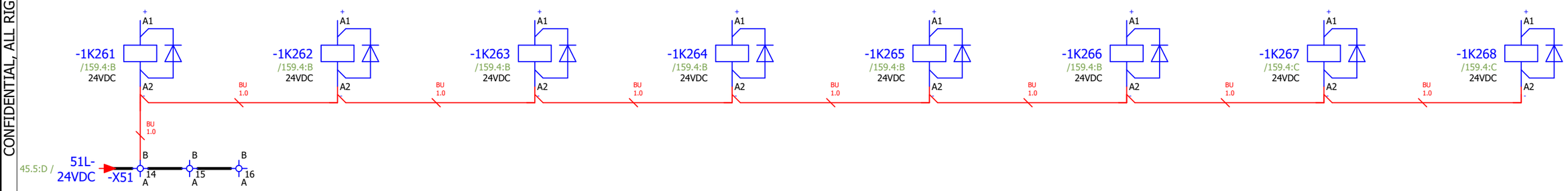
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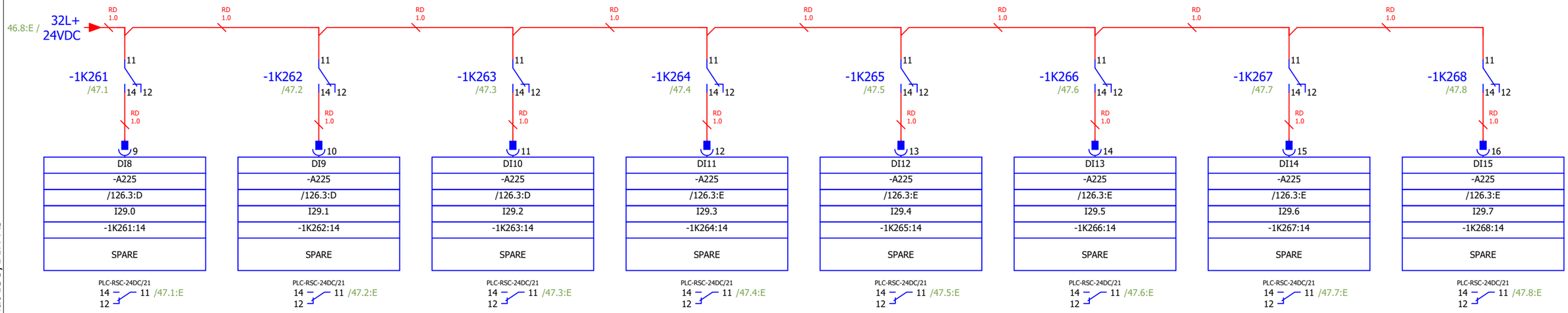
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 28.0 - 28.7 - SPARE	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 46
Checked by	Antonio Aleksić	01.04.2026.							PAGE 46 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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SPARE SPARE SPARE SPARE SPARE SPARE SPARE SPARE







Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 29.0 - 29.7 - SPARE	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 47
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 47 / 204

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL INPUTS: 30.0 - 30.7 - SPARE	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 48 PAGE 48 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

DUMMY

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska		PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				DIGITAL INPUTS: 31.0 - 31.7 - SPARE	Designer No.:	SALES No.:	PAGE 49
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 49 / 204

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ALARM ACKNOWLEDGE COMMAND	AUTOMATIC SYNCHRONIZATION START	AUTO START OR STOP ACTIVE	SPARE	ESD ACTIVATE	FIRE EXTINGUISHING SYSTEM ACITVATE	TRANSFORMER COOLING WATER VALVE CLOSE COMMAND	TRANSFORMER COOLING WATER VALVE OPEN COMMAND
-H01:X11:1	-2K201:A1	-2K202:A1	-2K203:A1	-2K204:A1	-2K205:A1	-X120:4	-X120:3
Q0.0	Q0.1	Q0.2	Q0.3	Q0.4	Q0.5	Q0.6	Q0.7
/128.3:C	/128.3:C	/128.3:C	/128.3:C	/128.3:C	/128.3:C	/128.3:D	/128.3:D
-A231	-A231	-A231	-A231	-A231	-A231	-A231	-A231
DQ0	DQ1	DQ2	DQ3	DQ4	DQ5	DQ6	DQ7
1	2	3	4	5	6	7	8

RD 1.0

-H01 /140.2:B
F1 X11:1
X11:3 -/N

-2K201 /161.4:C 24VDC
-2K202 /161.4:C 24VDC
-2K203 /161.4:C 24VDC
-2K204 /161.4:C 24VDC
-2K205 /161.4:C 24VDC
A1 A2

BU 1.0

33L- 24VDC / 51.3:D

Q0.6 / 66.5:C

Q0.7 / 66.4:C

BU 1.0

B

33L- 24VDC /15.5:E -X33 5 A

-X33 6 A 33L- 24VDC / 54.1:D

RXG22BD SE.RZM040W
14 11 /70.5:B
12
24 21
22

RXG22BD SE.RZM040W
14 11 /88.6:C
12
24 21
22

RXG22BD SE.RZM040W
14 11
12
24 21
22

RXG22BD SE.RZM040W
14 11 /76.3:D
12
24 21
22

RXG22BD SE.RZM040W
14 11 /76.7:C
12
24 21
22

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL OUTPUTS: 0.0 - 0.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 50
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 50 / 204

TRANSFORMER OIL PUMPS ON COMMAND

TRANSFORMER OIL PUMPS ON COMMAND
-X120:12
Q1.0
/128.3:D
-A231
DQ8

9

RD 1.0

Q1.0 / 69.5:C

SPARE

SPARE
Q1.1
/128.3:D
-A231
DQ9

10

GENERATOR HEATERS ON COMMAND

GENERATOR HEATERS ON COMMAND
-2K206:A1
Q1.2
/128.3:D
-A231
DQ10

11

RD 1.0

A1

-2K206
/161.4:C
24VDC

A2

50.6:D / 33L- 24VDC

BU 1.0

BU 1.0

33L- 24VDC / 52.3:D

GENERATOR COOLING WATER VALVE CLOSE COMMAND

GENERATOR COOLING WATER VALVE CLOSE COMMAND
-X120:6
Q1.3
/128.3:E
-A231
DQ11

12

RD 1.0

Q1.3 / 66.7:C

GENERATOR COOLING WATER VALVE OPEN COMMAND

GENERATOR COOLING WATER VALVE OPEN COMMAND
-X120:5
Q1.4
/128.3:E
-A231
DQ12

13

RD 1.0

Q1.4 / 66.6:C

DISCONNECTOR 1 OFF COMMAND

DISCONNECTOR 1 OFF COMMAND
=U6CJA10+GH100-2K09:A1
Q1.5
/128.3:E
-A231
DQ13

14

RD 1.0

Q1.5 / 68.2:C

DISCONNECTOR 1 ON COMMAND

DISCONNECTOR 1 ON COMMAND
=U6CJA10+GH100-2K10:A1
Q1.6
/128.3:E
-A231
DQ14

15

RD 1.0

Q1.6 / 68.3:C

DISCONNECTOR 2 OFF COMMAND

DISCONNECTOR 2 OFF COMMAND
-X120:9
Q1.7
/128.3:E
-A231
DQ15

16

RD 1.0

Q1.7 / 68.4:C

RXG22BD
SE.RZM040W
14
12
11 /89.2:B
24
22
21

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL OUTPUTS: 1.0 - 1.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 51
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 51 / 204

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DISCONNECTOR 2 ON COMMAND	SPARE	GENERATOR TEMPERATURE TRIP	ROTOR TEMPERATURE TRIP	TRANSFORMER OIL TEMPERATURE TRIP	GENERATOR CIRCUIT BREAKER OFF COMMAND	EXCITATION OFF COMMAND	EXCITATION ON COMMAND
DISCONNECTOR 2 ON COMMAND	SPARE	GENERATOR TEMPERATURE TRIP	ROTOR TEMPERATURE TRIP	TRANSFORMER OIL TEMPERATURE TRIP	GENERATOR CIRCUIT BREAKER OFF COMMAND	EXCITATION OFF COMMAND	EXCITATION ON COMMAND
-X120:10		-2K207:A1	-2K208:A1	-2K209:A1	-2K210:A1	-X120:2	-X120:1
Q2.0	Q2.1	Q2.2	Q2.3	Q2.4	Q2.5	Q2.6	Q2.7
/129.3:C	/129.3:C	/129.3:C	/129.3:C	/129.3:C	/129.3:C	/129.3:D	/129.3:D
-A232	-A232	-A232	-A232	-A232	-A232	-A232	-A232
DQ0	DQ1	DQ2	DQ3	DQ4	DQ5	DQ6	DQ7
1	2	3	4	5	6	7	8

RD 1.0

Q2.0 / 68.5:C

Q2.6 / 66.3:C

Q2.7 / 66.2:C

A1

-2K207 /161.4:C 24VDC

-2K208 /161.4:D 24VDC

-2K209 /161.4:D 24VDC

-2K210 /161.4:D 24VDC

A2

BU 1.0

51.4:D / 33L- 24VDC

33L- 24VDC / 53.1:D

RXG22BD SE.RZM040W

14 12 11 /95.2:B

24 22 21

RXG22BD SE.RZM040W

14 12 11 /95.3:B

24 22 21

RXG22BD SE.RZM040W

14 12 11 /95.4:B

24 22 21

RXG22BD SE.RZM040W

14 12 11 /68.6:C

24 22 21

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL OUTPUTS: 2.0 - 2.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 52 PAGE 52 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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SPARE (FCR MODE REQUEST)	SPARE (AVR MODE REQUEST)	SPARE (VAR MODE REQUEST)	SPARE (PF MODE REQUEST)	SPEED > EXCITATION ON LIMIT	SPARE (EXCITATION UNLOAD COMMAND)	SPARE (EXCITATION ISLAND MODE ON COMMAND)	EXCITATION RESET
-2K211:A1	-2K212:A1	=U7CJA10-2K21:A1	=U7CJA10-2K21:A2	-2K215:A1	-2K216:A1	-2K217:A1	-2K218:A1
Q3.0	Q3.1	Q3.2	Q3.3	Q3.4	Q3.5	Q3.6	Q3.7
/129.3:D	/129.3:D	/129.3:D	/129.3:E	/129.3:E	/129.3:E	/129.3:E	/129.3:E
-A232	-A232	-A232	-A232	-A232	-A232	-A232	-A232
DQ8	DQ9	DQ10	DQ11	DQ12	DQ13	DQ14	DQ15
9	10	11	12	13	14	15	16

RD 1.0

A1

-2K211 /161.4:D 24VDC

-2K212 /161.4:D 24VDC

-2K213 /161.4:D 24VDC

-2K214 /161.4:D 24VDC

-2K215 /161.4:D 24VDC

-2K216 /161.4:E 24VDC

-2K217 /161.4:E 24VDC

-2K218 /161.4:E 24VDC

A2

BU 1.0

52.6:D / 33L- 24VDC

RXG22BD SE.RZM040W	RXG22BD SE.RZM040W	RXG22BD SE.RZM040W	RXG22BD SE.RZM040W	RXG22BD SE.RZM040W	RXG22BD SE.RZM040W	RXG22BD SE.RZM040W	RXG22BD SE.RZM040W
14 — 11 /92.4:B	14 — 11 /92.5:B	14 — 11 /92.6:B	14 — 11 /92.7:B	14 — 11 /92.8:B	14 — 11 /93.2:B	14 — 11 /93.3:B	14 — 11 /93.4:B
12	12	12	12	12	12	12	12
24 — 21	24 — 21	24 — 21	24 — 21	24 — 21	24 — 21	24 — 21	24 — 21
22	22	22	22	22	22	22	22

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL OUTPUTS: 3.0 -3.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 53
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 53 / 204

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SPARE	SPARE	SYNCHRONIZATION ACTIVE	ELECTRICAL BRAKING ON RELEASE	SPEED < ELECTRICAL BRAKING STOP LIMIT	AUTOMATIC SYNCHRONIZER START	AC MAIN CB ON COMMAND	AC MAIN CB OFF COMAND
SPARE	SPARE	SYNCHRONIZATION ACTIVE	ELECTRICAL BRAKING ON RELEASE	SPEED < ELECTRICAL BRAKING STOP LIMIT	AUTOMATIC SYNCHRONIZER START	AC MAIN CB ON COMMAND	AC MAIN CB OFF COMAND
-2K219:A1	=U7CJA10-2K12:A1	-2K221:A1	-2K222:A1	-2K223:A1	-2K224:A1	-2K225:A1	-2K226:A1
Q4.0	Q4.1	Q4.2	Q4.3	Q4.4	Q4.5	Q4.6	Q4.7
/130.3:C	/130.3:C	/130.3:C	/130.3:C	/130.3:C	/130.3:C	/130.3:D	/130.3:D
-A233	-A233	-A233	-A233	-A233	-A233	-A233	-A233
DQ0	DQ1	DQ2	DQ3	DQ4	DQ5	DQ6	DQ7
1	2	3	4	5	6	7	8
RD 1.0	RD 1.0	RD 1.0	RD 1.0	RD 1.0	RD 1.0	RD 1.0	RD 1.0
A1	A1	A1	A1	A1	A1	A1	A1
-2K219 /161.4:E 24VDC	-2K220 /161.4:E 24VDC	-2K221 /161.4:E 24VDC	-2K222 /161.4:E 24VDC	-2K223 /161.4:F 24VDC	-2K224 /161.4:F 24VDC	-2K225 /161.4:F 24VDC	-2K226 /161.4:F 24VDC
A2	A2	A2	A2	A2	A2	A2	A2
BU 1.0	BU 1.0	BU 1.0	BU 1.0	BU 1.0	BU 1.0	BU 1.0	BU 1.0

33L- 24VDC / 55.1:D

BU 1.0

50.3:D / 33L- 24VDC -X33 B 7 A B 8 A

RXG22BD SE.RZM040W	RXG22BD SE.RZM040W	RXG22BD SE.RZM040W	RXG22BD SE.RZM040W	RXG22BD SE.RZM040W	RXG22BD SE.RZM040W	RXG22BD SE.RZM040W	RXG22BD SE.RZM040W
14 11	14 11	14 11 /93.7:B	14 11 /93.8:B	14 11 /94.2:B	14 11 /77.2:C	14 11 /96.2:B	14 11 /96.3:B
12	12	12	12	12	12	12	12
24 21	24 21	24 21	24 21	24 21	24 21	24 21	24 21
22	22	22	22	22	22	22	22

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	DIGITAL OUTPUTS: 4.0 - 4.7	Designer No.: 22044-DD-EE-08-07-100	SALES No.:
Checked by	Antonio Aleksić	01.04.2026.						
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			

= U7CJA10GH100
+ U7CJA10
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AC AUXILIARY CB ON COMMAND	AC AUXILIARY CB OFF COMMAND	BUSBAR DUCT FAN ON COMMAND	SPARE	GRAPHITE DUST SUCTION SYSTEM ON COMMAND	DRAINAGE UNDER GENERATOR ON COMMAND	DRAINAGE UNDER TURBINE A ON COMMAND	DRAINAGE UNDER TURBINE B ON COMMAND
AC AUXILIARY CB ON COMMAND	AC AUXILIARY CB OFF COMMAND	BUSBAR DUCT FAN ON COMMAND	SPARE	GRAPHITE DUST SUCTION SYSTEM ON COMMAND	DRAINAGE UNDER GENERATOR ON COMMAND	DRAINAGE UNDER TURBINE A ON COMMAND	DRAINAGE UNDER TURBINE B ON COMMAND
-2K227:A1	-2K228:A1	=U7CJA10-2K114:A1	=4UC01+CU03-G03.01.3:6+__DI	=4UC01+CU03-G03.01.4:8+__DI	=4UC01+CU03-G03.02.3:6+__DI	-2K233:A1	=4UC01+CU03-G03.02.2:4+__DI
Q5.0	Q5.1	Q5.2	Q5.3	Q5.4	Q5.5	Q5.6	Q5.7
/130.3:D	/130.3:D	/130.3:D	/130.3:E	/130.3:E	/130.3:E	/130.3:E	/130.3:E
-A233	-A233	-A233	-A233	-A233	-A233	-A233	-A233
DQ8	DQ9	DQ10	DQ11	DQ12	DQ13	DQ14	DQ15
9	10	11	12	13	14	15	16
RD 1.0	RD 1.0	RD 1.0	RD 1.0	RD 1.0	RD 1.0	RD 1.0	RD 1.0
A1	A1	A1	A1	A1	A1	A1	A1
-2K227 /161.4:F 24VDC	-2K228 /161.4:F 24VDC	-2K229 /162.4:C 24VDC	-2K230 /162.4:C 24VDC	-2K231 /162.4:C 24VDC	-2K232 /162.4:C 24VDC	-2K233 /162.4:C 24VDC	-2K234 /162.4:C 24VDC
A2	A2	A2	A2	A2	A2	A2	A2
BU 1.0	BU 1.0	BU 1.0	BU 1.0	BU 1.0	BU 1.0	BU 1.0	BU 1.0
33L- 24VDC							
54.8:D /							
RXG22BD SE.RZM040W 14 11 /96.4:B 12 24 21 22	RXG22BD SE.RZM040W 14 11 /96.5:B 12 24 21 22	RXG22BD SE.RZM040W 14 11 /96.6:B 12 24 21 22	RXG22BD SE.RZM040W 14 11 12 24 21 22	RXG22BD SE.RZM040W 14 11 /96.7:B 12 24 21 22	RXG22BD SE.RZM040W 14 11 /97.2:B 12 24 21 22	RXG22BD SE.RZM040W 14 11 /97.3:B 12 24 21 22	RXG22BD SE.RZM040W 14 11 /97.4:B 12 24 21 22

54 56

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL OUTPUTS: 5.0 -5.7	DRAWING No.: PERU-01-17-170-31-001
Approved by Designer	Grga Kolić	01.04.2026.					REV No.: 00
Checked by	Antonio Aleksić	01.04.2026.					= U7CJA10GH100 + U7CJA10
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER	Designer No.: 22044-DD-EE-08-07-100	SALES No.:
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DUMMY

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL OUTPUTS: 6.0 - 6.7	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 56
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 56 / 204

DUMMY

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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DIGITAL OUTPUTS: 7.0 -7.7 - SPARE	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 57 PAGE 57 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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TRANSFORMER OIL TEMPERATURE

TRANSFORMER OIL TEMPERATURE DOWN

GENERATOR CORE U TEMPERATURE

GENERATOR CORE V TEMPERATURE

=U7BAT10GH100
+U7BAT10

TRANSFORMER

=U7MKA10GH100
+U7MKA10

GENERATOR
AUXILIARY TERMINAL BOX

-X71 48 49

-X71 52 50 51 53

-X 1 2

-X 11 12

+U7BAT10-W7001
LiYCY 12x0.75 mm²
YE WH BN GN SH

+U7BAT10-W7002
LiYCY 12x0.75 mm²
BN WH YE GN SH

+U7CJA10-W7001
LiYCY 12x0.75 mm²
BN WH YE GN BK VT GY-PK RD-BU SH

-X43 A 23 24 25 26 B WH 0.75

-X43 A 27 28 29 30 B WH 0.75

A 31 32 33 34 B WH 0.75

A 35 36 37 38 B WH 0.75

-KF101 /166.4:D
-KF102 /166.4:D
-KF103 /166.4:D
-KF104 /166.4:D

4 3 In 2 1

PR electronics
3102
Pt100 converter

Out 5 6 - + Supply 7 8 +24V -

WH 0.75

10.7:D / 43L+ 24VDC
10.7:D / 43L- 24VDC

-X43 A B 2 3 14 15 16

43L+ 24VDC / 59.1:D
43L- 24VDC / 59.1:E

KF101 1 / 75.1:B
KF101 2 / 75.2:B
KF103 1 / 75.5:B
KF103 2 / 75.6:B
KF104 1 / 75.7:B
KF104 2 / 75.8:B

-X43a A C B D 1 2 3 4

1 5 9 13
1I0+ 1I0- UV0 2I0+
-A251
/132.3:C
PIW40
-X43a:1
TRANSFORMER OIL TEMPERATURE

2 6 10 14
1I1+ 1I1- UV1 2I1+
-A251
/132.3:C
PIW42
-X43a:2
TRANSFORMER OIL TEMPERATURE DOWN

3 7 11 15
1I2+ 1I2- UV2 2I2+
-A251
/132.3:D
PIW44
-X43a:3
GENERATOR CORE U TEMPERATURE

4 8 12 16
1I3+ 1I3- UV3 2I3+
-A251
/132.3:E
PIW46
-X43a:4
GENERATOR CORE V TEMPERATURE

57

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram ANALOG INPUTS: 1-4	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 58 PAGE 58 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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GENERATOR CORE W TEMPERATURE

GENERATOR WINDING U TEMPERATURE UP

GENERATOR WINDING V TEMPERATURE UP

GENERATOR WINDING W TEMPERATURE UP

=U7MKA10GH100
+U7MKA10

GENERATOR
AUXILIARY TERMINAL BOX

-X 17 10

+U7CJA10-W7001
LiYCY 12x0.75 mm²
GY PK BU RD SH

-X43 A 39 40 41 42 B
WH 0.75

-KF105 /166.4:E
4 3 In 2 1
PR electronics 3102 Pt100 converter
Out 5 6 - Supply 7 8 +24V -

-X 3 4

+U7CJA10-W7002
LiYCY 12x0.75 mm²
BN WH YE GN

A 43 44 45 46 B
WH 0.75

-KF106 /166.4:E
4 3 In 2 1
PR electronics 3102 Pt100 converter
Out 5 6 - Supply 7 8 +24V -

-X 5 6

GY PK BU RD

A 47 48 49 50 B
WH 0.75

-KF107 /166.4:E
4 3 In 2 1
PR electronics 3102 Pt100 converter
Out 5 6 - Supply 7 8 +24V -

-X 7 8

BK VT GY-PK RD-BU SH

A 51 52 53 54 B
WH 0.75

-KF108 /166.4:E
4 3 In 2 1
PR electronics 3102 Pt100 converter
Out 5 6 - Supply 7 8 +24V -

WH 0.75

58.8:D / 43L+ 24VDC
58.8:E / 43L- 24VDC

-X43 A B 6 7 8 9
-X43 A B 17 18 19 20

KF114+ / 61.3:E
KF114- / 61.3:E
KF115+ / 61.5:E
KF115- / 61.5:E
KF113+ / 61.1:E
KF113- / 61.1:E

43L+ 24VDC / 60.1:D
43L- 24VDC / 60.1:E

KF105 1 / 75.1:C
KF105 2 / 75.2:C
KF106 1 / 75.3:C
KF106 2 / 75.4:C
KF107 1 / 75.5:C
KF107 2 / 75.6:C
KF108 1 / 75.7:C
KF108 2 / 75.8:C

-X43a A C B D
5 6 7 8
WH 0.75

1 5 9 13 — 1I0+ 1I0- UV0 2I0+ — -A252 — /133.3:C — PIW48 — -X43a:5 — GENERATOR CORE W TEMPERATURE

2 6 10 14 — 1I1+ 1I1- UV1 2I1+ — -A252 — /133.3:C — PIW50 — -X43a:6 — GENERATOR WINDING U TEMPERATURE UP

3 7 11 15 — 1I2+ 1I2- UV2 2I2+ — -A252 — /133.3:D — PIW52 — -X43a:7 — GENERATOR WINDING V TEMPERATURE UP

4 8 12 16 — 1I3+ 1I3- UV3 2I3+ — -A252 — /133.3:E — PIW54 — -X43a:8 — GENERATOR WINDING W TEMPERATURE UP

58

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram ANALOG INPUTS: 5-8	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 59 PAGE 59 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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GENERATOR HOT AIR TEMPERATURE

GENERATOR COLD AIR TEMP TEMPERATURE

GENERATOR WINDING U TEMPERATURE DOWN

GENERATOR WINDING V TEMPERATURE DOWN

=U7MKA10GH100
+U7MKA10

GENERATOR AUXILIARY TERMINAL BOX

-X 23 24

-X 25 26

-X 13 14

-X 15 16

+U7CJA10-W7003
LiYCY 12x0.75 mm²
BN WH YE GN GY PK BU RD BK VT GY-PK RD-BU SH

+U7CJA10-W7004
LiYCY 12x0.75 mm²
BN WH YE GN SH

-X43 A 55 56 57 58 B WH 0.75

-X60 A 1 2 3 4 B WH 0.75

-X60 A 5 6 7 8 B WH 0.75

-KF109 /166.4:F
-KF110 /166.4:F
-KF111 /167.4:B
-KF112 /167.4:C

4 3 2 1 In

PR electronics 3102 Pt100 converter

Out 5 6 + - Supply 7 8 +24V -

WH 0.75

59.8:D / 43L+ 24VDC
59.8:E / 43L- 24VDC

-X43 A B 10 11
-X43 A B 21 22

KF112+ / 60.7:E
KF112- / 60.7:E

60.2:E / KF112+
60.2:E / KF112-

KF109 1 / 75.1:E
KF109 2 / 75.2:E
KF110 1 / 75.3:E
KF110 2 / 75.4:E
KF111 1 / 75.5:E
KF111 2 / 75.6:E
KF112 1 / 75.7:E
KF112 2 / 75.8:E

-X43a A C B D 9 10 21 22

WH 0.75

1	5	9	13
1I0+	1I0-	UV0	2I0+
-A253			
/134.3:C			
PIW56			
-X43a:9			
GENERATOR HOT AIR TEMPERATURE			

2	6	10	14
1I1+	1I1-	UV1	2I1+
-A253			
/134.3:C			
PIW58			
-X43a:10			
GENERATOR COLD AIR TEMP TEMPERATURE			

3	7	11	15
1I2+	1I2-	UV2	2I2+
-A253			
/134.3:D			
PIW60			
-X43a:21			
GENERATOR WINDING U TEMPERATURE DOWN			

4	8	12	16
1I3+	1I3-	UV3	2I3+
-A253			
/134.3:E			
PIW62			
-X43a:22			
GENERATOR WINDING V TEMPERATURE DOWN			

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram ANALOG INPUTS: 9-12	DRAWING No.: PERU-01-17-170-31-001
Approved by Designer	Grga Kolić	01.04.2026.					REV No.: 00
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		SALES No.:

= U7CJA10GH100	
+ U7CJA10	
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GENERATOR WINDING W TEMPERATURE DOWN

GENERATOR LEFT SIDE TEMEPERATURE

GENERATOR RIGHT SIDE TEMEPERATURE

SPARE

=U7MKA10GH100
+U7MKA10

GENERATOR AUXILIARY TERMINAL BOX

=??
+??

-X 17 18

-X 19 20

-X 21 22

-X? 25 27 29 31

+U7CJA10-W7004
LiYCY 12x0.75 mm²
GY PK BU RD

+U7CJA10-W7005
LiYCY 12x0.75 mm²
GY PK BU RD

+U7CJA10-W7014
LiYCY 4x0.75 mm²
BN WH YE GN SH

-X61 A1 B A2 B A3 B A4 B
WH 0.75

-X61 A9 B A11 B A13 B A15 B
WH 0.75

-X61 A17 B A19 B A21 B A23 B
WH 0.75

-X61 A25 B A27 B A29 B A31 B

-KF113 /167.4:C
-KF114 /167.4:C
-KF115 /167.4:C

4 3 In 2 1

PR electronics
3102
Pt100 converter

Out 5 6 - Supply 7 8 +24V -

WH 0.75

59.6:E / KF113+
59.6:E / KF113-

-X61 A5 B A6 B

KF113 1 / 75.a.1:B
KF113 2 / 75.a.2:B

-X43a A C 23 B D

59.2:E / KF114+
59.2:E / KF114-

-X61 A7 B A8 B

-X61 A10 B A12 B A14 B A16 B

59.4:E / KF115+
59.4:E / KF115-

-X61 A18 B A20 B A22 B A24 B

-X61 A26 B A28 B A30 B A32 B

1 5 9 13
1I0+ 1I0- UV0 2I0+
-A254
/135.3:C
PIW64
-X43a:23
GENERATOR WINDING W TEMPERATURE DOWN

2 6 10 14
1I1+ 1I1- UV1 2I1+
-A254
/135.3:C
PIW66
-KF114:5
GENERATOR LEFT SIDE TEMEPERATURE

3 7 11 15
1I2+ 1I2- UV2 2I2+
-A254
/135.3:D
PIW68
-KF115:5
GENERATOR RIGHT SIDE TEMEPERATURE

4 8 12 16
1I3+ 1I3- UV3 2I3+
-A254
/135.3:E
PIW70
-X61:26
SPARE

60

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram ANALOG INPUTS: 13-16	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 61 PAGE 61 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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SPARE

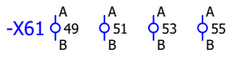
SPARE

SPARE

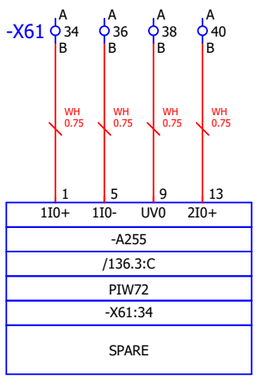
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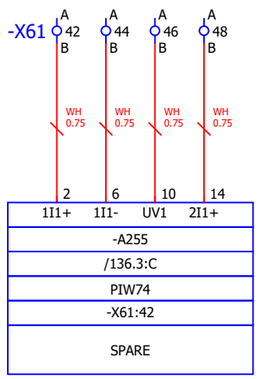


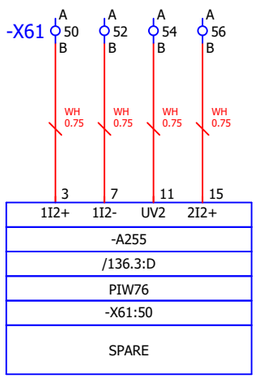


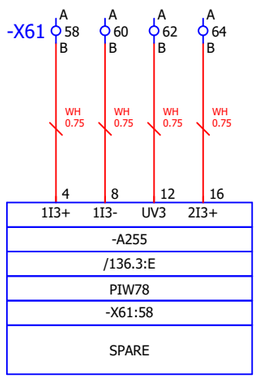












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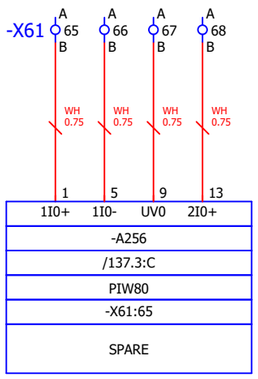
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska		PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				ANALOG INPUTS: 17-20	Designer No.:	SALES No.: PAGE 62
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 62 / 204

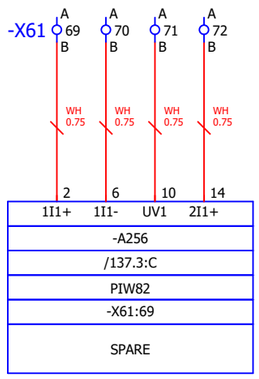
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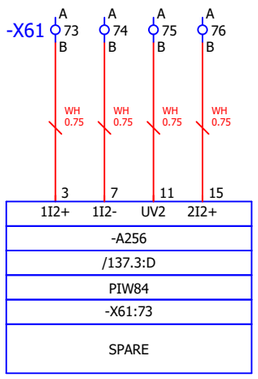
SPARE

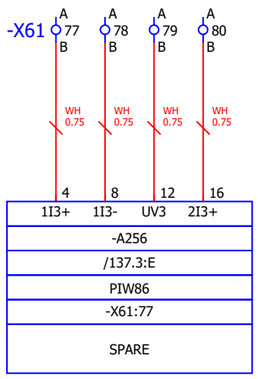
SPARE

SPARE









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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram ANALOG INPUTS: 21-24 - SPARE	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 63
Checked by	Antonio Aleksić	01.04.2026.							PAGE 63 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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DUMMY

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram ANALOG INPUTS: 25-28 - SPARE	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 64 PAGE 64 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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Format ISO/DIN A3

ENABLE LOCAL CONTROL

AUTOMATIC/TEST SYNCHRONIZATION ON

9.8:D / 37L+ 24VDC

-X37

CTRL / 69.1:B

37L / 74.1:B

2K210 24VDC / 68.6:C

LOC/MAN / 66.1:B

-S103 /18.1 REMOTE CONTROL

-S104 /169.4:C AUTOMATIC CONTROL

-K01 /70.5 SYNCHRONIZATION RUN

-K107 /71.4 SYNCHRONIZATION AUTO

-K108 /71.5 SYNCHRONIZATION TEST

-K100 /65.4 LOCAL CONTROL ENABLED

-K100 24VDC LOCAL CONTROL ENABLED

-K101 24VDC AUTOMATIC/TEST SYNCHRONIZATION ON

9.8:D / 37L- 24VDC

-X37

37L- 24VDC / 66.1:F

RXM4AB2BD RXZE2S114M

14 12 11 /65.6:B

24 22 21 /70.4:B

34 32 31

44 42 41

RXM4AB2BD RXZE2S114M

14 12 11 /78.2:B

24 22 21 /78.6:B

34 32 31 /77.3:C

44 42 41

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram LOCAL CONTROL	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 65 PAGE 65 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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EXCITATION OFF COMMAND | EXCITATION ON COMMAND | TRANSFORMER COOLING WATER VALVE OPEN COMMAND | TRANSFORMER COOLING WATER VALVE CLOSE COMMAND | GENERATOR COOLING WATER VALVE OPEN COMMAND | GENERATOR COOLING WATER VALVE CLOSE COMMAND

65.7:C / LOC/MAN — LOC/MAN / 67.1:B

-S223 /177.4:F RED — -S224 /178.4:C GREEN — -S151 /174.4:C RED — -2K102 /66.4 — -S152 /174.4:E — -S152 /174.4:E GREEN — -2K101 /66.5 — -S151 /174.4:C — -S161 /175.4:C RED — -S162 /175.4:E GREEN

-X120

Q2.7 / 52.8:C — Q2.6 / 52.7:C — Q0.7 / 50.8:C — Q0.6 / 50.7:C — Q1.4 / 51.5:C — Q1.3 / 51.4:C

-2K106 24VDC EXCITATION OFF COMMAND — -2K105 24VDC EXCITATION ON COMMAND — -2K102 24VDC TRANSFORMER COOLING WATER VALVE OPEN COMMAND — -2K101 24VDC TRANSFORMER COOLING WATER VALVE CLOSE COMMAND — -2K104 24VDC GENERATOR COOLING WATER VALVE OPEN COMMAND — -2K103 24VDC GENERATOR COOLING WATER VALVE CLOSE COMMAND

65.5:F / 37L- 24VDC — -X37 — 37L- 24VDC / 67.1:F

RXG12BD SE.RZM040W 14 12 — 11 /92.2:B
RXG12BD SE.RZM040W 14 12 — 11 /92.3:B
RXG22BD SE.RZM040W 14 12 — 11 /90.5:B; 24 22 — 21 /66.4:B
RXG22BD SE.RZM040W 14 12 — 11 /90.6:B; 24 22 — 21 /66.5:B
RXG12BD SE.RZM040W 14 12 — 11 /89.3:B
RXG12BD SE.RZM040W 14 12 — 11 /89.4:B

65 67

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram LOCAL CONTROL	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 66
Checked by	Antonio Aleksić	01.04.2026.							PAGE 66 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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DISCONNECTOR 1 SELECTED

DISCONNECTOR 2 SELECTED

GENERATOR CIRCUIT BREAKER SELECTED

66.8:B / LOC/MAN — RD 1.0 — LOC/MAN / 68.1:B

-S111 /169.4:D WHITE — NO 13 — 14 NO

-S112 /169.4:E WHITE — NO 13 — 14 NO

-S113 /169.4:E WHITE — NO 13 — 14 NO

-K102 /164.4:C 24VDC DISCONNECTOR 1 SELECTED — A1 (+) / A2 (-)

-K103 /164.4:C 24VDC DISCONNECTOR 2 SELECTED — A1 (+) / A2 (-)

-K104 /164.4:C 24VDC GENERATOR CIRCUIT BREAKER SELECTED — A1 (+) / A2 (-)

BU 1.0

66.3:F / 37L- 24VDC — -X37 B 9 A — 37L- 24VDC / 69.1:F

37L- 24VDC / 68.1:E

RXM4AB2BD RXZE2S114M

14	11	/68.2:B
12		
24	21	/18.7:D
22		
34	31	/81.6:C
32		
44	41	
42		

RXM4AB2BD RXZE2S114M

14	11	/68.4:B
12		
24	21	/18.8:D
22		
34	31	/81.7:C
32		
44	41	
42		

RXM4AB2BD RXZE2S114M

14	11	/68.6:B
12		
24	21	/19.1:D
22		
34	31	/81.8:C
32		
44	41	/72.2:C
42		

66 68

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska		PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				110 KV CONTROL	Designer No.:	SALES No.: PAGE 67
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 67 / 204

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DISCONNECTOR 1 OFF COMMAND

DISCONNECTOR 1 ON COMMAND

DISCONNECTOR 2 OFF COMMAND

DISCONNECTOR 2 ON COMMAND

GENERATOR CIRCUIT BREAKER OFF COMMAND

67.8:B / LOC/MAN — RD 1.0

LOC/MAN / 69.1:B

-K102 /67.2 DISCONNECTOR 1 SELECTED — 11, 14, 12

-K103 /67.4 DISCONNECTOR 2 SELECTED — 11, 14, 12

-K104 /67.6 GENERATOR CIRCUIT BREAKER SELECTED — 11, 14, 12

-S114 /19.2 OFF COMMAND — 23, 24

-S115 /19.3 ON COMMAND — 23, 24

-S114 /19.2 OFF COMMAND — 33, 34

-S115 /19.3 ON COMMAND — 33, 34

-S114 /19.2 OFF COMMAND — 43, 44

-2K210 /52.6 — 11, 14, 12

2K210 24VDC / 65.7:B

Q1.5 / 51.6:C

Q1.6 / 51.7:C

Q1.7 / 51.8:C

Q2.0 / 52.1:C

-X120 — 7, 8, 9, 10, 11

-2K01 /159.4:D 24VDC DISCONNECTOR 1 OFF COMMAND — A1, A2

-2K02 /159.4:D 24VDC DISCONNECTOR 1 ON COMMAND — A1, A2

-2K03 /159.4:D 24VDC DISCONNECTOR 2 OFF COMMAND — A1, A2

-2K04 /159.4:D 24VDC DISCONNECTOR 2 ON COMMAND — A1, A2

-2K05 /159.4:E 24VDC GENERATOR CIRCUIT BREAKER OFF COMMAND — A1, A2

67.8:E / 37L- 24VDC — BU 1.0

3RH2140-1VB40
13–14 /91.1:B
23–24 /91.1:E
33–34
43–44

3RH2140-1VB40
13–14 /91.2:B
23–24 /91.2:E
33–34
43–44

3RH2140-1VB40
13–14 /91.3:B
23–24 /91.3:E
33–34
43–44

3RH2140-1VB40
13–14 /91.4:B
23–24 /91.4:E
33–34
43–44

3TH4253-0BB4
3TX7402-3A
13–14 /91.5:B
23–24 /91.5:E
33–34 /91.6:B
43–44 /91.6:B
53–54
61–62
71–72
81–82

67

69

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram 110 KV CONTROL	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 68
Checked by	Antonio Aleksić	01.04.2026.							PAGE 68 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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TRANSFORMER OIL PUMP 1 SELECTED

TRANSFORMER OIL PUMP 2 SELECTED

TRANSFORMER OIL PUMPS 1&2 SELECTED

TRANSFORMER OIL PUMPS RUN COMMAND

68.8:B / LOC/MAN

65.7:A / CTRL

RD 1.0

-S141 /19.4 LEFT 33 34

-S141 /19.4 RIGHT 43 44

-K203 /69.2 TRANSFORMER OIL PUMP 1 SELECTED 11 12 14

-K105 /69.4 TRANSFORMER OIL PUMPS 1&2 SELECTED 11 12 14

-S142 /173.4:C RED NO 13 14 NO

-2K401 /69.5 31 34 32

-S143 /173.4:D GREEN NC 11 12 NC

Q1.0 / 51.1:C

-X120 A B C D 12

-K203 /165.4:C 24VDC TRANSFORMER OIL PUMP 1 SELECTED A1 A2

-K204 /165.4:C 24VDC TRANSFORMER OIL PUMP 2 SELECTED A1 A2

-K105 /164.4:D 24VDC TRANSFORMER OIL PUMPS 1&2 SELECTED A1 A2

-2K401 /162.4:E 24VDC TRANSFORMER OIL PUMPS RUN COMMAND A1 A2

BU 1.0

67.3:F / 37L- 24VDC

-X37 B 10 A

37L- 24VDC / 74.1:F

RXG22BD SE.RZM040W
14 12 11 /69.3:B
24 22 21 /90.2:C

RXG22BD SE.RZM040W
14 12 11 /90.3:C
24 22 21

RXM4AB2BD RXZE2S114M
14 12 11 /69.3:C
24 22 21 /90.2:C
34 32 31 /90.4:C
44 42 41

RXM4AB2BD RXZE2S114M
14 12 11 /90.2:B
24 22 21
34 32 31 /69.6:B
44 42 41

68

70

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram TRANSFORMATOR OIL PUMPS CONTROL	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 69
Checked by	Antonio Aleksić	01.04.2026.							PAGE 69 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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SYNCHRONIZATION
RUN

10.3:D / 39L+ 24VDC

-X39 A B 2

RD 1.0

21

-K100
/65.4
LOCAL CONTROL ENABLED

24 22

RD 1.0

NC 11

-S131
/172.4:C
SYNCHRONIZATION STOP
GREEN

12 NC

RD 1.0

RD 1.0

NO 13

-S130
/171.4:E
SYNCHRONIZATION START
RED

14 NO

73

-K01
/70.5
SYNCHRONIZATION
RUN

74

RD 1.0

RD 1.0

RD 1.0

-X39 A B 3

RD 1.0

11

-2K201
/50.2
AUTOMATIC
SYNCHRONIZATION START

14 12

RD 1.0

11

-K206
/73.2
GENERATOR CIRCUIT
BREAKER ON

14 12

RD 1.0

+ A1

-K01
/163.4:B
SYNCHRONIZATION
RUN

A2 -

BU 1.0

10.3:D / 39L- 24VDC

-X39 B 7 A

3RH2140-1FB40
3RH2911-1GA31

13 — 14 /79.4:C
23 — 24 /79.4:C
33 — 34 /79.5:C
43 — 44 /79.5:C
53 — 54 /81.5:C
61 — 62 /81.4:C
73 — 74 /70.4:C
83 — 84 /65.5:B

39L+ 24VDC / 71.1:B

39L- 24VDC / 71.1:F

69

71

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram SYNCHRONIZATION CONTROL	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.							+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 70
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 70 / 204

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SYNCHRONIZATION MANUAL

SYNCHRONIZATION AUTO

SYNCHRONIZATION TEST

70.8:B / 39L+ 24VDC

-X39 A B 4

39L+ 24VDC / 72.1:B

RD 1.0

MAN TEST AUTO

-S121 /171.4:B MANUAL NO 13 14 NO

AUTO NO 23 24 NO

-S121 /171.4:B MANUAL NC 11 12 NC

-S121 /171.4:B AUTO NC 21 22 NC

-K106 /164.4:D 24VDC SYNCHRONIZATION MANUAL A1 + A2 -

-K107 /164.4:D 24VDC SYNCHRONIZATION AUTO A1 + A2 -

-K108 /164.4:D 24VDC SYNCHRONIZATION TEST A1 + A2 -

BU 1.0

70.8:F / 39L- 24VDC

-X39 B A 8

39L- 24VDC / 72.1:F

RXM4AB2BD RXZE2S114M

14 12 11 /18.5:D; 24 22 21; 34 32 31 /72.2:B; 44 42 41

RXM4AB2BD RXZE2S114M

14 12 11 /18.6:D; 24 22 21 /65.5:C; 34 32 31 /72.3:B; 44 42 41

RXM4AB2BD RXZE2S114M

14 12 11 /65.6:C; 24 22 21 /72.4:B; 34 32 31 /72.2:E; 44 42 41 /21.6:D

70

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska		PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				SYNCHRONIZATION CONTROL	Designer No.:	SALES No.: PAGE 71
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 71 / 204

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GENERATOR CB CLOSING

71.8:B / 39L+ 24VDC

-X39 A B 5

RD 1.0

-K106 /71.4 SYNCHRONIZATION MANUAL 31 34 32

-K107 /71.4 SYNCHRONIZATION AUTO 31 34 32

-K108 /71.5 SYNCHRONIZATION TEST 21 24 22

-K104 /67.6 GENERATOR CIRCUIT BREAKER SELECTED 41 44 42

-S115 /19.3 ON COMMAND 43 44

-F25G SPM-D10 /143.2:B 15 BO1 14

-P411 /168.4:B 5 6

-K108 /71.5 SYNCHRONIZATION TEST 31 34 32

SYNC_TEST 83.7:E

-K401 /166.4:B A1 A2

71.8:F / 39L- 24VDC

-X39 B 9 A

SIE.3TH4253-0BB4 3TH4253-0BB4 13 14 /72.6:B 23 24 /72.6:B 33 34 /72.6:F 43 44 /72.6:E 53 54 61 62 71 72 81 82

71

GENERATOR CB CLOSING

OG 1.5

-K401 /166.4:B 14 13

-K401 /166.4:B 24 23

-X71 B 1 A B 3 A

+U7AEA10-W3051 NYCY 4x2.5 mm² BN GY

=U7AEA10GH100
+U7AEA10

-X04 34 36

-X04 31 38

MARSALLING KIOSK

+U7AEA10-W3051 NYCY 4x2.5 mm² BK BU

-X71 A 2 B A 4 B

-K401 /72.2 43 44

-K401 /72.2 33 34

73

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram GENERATOR CB CLOSING CONTROL	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 72
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 72 / 204

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GENERATOR CIRCUIT BREAKER OFF

GENERATOR CIRCUIT BREAKER ON

DISCONNECTOR 1 ON

DISCONNECTOR 2 ON

34.8:B /
45L+
24VDC
-X45
A
B12
B13

RD 1.0
BU 1.0

-1K87
/29.7
GENERATOR CIRCUIT BREAKER OFF
21
24
22

-1K88
/29.8
GENERATOR CIRCUIT BREAKER ON

-1K84
/29.4
DISCONNECTOR 1 ON

-1K86
/29.6
DISCONNECTOR 2 ON

-K205
/165.4:C
24VDC GENERATOR CIRCUIT BREAKER OFF
A1
A2

-K206
/165.4:C
24VDC GENERATOR CIRCUIT BREAKER ON

-K208
/165.4:D
24VDC GENERATOR CIRCUIT BREAKER ON

-K150
/164.4:E
DC 24V DISCONNECTOR 1 ON

-K151
/164.4:E
24VDC DISCONNECTOR 2 ON

33.1:D /
45L-
24VDC
-X45
B18
A19

RXG22BD
SE.RZM040W
14 11 /77.3:C
12
24 21 /83.6:C
22

RXG22BD
SE.RZM040W
14 11 /70.5:D
12
24 21 /83.7:C
22

RXG22BD
SE.RZM040W
14 11 /88.5:C
12
24 21
22

3RH2122-1FB40
13 14 /101.4:D
21 22
32 31 /101.4:D
43 44 /83.3:C

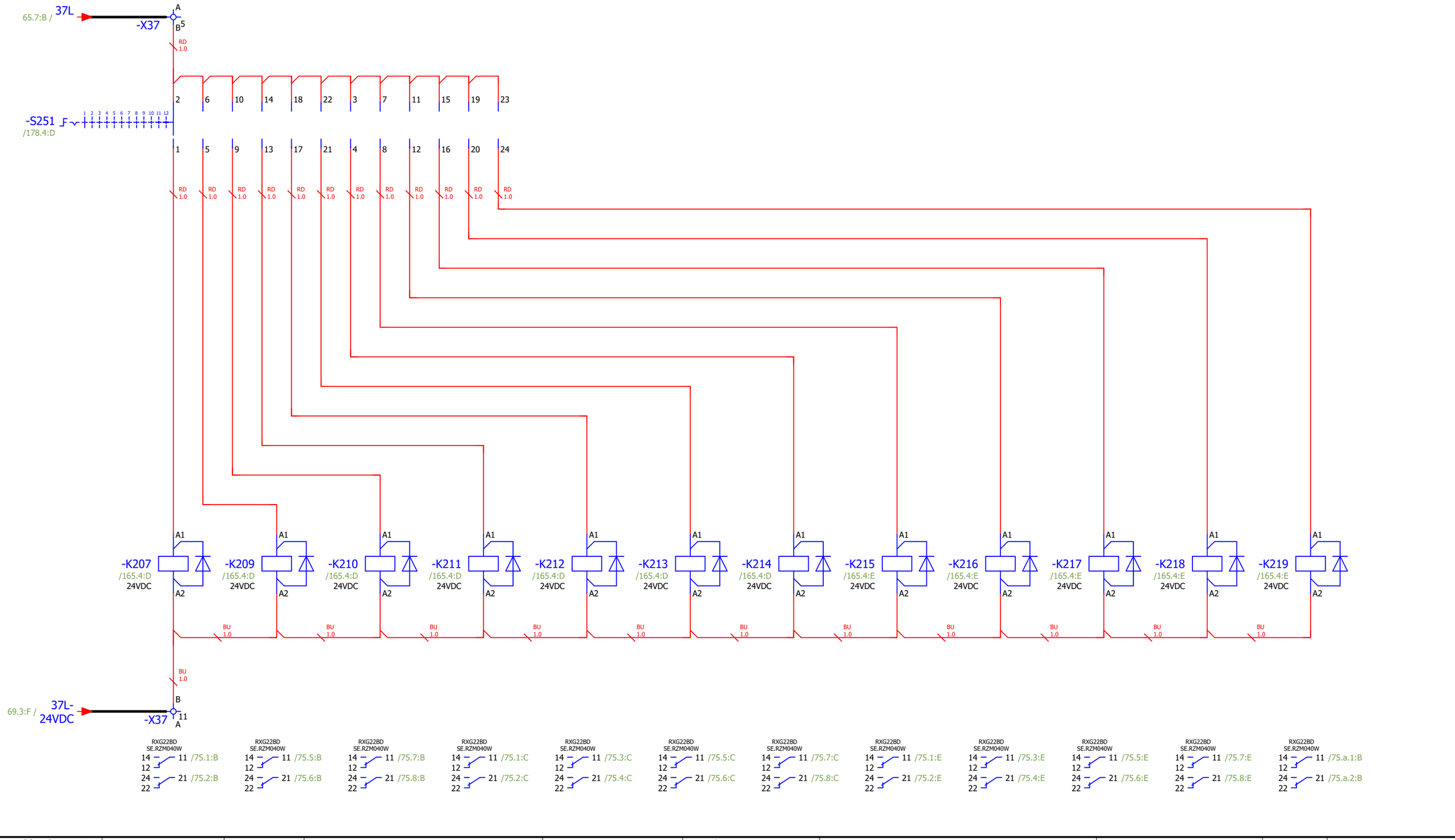
3RH2122-1FB40
13 14 /101.4:D
21 22
32 31 /101.4:D
43 44 /83.5:C

72

74

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram GENERATOR CB AND DISCONNECTORS SIGNALS	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00 / SALES No.:	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100		PAGE 73 PAGE 73 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

TEMPERATURE DISPLAY SWITCH



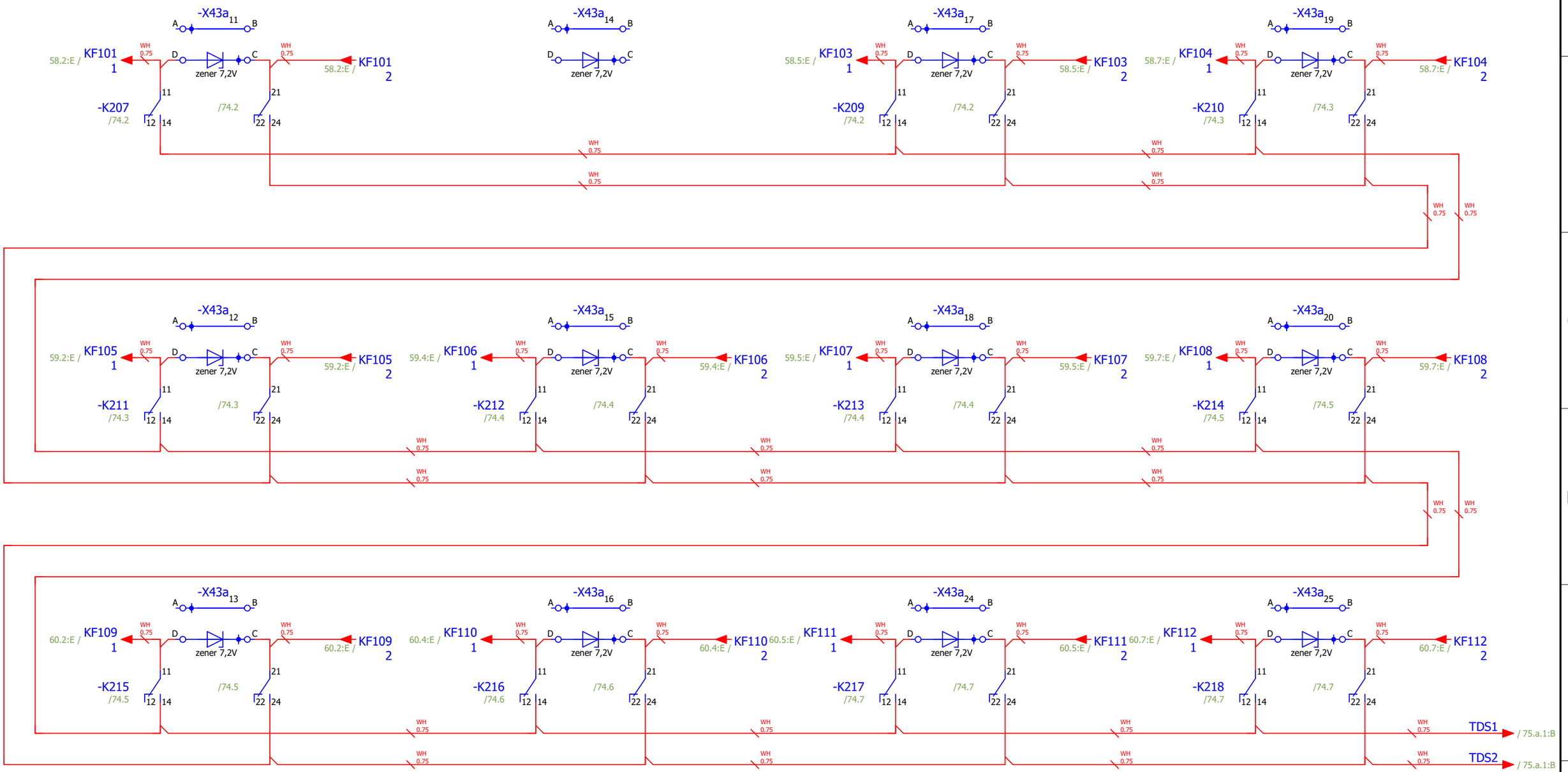
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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram TEMPERATURE DISPLAY SWITCH	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 74 PAGE 74 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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TEMPERATURE DISPLAY SWITCH

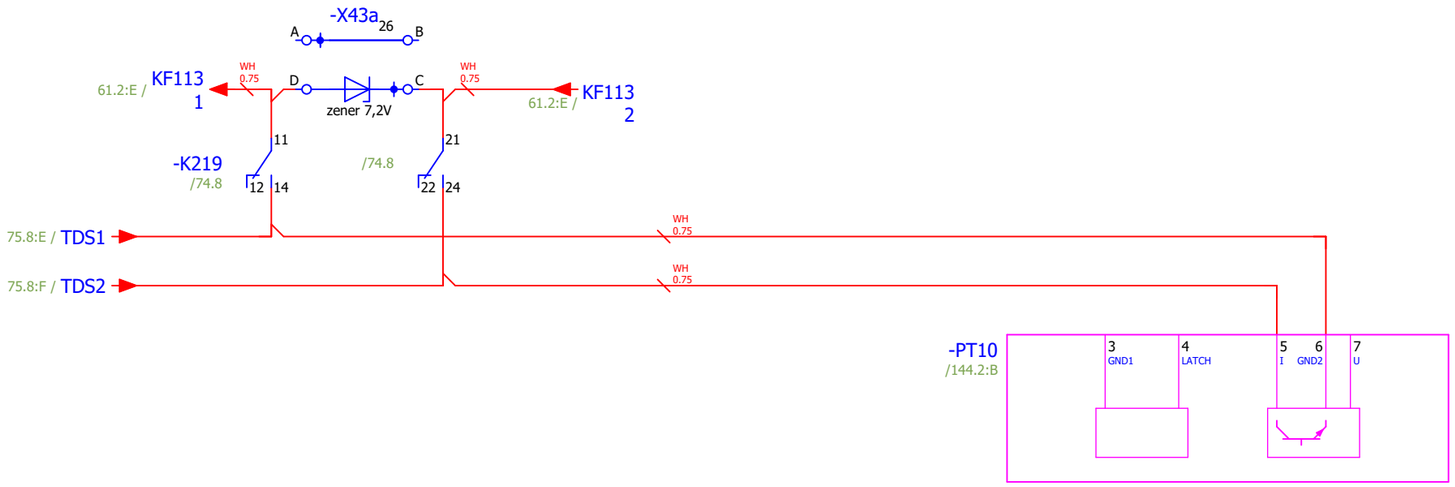


Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	TEMPERATURE DISPLAY SWITCH			+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 75
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 75 / 204

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TEMPERATURE DISPLAY SWITCH



Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska		PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				TEMPERATURE DISPLAY SWITCH	Designer No.:	SALES No.:	PAGE 75.a
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 76 / 204

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ESD ACTIVATION

-S301
/19.6
EMERGENCY SHUTDOWN
11
12

OG
1.5

-S302
/19.7
FIRE
EXTINGUISHING
SYSTEM
11
12

OG
1.5

-2K204
/50.5
ESD ACTIVATE
11
12
14

OG
1.5

OG
1.5

-X100
B
1
A

B
2
A

+U7CBC10-W5025
NYCY 10x1.5 mm²
4
5

=U7CBC10GH100
+U7CBC10

-X341
55
56

TURBINE GOVERNOR

75.a

FIRE EXTINGUISHING
SYSTEM ACTIVATION

OG
1.5

-S302
/19.7
23
24

-2K205
/50.6
FIRE EXTINGUISHING
SYSTEM ACITVATE
11
12
14

OG
1.5

OG
1.5

OG
1.5

-X100
B
3
A

B
4
A

B
5
A

+U7CJA10-W5207
NYCY 7x1.5 mm²
1
2

=FES
+FES

-X?

-X?

FIRE EXTINGUISHING SYSTEM

76.a

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram EMERGENCY STOP	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 76
Checked by	Antonio Aleksić	01.04.2026.							PAGE 77 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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TURBINE GOVERNOR
ESD

=U7CBC10GH100
+U7CBC10

-X110 5 6

TURBINE GOVERNOR

+U7CBC10-W5025
NYCY 10x1.5 mm²

8 9

-X110 B 1 A B 2 A

OG 1.5 OG 1.5

OG 1.5

A1 -K30.1 /163.4:C A2

+ A1 -K30.2 /163.4:D A2 -

OG 1.5

3RH2122-1FB40
3RH2911-1GA04

13 14 /38.6:C
21 22 /94.6:B
31 32 /94.7:B
43 44
51 52 /94.3:B
61 62 /94.4:B
71 72 /76.b.5:B
81 82

3TH4244-0BB4
3TX7402-3A

13 14
23 24
33 34
43 44
51 52 /91.7:B
61 62 /91.7:B
71 72 /91.8:B
81 82 /76.b.5:C

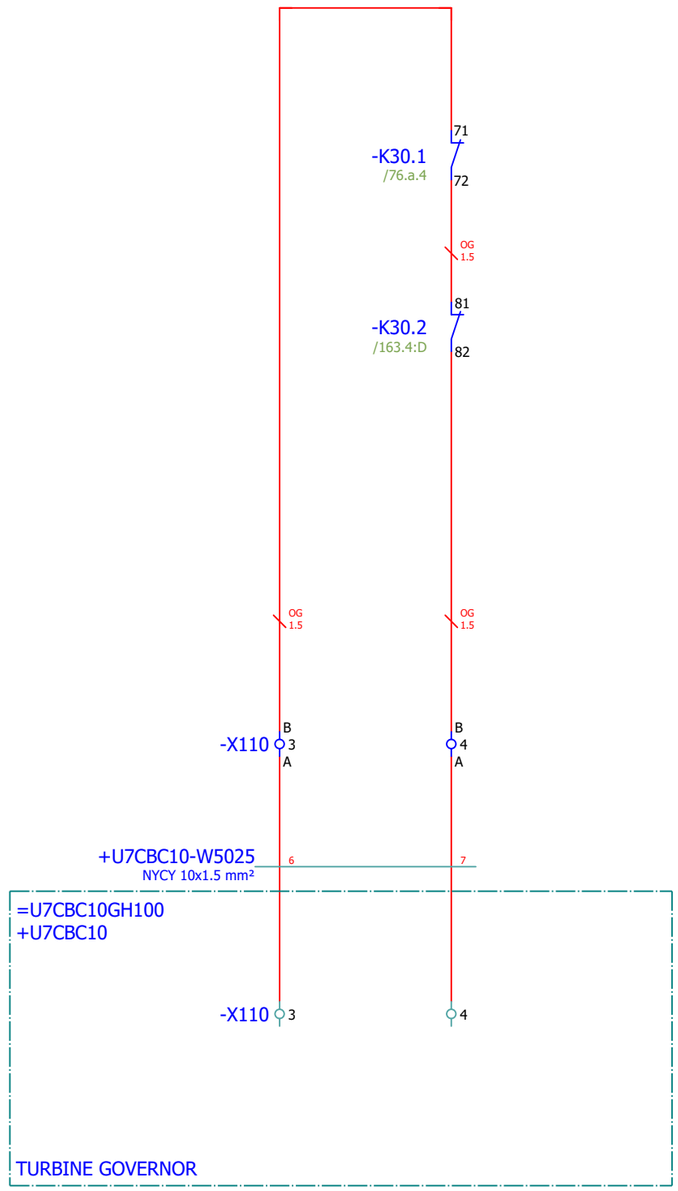
76

76.b

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram EMERGENCY STOP	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 76.a PAGE 78 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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TURBINE GOVERNOR ESD
FEEDBACK



Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram EMERGENCY STOP	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 76.b PAGE 79 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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AUTOMATIC SYNCHRONIZER START

GENERATOR CB OPEN

SPARE

SPARE

10.4:D / 40L+ 24VDC

-X40 A B 2

-X40 A B 3

RD 1.0

-2K224 /54.6 AUTOMATIC SYNCHRONIZER START 11 12 14

-K101 /65.5 AUTOMATIC/TEST SYNCHRONIZATION ON 31 32 34

-K205 /73.2 GENERATOR CIRCUIT BREAKER OFF 11 12 14

-X40 A 6 B

-X40 A 7 B

-X40 A 8 B

-X40 A 9 B

-X40 A 10 B

-X40 A 11 B

-F25G SPM-D2-10 /143.2:B

3 BI1 Int

4 BI2 Int

5 BI3 Int

6 BI4 7

BU 1.0

10.4:D / 40L- 24VDC

-X40 B 5 A

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska		PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				SYNCHRONIZATION	Designer No.:	SALES No.:	PAGE 77
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 80 / 204

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EXCITATION SETPOINT
LOWER/HIGHER

-K101
/65.5
AUTOMATIC/TEST
SYNCHRONIZATION
ON

-S103
/168.4:E
REMOTE

-S222
/20.2
LOWER

-S222
/20.2
HIGHER

-F25G
SPM-D2-10
/143.2:B

-X90
-X030
-X040
-X040

+U7CJA10-W5206
NYCY 10x1.5 mm²

=U7MKC10GH100
+U7MKC11

EXCITATION

TURBINE GOVERNOR SETPOINT
LOWER/HIGHER

-K101
/65.5
AUTOMATIC/TEST
SYNCHRONIZATION
ON

-S103
/168.4:E
REMOTE

-S221
/19.8
LOWER

-S221
/19.8
HIGHER

-F25G
SPM-D2-10
/143.2:B

-X90
-X332

-1K133
24VDC

-1K132
24VDC

+U7CBC10-W5025
NYCY 10x1.5 mm²

=U7CBC10GH100
+U7CBC10

TURBINE GOVERNOR

77

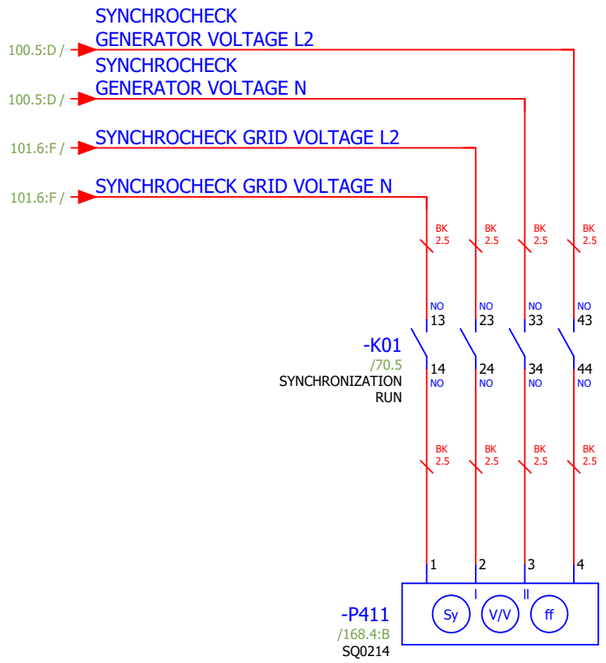
79

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.							+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				SYNCHRONIZATION	Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 78
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 81 / 204

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SYNCHROCHECK
VOLTAGE MEASUREMENT



Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram SYNCHRONIZATION	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 79
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 82 / 204

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LAMP TEST ON | EXCITATION ON | EXCITATION OFF | GENERATOR COOLING WATER VALVE OPEN | GENERATOR COOLING WATER VALVE CLOSED | TRANSFORMER OIL PUMPS ON | TRANSFORMER OIL PUMPS OFF

10.6:D / 42L+ 24VDC → -X42 A/B2 → A/B3 → A/B4 → 42L4 / 98.6:B; 42L+ 24VDC / 83.1:B

-S203 /176.4:B BLACK (NO 13/14) — -K220 /80.2 LAMP TEST ON (11/12/14) — -1K152 /36.2 (21/24/22) — -1K153 /36.3 — -1K192 /40.2 — -1K193 /40.3 — -1K202 /41.2 — -1K203 /41.3

42L+ 24VDC / 81.1:B; 42L1 / 98.6:A; 42L+ 24VDC / 81.1:C; 42L3 / 98.6:C

-K220 /165.4:E 24VDC LAMP TEST ON (A1/A2)

-S223 /177.4:F — -S224 /178.4:C — -S161 /175.4:C — -S162 /175.4:E — -S142 /173.4:C — -S143 /173.4:D (Sign_IN / Test_IN / Out)

-S223 /177.4:F 24 V AC/DC RED — -S224 /178.4:C 24 V AC/DC GREEN — -S161 /175.4:C 24 V AC/DC RED — -S162 /175.4:E 24 V AC/DC GREEN — -S142 /173.4:C 24 V AC/DC RED — -S143 /173.4:D 24 V AC/DC GREEN (X1/X2)

42L2 / 98.6:C; 42L- 24VDC / 81.1:E

10.6:D / 42L- 24VDC → -X42 B/11 A → B/12 A → 42L- 24VDC / 83.1:F; 42L1- 24VDC / 98.6:B

RXG22BD SE.RZM040W (14/11 /80.2:C; 12; 24/21 /83.1:C; 22)

79 81

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				SIGNALIZATION	Designer No.:	SALES No.: PAGE 80
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 83 / 204

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TRANSFORMER COOLING WATER VALVE OPEN

TRANSFORMER COOLING WATER VALVE CLOSED

SYNCHRONIZATION STOP

SYNCHRONIZATION RUN

DISCONNECTOR 1 SELECTED

DISCONNECTOR 2 SELECTED

GENERATOR CIRCUIT BREAKER SELECTED

80.8:B / 42L+ 24VDC

80.8:C / 42L+ 24VDC

80.8:E / 42L- 24VDC

-1K205 /41.5 — 21, 24, 22

-1K204 /41.4 — 21, 24, 22

-K01 /70.5 — 61, 62

-K01 /70.5 — 53, 54

-K102 /67.2 — 31, 34, 32

-K103 /67.4 — 31, 34, 32

-K104 /67.6 — 31, 34, 32

-S151 /174.4:C — Sign_IN, Test_IN, Out

-S152 /174.4:E — Sign_IN, Test_IN, Out

-S131 /172.4:C — Sign_IN, Test_IN, Out

-S130 /171.4:E — Sign_IN, Test_IN, Out

-S111 /169.4:D — Sign_IN, Test_IN, Out

-S112 /169.4:E — Sign_IN, Test_IN, Out

-S113 /169.4:E — Sign_IN, Test_IN, Out

-S151 /174.4:C 24 V AC/DC RED

-S152 /174.4:E 24 V AC/DC GREEN

-S131 /172.4:C 24 V AC/DC GREEN

-S130 /171.4:E 24 V AC/DC RED

-S111 /169.4:D 24 V AC/DC WHITE

-S112 /169.4:E 24 V AC/DC WHITE

-S113 /169.4:E 24 V AC/DC WHITE

RD 1.0, BU 1.0, X1, X2

42L+ 24VDC / 82.1:B

42L+ 24VDC / 82.1:C

42L- 24VDC / 82.1:E

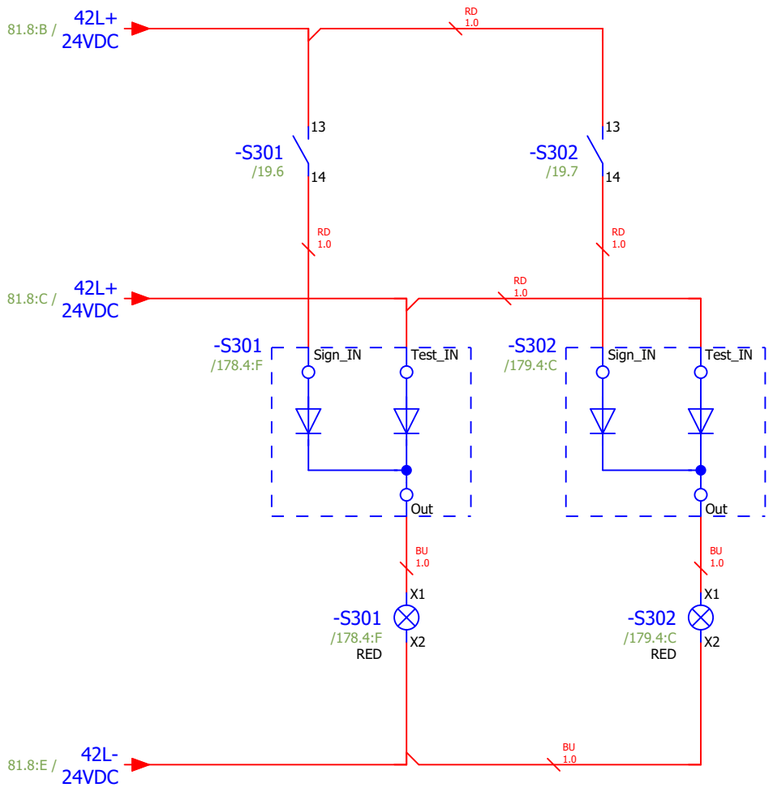
80

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				SIGNALIZATION	Designer No.:	SALES No.:	PAGE 81
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 84 / 204

EMERGENCY SHUTDOWN

FIRE EXTINGUISHING SYSTEM



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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram SIGNALIZATION	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 82 PAGE 85 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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DISCONNECTOR 1 OFF | DISCONNECTOR 1 ON | DISCONNECTOR 2 OFF | DISCONNECTOR 2 ON | GENERATOR CIRCUIT BREAKER OFF | GENERATOR CIRCUIT BREAKER ON

80.4:B / 42L+ 24VDC -X42 A B5 — A B6 42L+ 24VDC / 85.1:B

42L+ 24VDC / 84.1:B

-K220 /80.2 LAMP TEST ON: 21 22 24
-1K83 /29.3: 21 24 22
-K150 /164.4:E: 44 43
-1K85 /29.5: 21 24 22
-K151 /164.4:E: 44 43
-K205 /73.2: 21 24 22
-K206 /73.2: 21 24 22

RD 1.0

-X120: C A / B D — 13, 14, 15, 16, 17, 18

42L+ 24VDC / 84.1:D

-H101 /148.4:D
-H102 /148.4:D
-H103 /148.4:E
green X2, red X1, X0

BU 1.0

SYNC_TEST 72.1:E

42L- 24VDC / 84.1:E

80.4:F / 42L- 24VDC -X42 B 13 A

82 84

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska		PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				POSITION INDICATORS	Designer No.:	SALES No.:	PAGE 83
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 86 / 204

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BRAKING DISCONNECTOR OFF

BRAKING DISCONNECTOR ON

BRAKING EARTH SWITCH OFF

BRAKING EARTH SWITCH ON

TRANSFORMER NEUTRAL POINT DISCONNECTOR OFF

TRANSFORMER NEUTRAL POINT DISCONNECTOR ON

83.7:B / 42L+ 24VDC

-1K91 /30.1

-1K92 /30.2

-1K93 /30.3

-1K94 /30.4

-1K95 /30.5

-1K96 /30.6

83.7:D / 42L+ 24VDC

-X120

/ 85.1:D 42L+ 24VDC

-H104 /148.4:E

-H105 /148.4:E

-H106 /148.4:E

83.7:E / 42L- 24VDC

/ 85.1:E 42L- 24VDC

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram POSITION INDICATORS	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 84
Checked by	Antonio Aleksić	01.04.2026.							PAGE 87 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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EXCITATION MAIN CIRCUIT BREAKER

EXCITATION MAIN CIRCUIT BREAKER

EXCITATION AUXILIARY CIRUCIT BREAKER

EXCITATION AUXILIARY CIRUCIT BREAKER

ELECTRICAL BRAKING TRANSFORMER CIRCUIT BREAKER

ELECTRICAL BRAKING TRANSFORMER CIRCUIT BREAKER

83.2:B /
42L+
24VDC

-X42

+U7CJA10-W5202
NYCY 10x1.5 mm²

+U7BJA13-W5102
NYCY 2x1.5/1.5 mm²

=U7MKC10GH100
+U7MKC11

-X081

=U7BJA10GH100
+U7BJA13.QA001

400VAC
SUB-DISTRIBUTION
ELECTRICAL
BRAKING
TRANSFORMER

EXCITATION

+U7CJA10-W5202
NYCY 10x1.5 mm²

+U7BJA13-W5103
NYCY 2x1.5/1.5 mm²

84.7:D /
42L+
24VDC

-X120

-H107
/148.4:E

-H108
/148.4:E

-H109
/148.4:E

84.7:E /
42L-
24VDC

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram POSITION INDICATORS	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00
Approved by Designer	Grga Kolić	01.04.2026.						= U7CJA10GH100 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			PAGE 85 PAGE 88 / 204

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TURBINE GOVERNOR ALARM

EXCITATION ALARM OK

GENERATOR CIRCUIT BREAKER SF6 ALARM

24V DC POWER SUPPLY DEVICES ALARM

GENERATOR COOLING WATER VALVE OVERLOAD/BIMETAL WARNING

TRANSFORMER TEMPERATURE ALARM

TRANSFORMER COOLING VALVE READY

220VDC SYSTEM MCB TRIP GROUP - OK

10.5:D / 41L+ 24VDC -X41 A B 2 41L+ 24VDC / 88.1:B

RD 1.0

41L1 / 87.1:B

-1K174 /38.4

-1K151 /36.1

-1K97 /30.7

-K201 /17.2

-K202 /17.3

-1K197 /40.7

-1K208 /41.8

-1K228 /43.8

-1K75 /28.5

-1K76 /28.6

21 24 22

BU 1.0

10.5:D / 41L1- 24VDC 41L1- 24VDC / 87.7:D

-H01 /140.2:B

-H01:

E1 X10:1

E2 X10:2

E3 X10:3

E4 X10:4

E5 X10:5

E6 X10:6

E7 X10:7

E8 X10:8

-/N X10:9

Vaux +24V_1

+Vin

Vaux 0V_1

Lo

RD 1.5

BU 1.5

*Note: Must be wired

*Note: Must be wired

85

87

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska		PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				ALARM ANNUCIATOR - ALARM	Designer No.:	SALES No.: PAGE 86
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 89 / 204

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TURBINE GOVERNOR TRIP

EXCITATION THYRICON TRIP

GENERATOR CIRCUIT BREAKER SF6 TRIP

PROTECTION SYSTEM ACTIVATED

GENERATOR FIRE EXTINGUISHING SYSTEM TRIP

TRANSFORMER BUCHHOLZ TRIP

TRANSFORMER TEMPERATURE TRIP

PROTECTION SYSTEM INTERNAL FAULT

86.8:B / 41L1

RD 1.0

-1K175 /38.5

-1K155 /36.5

-1K98 /30.8

-1K141 /35.1

-1K144 /35.4

-1K198 /40.8

-1K207 /41.7

-1K211 /42.1

-1K142 /35.2

-1K145 /35.5

21 24 22

BU 1.0

86.8:D / 41L1- 24VDC

-H01 /140.2:B

E1 X12:1

E2 X12:2

E3 X12:3

E4 X12:4

E5 X12:5

E6 X12:6

E7 X12:7

E8 X12:8

-/N X12:9

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram ALARM ANNUCIATOR - TRIP	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 87
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 90 / 204

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SIREN WARNING ACTIVE ALARM ACTIVE UNIT RUNNING

86.2:B / 41L+ 24VDC -X41 A B 3

RD 1.0

-H01 /140.2:B
X2:8 X2:2 X2:5
RELAY 1 RELAY 2 RELAY 3
X2:7 X2:9 X2:1 X2:3 X2:4 X2:6

-K208 /73.3 UNIT RUNNING
11 14 12

-2K202 /50.3 AUTO START OR STOP ACTIVE
11 14 12

-X71 B A 5 13 14 15 16
+U7CJA10-W5005 NYCY 7x1.5 mm²
1 2 3 4

=OUT
+OUT
-H02
4 2 3 1 5
1 1 1 1
YELLOW RED GREEN
0 0 0 0 0

SIEMENS
8WD4408-0AA

OUTSIDE
+U7CJA10-W5005 NYCY 7x1.5 mm² 5
-X71 A B 6
BU 1.0

10.5:D / 41L- 24VDC -X41 B A 5 B A 6

87 89

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram SIGNAL COLUMNS	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00
Approved by Designer	Grga Kolić	01.04.2026.						= U7CJA10GH100
Checked by	Antonio Aleksić	01.04.2026.						+ U7CJA10
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		Designer No.: 22044-DD-EE-08-07-100	SALES No.:
								PAGE 88
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GENERATOR HEATERS ON COMMAND

GENERATOR COOLING WATER VALVE OPEN COMMAND

GENERATOR COOLING WATER VALVE CLOSE COMMAND

-2K206 /51.3

-2K104 /66.6

-2K103 /66.7

11 14 12

OG 1.5

-X71 B 7 A, B 8 A, B 9 A, B 10 A, B 11 A, B 12 A

+U7BJA13-W5092 NYCY 2x1.5/1.5 mm² BN BU

+U7CJA10-W5101 NYCY 12x1.5 mm² 7 8 9 10

=U7BJA10GH100
+U7BJA13.NB001

12 16

400VAC
SUB-DISTRIBUTION
GENERATOR
HEATER
CONTROL
CUBICLE

=7NU01
+SC44

-X3 9 12 10 11

GENERATOR COOLING

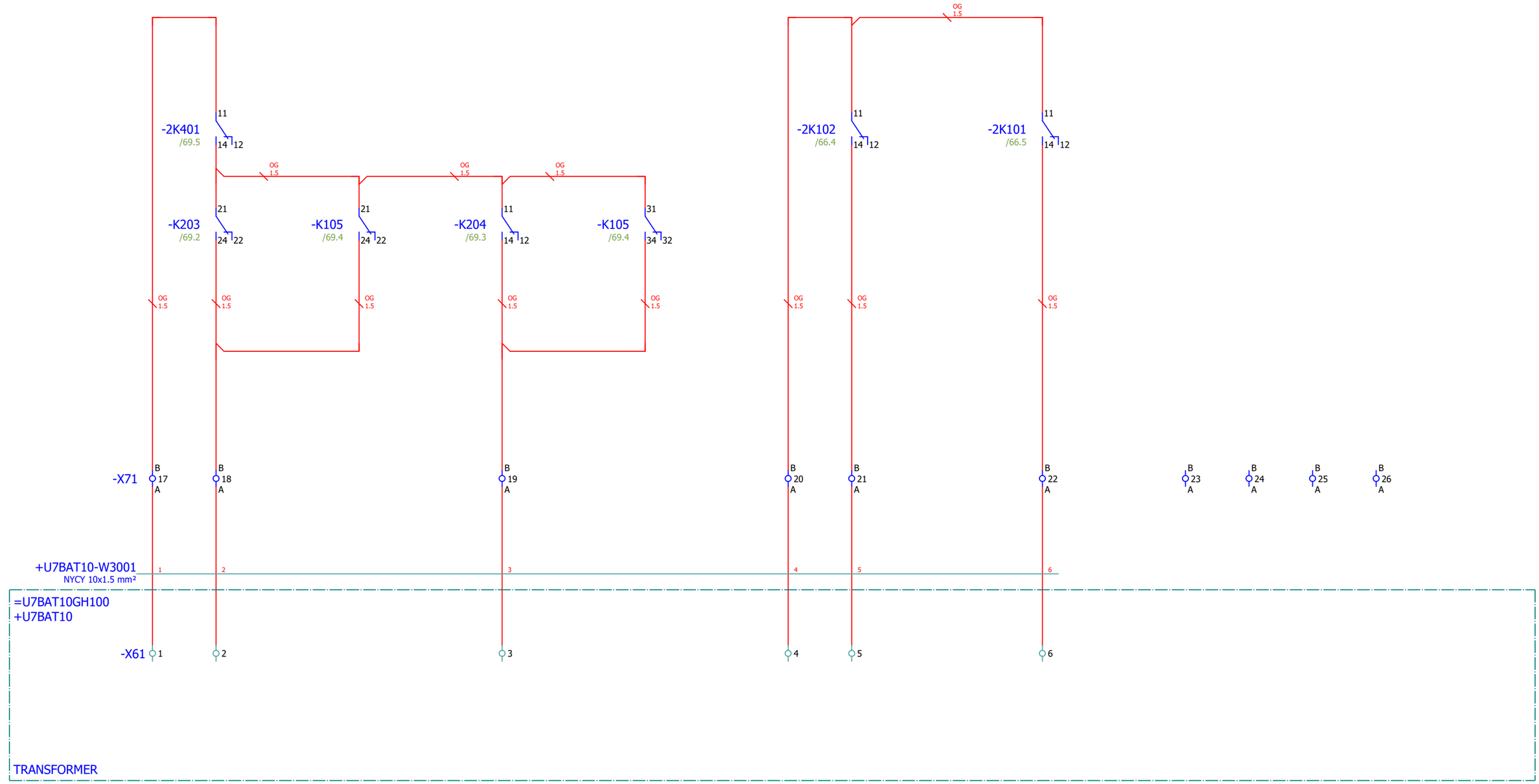
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska		PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				EXTERNAL CONTROL - GENERATOR	Designer No.:	SALES No.:	PAGE 89
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 92 / 204

TRANSFORMER OIL PUMP I
RUN COMMAND

TRANSFORMER OIL PUMP II
RUN COMMAND

TRANSFORMER
COOLING WATER VALVE
OPEN COMMAND

TRANSFORMER
COOLING WATER VALVE
CLOSE COMMAND



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89

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				EXTERNAL CONTROL - TRANSFORMATOR	Designer No.:	SALES No.: PAGE 90
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 93 / 204

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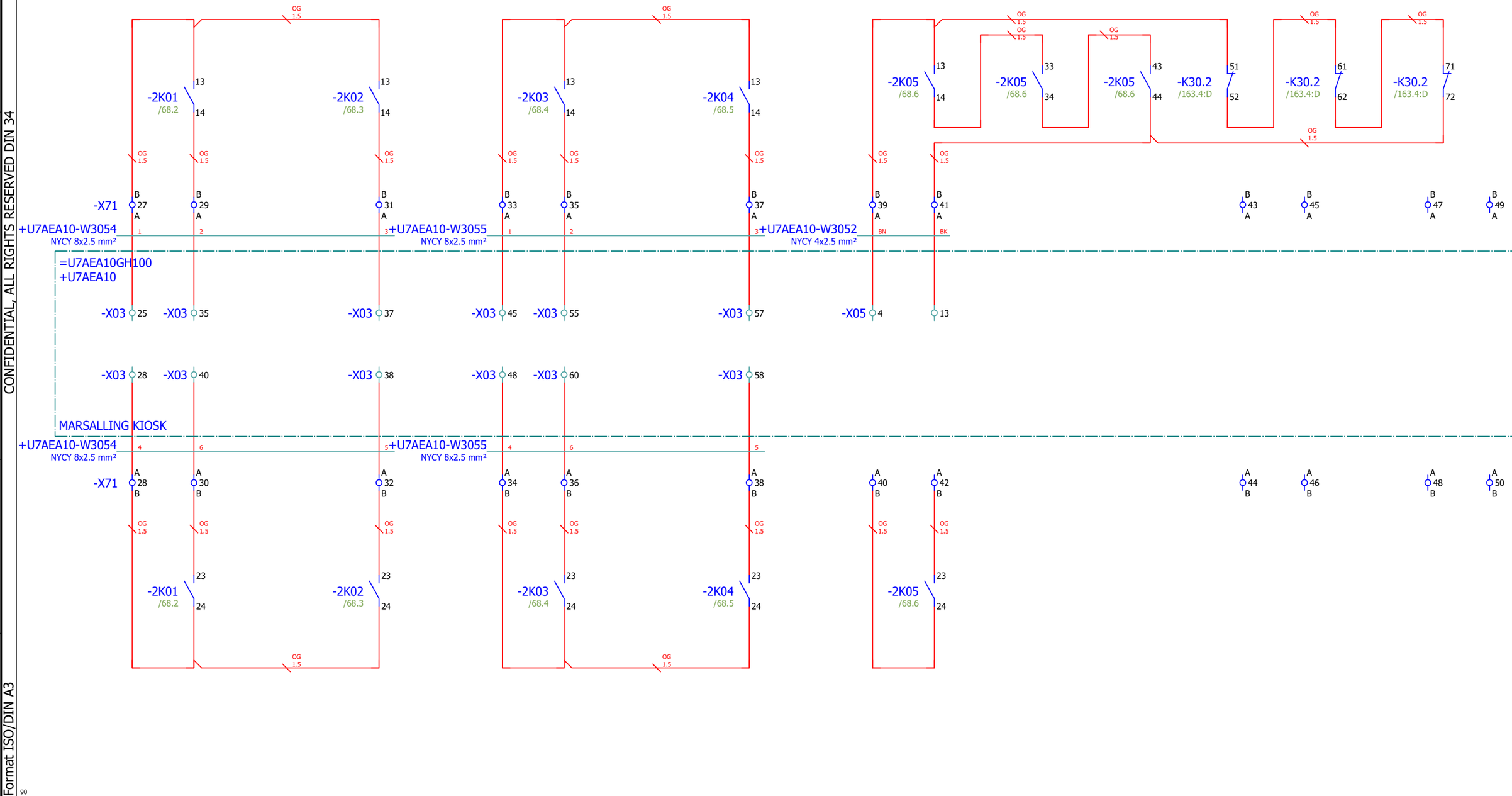
DISCONNECTOR 1 OFF COMMAND

DISCONNECTOR 1 ON COMMAND

DISCONNECTOR 2 OFF COMMAND

DISCONNECTOR 2 ON COMMAND

GENERATOR CIRCUIT BREAKER OFF COMMAND



Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram EXTERNAL CONTROL - MARSALLING KIOSK	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00
Approved by Designer	Grga Kolić	01.04.2026.						= U7CJA10GH100 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			PAGE 91 PAGE 94 / 204

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EXCITATION OFF COMMAND

EXCITATION ON COMMAND

SPARE (FCR MODE REQUEST)

SPARE (AVR MODE REQUEST)

SPARE (VAR MODE REQUEST)

SPARE (PF MODE REQUEST)

SPEED > EXCITATION ON LIMIT

-2K106 /66.2 — 11, 14, 12 — OG 1.5

-2K105 /66.3 — 11, 14, 12 — OG 1.5

-2K211 /53.1 — 11, 14, 12 — OG 1.5

-2K212 /53.2 — 11, 14, 12 — OG 1.5

-2K213 /53.3 — 11, 14, 12 — OG 1.5

-2K214 /53.4 — 11, 14, 12 — OG 1.5

-2K215 /53.5 — 11, 14, 12 — OG 1.5

-X71: B 51 A, B 52 A, B 53 A, B 54 A, B 55 A, B 56 A, B 57 A, B 58 A, B 59 A, B 60 A, B 61 A, B 62 A, B 63 A, B 64 A

+U7CJA10-W5203 NYCY 10x1.5 mm² — 4, 5, 6, 7, 8, 9

+U7CJA10-W5204 NYCY 10x1.5 mm² — 1, 2, 3, 4, 5, 6, 7, 8

=U7MKC10GH100
+U7MKC11

-X030 2, -X040 2, -X030 1, -X040 1, -X030 3, -X040 3, -X030 4, -X040 4, -X030 5, -X040 5, -X030 6, -X040 6, -X030 7, -X040 7

EXCITATION

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00 / = U7CJA10GH100 / + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company, Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	EXTERNAL CONTROL - EXCITATION	Designer No.: 22044-DD-EE-08-07-100	SALES No.: / PAGE 92 / PAGE 95 / 204
Checked by	Antonio Aleksić	01.04.2026.						
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			

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SPARE (EXCITATION UNLOAD COMMAND)

SPARE (EXCITATION ISLAND MODE ON COMMAND)

EXCITATION RESET

GRID VT MCB OK - SYNCHRONIZATION

SPARE (VT MCB FOR CONTROL SYSTEM (SYNCH) OK)

SYNCHRONIZATION ACTIVE

ELECTRICAL BRAKING ON RELEASE

-2K216 /53.6 — 11, 14, 12
-2K217 /53.7 — 11, 14, 12
-2K218 /53.8 — 11, 14, 12
-1K134 /34.4 — 21, 24, 22
-1K123 /33.3 — 21, 24, 22
-2K221 /54.3 — 11, 14, 12
-2K222 /54.4 — 11, 14, 12

OG 1.5

-X71: B 65 A, B 66 A, B 67 A, B 68 A, B 69 A, B 70 A, B 71 A, B 72 A, B 73 A, B 74 A, B 75 A, B 76 A, B 77 A, B 78 A

+U7CJA10-W5204 NYCY 10x1.5 mm² — 9, 10, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 — +U7CJA10-W5205 NYCY 10x1.5 mm²

=U7MKC10GH100
+U7MKC11

-X030 8, -X040 8
-X030 9, -X040 9
-X030 10, -X040 10
-X030 14, -X040 14
-X030 15, -X040 15
-X030 17, -X040 17

EXCITATION

92

94

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska		PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				EXTERNAL CONTROL - EXCITATION	Designer No.:	SALES No.: PAGE 93
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 96 / 204

SPARE (SPEED < ELECTRICAL BRAKING STOP LIMIT)

EXCITATION MAIN AND AUXILIARY CIRCUIT BREAKER TRIP COIL 1

DEEXCITATION FORCING

EXCITATION MAIN AND AUXILIARY CIRCUIT BREAKER TRIP COIL 2

DEEXCITATION FORCING

-2K223 /54.5 — 11, 14, 12

-K30.1 /76.a.4 EXCITATION MAIN AND AUXILIARY CIRCUIT BREAKER TRIP COIL 1 — 51, 52

-K30.1 /76.a.4 DEEXCITATION FORCING — 61, 62

-K30.1 /76.a.4 EXCITATION MAIN AND AUXILIARY CIRCUIT BREAKER TRIP COIL 2 — 21, 22

-K30.1 /76.a.4 DEEXCITATION FORCING — 31, 32

OG 1.5

-X71 B 79 A, B 80 A, B 81 A, B 82 A, B 83 A, B 84 A, B 85 A, B 86 A

+U7CJA10-W5206
NYCY 10x1.5 mm²
6, 7, 8, 9, 4, 5

=U7MKC10GH100
+U7MKC11

-X030 20, -X040 20, -X080 2, 5, 6, 10, 13, 14

EXCITATION

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93

95

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00 / = U7CJA10GH100 / + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company, Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	EXTERNAL CONTROL - EXCITATION	Designer No.: 22044-DD-EE-08-07-100	SALES No.:
Checked by	Antonio Aleksić	01.04.2026.					PAGE 94	
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		PAGE 97 / 204	

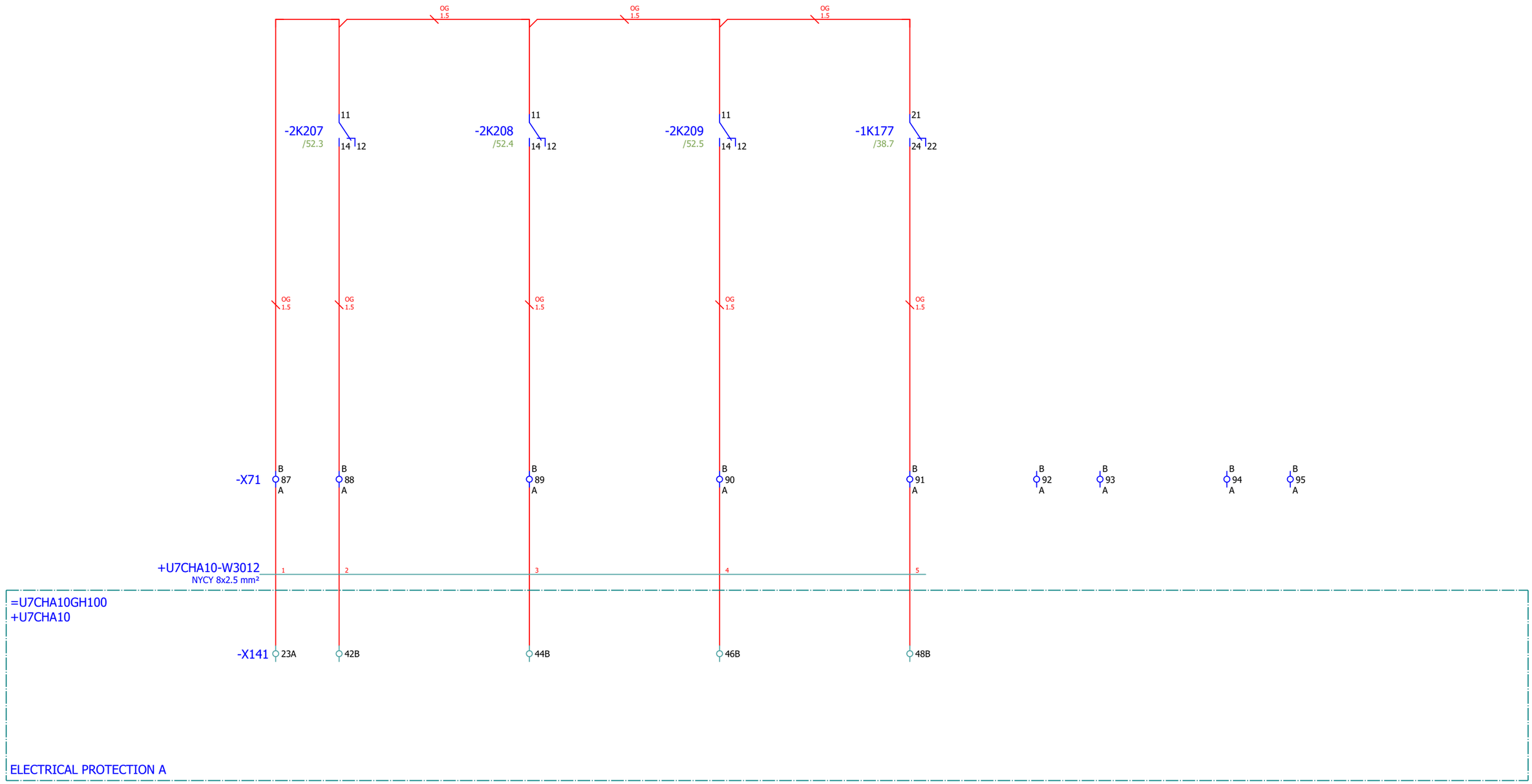
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GENERATOR TEMPERATURE TRIP

ROTOR TEMPERATURE TRIP

TRANSFORMER OIL TEMPERATURE TRIP

VIBRATION MONITORING TRIP



Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00 / SALES No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska				+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				EXTERNAL CONTROL - GENERATOR PROTECTION	Designer No.: 22044-DD-EE-08-07-100		PAGE 95
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 98 / 204

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AC MAIN CB ON COMMAND

AC MAIN CB OFF COMAND

AC AUXILIARY CB ON COMMAND

AC AUXILIARY CB OFF COMMAND

BUSBAR DUCT FAN ON COMMAND

GRAPHITE DUST SUCTION SYSTEM ON COMMAND

-2K225 /54.7 — 11, 14, 12

-2K226 /54.8 — 11, 14, 12

-2K227 /55.1 — 11, 14, 12

-2K228 /55.2 — 11, 14, 12

-2K229 /55.3 — 11, 14, 12

-2K231 /55.5 — 11, 14, 12

OG 1.5

-X71: B 96 A, B 97 A, B 98 A, B 99 A, B 100 A, B 101 A, B 102 A, B 103 A, B 104 A, B 105 A, B 106 A, B 107 A

+U7BJA11-W5022 NYCY 7x1.5 mm² — 1, 2, 3, 4

=U7BJA10GH100
+U7BJA11.EB001

-X21: 10, 11, 12, 13

MAIN INCOMING SUPPLY +NE4

+U7BJA11-W5032 NYCY 7x1.5 mm² — 1, 2, 3, 4

=U7BJA10GH100
+U7BJA11.LB001

-X31: 6, 7, 8, 9

AUXILIARY INCOMING SUPPLY +NE16

+U7CJA10-W5004 NYCY 10x1.5 mm² — 6, 7

=U7SAA10GH100
+U7SAA10

-X61: 3, 4

BUSBAR DUCT FAN
CONTROL CUBICLE

+U7BJA13-W5132 NYCY 2x1.5/1.5 mm² — BN, BU

=U7BJA10GH100
+U7BJA13.FA002

17, 18

400VAC SUB-DISTRIBUTION
GRAPHITE DUST SUCTION
SYSTEM

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska		PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				EXTERNAL CONTROL - 400VAC SUB-DISTRIBUTION	Designer No.:	SALES No.: PAGE 96
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 99 / 204

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DRAINAGE UNDER GENERATOR ON COMMAND

DRAINAGE UNDER TURBINE A ON COMMAND

DRAINAGE UNDER TURBINE B ON COMMAND

-2K232 /55.6
11
14 12

-2K233 /55.7
11
14 12

-2K234 /55.8
11
14 12

OG 1.5
OG 1.5

-X71
B 108 A
B 109 A
B 110 A
B 111 A
B 112 A
B 113 A
B 114 A
B 115 A
B 116 A
B 117 A

+U7BJA13-W5142
NYCY 2x1.5/1.5 mm²
BN BU

+U7BJA13-W5152
NYCY 2x1.5/1.5 mm²
BN BU

+U7BJA13-W5162
NYCY 2x1.5/1.5 mm²
BN BU

=U7BJA10GH100
+U7BJA13.GA002
15 16
400VAC SUB-DISTRIBUTION DRAINAGE UNDER GENERATOR

=U7BJA10GH100
+U7BJA13.HA002
15 16
400VAC SUB-DISTRIBUTION DRAINAGE UNDER TURBINE A

=U7BJA10GH100
+U7BJA13.JB002
15 16
400VAC SUB-DISTRIBUTION DRAINAGE UNDER TURBINE B

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram EXTERNAL CONTROL - 400VAC SUB-DISTRIBUTION	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00
Approved by Designer	Grga Kolić	01.04.2026.						= U7CJA10GH100 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			PAGE 97 PAGE 100 / 204

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ANY TRANSFORMER OIL PUMP
ON

-X61 A/B 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96

-X110 A/B 5 6 7 8 9 10 11 12

-X111 A/B 1 2 3 4 5 6 7 8 9 10 11 12

80.8:C / 42L1 → A1 -K169 /165.4:C 24VDC A2 ← 80.4:F / 42L1- 24VDC

80.4:A / 42L4 → 11 -K169 /98.6 14 12 → 80.8:E / 42L2

12 → 80.8:C / 42L3

RXG22BD
SE.RZM040W
14 — 11 /98.6:C
12
24 — 21
22

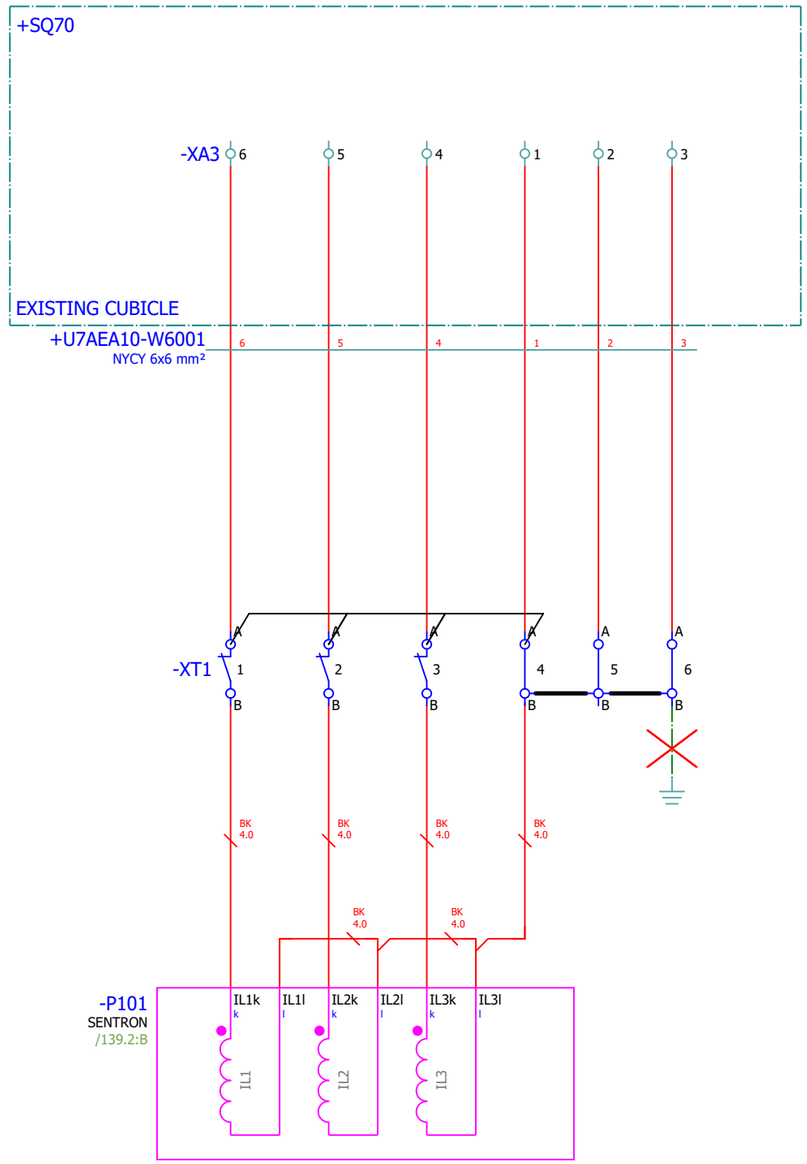
97 99

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska		PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				SPARE CONTACTS	Designer No.:	SALES No.: PAGE 98
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 101 / 204

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MULTIMETER



Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska		PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				CT CIRCUITS	Designer No.:	SALES No.: PAGE 99
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 102 / 204

Format ISO/DIN A3

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SYNCHRONIZATION
110kV GENERATOR SIDE VOLTAGE

=U7AEA10GH100
+U7AEA10

-XU3 4 5 6 -XU1 11

MARSALLING KIOSK
+U7AEA10-W6504
NYCY 4x2.5 mm²

BN BK GY BU

-XT2 1 2 3 4 5 6 7 8 9 10 11 12

BK 2.5

SENTRON VOLTAGE L1 103.2:C
SENTRON VOLTAGE L2 103.2:D
SENTRON VOLTAGE L3 103.2:D
SENTRON VOLTAGE N 103.2:D
SYNCHROCHECK GENERATOR VOLTAGE L2 79.3:B
SYNCHROCHECK GENERATOR VOLTAGE N 79.3:C

101.6:F / SYNCHRONIZER MAINS VOLTAGE L
101.6:F / SYNCHRONIZER MAINS VOLTAGE N

-F25G
SPM-D2-10
/143.2:B

20 UG 21 23 UB 24

99 101

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram VT CIRCUITS	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 100
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 103 / 204

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110kV BUSBAR SIDE VOLTAGE

=U7AEA10GH100
+U7AEA10

MARSALLING KIOSK

-XU5 21 22 2

+U7AEA10-W6507
NYCY 4x2.5 mm²

BU BK BN

-XT3 1 2 3 4 5 6 7 8 9 10 11 12 13

BK 2.5

=U7CBC10GH100
+U7CBC10

TURBINE GOVERNOR

-XT7 5 6

+U7CBC10-W6501
NYCY 4x4 mm²

BN BU

BUS_N 101.3:E

VOLTMETER BUSBAR I L 102.3:C

VOLTMETER BUSBAR I N 102.3:D

VOLTMETER BUSBAR II L 102.5:C

VOLTMETER BUSBAR II N 102.5:D

-K150 /164.4:E 14 13

-K151 /164.4:E 14 13

-K151 /164.4:E 31 32

-K150 /164.4:E 31 32

-XT4 1 2 3 4

BUS_N 101.6:C

EXCITATION L 103.6:C

EXCITATION N 103.6:D

SYNCHROCHECK GRID VOLTAGE L2 79.3:C

SYNCHROCHECK GRID VOLTAGE N 79.3:C

SYNCHRONIZER MAINS VOLTAGE L 100.5:E

SYNCHRONIZER MAINS VOLTAGE N 100.5:E

1	2	3
UL1	UL2	N
-A257	-A257	-A257
/138.3:B	/138.3:B	/138.3:C
110kV BUSBAR 1 VOLTAGE - L2	110kV BUSBAR 2 VOLTAGE - L2	110kV BUSBAR VOLTAGE - N

110kV BUSBAR 1 VOLTAGE - L2

110kV BUSBAR 2 VOLTAGE - L2

110kV BUSBAR VOLTAGE - N

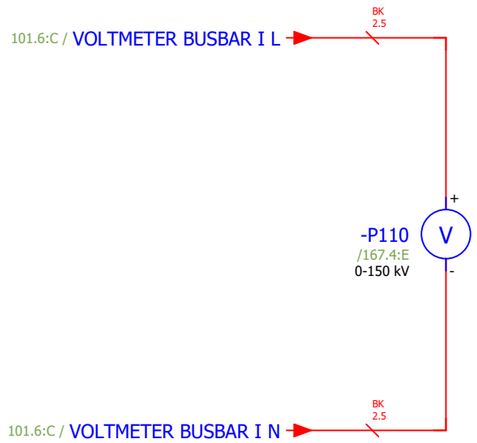
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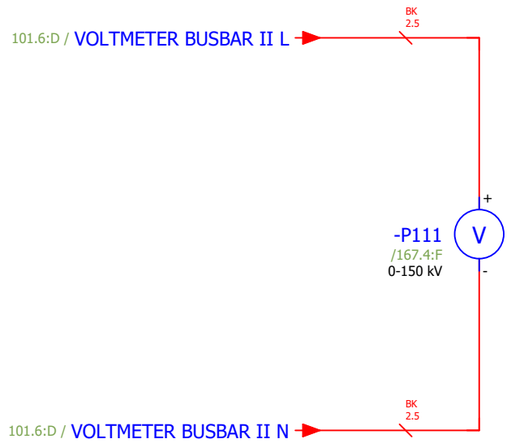
102

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				VT CIRCUITS	Designer No.:	SALES No.:	PAGE 101
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 104 / 204

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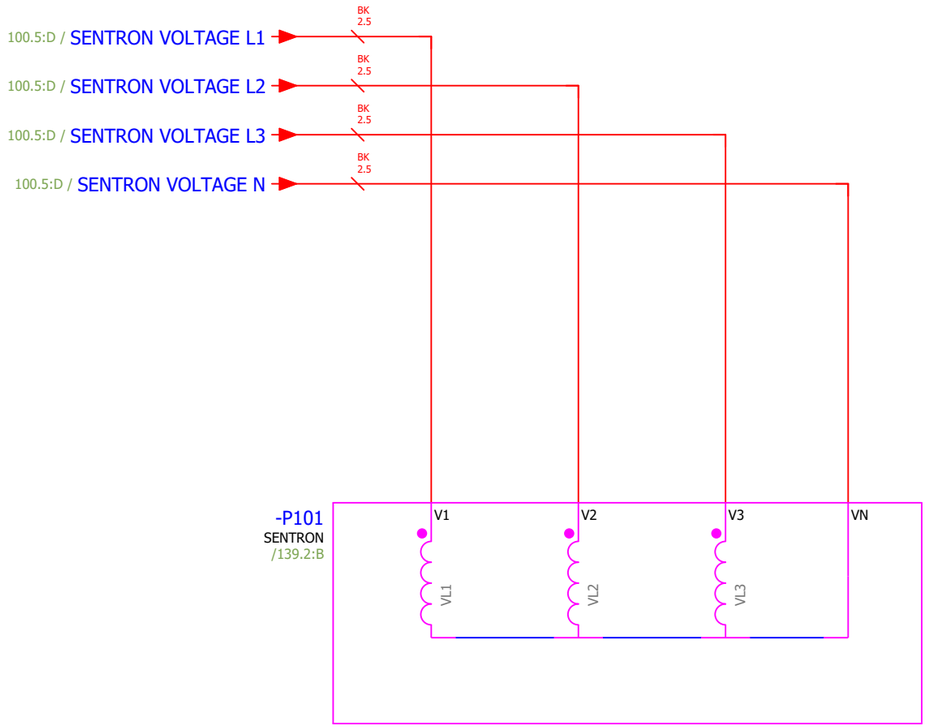


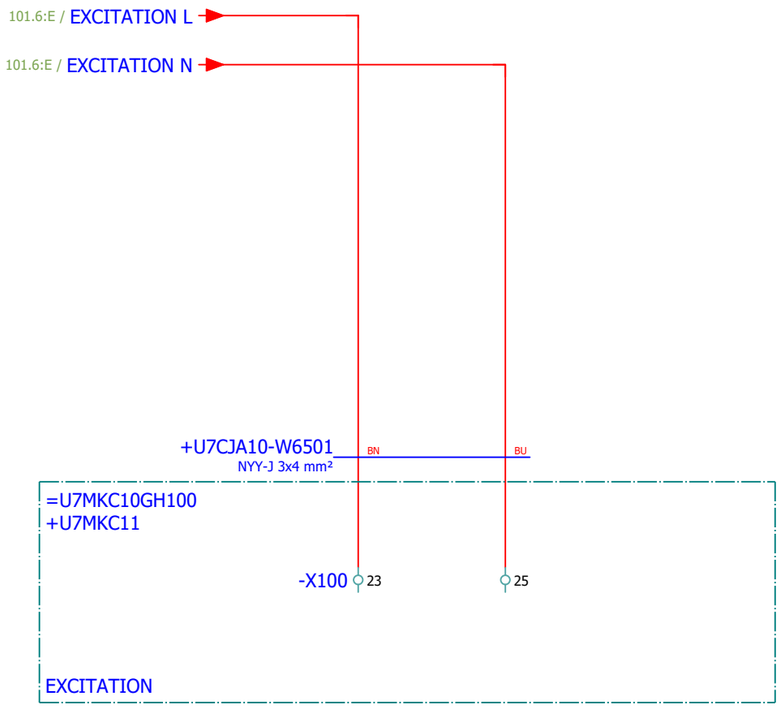


Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram VT CIRCUITS	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.					Designer No.:	SALES No.:	PAGE 102
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 105 / 204

Format ISO/DIN A3

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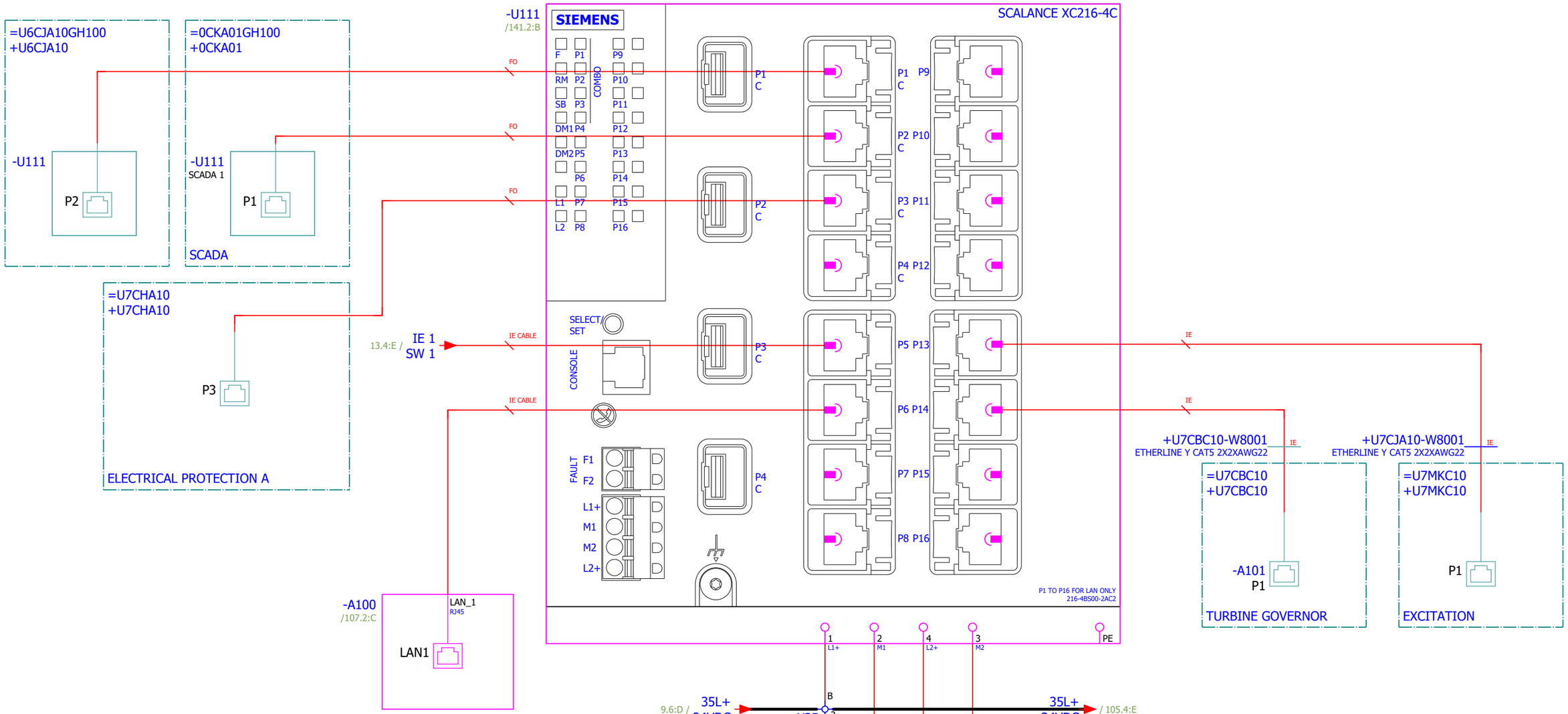




Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska		PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				VT CIRCUITS	Designer No.:	SALES No.: PAGE 103
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 106 / 204

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00
Approved by Designer	Grga Kolić	01.04.2026.						= U7CJA10GH100
Checked by	Antonio Aleksić	01.04.2026.				COMMUNICATION	Designer No.: 22044-DD-EE-08-07-100	+ U7CJA10
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		SALES No.:	PAGE 104
								PAGE 107 / 204

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=U6CJA10GH100
+U6CJA10

-U112
P2

=0CKA01GH100
+0CKA01

-U112
SCADA 2
P1

SCADA

=U7CHB10
+U7CHB10

P3

ELECTRICAL PROTECTION B

-U112
/142.2:B

SIEMENS

SCALANCE XC216-4C

F P1 P9
RM P2 COMBO P10
SB P3 P11
DM1 P4 P12
DM2 P5 P13
P6 P14
L1 P7 P15
L2 P8 P16

FO
FO
FO

P1 C
P2 C
P3 C
P4 C

P1 P9 C
P2 P10 C
P3 P11 C
P4 P12 C
P5 P13 C
P6 P14 C
P7 P15 C
P8 P16 C

SELECT/SET
CONSOLE

IE 2
SW 2
13.5:E /

IE CABLE
IE CABLE

FAULT F1
F2

L1+
M1
M2
L2+

P1 TO P16 FOR LAN ONLY
216-4BS00-2AC2

IE
IE

+U7CBC10-W8002
ETHERLINE Y CAT5 2X2XAWG22
IE

+U7CJA10-W8002
ETHERLINE Y CAT5 2X2XAWG22
IE

=U7CBC10
+U7CBC10

-A102
P1

TURBINE GOVERNOR

=U7MKC10
+U7MKC10

P1

EXCITATION

-A100
/107.2:C

LAN_2
RJ45

LAN2

PE

1 L1+
2 M1
4 L2+
3 M2

104.6:E / 35L+ 24VDC
-X35 B 3 A

104.6:E / 35L- 24VDC
-X35 B 6 A

104.6:F / 36L+ 24VDC
-X36 B 3 A

104.6:F / 36L- 24VDC
-X36 B 6 A

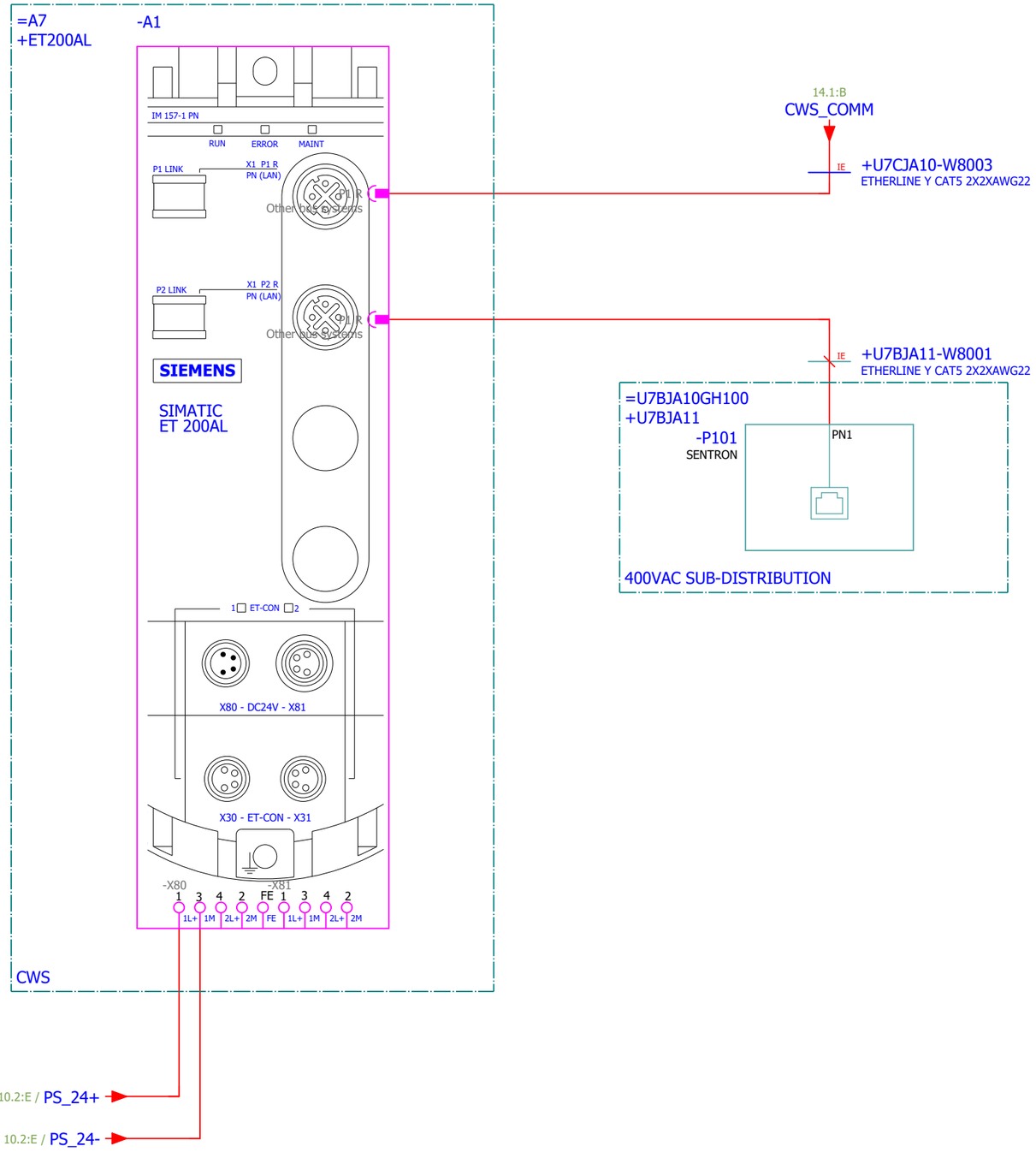
104

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00
Approved by Designer	Grga Kolić	01.04.2026.						= U7CJA10GH100
Checked by	Antonio Aleksić	01.04.2026.				COMMUNICATION	Designer No.: 22044-DD-EE-08-07-100	+ U7CJA10
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		SALES No.:	PAGE 105
								PAGE 108 / 204

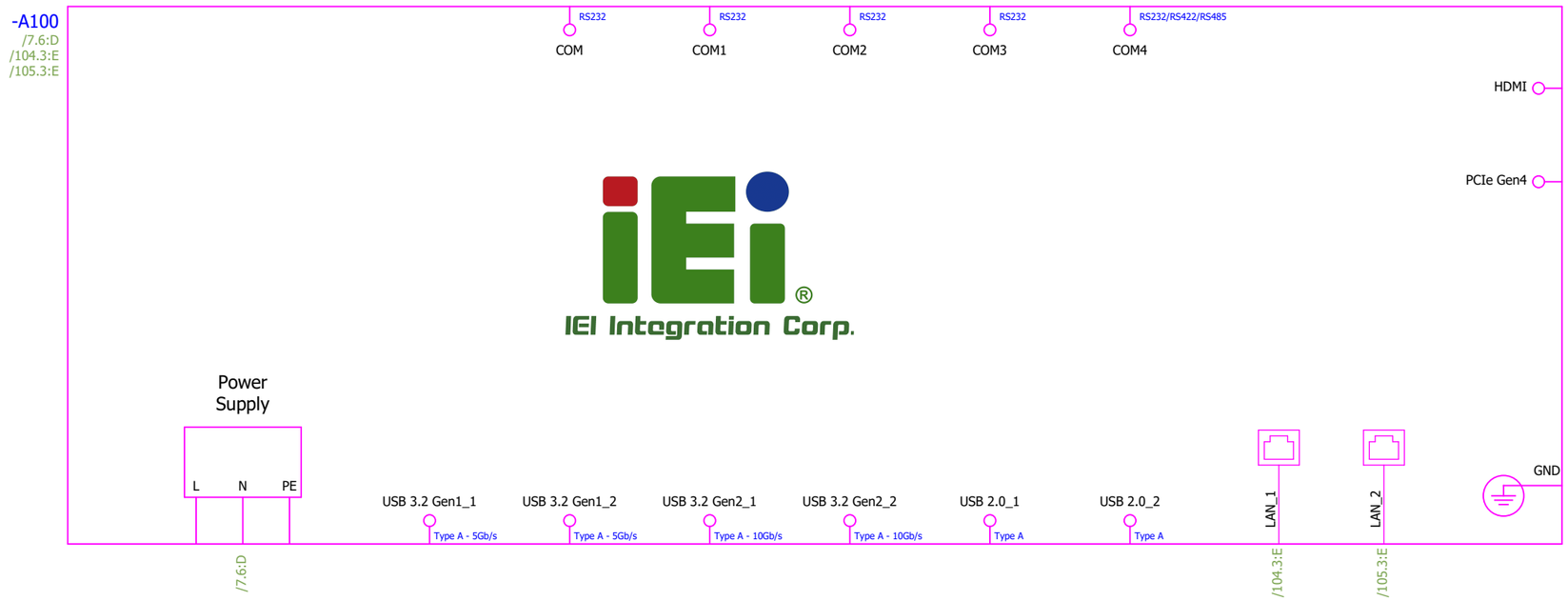
Format ISO/DIN A3

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.							+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				COMMUNICATION	Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 106
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 109 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	21.5” High performance Panel PC 21.5" 350 cd/m² 1920x1280 panel PC with 12th Intel® Core™ i5-12400 processor (18M cache, up to 4.40 GHz), TDP 65W, 8GB DDR4, HDMI output, 1 x PCIe Gen4 x16 slot, 1 x M.2 E key, 1 x M.2 M key, ATX/AT power supply, PCAP touchscreen iEi PPC2-CW22-ADL-i5/4G-R10	Integration Corp.	+U7CJA10	-A100	



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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram PANEL PC	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 107
Checked by	Antonio Aleksić	01.04.2026.							PAGE 110 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Power supply module, Simatic S7-1500 PS 25W Order No.: 6ES7505-0KA00-0AB0	Siemens	+U7CJA10	-A101	

-A101
/13.2:A

-X80:1 L+ /13.2:D

-X80:2 M /13.2:D

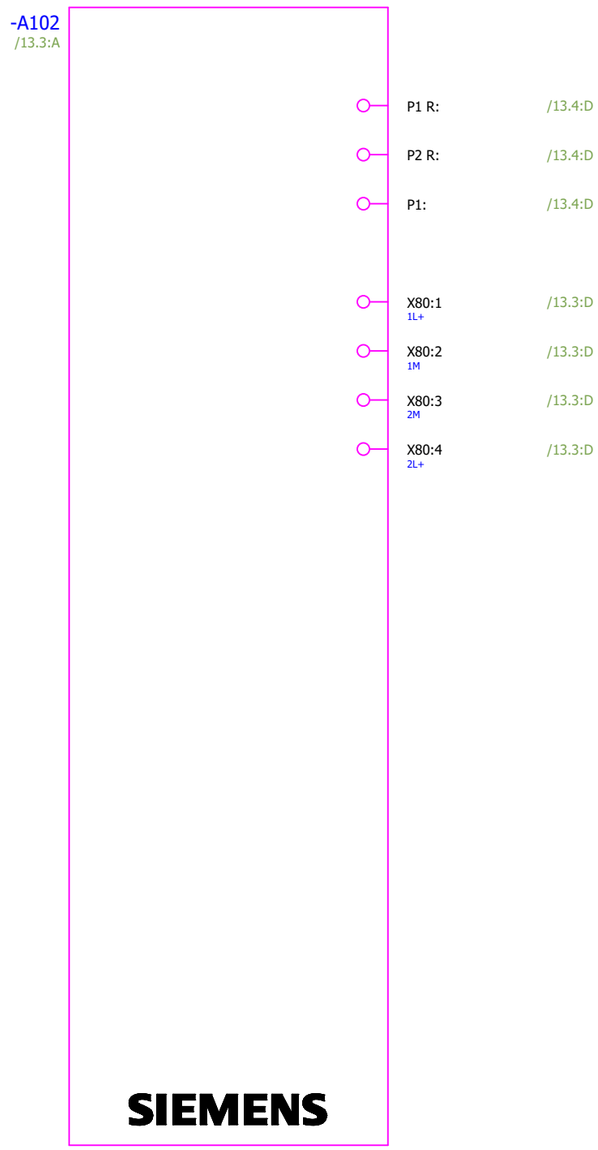
-X80:3 PE /13.3:D

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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram PLC OVERVIEW	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 108
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 111 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	CPU 1515-2 PN, 1MB Prog., 4,5MB Data CPU 1515-2 PN, 1MB for prog./ 4.5 MB Data PROFINET IRT with 2-port switch 6 ns bit performance 6ES7515-2AN03-0AB0 Memory card 24 MB for S7-1X00 CPU Order No.: 6ES7954-8LF03-0AA0	Siemens	+U7CJA10	-A102	

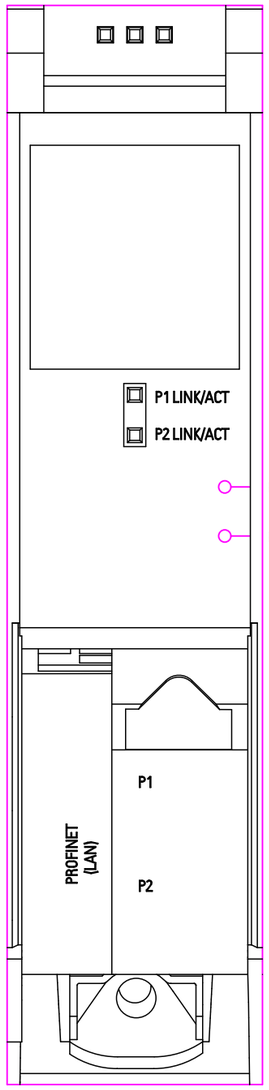


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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				PLC OVERVIEW	Designer No.:	SALES No.:	PAGE 109
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 112 / 204

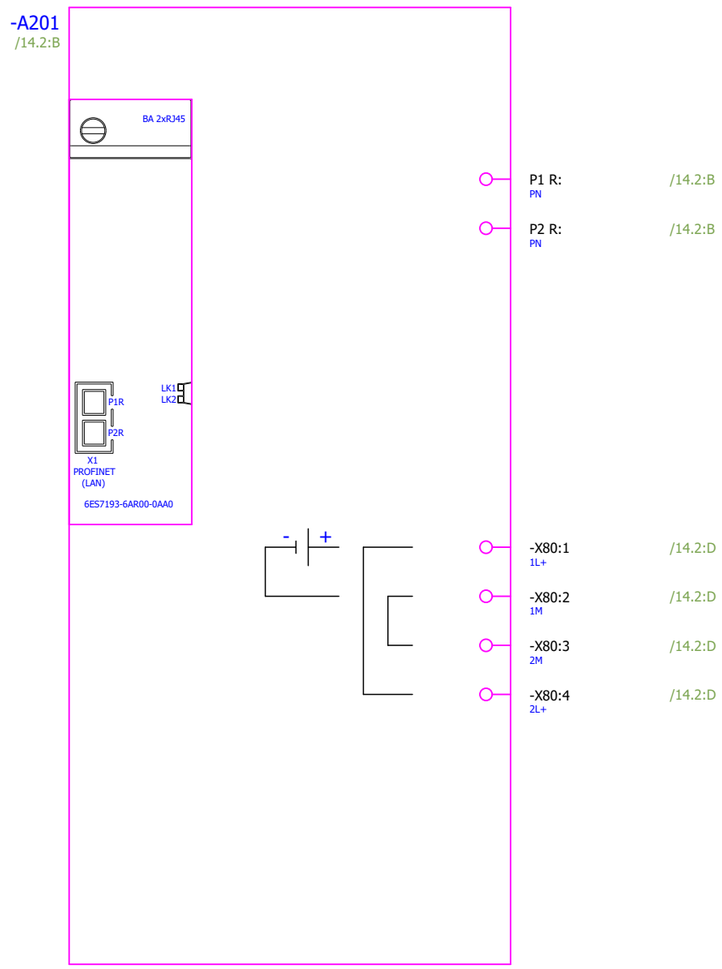
Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	COMMUNICATIONS MODULE CM 1542-1 COMMUNICATIONS MODULE CM 1542-1 FOR CONNECTING S7-1500 TO PROFINET AS IO-CONTROLLER: TCP/IP, ISO-ON-TCP, UDP, S7-COMMUNICATION, IP-BROADCAST/ MULTICAST, SNMPV1, DHCP, 6GK7542-1AX00-0XE0	Siemens	+U7CJA10	-A103	



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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				PLC OVERVIEW	Designer No.:	SALES No.:	PAGE 110
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 113 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Interface module ET200SP PROFINET, 2-port interface module IM 155-6PN/2 1 BusAdapter slot max. 64 I/O modules 6ES7155-6AU01-0CN0 ET 200SP, BA 2XRJ45 6ES7193-6AR00-0AA0	Siemens	+U7CJA10	-A201	



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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				PLC OVERVIEW	Designer No.:	SALES No.:	PAGE 111
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 114 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: DI 16X24V DC Standard Order No.: 6ES7131-6BH01-0BA0 Baseunit Type A0, BU15-P16+A0+2D 6ES7193-6BP00-0DA0	Siemens	+U7CJA10	-A211	

-A211
/14.3:B

I0.0	1 DI0	/18.1:E	REMOTE CONTROL
I0.1	2 DI1	/18.2:E	MANUAL CONTROL
I0.2	3 DI2	/18.3:E	AUTOMATIC CONTROL
I0.3	4 DI3	/18.4:E	AUTOMATIC SYNCHRONIZER READY
I0.4	5 DI4	/18.5:E	SYNCHRONIZATION MANUAL
I0.5	6 DI5	/18.6:E	SYNCHRONIZATION AUTOMATIC
I0.6	7 DI6	/18.7:E	DISCONNECTOR 1 SELECTED
I0.7	8 DI7	/18.8:E	DISCONNECTOR 2 SELECTED
I1.0	9 DI8	/19.1:E	GENERATOR CIRCUIT BREAKER SELECTED
I1.1	10 DI9	/19.2:E	OFF COMMAND
I1.2	11 DI10	/19.3:E	ON COMMAND
I1.3	12 DI11	/19.4:E	TRANSFORMER OIL PUMP 1 SELECTED
I1.4	13 DI12	/19.5:E	TRANSFORMER OIL PUMPS 1 & 2 SELECTED
I1.5	14 DI13	/19.6:E	ESD PUSHBUTTON OK
I1.6	15 DI14	/19.7:E	FIRE EXTINGUISHING SYSTEM PUSHBUTTON OK
I1.7	16 DI15	/19.8:E	SPEED LOWER
	L+ 24VDC	/14.3:D	
	M 0VDC	/14.3:D	

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00
Approved by Designer	Grga Kolić	01.04.2026.						= U7CJA10GH100
Checked by	Antonio Aleksić	01.04.2026.				PLC OVERVIEW	Designer No.: 22044-DD-EE-08-07-100	+ U7CJA10
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		SALES No.:	PAGE 112
								PAGE 115 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: DI 16X24V DC Standard Order No.: 6ES7131-6BH01-0BA0 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A212	

-A212
/14.4:B

I2.0	1 DI0	/20.1:E	SPEED HIGHER
I2.1	2 DI1	/20.2:E	VOLTAGE LOWER
I2.2	3 DI2	/20.3:E	VOLTAGE HIGHER
I2.3	4 DI3	/20.4:E	DC CONVERTER 1 OK
I2.4	5 DI4	/20.5:E	DC CONVERTER 2 OK
I2.5	6 DI5	/20.6:E	SPARE
I2.6	7 DI6	/20.7:E	SWITCH 1 OK
I2.7	8 DI7	/20.8:E	SWITCH 2 OK
I3.0	9 DI8	/21.1:E	230 VAC MCB OK - ALARM
I3.1	10 DI9	/21.2:E	230 VAC UPS MCB OK - ALARM
I3.2	11 DI10	/21.3:E	220 VDC MCB ALARM GROUP - OK
I3.3	12 DI11	/21.4:E	24 VDC MCB ALARM GROUP - OK
I3.4	13 DI12	/21.5:E	24 VDC MCB TRIP GROUP - OK
I3.5	14 DI13	/21.6:E	SYNCHRONIZATION TEST
I3.6	15 DI14	/21.7:E	SPARE
I3.7	16 DI15	/21.8:E	SPARE
	L+ 24VDC	/14.4:D	
	M 0VDC	/14.4:D	

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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				PLC OVERVIEW	Designer No.:	SALES No.: PAGE 113
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 116 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: DI 16X24V DC Standard Order No.: 6ES7131-6BH01-0BA0 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A213	

-A213
/14.4:B

I4.0	1 DI0	/22.1:E	AC MAIN SUPPLY REMOTE
I4.1	2 DI1	/22.2:E	AC MAIN SUPPLY CB OFF
I4.2	3 DI2	/22.3:E	AC MAIN SUPPLY CB INSERTED
I4.3	4 DI3	/22.4:E	AC MAIN SUPPLY CB READY
I4.4	5 DI4	/22.5:E	AC AUXILIARY SUPPLY CB OFF
I4.5	6 DI5	/22.6:E	AC AUXILIARY SUPPLY CB INSERTED
I4.6	7 DI6	/22.7:E	AC VOLTAGE OK
I4.7	8 DI7	/22.8:E	AC VOLTAGE OK MCB OK
I5.0	9 DI8	/23.1:E	AC SURGE ARRESTER OK
I5.1	10 DI9	/23.2:E	230 VAC MARSHALLING KIOSK MCB ALARM GROUP - OK
I5.2	11 DI10	/23.3:E	GENERATOR COOLING WATER VALVE FEEDER INSERTED
I5.3	12 DI11	/23.4:E	GENERATOR COOLING WATER VALVE READY
I5.4	13 DI12	/23.5:E	TRANSFORMER COOLING WATER VALVE FEEDER INSERTED
I5.5	14 DI13	/23.6:E	TRANSFORMER COOLING WATER VALVE READY
I5.6	15 DI14	/23.7:E	EXCITATION COOLING FANS & AC FIELD FLASHING FEEDER INSERTED
I5.7	16 DI15	/23.8:E	EXCITATION COOLONG FANS & AC FIELD FLASHING READY
L+	24VDC	/14.4:D	
M	0VDC	/14.4:D	

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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram PLC OVERVIEW	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 114 PAGE 117 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: DI 16X24V DC Standard Order No.: 6ES7131-6BH01-0BA0 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A214	

-A214
/14.4:B

I6.0	1 DI0	/24.1:E	BUS DUCT FAN FEEDER INSERTED
I6.1	2 DI1	/24.2:E	BUS DUCT FAN REMOTE
I6.2	3 DI2	/24.3:E	BUS DUCT FAN AUTO
I6.3	4 DI3	/24.4:E	BUS DUCT FAN ON
I6.4	5 DI4	/24.5:E	BUS DUCT FAN READY 1
I6.5	6 DI5	/24.6:E	BUS DUCT FAN READY 2
I6.6	7 DI6	/24.7:E	GRAPHITE DUST SUCTION SYSTEM FEEDER INSERTED
I6.7	8 DI7	/24.8:E	GRAPHITE DUST SUCTION SYSTEM REMOTE
I7.0	9 DI8	/25.1:E	BUSBAR DUCT VOLTAGE OK
I7.1	10 DI9	/25.2:E	GRAPHITE DUST SUCTION SYSTEM ON
I7.2	11 DI10	/25.3:E	GRAPHITE DUST SUCTION SYSTEM READY
I7.3	12 DI11	/25.4:E	GRAPHITE DUST SUCTION SYSTEM SPARE
I7.4	13 DI12	/25.5:E	GENERATOR HEATERS FEEDER INSERTED
I7.5	14 DI13	/25.6:E	GENERATOR HEATERS REMOTE
I7.6	15 DI14	/25.7:E	GENERATOR HEATERS AUTO
I7.7	16 DI15	/25.8:E	GENERATOR HEATERS ON
	L+ 24VDC	/14.5:D	
	M 0VDC	/14.5:D	

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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram PLC OVERVIEW	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 115
Checked by	Antonio Aleksić	01.04.2026.							PAGE 118 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

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Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: DI 16X24V DC Standard Order No.: 6ES7131-6BH01-0BA0 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A215	

-A215
/14.5:B

I8.0	1 DI0	/26.1:E	GENERATOR HEATER READY
I8.1	2 DI1	/26.2:E	GENERATOR HEATERS SPARE
I8.2	3 DI2	/26.3:E	DRAINAGE UNDER GENERATOR FEEDER INSERTED
I8.3	4 DI3	/26.4:E	SPARE
I8.4	5 DI4	/26.5:E	DRAINAGE UNDER GENERATOR AUTO
I8.5	6 DI5	/26.6:E	DRAINAGE UNDER GENERATOR ON
I8.6	7 DI6	/26.7:E	DRAINAGE UNDER GENERATOR READY
I8.7	8 DI7	/26.8:E	DRAINAGE UNDER GENERATOR SPARE
I9.0	9 DI8	/27.1:E	DRAINAGE UNDER TURBINE A FEEDER INSERTED
I9.1	10 DI9	/27.2:E	SPARE
I9.2	11 DI10	/27.3:E	DRAINAGE UNDER TURBINE A AUTO
I9.3	12 DI11	/27.4:E	DRAINAGE UNDER TURBINE A ON
I9.4	13 DI12	/27.5:E	DRAINAGE UNDER TURBINE A READY
I9.5	14 DI13	/27.6:E	DRAINAGE UNDER TURBINE A SPARE
I9.6	15 DI14	/27.7:E	DRAINAGE UNDER TURBINE B FEEDER INSERTED
I9.7	16 DI15	/27.8:E	SPARE
	L+ 24VDC	/14.5:D	
	M 0VDC	/14.5:D	

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram PLC OVERVIEW	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 116 PAGE 119 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

Format ISO/DIN A3

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: DI 16X24V DC Standard Order No.: 6ES7131-6BH01-0BA0 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A216	

-A216
/14.5:B

I10.0	1 DI0	/28.1:E	DRAINAGE UNDER TURBINE B AUTO
I10.1	2 DI1	/28.2:E	DRAINAGE UNDER TURBINE B ON
I10.2	3 DI2	/28.3:E	DRAINAGE UNDER TURBINE B READY
I10.3	4 DI3	/28.4:E	DRAINAGE UNDER TURBINE B SPARE
I10.4	5 DI4	/28.5:E	220 VDC SYSTEM 1 MCB TRIP GROUP - OK
I10.5	6 DI5	/28.6:E	220 VDC SYSTEM 2 MCB TRIP GROUP - OK
I10.6	7 DI6	/28.7:E	AC AUXILIARY SUPPLY CB READY
I10.7	8 DI7	/28.8:E	24 VDC MCB TRIP GROUP - OK
I11.0	9 DI8	/29.1:E	DC CONVERTER 1 OK
I11.1	10 DI9	/29.2:E	DC CONVERTER 2 OK
I11.2	11 DI10	/29.3:E	DISCONNECTOR 1 OFF
I11.3	12 DI11	/29.4:E	DISCONNECTOR 1 ON
I11.4	13 DI12	/29.5:E	DISCONNECTOR 2 OFF
I11.5	14 DI13	/29.6:E	DISCONNECTOR 2 ON
I11.6	15 DI14	/29.7:E	GENERATOR CIRCUIT BREAKER OFF
I11.7	16 DI15	/29.8:E	GENERATOR CIRCUIT BREAKER ON
	L+ 24VDC	/14.6:D	
	M 0VDC	/14.6:D	

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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				PLC OVERVIEW	Designer No.:	SALES No.: PAGE 117
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 120 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: DI 16X24V DC Standard Order No.: 6ES7131-6BH01-0BA0 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A217	

-A217
/14.6:B

I12.0	1 DI0	/30.1:E	BRAKING DISCONNECTOR OFF
I12.1	2 DI1	/30.2:E	BRAKING DISCONNECTOR ON
I12.2	3 DI2	/30.3:E	BRAKING EARTH SWITCH OFF
I12.3	4 DI3	/30.4:E	BRAKING EARTH SWITCH ON
I12.4	5 DI4	/30.5:E	TRANSFORMER NEUTRAL POINT DISCONNECTOR OFF
I12.5	6 DI5	/30.6:E	TRANSFORMER NEUTRAL POINT DISCONNECTOR ON
I12.6	7 DI6	/30.7:E	GENERATOR CIRCUIT BREAKER SF6 ALARM
I12.7	8 DI7	/30.8:E	GENERATOR CIRCUIT BREAKER SF6 TRIP
I13.0	9 DI8	/31.1:E	GENERATOR CIRCUIT BREAKER MOTOR NOT READY 1
I13.1	10 DI9	/31.2:E	GENERATOR CIRCUIT BREAKER MCB ALARM GROUP NOT OK
I13.2	11 DI10	/31.3:E	GENERATOR CIRCUIT BREAKER REMOTE
I13.3	12 DI11	/31.4:E	GENERATOR CIRCUIT BREAKER ON/OFF RELEASE
I13.4	13 DI12	/31.5:E	GENERATOR CIRCUIT BREAKER NOT READY ON (SPRING NOT CHARGED)
I13.5	14 DI13	/31.6:E	DISCONNECTOR 1 REMOTE
I13.6	15 DI14	/31.7:E	DISCONNECTOR 1 MOTOR READY
I13.7	16 DI15	/31.8:E	DISCONNECTOR 1 DOOR OPEN
	L+ 24VDC	/14.6:D	
	M 0VDC	/14.6:D	

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram PLC OVERVIEW	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 118 PAGE 121 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: DI 16X24V DC Standard Order No.: 6ES7131-6BH01-0BA0 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A218	

-A218
/14.6:B

I14.0	1 DI0	/32.1:E	DISCONNECTOR 1 HEATER MCB OK
I14.1	2 DI1	/32.2:E	DISCONNECTOR 2 REMOTE
I14.2	3 DI2	/32.3:E	DISCONNECTOR 2 MOTOR READY
I14.3	4 DI3	/32.4:E	DISCONNECTOR 2 DOOR OPEN
I14.4	5 DI4	/32.5:E	DISCONNECTOR 2 HEATER MCB OK
I14.5	6 DI5	/32.6:E	BRAKING EARTHING SWITCH REMOTE
I14.6	7 DI6	/32.7:E	BRAKING EARTHING SWITCH MOTOR READY
I14.7	8 DI7	/32.8:E	BRAKING EARTHING SWITCH DOOR OPEN
I15.0	9 DI8	/33.1:E	BRAKING DISCONNECTOR EARTHING SWITCH HEATER READY
I15.1	10 DI9	/33.2:E	VT MCB OK - MEASUREMENT
I15.2	11 DI10	/33.3:E	VT MCB FOR CONTROL SYSTEM (SYNCH) OK
I15.3	12 DI11	/33.4:E	VT MCB OK - 110kV MEASUREMENT DEVICES
I15.4	13 DI12	/33.5:E	GENERATOR CIRCUIT BREAKER MOTOR READY
I15.5	14 DI13	/33.6:E	DISCONNECTOR MOTORS READY
I15.6	15 DI14	/33.7:E	MARSALLING KIOSK MCB ALARM GROUP - OK
I15.7	16 DI15	/33.8:E	MARSALLING KIOSK LOCAL
	L+ 24VDC	/14.7:D	
	M 0VDC	/14.7:D	

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				PLC OVERVIEW	Designer No.:	SALES No.: PAGE 119
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 122 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: DI 16X24V DC Standard Order No.: 6ES7131-6BH01-0BA0 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A219	

-A219
/14.7:B

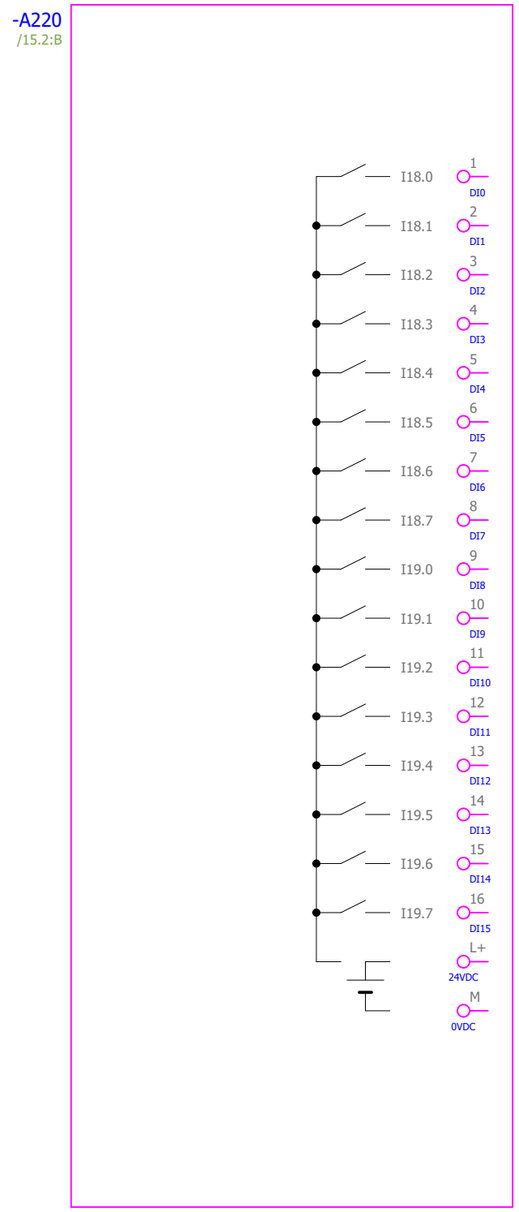
I16.0	1 DI0	/34.1:E	MARSALLING KIOSK REMOTE
I16.1	2 DI1	/34.2:E	SPARE
I16.2	3 DI2	/34.3:E	SYNHRONSIATION VOLTAGE MCB of VTs ON BUSBAR SYTEM OK
I16.3	4 DI3	/34.4:E	GRID VT MCB OK - SYNCHRONIZATION
I16.4	5 DI4	/34.5:E	SYNHRONSIATION VOLTAGE MCB of VTs ON BUSBAR SYTEM OK
I16.5	6 DI5	/34.6:E	MARSHALLING KIOSK MOTORS POWER SUPPLY ON
I16.6	7 DI6	/34.7:E	SPARE
I16.7	8 DI7	/34.8:E	SPARE
I17.0	9 DI8	/35.1:E	GENERATOR PROTECTION 1 RELAY TRIP
I17.1	10 DI9	/35.2:E	GENERATOR PROTECTION 1 INTERNAL FAULT
I17.2	11 DI10	/35.3:E	SPARE
I17.3	12 DI11	/35.4:E	GENERATOR PROTECTION 2 RELAY TRIP OK
I17.4	13 DI12	/35.5:E	GENERATOR PROTECTION 2 INTERNAL FAULT
I17.5	14 DI13	/35.6:E	SPARE
I17.6	15 DI14	/35.7:E	EXCITATION LOCAL
I17.7	16 DI15	/35.8:E	EXCITATION ON RELEASE
	L+ 24VDC	/14.7:D	
	M 0VDC	/14.7:D	

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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				PLC OVERVIEW	Designer No.:	SALES No.: PAGE 120
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 123 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: DI 16X24V DC Standard Order No.: 6ES7131-6BH01-0BA0 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A220	



/36.1:E	EXCITATION ALARM OK
/36.2:E	EXCITATION ON
/36.3:E	EXCITATION OFF
/36.4:E	EXCITATION EXTERNAL TRIP
/36.5:E	EXCITATION THYRICON TRIP
/36.6:E	EXCITATION FCR MODE SELECTED
/36.7:E	EXCITATION AVR MODE SELECTED
/36.8:E	EXCITATION VAR MODE SELECTED
/37.1:E	EXCITATION PF MODE SELECTED
/37.2:E	EXCITATION ELECTRICAL BRAKING READY
/37.3:E	EXCITATION ELECTRICAL BRAKING ON
/37.4:E	SPARE
/37.5:E	GENERATOR POWER LOWER THEN MINIMAL
/37.6:E	SPARE (TURBINE SPEED < ELECTRICAL BRAKING STOP LIMIT)
/37.7:E	TURBINE SPEED > EXCITATION START LIMIT
/37.8:E	SPARE
/15.2:D	
/15.2:D	

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram PLC OVERVIEW	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 121 PAGE 124 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: DI 16X24V DC Standard Order No.: 6ES7131-6BH01-0BA0 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A221	

-A221
/15.2:B

I20.0	1 DI0	/38.1:E	TURBINE SPEED > SYNCHRONIZATION START LIMIT
I20.1	2 DI1	/38.2:E	TURBINE SPEED > OVERSPEED LIMIT
I20.2	3 DI2	/38.3:E	TURBINE GOVERNOR SPARE
I20.3	4 DI3	/38.4:E	TURBINE GOVERNOR ALARM
I20.4	5 DI4	/38.5:E	TURBINE GOVERNOR TRIP
I20.5	6 DI5	/38.6:E	TURBINE GOVERNOR EMERGENCY SHUTDOWN OK
I20.6	7 DI6	/38.7:E	VIBRATION MONITORING TRIP
I20.7	8 DI7	/38.8:E	MONITORING SYSTEM ALARM 2
I21.0	9 DI8	/39.1:E	MONITORING SYSTEM ALARM 3
I21.1	10 DI9	/39.2:E	MONITORING SYSTEM ALARM 4
I21.2	11 DI10	/39.3:E	MONITORING SYSTEM ALARM 5
I21.3	12 DI11	/39.4:E	MONITORING SYSTEM ALARM 6
I21.4	13 DI12	/39.5:E	MONITORING SYSTEM ALARM 7
I21.5	14 DI13	/39.6:E	MONITORING SYSTEM ALARM 8
I21.6	15 DI14	/39.7:E	SHAFT CURRENT ALARM
I21.7	16 DI15	/39.8:E	SHAFT CURRENT TRIP
	L+ 24VDC	/15.2:D	
	M 0VDC	/15.2:D	

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram PLC OVERVIEW	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 122 PAGE 125 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: DI 16X24V DC Standard Order No.: 6ES7131-6BH01-0BA0 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A222	

-A222
/15.2:B

I22.0	1 DI0	/40.1:E	SHAFT CURRENT SYSTEM OK
I22.1	2 DI1	/40.2:E	GENERATOR COOLING WATER VALVE OPEN
I22.2	3 DI2	/40.3:E	GENERATOR COOLING WATER VALVE CLOSED
I22.3	4 DI3	/40.4:E	TURBINE A LEAKAGE TANK LEVEL HIGH
I22.4	5 DI4	/40.5:E	TURBINE B LEAKAGE TANK LEVEL HIGH
I22.5	6 DI5	/40.6:E	GENERAL LOCKOUT LOSS OF SF6
I22.6	7 DI6	/40.7:E	GENERATOR COOLING WATER VALVE OVERLOAD/BIMETAL WARNING
I22.7	8 DI7	/40.8:E	GENERATOR FIRE EXTINGUISHING SYSTEM TRIP
I23.0	9 DI8	/41.1:E	CLOSING LOCKOUT OPERATING MECHANISAM
I23.1	10 DI9	/41.2:E	TRANSFORMER OIL PUMP 1 ON
I23.2	11 DI10	/41.3:E	TRANSFORMER OIL PUMP 2 ON
I23.3	12 DI11	/41.4:E	TRANSFORMER COOLING WATER VALVE CLOSED
I23.4	13 DI12	/41.5:E	TRANSFORMER COOLING WATER VALVE OPEN
I23.5	14 DI13	/41.6:E	TRANSFORMER BUCHHOLZ ALARM
I23.6	15 DI14	/41.7:E	TRANSFORMER BUCHHOLZ TRIP
I23.7	16 DI15	/41.8:E	TRANSFORMER TEMPERATURE ALARM
	L+ 24VDC	/15.3:D	
	M 0VDC	/15.3:D	

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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				PLC OVERVIEW	Designer No.:	SALES No.: PAGE 123
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 126 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: DI 16X24V DC Standard Order No.: 6ES7131-6BH01-0BA0 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A223	

-A223
/15.3:B

I24.0	1 DI0	/42.1:E	TRANSFORMER TEMPERATURE TRIP
I24.1	2 DI1	/42.2:E	TRANSFORMER OIL LEVEL LOW
I24.2	3 DI2	/42.3:E	TRANSFORMER OIL LEVEL HIGH
I24.3	4 DI3	/42.4:E	TRANSFORMER COOLER 1 OIL FLOW OK
I24.4	5 DI4	/42.5:E	TRANSFORMER COOLER 1 PRESSURE LOW
I24.5	6 DI5	/42.6:E	TRANSFORMER COOLER 1 WATER FLOW OK
I24.6	7 DI6	/42.7:E	TRANSFORMER COOLER 1 TEMPARTURE ALARM
I24.7	8 DI7	/42.8:E	TRANSFORMER COOLER 2 TEMPARTURE ALARM
I25.0	9 DI8	/43.1:E	TRANSFORMER COOLER 2 WATER FLOW OK
I25.1	10 DI9	/43.2:E	TRANSFORMER COOLER 2 PRESSURE LOW
I25.2	11 DI10	/43.3:E	TRANSFORMER COOLER 2 OIL FLOW OK
I25.3	12 DI11	/43.4:E	TRANSFORMER COOLING CONTROL CUBICLE AC VOLTAGE OK
I25.4	13 DI12	/43.5:E	TRANSFORMER OIL PUMP 1 READY
I25.5	14 DI13	/43.6:E	TRANSFORMER OIL PUMP 2 READY
I25.6	15 DI14	/43.7:E	TRANSFORMER COOLING SYSTEM REMOTE
I25.7	16 DI15	/43.8:E	TRANSFORMER COOLING VALVE READY
	L+ 24VDC	/15.3:D	
	M 0VDC	/15.3:D	

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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram PLC OVERVIEW	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 124 PAGE 127 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: DI 16X24V DC Standard Order No.: 6ES7131-6BH01-0BA0 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A224	

-A224
/15.3:B

I26.0	1 DI0	/44.1:E	TRANSFORMER COOLING VALVE FAULT
I26.1	2 DI1	/44.2:E	TRANSFORMER COOLING WATER VALVE TORQUE SWITCH OPEN
I26.2	3 DI2	/44.3:E	TRANSFORMER COOLING WATER VALVE TORQUE SWITCH CLOSED
I26.3	4 DI3	/44.4:E	TRANSFORMER 230 VAC MCB GROUP OK - ALARM
I26.4	5 DI4	/44.5:E	TRANSFORMER COOLING SYSTEM 220 VDC MCBs OK - TRIP
I26.5	6 DI5	/44.6:E	SPARE
I26.6	7 DI6	/44.7:E	SPARE
I26.7	8 DI7	/44.8:E	SPARE
I27.0	9 DI8	/45.1:E	GENERATOR COOLING WATER VALVE REMOTE
I27.1	10 DI9	/45.2:E	SPARE
I27.2	11 DI10	/45.3:E	SPARE
I27.3	12 DI11	/45.4:E	SPARE
I27.4	13 DI12	/45.5:E	SPARE
I27.5	14 DI13	/45.6:E	SPARE
I27.6	15 DI14	/45.7:E	SPARE
I27.7	16 DI15	/45.8:E	SPARE
	L+ 24VDC	/15.4:D	
	M 0VDC	/15.4:D	

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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				PLC OVERVIEW	Designer No.:	SALES No.: PAGE 125
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 128 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: DI 16X24V DC Standard Order No.: 6ES7131-6BH01-0BA0 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A225	

-A225
/15.4:B

I28.0	1 DI0	/46.1:E	SPARE
I28.1	2 DI1	/46.2:E	SPARE
I28.2	3 DI2	/46.3:E	SPARE
I28.3	4 DI3	/46.4:E	SPARE
I28.4	5 DI4	/46.5:E	SPARE
I28.5	6 DI5	/46.6:E	SPARE
I28.6	7 DI6	/46.7:E	SPARE
I28.7	8 DI7	/46.8:E	SPARE
I29.0	9 DI8	/47.1:E	SPARE
I29.1	10 DI9	/47.2:E	SPARE
I29.2	11 DI10	/47.3:E	SPARE
I29.3	12 DI11	/47.4:E	SPARE
I29.4	13 DI12	/47.5:E	SPARE
I29.5	14 DI13	/47.6:E	SPARE
I29.6	15 DI14	/47.7:E	SPARE
I29.7	16 DI15	/47.8:E	SPARE
	L+ 24VDC	/15.4:D	
	M 0VDC	/15.4:D	

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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				PLC OVERVIEW	Designer No.:	SALES No.:	PAGE 126
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 129 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
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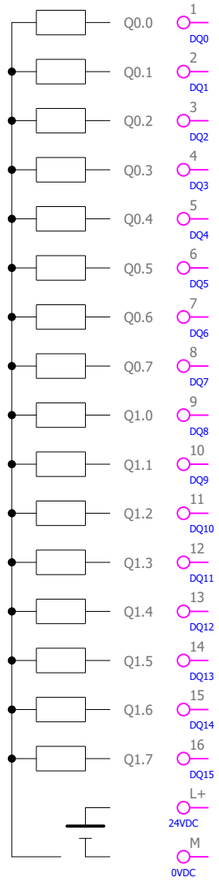
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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska		PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				PLC OVERVIEW	Designer No.:	SALES No.:	PAGE 127
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 130 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: DQ 16x 24V DC/0,5A Order No.: 6ES7132-6BH01-0BA0 Baseunit Type A0, BU15-P16+A0+2D 6ES7193-6BP00-0DA0	Siemens	+U7CJA10	-A231	

-A231
/15.4:B



/50.1:B	ALARM ACKNOWLEDGE COMMAND
/50.2:B	AUTOMATIC SYNCHRONIZATION START
/50.3:B	AUTO START OR STOP ACTIVE
/50.4:B	SPARE
/50.5:B	ESD ACTIVATE
/50.6:B	FIRE EXTINGUISHING SYSTEM ACITVATE
/50.7:B	TRANSFORMER COOLING WATER VALVE CLOSE COMMAND
/50.8:B	TRANSFORMER COOLING WATER VALVE OPEN COMMAND
/51.1:B	TRANSFORMER OIL PUMPS ON COMMAND
/51.2:B	SPARE
/51.3:B	GENERATOR HEATERS ON COMMAND
/51.4:B	GENERATOR COOLING WATER VALVE CLOSE COMMAND
/51.5:B	GENERATOR COOLING WATER VALVE OPEN COMMAND
/51.6:B	DISCONNECTOR 1 OFF COMMAND
/51.7:B	DISCONNECTOR 1 ON COMMAND
/51.8:B	DISCONNECTOR 2 OFF COMMAND
/15.5:D	
/15.5:D	

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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				PLC OVERVIEW	Designer No.:	SALES No.:	PAGE 128
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 131 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: DQ 16x 24V DC/0,5A Order No.: 6ES7132-6BH01-0BA0 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A232	

-A232
/15.5:B

Q2.0	1 DQ0	/52.1:B	DISCONNECTOR 2 ON COMMAND
Q2.1	2 DQ1	/52.2:B	SPARE
Q2.2	3 DQ2	/52.3:B	GENERATOR TEMPERATURE TRIP
Q2.3	4 DQ3	/52.4:B	ROTOR TEMPERATURE TRIP
Q2.4	5 DQ4	/52.5:B	TRANSFORMER OIL TEMPERATURE TRIP
Q2.5	6 DQ5	/52.6:B	GENERATOR CIRCUIT BREAKER OFF COMMAND
Q2.6	7 DQ6	/52.7:B	EXCITATION OFF COMMAND
Q2.7	8 DQ7	/52.8:B	EXCITATION ON COMMAND
Q3.0	9 DQ8	/53.1:B	SPARE (FCR MODE REQUEST)
Q3.1	10 DQ9	/53.2:B	SPARE (AVR MODE REQUEST)
Q3.2	11 DQ10	/53.3:B	SPARE (VAR MODE REQUEST)
Q3.3	12 DQ11	/53.4:B	SPARE (PF MODE REQUEST)
Q3.4	13 DQ12	/53.5:B	SPEED > EXCITATION ON LIMIT
Q3.5	14 DQ13	/53.6:B	SPARE (EXCITATION UNLOAD COMMAND)
Q3.6	15 DQ14	/53.7:B	SPARE (EXCITATION ISLAND MODE ON COMMAND)
Q3.7	16 DQ15	/53.8:B	EXCITATION RESET
	L+ 24VDC	/15.5:D	
	M 0VDC	/15.5:D	

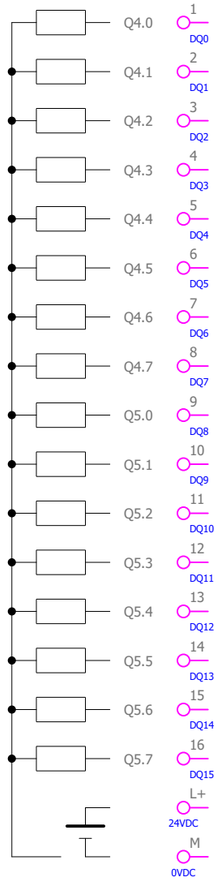
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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00 + U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				PLC OVERVIEW	Designer No.:	SALES No.: PAGE 129
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 132 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: DQ 16x 24V DC/0,5A Order No.: 6ES7132-6BH01-0BA0 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A233	

-A233
/15.5:B



/54.1:B	SPARE	
/54.2:B	SPARE	
/54.3:B	SYNCHRONIZATION ACTIVE	
/54.4:B	ELECTRICAL BRAKING ON RELEASE	
/54.5:B	SPEED < ELECTRICAL BRAKING STOP LIMIT	
/54.6:B	AUTOMATIC SYNCHRONIZER START	
/54.7:B	AC MAIN CB ON COMMAND	
/54.8:B	AC MAIN CB OFF COMAND	
/55.1:B	AC AUXILIARY CB ON COMMAND	
/55.2:B	AC AUXILIARY CB OFF COMMAND	
/55.3:B	BUSBAR DUCT FAN ON COMMAND	
/55.4:B	SPARE	
/55.5:B	GRAPHITE DUST SUCTION SYSTEM ON COMMAND	
/55.6:B	DRAINAGE UNDER GENERATOR ON COMMAND	
/55.7:B	DRAINAGE UNDER TURBINE A ON COMMAND	
/55.8:B	DRAINAGE UNDER TURBINE B ON COMMAND	
/15.6:D		
/15.6:D		

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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram PLC OVERVIEW	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 130 PAGE 133 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1					

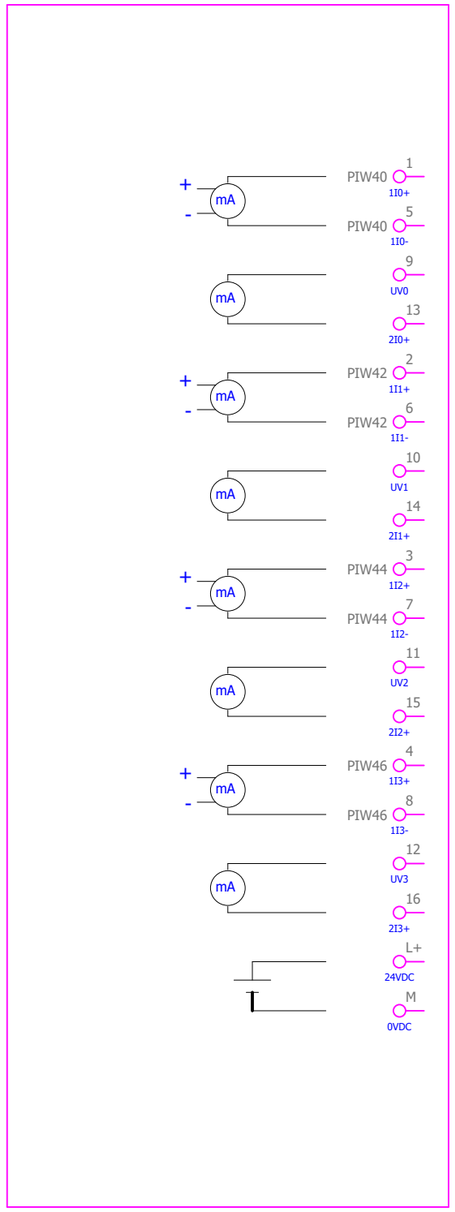
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CONFIDENTIAL, ALL RIGHTS RESERVED DIN 34

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram PLC OVERVIEW	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 131
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 134 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: AI 4XI 2-/4-WIRE Standard Order No.: 6ES7134-6GD01-0BA1 Baseunit Type A0, BU15-P16+A0+2D 6ES7193-6BP00-0DA0	Siemens	+U7CJA10	-A251	



/58.2:F	TRANSFORMER OIL TEMPERATURE		
/58.2:F			
/58.2:F			
/58.2:F			
/58.3:F	TRANSFORMER OIL TEMPERATURE DOWN		
/58.3:F			
/58.4:F			
/58.4:F			
/58.5:F	GENERATOR CORE U TEMPERATURE		
/58.5:F			
/58.5:F			
/58.5:F			
/58.7:F	GENERATOR CORE V TEMPERATURE		
/58.7:F			
/58.7:F			
/58.7:F			
/16.2:D			
/16.2:D			

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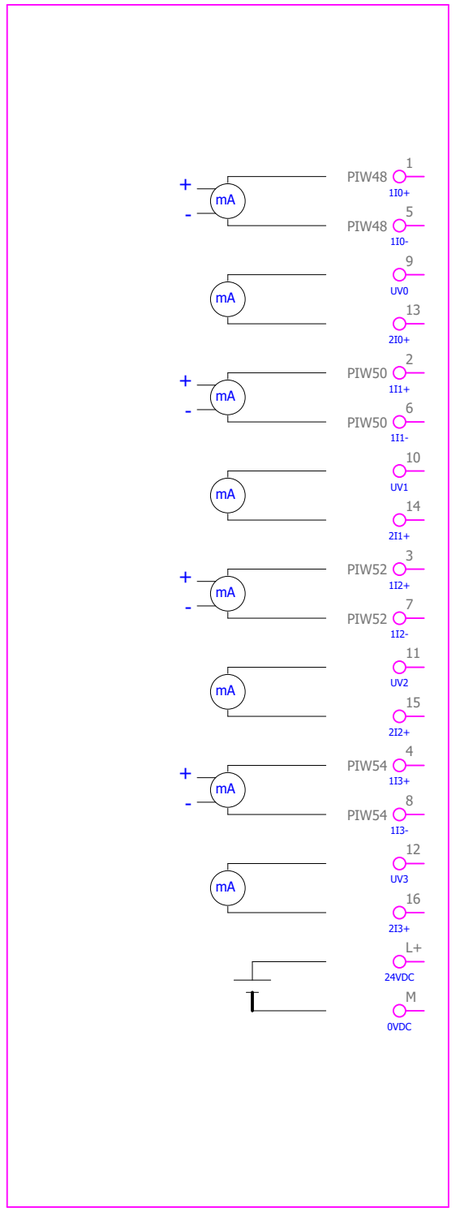
Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00
Approved by Designer	Grga Kolić	01.04.2026.						= U7CJA10GH100
Checked by	Antonio Aleksić	01.04.2026.				PLC OVERVIEW		+ U7CJA10
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		Designer No.: 22044-DD-EE-08-07-100	SALES No.:
								PAGE 132
								PAGE 135 / 204

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Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: AI 4XI 2-/4-WIRE Standard Order No.: 6ES7134-6GD01-0BA1 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A252	

-A252
/16.2:B



GENERATOR CORE W TEMPERATURE

GENERATOR WINDING U TEMPERATURE UP

GENERATOR WINDING V TEMPERATURE UP

GENERATOR WINDING W TEMPERATURE UP

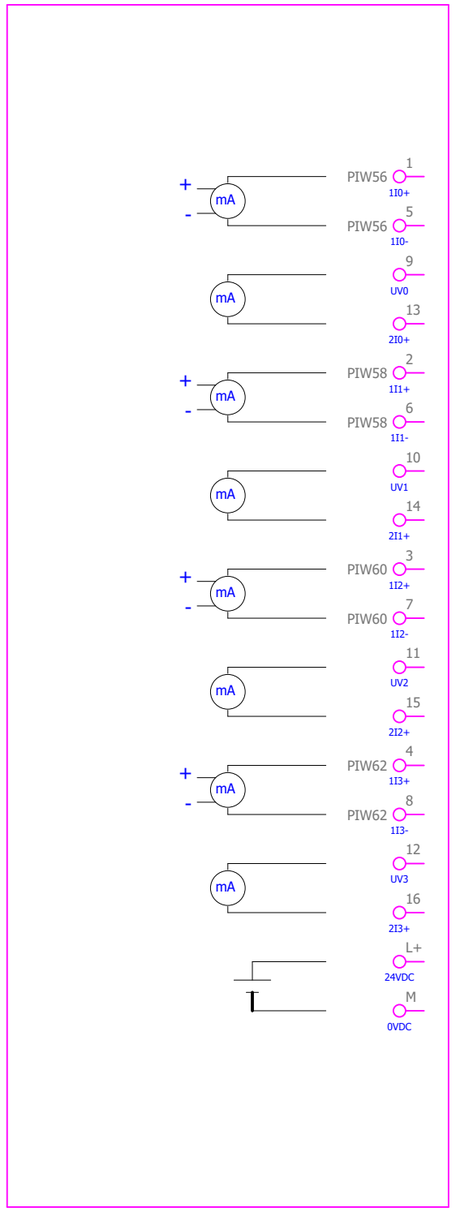
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00 / = U7CJA10GH100 / + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.				PLC OVERVIEW	Designer No.: 22044-DD-EE-08-07-100	SALES No.:
Checked by	Antonio Aleksić	01.04.2026.					PAGE	133
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		PAGE	136 / 204

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Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: AI 4XI 2-/4-WIRE Standard Order No.: 6ES7134-6GD01-0BA1 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A253	

-A253
/16.2:B

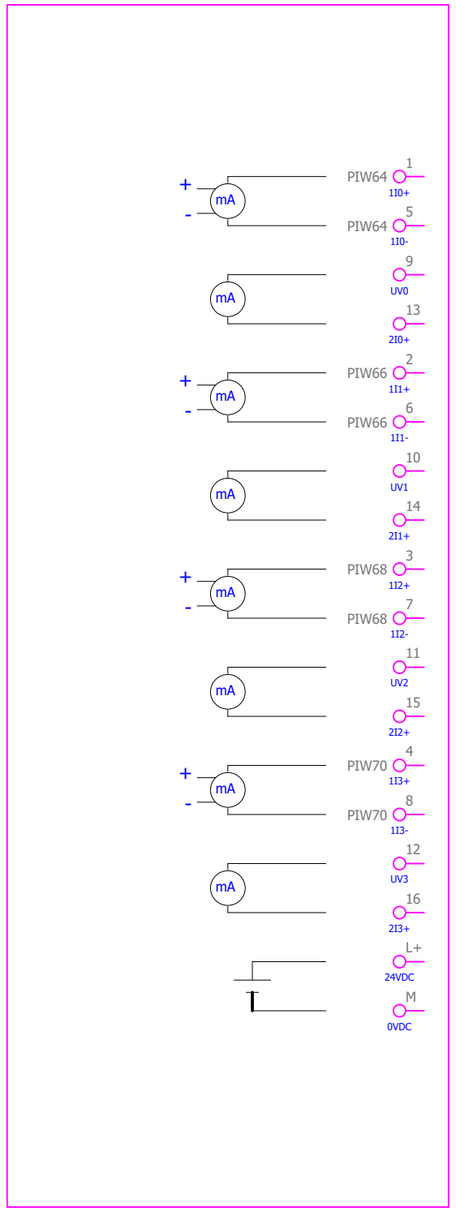


Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00 / = U7CJA10GH100 / + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.				PLC OVERVIEW	Designer No.: 22044-DD-EE-08-07-100	SALES No.:
Checked by	Antonio Aleksić	01.04.2026.					PAGE	134
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		PAGE	137 / 204

Format ISO/DIN A3

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: AI 4XI 2-/4-WIRE Standard Order No.: 6ES7134-6GD01-0BA1 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A254	

-A254
/16.3:B



/61.2:F	GENERATOR WINDING W TEMPERATURE DOWN	
/61.2:F		
/61.2:F		
/61.2:F		
/61.3:F	GENERATOR LEFT SIDE TEMEPERATURE	
/61.3:F		
/61.4:F		
/61.4:F		
/61.5:F	GENERATOR RIGHT SIDE TEMEPERATURE	
/61.5:F		
/61.5:F		
/61.5:F		
/61.7:F	SPARE	
/61.7:F		
/61.7:F		
/61.7:F		
/16.3:D		
/16.3:D		

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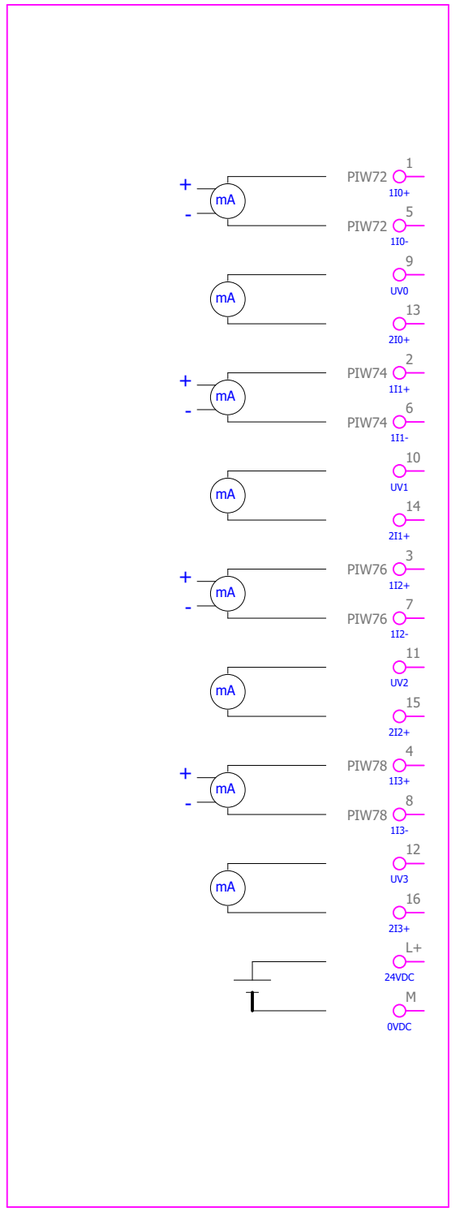
Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.							+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				PLC OVERVIEW	Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 135
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 138 / 204

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Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: AI 4XI 2-/4-WIRE Standard Order No.: 6ES7134-6GD01-0BA1 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A255	

-A255
/16.3:B



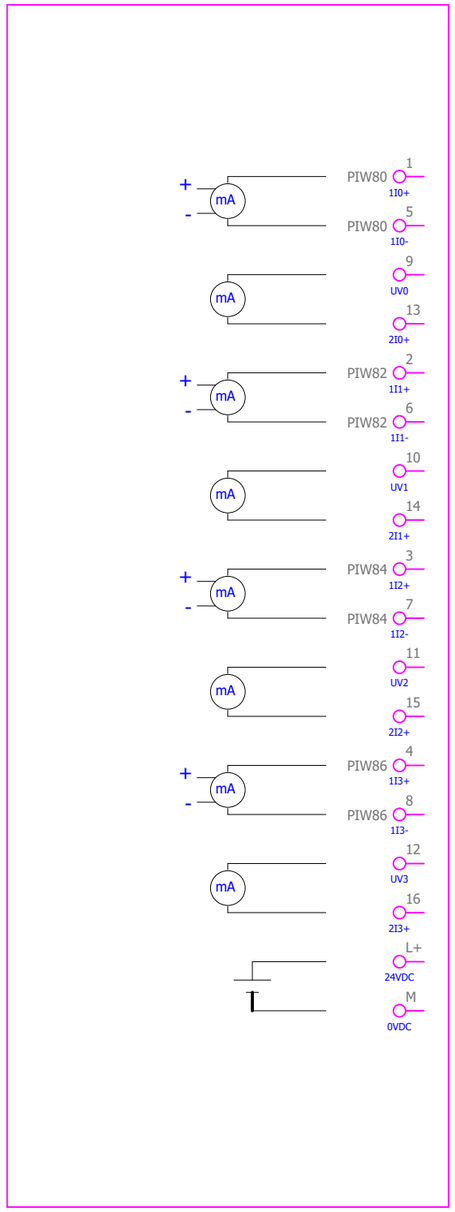
/62.2:F	SPARE	
/62.2:F		
/62.2:F		
/62.2:F		
/62.3:F	SPARE	
/62.3:F		
/62.4:F		
/62.4:F		
/62.5:F	SPARE	
/62.5:F		
/62.5:F		
/62.5:F		
/62.7:F	SPARE	
/62.7:F		
/62.7:F		
/62.7:F		
/16.4:D		
/16.4:D		

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.							+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				PLC OVERVIEW	Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 136
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 139 / 204

CONFIDENTIAL, ALL RIGHTS RESERVED DIN 34

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Type: AI 4XI 2-/4-WIRE Standard Order No.: 6ES7134-6GD01-0BA1 Baseunit Type A0, BU15-P16+A0+2B 6ES7193-6BP00-0BA0	Siemens	+U7CJA10	-A256	

-A256
/16.4:B

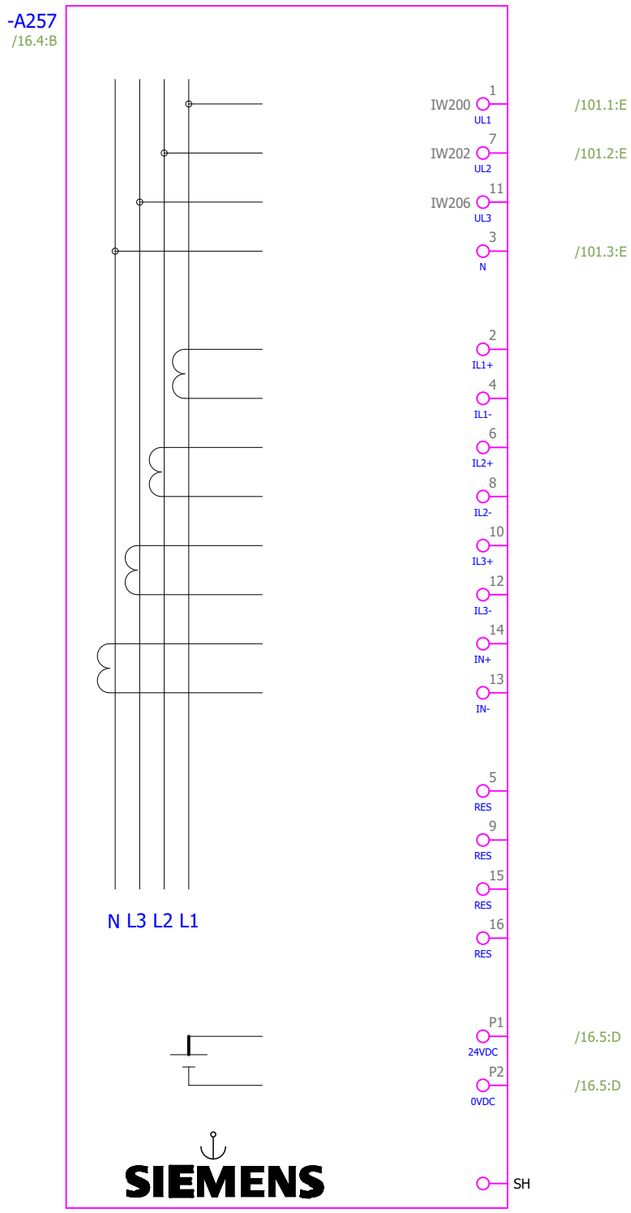


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/63.3:F	SPARE	
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/63.5:F	SPARE	
/63.5:F		
/63.5:F		
/63.5:F		
/63.7:F	SPARE	
/63.7:F		
/63.7:F		
/63.7:F		
/16.4:D		
/16.4:D		

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				PLC OVERVIEW	Designer No.:	SALES No.:	PAGE 137
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 140 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	SIMATIC ET 200SP, analog input module, AI Energy Meter CT ST, for 1A or 5A current transformer, suitable for BU type U0, channel diagnostics	Siemens	+U7CJA10	-A257	



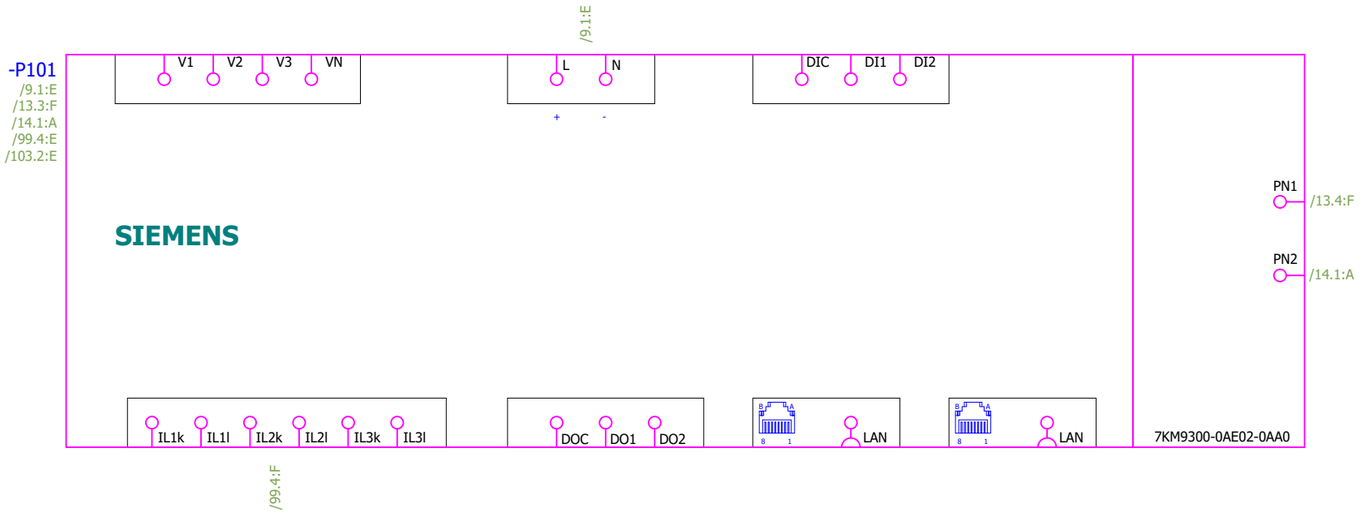
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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	PLC OVERVIEW	Designer No.: 22044-DD-EE-08-07-100	SALES No.:
Checked by	Antonio Aleksić	01.04.2026.						
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			

= U7CJA10GH100
+ U7CJA10
PAGE 138
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Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Power supply: 24 ... 60 VDC LCD 96x96mm Type: SENTRON PAC3220 Switched Ethernet PROFINET Module V3 7KM9300-0AE02-0AA0	Siemens	+U7CJA10	-P101	



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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				MULTIMETER	Designer No.:	SALES No.:	PAGE 139
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 142 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Panel mounted fault annunciator The flexible fault annunciator series BSM-P		+U7CJA10	-H01	

-H01

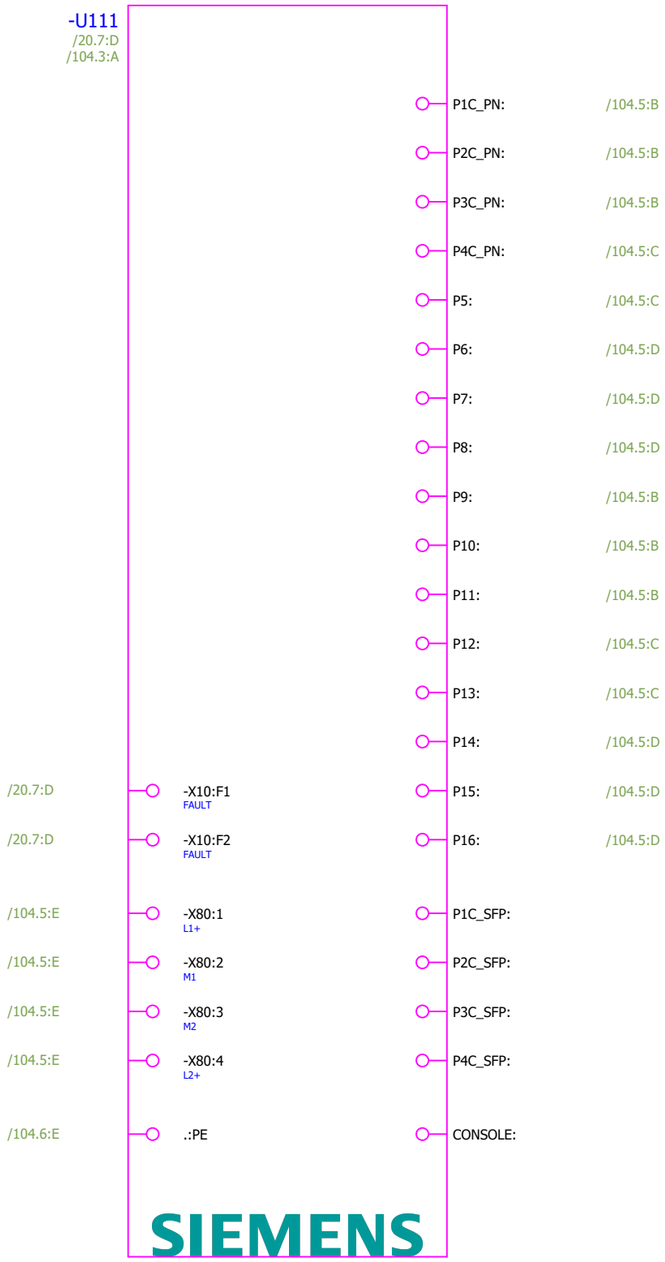
EES
BSM-C 59B1611CR000

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Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				PANEL MOUNTED FAULT ANNUCIATOR	Designer No.:	SALES No.:	PAGE 140
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 143 / 204

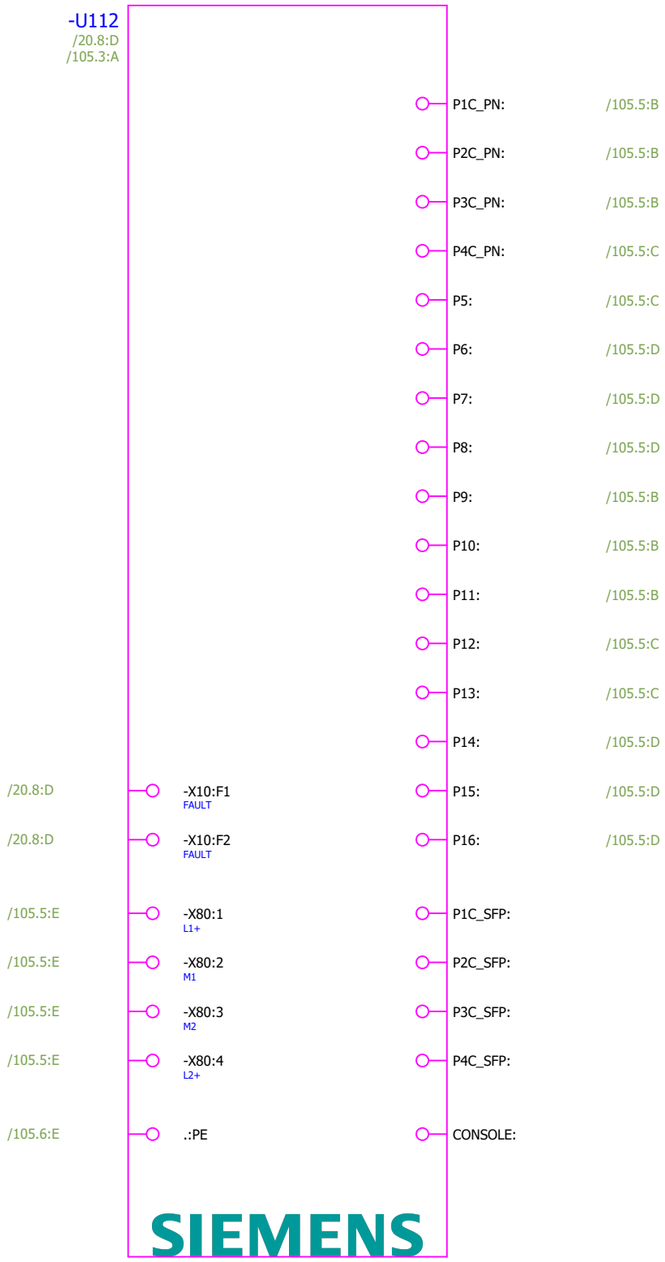
Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	SCALANCE XC216-4C managed Layer 2 IE switch; IEC 62443-4-2 certified; 12x 10/100 Mbps RJ45 ports; 4x 1000 Mbps combo ports (either 1000 Mbps/ SFPs or 10/100/1000 Mbps RJ45 ports can be used); 1x console port; diagnostics LED; redundant power supply; temp. range -40 °C to +70 °C mounting: DIN rail/S7 mounting rail/wall; redundancy 6GK5216-4BS00-2AC2	Siemens	+U7CJA10	-U111	



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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska				+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				ETHERNET SWITCH	Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 141
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 144 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	SCALANCE XC216-4C managed Layer 2 IE switch; IEC 62443-4-2 certified; 12x 10/100 Mbps RJ45 ports; 4x 1000 Mbps combo ports (either 1000 Mbps/ SFPs or 10/100/1000 Mbps RJ45 ports can be used); 1x console port; diagnostics LED; redundant power supply; temp. range -40 °C to +70 °C mounting: DIN rail/S7 mounting rail/wall; redundancy 6GK5216-4BS00-2AC2	Siemens	+U7CJA10	-U112	

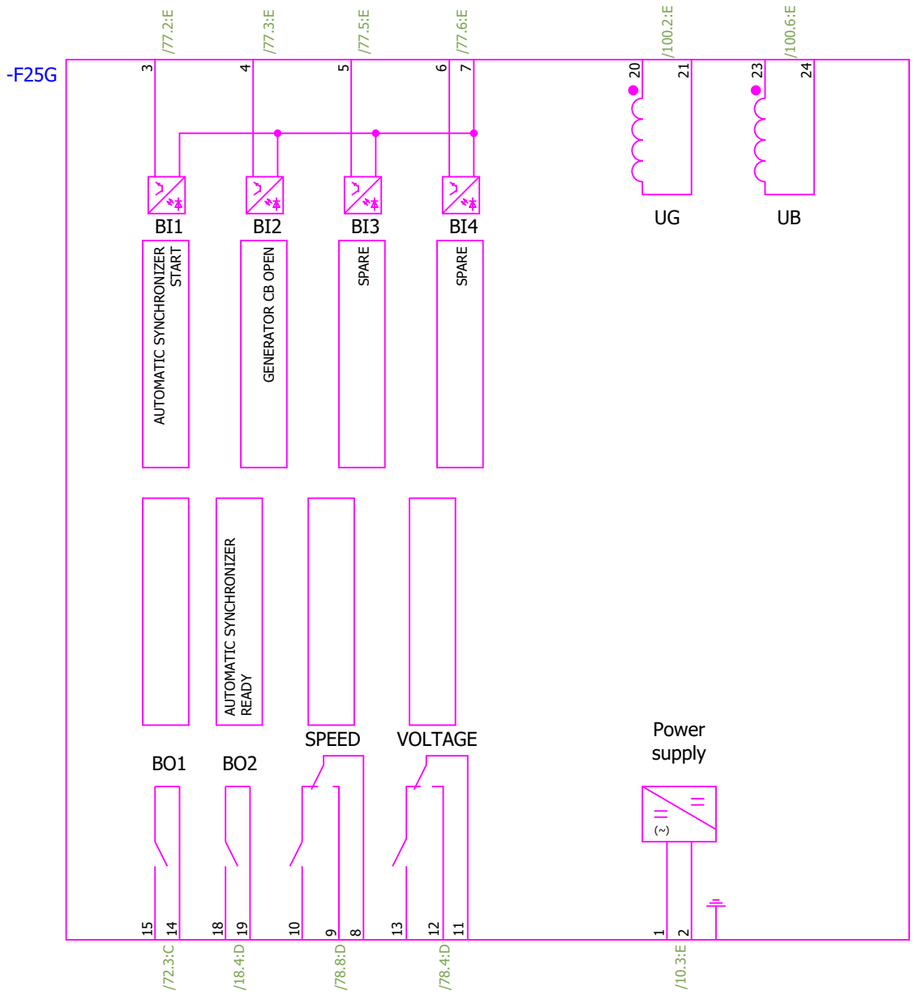


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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.							+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				ETHERNET SWITCH	Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 142
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 145 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	Power supply: 24 Vdc Discrete raise/lower: Speed Discrete raise/lower: Voltage Discrete alarm inputs: 4 Discrete outputs: 2 Type: SPM-D2-10	Woodward	+U7CJA10	-F25G	

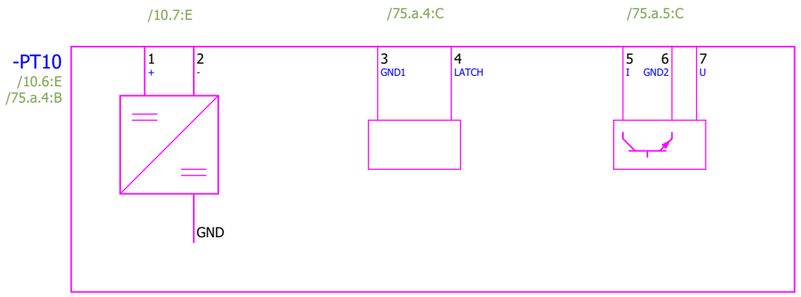


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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: 00
Approved by Designer	Grga Kolić	01.04.2026.						= U7CJA10GH100
Checked by	Antonio Aleksić	01.04.2026.				AUTOMATIC SYNCHRONIZER		+ U7CJA10
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		Designer No.: 22044-DD-EE-08-07-100	SALES No.:
								PAGE 143
								PAGE 146 / 204

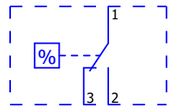
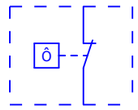
Qty. (pcs)	Equipment technical description	Manufacturer	Location (= / +)	Designation (-)	Symbol Position in shematics (Page, column)
1	MCR digital display, for measurement and display of standard signals, 5-fig. display	Phoenix Contact	+U7CJA10	-PT10	



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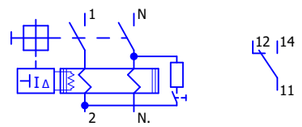
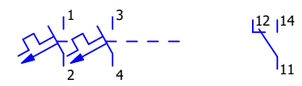
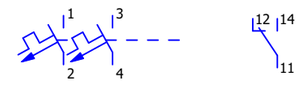
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.			**Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska		PERU-01-17-170-31-001	00	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				TEMPERATURE DISPLAY	Designer No.:	SALES No.:	PAGE 144
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 147 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)
	Termostat 1 NC kontakt 0-60 °C 10A Narudžbeni br.: THR02	Alfa Electric			
1			+U7CJA10	-B101	/6.5:E /6.5:E
	Higrostat Higrostat, Nazivni napon 120-240 V AC, Nazivna struja 10-5 A 10-90 %RF Broj narudžbe: IGR%6F Narudžbeni br.: IGR35F	Fanndis			
1			+U7CJA10	-B102	/6.5:E
	Termostat NO kontakt 0...+60°C Tip: THV02 Narudžbeni br.: THV02	Alfa Electric			
1			+U7CJA10	-B103	/6.6:C
	Rasvjeta: LED svjetiljka za tanko kućište, 9 W, 100 do 230 V, 50H, 538×23×36 mm Narudžbeni br.: LL49W-100-230V	ALFA ELECTRIC			
			+U7CJA10	-E101	/6.3:F
2			+U7CJA10	-E102	/6.4:F

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Format ISO/DIN A3

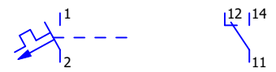
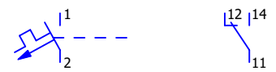
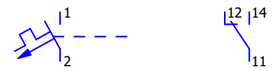
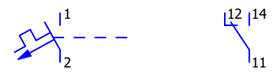
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 145
Checked by	Antonio Aleksić	01.04.2026.						PAGE 148 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)	
	RCBO RCBO, 10 kA, 1P+N, tip AC, 30 mA, C-char, In: 13 A, Un AC: 230 V Narudžbeni br.: 5SU1354-1KK13 Pomoćni prekidač Tip: 5ST3016 1 C/O 5ST3805-1 Spojnica ručke za minijaturni prekidač diferencijalne struje D=70 mm za montažu dodatnih komponenti (1 set=5 jedinica u polietilenskoj vrećici)	Siemens				
1			+U7CJA10	-F01	/6.2:C	
	10A, 2P, 440 VDC, B krivulja MCB - 440 VDC Broj polova: 2P Klasa: B Maksimalna struja: 10A Narudžbeni br.: 5SY5210-6 Pomoćni prekidač Tip: 5ST3016 1 C/O	Siemens				
			+U7CJA10	-F11	/7.4:D	/17.2:B
2			+U7CJA10	-F12	/7.5:D	/17.3:B
	6A, 2P, 440 VDC, B krivulja MCB - 440 VDC Broj polova: 2P Klasa: B Maksimalna struja: 6A Narudžbeni br.: 5SY5206-6 Pomoćni prekidač Tip: 5ST3016 1 C/O	Siemens				
			+U7CJA10	-F13	/7.6:C	/21.3:D
2			+U7CJA10	-F14	/7.7:C	
	Minijaturni prekidač Narudžbeni br.: 5SY4106-6 Pomoćni prekidač Tip: 5ST3016 1 C/O	Siemens				
			+U7CJA10	-F31	/9.1:B	/21.4:B
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Format ISO/DIN A3

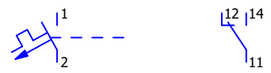
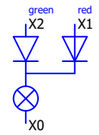
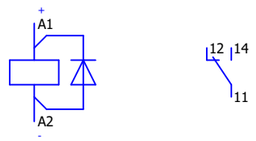
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.				DEVICE TAG LIST		
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 146
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			PAGE 149 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)	
	Minijaturni prekidač Narudžbeni br.: 5SY4106-6 Pomoćni prekidač Tip: 5ST3016 1 C/O	Siemens				
3			+U7CJA10	-F33	/9.3:B	/21.4:B
	Zaštitni prekidač 230/400V 10kA 1-pol B karakteristika 2A Narudžbeni br.: 5SY4102-6 Pomoćni prekidač Tip: 5ST3016 1 C/O	Siemens				
1			+U7CJA10	-F34	/9.4:B	/21.4:C
	Minijaturni prekidač Narudžbeni br.: 5SY4106-6 Pomoćni prekidač Tip: 5ST3016 1 C/O	Siemens				
			+U7CJA10	-F35	/9.6:B	/21.4:C
			+U7CJA10	-F36	/9.7:B	/21.4:C
			+U7CJA10	-F37	/9.8:B	/21.4:C
			+U7CJA10	-F38	/10.1:B	/21.4:D
			+U7CJA10	-F39	/10.2:B	/21.4:D
			+U7CJA10	-F40	/10.3:B	/21.4:D
7			+U7CJA10	-F41	/10.4:B	/21.4:B
	Zaštitni prekidač 230/400V 10kA 1-pol B karakteristika 2A Narudžbeni br.: 5SY4102-6 Pomoćni prekidač Tip: 5ST3016 1 C/O	Siemens				
			+U7CJA10	-F42	/10.6:B	/21.4:C
			+U7CJA10	-F43	/10.7:B	/21.4:C
			+U7CJA10	-F44	/10.8:B	/21.4:C

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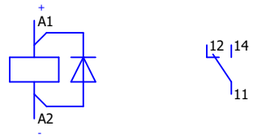
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.						
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 147
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			PAGE 150 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)	
	Zaštitni prekidač 230/400V 10kA 1-pol B karakteristika 2A Narudžbeni br.: 5SY4102-6 Pomoćni prekidač Tip: 5ST3016 1 C/O	Siemens				
			+U7CJA10	-F45	/11.1:B	/21.4:D
			+U7CJA10	-F46	/11.2:B	/21.4:D
			+U7CJA10	-F47	/11.3:B	/21.4:D
			+U7CJA10	-F48	/11.4:B	/21.5:B
			+U7CJA10	-F49	/11.6:B	/21.5:C
			+U7CJA10	-F50	/11.7:B	/21.4:C
			+U7CJA10	-F51	/11.8:B	
11			+U7CJA10	-F52	/12.1:B	
	Svjetlo indikatora Indikatori svijetle u dvije boje: crveno ili zeleno. Napajanje je spojeno na sljedeći način: X0 - zajednički terminal (zemlja) X1 - crvena boja (napon) X2 - zelena boja (napon) Promjer -22 mm 24 AC/DC Narudžbeni br.: PK22-WP/24					
			+U7CJA10	-H101	/83.2:D	
			+U7CJA10	-H102	/83.4:D	
			+U7CJA10	-H103	/83.6:D	
			+U7CJA10	-H104	/84.2:D	
			+U7CJA10	-H105	/84.4:D	
			+U7CJA10	-H106	/84.6:D	
			+U7CJA10	-H107	/85.2:D	
			+U7CJA10	-H108	/85.4:D	
9			+U7CJA10	-H109	/85.6:D	
	Pomoćni relej Narudžbeni br.: 2966171	Phoenix Contact				
			+U7CJA10	-1K11	/22.1:D	/22.1:E

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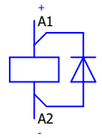
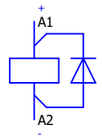
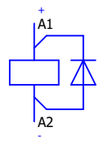
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.:	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 148 PAGE 151 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)	
	Pomoćni relej Narudžbeni br.: 2966171	Phoenix Contact				
			+U7CJA10	-1K12	/22.2:D	/22.2:E
			+U7CJA10	-1K13	/22.3:D	/22.3:E
			+U7CJA10	-1K14	/22.4:D	/22.4:E
			+U7CJA10	-1K15	/22.5:D	/22.5:E
			+U7CJA10	-1K16	/22.6:D	/22.6:E
			+U7CJA10	-1K17	/22.7:D	/22.7:E
			+U7CJA10	-1K18	/22.8:D	/22.8:E
			+U7CJA10	-1K21	/23.1:D	/23.1:E
			+U7CJA10	-1K22	/23.2:D	/23.2:E
			+U7CJA10	-1K23	/23.3:D	/23.3:E
			+U7CJA10	-1K24	/23.4:D	/23.4:E
			+U7CJA10	-1K25	/23.5:D	/23.5:E
			+U7CJA10	-1K26	/23.6:D	/23.6:E
			+U7CJA10	-1K27	/23.7:D	/23.7:E
			+U7CJA10	-1K28	/23.8:D	/23.8:E
			+U7CJA10	-1K31	/24.1:D	/24.1:E
			+U7CJA10	-1K32	/24.2:D	/24.2:E
			+U7CJA10	-1K33	/24.3:D	/24.3:E
			+U7CJA10	-1K34	/24.4:D	/24.4:E
			+U7CJA10	-1K35	/24.5:D	/24.5:E
			+U7CJA10	-1K36	/24.6:D	/24.6:E
			+U7CJA10	-1K37	/24.7:D	/24.7:E
			+U7CJA10	-1K38	/24.8:D	/24.8:E
			+U7CJA10	-1K41	/25.1:D	/25.1:E
			+U7CJA10	-1K42	/25.2:D	/25.2:E
			+U7CJA10	-1K43	/25.3:D	/25.3:E
			+U7CJA10	-1K44	/25.4:D	/25.4:E
			+U7CJA10	-1K45	/25.5:D	/25.5:E
			+U7CJA10	-1K46	/25.6:D	/25.6:E
			+U7CJA10	-1K47	/25.7:D	/25.7:E
			+U7CJA10	-1K48	/25.8:D	/25.8:E
			+U7CJA10	-1K51	/26.1:D	/26.1:E
			+U7CJA10	-1K52	/26.2:D	/26.2:E
			+U7CJA10	-1K53	/26.3:D	/26.3:E

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.				DEVICE TAG LIST		
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 149
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			PAGE 152 / 204

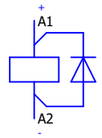
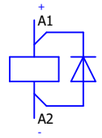
Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)		
	Pomoćni relej Narudžbeni br.: 2966171	Phoenix Contact					
			+U7CJA10	-1K54	/26.4:D	/26.4:E	
			+U7CJA10	-1K55	/26.5:D	/26.5:E	
			+U7CJA10	-1K56	/26.6:D	/26.6:E	
			+U7CJA10	-1K57	/26.7:D	/26.7:E	
			+U7CJA10	-1K58	/26.8:D	/26.8:E	
			+U7CJA10	-1K61	/27.1:D	/27.1:E	
			+U7CJA10	-1K62	/27.2:D	/27.2:E	
			+U7CJA10	-1K63	/27.3:D	/27.3:E	
			+U7CJA10	-1K64	/27.4:D	/27.4:E	
			+U7CJA10	-1K65	/27.5:D	/27.5:E	
			+U7CJA10	-1K66	/27.6:D	/27.6:E	
			+U7CJA10	-1K67	/27.7:D	/27.7:E	
			+U7CJA10	-1K68	/27.8:D	/27.8:E	
			+U7CJA10	-1K71	/28.1:D	/28.1:E	
			+U7CJA10	-1K72	/28.2:D	/28.2:E	
			+U7CJA10	-1K73	/28.3:D	/28.3:E	
52			+U7CJA10	-1K74	/28.4:D	/28.4:E	
	Relej sučelja 24VDC, 2CO Pomoćni relej Nazivni napon svitka: 24VDC 2 CO Narudžbeni br.: 2967060	Phoenix Contact					
			+U7CJA10	-1K75	/28.5:D	/28.5:E	/86.7:C
2			+U7CJA10	-1K76	/28.6:D	/28.6:E	/86.8:C
	Pomoćni relej Narudžbeni br.: 2966171	Phoenix Contact					
			+U7CJA10	-1K77	/28.7:D	/28.7:E	
			+U7CJA10	-1K78	/28.8:D	/28.8:E	
			+U7CJA10	-1K81	/29.1:D	/29.1:E	
4			+U7CJA10	-1K82	/29.2:D	/29.2:E	

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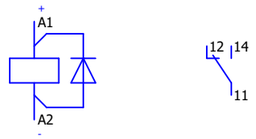
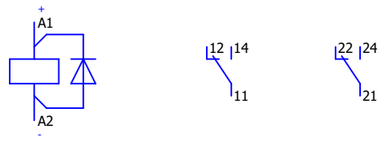
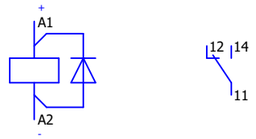
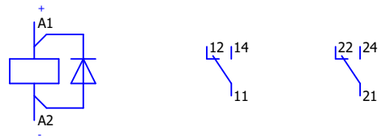
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.				DEVICE TAG LIST	Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 150
Checked by	Antonio Aleksić	01.04.2026.						PAGE 153 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			

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Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)		
	Relej sučelja 24VDC, 2CO Pomoćni relej Nazivni napon svitka: 24VDC 2 CO Narudžbeni br.: 2967060	Phoenix Contact					
			+U7CJA10	-1K83	/29.3:D	/29.3:E	/83.2:C
			+U7CJA10	-1K84	/29.4:D	/29.4:E	/73.5:C
			+U7CJA10	-1K85	/29.5:D	/29.5:E	/83.4:C
			+U7CJA10	-1K86	/29.6:D	/29.6:E	/73.6:C
			+U7CJA10	-1K87	/29.7:D	/29.7:E	/73.2:C
			+U7CJA10	-1K88	/29.8:D	/29.8:E	/73.2:C
			+U7CJA10	-1K91	/30.1:D	/30.1:E	/84.2:C
			+U7CJA10	-1K92	/30.2:D	/30.2:E	/84.3:C
			+U7CJA10	-1K93	/30.3:D	/30.3:E	/84.4:C
			+U7CJA10	-1K94	/30.4:D	/30.4:E	/84.5:C
			+U7CJA10	-1K95	/30.5:D	/30.5:E	/84.6:C
			+U7CJA10	-1K96	/30.6:D	/30.6:E	/84.7:C
			+U7CJA10	-1K97	/30.7:D	/30.7:E	/86.3:C
14			+U7CJA10	-1K98	/30.8:D	/30.8:E	/87.3:C
	Pomoćni relej Narudžbeni br.: 2966171	Phoenix Contact					
			+U7CJA10	-1K101	/31.1:D	/31.1:E	
			+U7CJA10	-1K102	/31.2:D	/31.2:E	
			+U7CJA10	-1K103	/31.3:D	/31.3:E	
			+U7CJA10	-1K104	/31.4:D	/31.4:E	
			+U7CJA10	-1K105	/31.5:D	/31.5:E	
			+U7CJA10	-1K106	/31.6:D	/31.6:E	
			+U7CJA10	-1K107	/31.7:D	/31.7:E	
			+U7CJA10	-1K108	/31.8:D	/31.8:E	
			+U7CJA10	-1K111	/32.1:D	/32.1:E	
			+U7CJA10	-1K112	/32.2:D	/32.2:E	
			+U7CJA10	-1K113	/32.3:D	/32.3:E	
			+U7CJA10	-1K114	/32.4:D	/32.4:E	
			+U7CJA10	-1K115	/32.5:D	/32.5:E	
			+U7CJA10	-1K116	/32.6:D	/32.6:E	
			+U7CJA10	-1K117	/32.7:D	/32.7:E	

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.: = U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001	+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				DEVICE TAG LIST	Designer No.:	SALES No.: PAGE 151
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100	PAGE 154 / 204

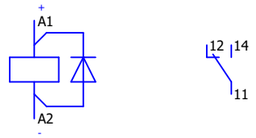
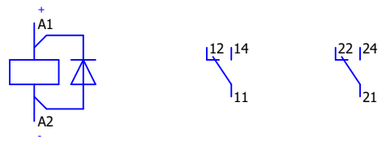
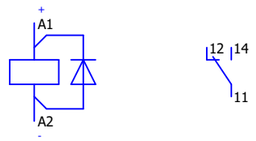
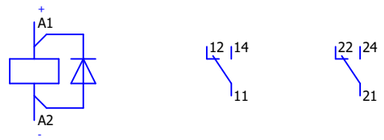
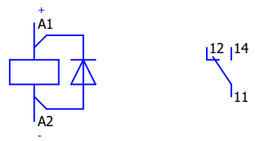
Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)		
18	Pomoćni relej Narudžbeni br.: 2966171	Phoenix Contact					
			+U7CJA10	-1K118	/32.8:D	/32.8:E	
			+U7CJA10	-1K121	/33.1:D	/33.1:E	
			+U7CJA10	-1K122	/33.2:D	/33.2:E	
1	Relej sučelja 24VDC, 2CO Pomoćni relej Nazivni napon svitka: 24VDC 2 CO Narudžbeni br.: 2967060	Phoenix Contact					
			+U7CJA10	-1K123	/33.3:D	/33.3:E	/93.6:B
8	Pomoćni relej Narudžbeni br.: 2966171	Phoenix Contact					
			+U7CJA10	-1K124	/33.4:D	/33.4:E	
			+U7CJA10	-1K125	/33.5:D	/33.5:E	
			+U7CJA10	-1K126	/33.6:D	/33.6:E	
			+U7CJA10	-1K127	/33.7:D	/33.7:E	
			+U7CJA10	-1K128	/33.8:D	/33.8:E	
			+U7CJA10	-1K131	/34.1:D	/34.1:E	
			+U7CJA10	-1K132	/34.2:D	/34.2:E	
			+U7CJA10	-1K133	/34.3:D	/34.3:E	
1	Relej sučelja 24VDC, 2CO Pomoćni relej Nazivni napon svitka: 24VDC 2 CO Narudžbeni br.: 2967060	Phoenix Contact					
			+U7CJA10	-1K134	/34.4:D	/34.4:E	/93.5:B

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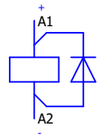
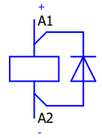
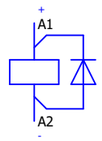
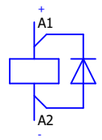
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001		+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				DEVICE TAG LIST	Designer No.:	SALES No.:	PAGE 152
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 155 / 204

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Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)		
	Pomoćni relej Narudžbeni br.: 2966171	Phoenix Contact					
			+U7CJA10	-1K135	/34.5:D	/34.5:E	
			+U7CJA10	-1K136	/34.6:D	/34.6:E	
			+U7CJA10	-1K137	/34.7:D	/34.7:E	
4			+U7CJA10	-1K138	/34.8:D	/34.8:E	
	Relej sučelja 24VDC, 2CO Pomoćni relej Nazivni napon svitka: 24VDC 2 CO Narudžbeni br.: 2967060	Phoenix Contact					
			+U7CJA10	-1K141	/35.1:D	/35.1:E	/87.4:C
2			+U7CJA10	-1K142	/35.2:D	/35.2:E	/87.7:C
	Pomoćni relej Narudžbeni br.: 2966171	Phoenix Contact					
			+U7CJA10	-1K143	/35.3:D	/35.3:E	
1							
	Relej sučelja 24VDC, 2CO Pomoćni relej Nazivni napon svitka: 24VDC 2 CO Narudžbeni br.: 2967060	Phoenix Contact					
			+U7CJA10	-1K144	/35.4:D	/35.4:E	/87.4:C
2			+U7CJA10	-1K145	/35.5:D	/35.5:E	/87.7:C
	Pomoćni relej Narudžbeni br.: 2966171	Phoenix Contact					
			+U7CJA10	-1K146	/35.6:D	/35.6:E	
1							

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.						
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 153
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			PAGE 156 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)		
5	Relej sučelja 24VDC, 2CO Pomoćni relej Nazivni napon svitka: 24VDC 2 CO Narudžbeni br.: 2967060	Phoenix Contact					
			+U7CJA10	-1K147	/35.7:D	/35.7:E	
			+U7CJA10	-1K148	/35.8:D	/35.8:E	
			+U7CJA10	-1K151	/36.1:D	/36.1:E	/86.2:C
			+U7CJA10	-1K152	/36.2:D	/36.2:E	/80.3:C
			+U7CJA10	-1K153	/36.3:D	/36.3:E	/80.4:C
1	Pomoćni relej Narudžbeni br.: 2966171	Phoenix Contact					
			+U7CJA10	-1K154	/36.4:D	/36.4:E	
1	Relej sučelja 24VDC, 2CO Pomoćni relej Nazivni napon svitka: 24VDC 2 CO Narudžbeni br.: 2967060	Phoenix Contact					
			+U7CJA10	-1K155	/36.5:D	/36.5:E	/87.2:C
1	Pomoćni relej Narudžbeni br.: 2966171	Phoenix Contact					
			+U7CJA10	-1K156	/36.6:D	/36.6:E	
			+U7CJA10	-1K157	/36.7:D	/36.7:E	
			+U7CJA10	-1K158	/36.8:D	/36.8:E	
			+U7CJA10	-1K161	/37.1:D	/37.1:E	
			+U7CJA10	-1K162	/37.2:D	/37.2:E	
			+U7CJA10	-1K163	/37.3:D	/37.3:E	
			+U7CJA10	-1K164	/37.4:D	/37.4:E	
			+U7CJA10	-1K165	/37.5:D	/37.5:E	
			+U7CJA10	-1K166	/37.6:D	/37.6:E	
			+U7CJA10	-1K167	/37.7:D	/37.7:E	
			+U7CJA10	-1K168	/37.8:D	/37.8:E	
			+U7CJA10	-1K171	/38.1:D	/38.1:E	

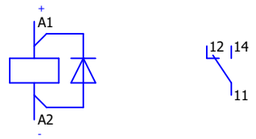
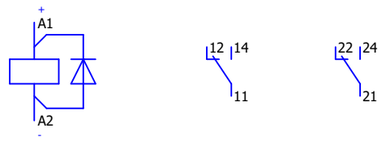
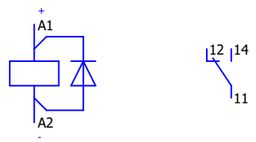
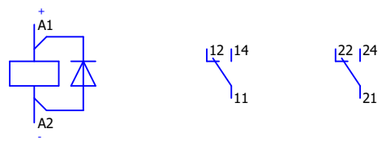
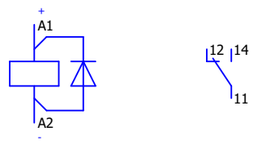
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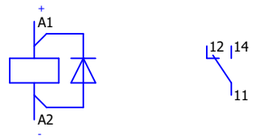
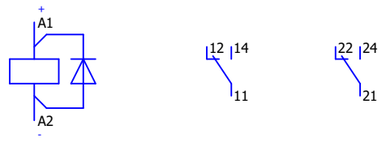
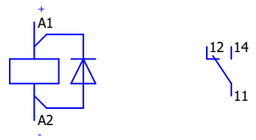
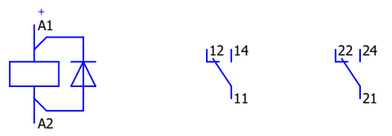
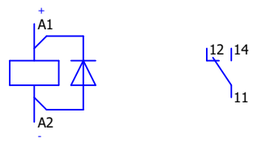
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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.:	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 154
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 157 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)		
	Pomoćni relej	Phoenix Contact					
	Narudžbeni br.: 2966171						
			+U7CJA10	-1K172	/38.2:D	/38.2:E	
14			+U7CJA10	-1K173	/38.3:D	/38.3:E	
	Relej sučelja 24VDC, 2CO	Phoenix Contact					
	Pomoćni relej Nazivni napon svitka: 24VDC 2 CO Narudžbeni br.: 2967060						
			+U7CJA10	-1K174	/38.4:D	/38.4:E	/86.2:C
2			+U7CJA10	-1K175	/38.5:D	/38.5:E	/87.1:C
	Pomoćni relej	Phoenix Contact					
	Narudžbeni br.: 2966171						
1			+U7CJA10	-1K176	/38.6:D	/38.6:E	
	Relej sučelja 24VDC, 2CO	Phoenix Contact					
	Pomoćni relej Nazivni napon svitka: 24VDC 2 CO Narudžbeni br.: 2967060						
1			+U7CJA10	-1K177	/38.7:D	/38.7:E	/95.5:B
	Pomoćni relej	Phoenix Contact					
	Narudžbeni br.: 2966171						
			+U7CJA10	-1K178	/38.8:D	/38.8:E	
			+U7CJA10	-1K181	/39.1:D	/39.1:E	
			+U7CJA10	-1K182	/39.2:D	/39.2:E	
			+U7CJA10	-1K183	/39.3:D	/39.3:E	
			+U7CJA10	-1K184	/39.4:D	/39.4:E	
			+U7CJA10	-1K185	/39.5:D	/39.5:E	
			+U7CJA10	-1K186	/39.6:D	/39.6:E	

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.						
Checked by	Antonio Aleksić	01.04.2026.				DEVICE TAG LIST	Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 155
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			PAGE 158 / 204

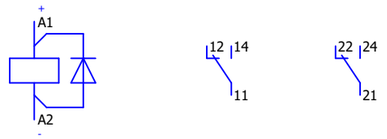
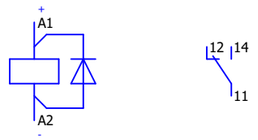
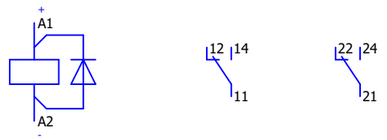
Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)		
10	Pomoćni relej Narudžbeni br.: 2966171	Phoenix Contact					
			+U7CJA10	-1K187	/39.7:D	/39.7:E	
			+U7CJA10	-1K188	/39.8:D	/39.8:E	
			+U7CJA10	-1K191	/40.1:D	/40.1:E	
2	Relej sučelja 24VDC, 2CO Pomoćni relej Nazivni napon svitka: 24VDC 2 CO Narudžbeni br.: 2967060	Phoenix Contact					
			+U7CJA10	-1K192	/40.2:D	/40.2:E	/80.5:C
			+U7CJA10	-1K193	/40.3:D	/40.3:E	/80.6:C
3	Pomoćni relej Narudžbeni br.: 2966171	Phoenix Contact					
			+U7CJA10	-1K194	/40.4:D	/40.4:E	
			+U7CJA10	-1K195	/40.5:D	/40.5:E	
			+U7CJA10	-1K196	/40.6:D	/40.6:E	
2	Relej sučelja 24VDC, 2CO Pomoćni relej Nazivni napon svitka: 24VDC 2 CO Narudžbeni br.: 2967060	Phoenix Contact					
			+U7CJA10	-1K197	/40.7:D	/40.7:E	/86.5:C
			+U7CJA10	-1K198	/40.8:D	/40.8:E	/87.4:C
1	Pomoćni relej Narudžbeni br.: 2966171	Phoenix Contact					
			+U7CJA10	-1K201	/41.1:D	/41.1:E	

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001		+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				DEVICE TAG LIST	Designer No.:	SALES No.:	PAGE 156
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 159 / 204

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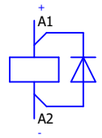
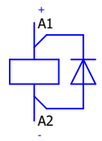
Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)		
8	Relej sučelja 24VDC, 2CO Pomoćni relej Nazivni napon svitka: 24VDC 2 CO Narudžbeni br.: 2967060	Phoenix Contact					
			+U7CJA10	-1K202	/41.2:D	/41.2:E	/80.7:C
			+U7CJA10	-1K203	/41.3:D	/41.3:E	/80.8:C
			+U7CJA10	-1K204	/41.4:D	/41.4:E	/81.3:C
			+U7CJA10	-1K205	/41.5:D	/41.5:E	/81.2:C
			+U7CJA10	-1K206	/41.6:D	/41.6:E	
			+U7CJA10	-1K207	/41.7:D	/41.7:E	/87.5:C
			+U7CJA10	-1K208	/41.8:D	/41.8:E	/86.6:C
			+U7CJA10	-1K211	/42.1:D	/42.1:E	/87.6:C
14	Pomoćni relej Narudžbeni br.: 2966171	Phoenix Contact					
			+U7CJA10	-1K212	/42.2:D	/42.2:E	
			+U7CJA10	-1K213	/42.3:D	/42.3:E	
			+U7CJA10	-1K214	/42.4:D	/42.4:E	
			+U7CJA10	-1K215	/42.5:D	/42.5:E	
			+U7CJA10	-1K216	/42.6:D	/42.6:E	
			+U7CJA10	-1K217	/42.7:D	/42.7:E	
			+U7CJA10	-1K218	/42.8:D	/42.8:E	
			+U7CJA10	-1K221	/43.1:D	/43.1:E	
			+U7CJA10	-1K222	/43.2:D	/43.2:E	
			+U7CJA10	-1K223	/43.3:D	/43.3:E	
			+U7CJA10	-1K224	/43.4:D	/43.4:E	
			+U7CJA10	-1K225	/43.5:D	/43.5:E	
			+U7CJA10	-1K226	/43.6:D	/43.6:E	
			+U7CJA10	-1K227	/43.7:D	/43.7:E	
1	Relej sučelja 24VDC, 2CO Pomoćni relej Nazivni napon svitka: 24VDC 2 CO Narudžbeni br.: 2967060	Phoenix Contact					
			+U7CJA10	-1K228	/43.8:D	/43.8:E	/86.6:C

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001
Approved by Designer	Grga Kolić	01.04.2026.					REV No.:
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		SALES No.:

= U7CJA10GH100
+ U7CJA10
PAGE 157
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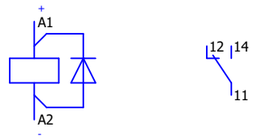
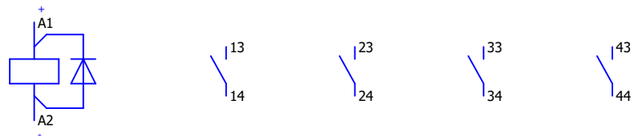
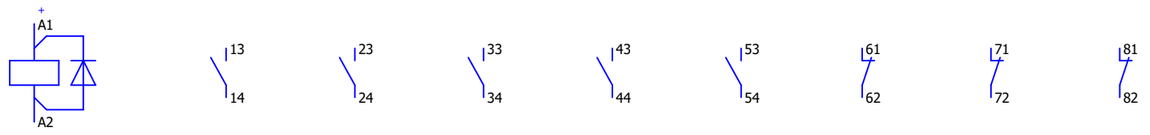
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Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)	
	Pomoćni relej Narudžbeni br.: 2966171	Phoenix Contact			 	
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			+U7CJA10	-1K232	/44.2:D	/44.2:E
			+U7CJA10	-1K233	/44.3:D	/44.3:E
			+U7CJA10	-1K234	/44.4:D	/44.4:E
			+U7CJA10	-1K235	/44.5:D	/44.5:E
			+U7CJA10	-1K236	/44.6:D	/44.6:E
			+U7CJA10	-1K237	/44.7:D	/44.7:E
			+U7CJA10	-1K238	/44.8:D	/44.8:E
			+U7CJA10	-1K241	/45.1:D	/45.1:E
			+U7CJA10	-1K242	/45.2:D	/45.2:E
			+U7CJA10	-1K243	/45.3:D	/45.3:E
			+U7CJA10	-1K244	/45.4:D	/45.4:E
			+U7CJA10	-1K245	/45.5:D	/45.5:E
			+U7CJA10	-1K246	/45.6:D	/45.6:E
			+U7CJA10	-1K247	/45.7:D	/45.7:E
16			+U7CJA10	-1K248	/45.8:D	/45.8:E
	Relej sučelja 24VDC, 2CO Pomoćni relej Nazivni napon svitka: 24VDC 2 CO Narudžbeni br.: 2967060	Phoenix Contact			  	
			+U7CJA10	-1K251	/46.1:D	/46.1:E
			+U7CJA10	-1K252	/46.2:D	/46.2:E
			+U7CJA10	-1K253	/46.3:D	/46.3:E
			+U7CJA10	-1K254	/46.4:D	/46.4:E
			+U7CJA10	-1K255	/46.5:D	/46.5:E
			+U7CJA10	-1K256	/46.6:D	/46.6:E
			+U7CJA10	-1K257	/46.7:D	/46.7:E
8			+U7CJA10	-1K258	/46.8:D	/46.8:E

Format ISO/DIN A3

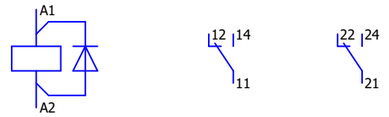
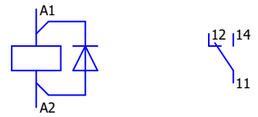
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.						
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 158
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			PAGE 161 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)							
8	Pomoćni relej Narudžbeni br.: 2966171	Phoenix Contact										
			+U7CJA10	-1K261	/47.1:D	/47.1:E						
			+U7CJA10	-1K262	/47.2:D	/47.2:E						
			+U7CJA10	-1K263	/47.3:D	/47.3:E						
			+U7CJA10	-1K264	/47.4:D	/47.4:E						
			+U7CJA10	-1K265	/47.5:D	/47.5:E						
			+U7CJA10	-1K266	/47.6:D	/47.6:E						
			+U7CJA10	-1K267	/47.7:D	/47.7:E						
			+U7CJA10	-1K268	/47.8:D	/47.8:E						
4	PAR.KONT.RELEJ,4NO,DC24V,DIOD SPOJNI KONTAKTOR RELEJ, 4NO, DC 24V, 0.85...1.85*US, W/ INTEGRIRANA DIODA, VELIČINA S00, NAVIJAČNI STEZALJKA Narudžbeni br.: 3RH2140-1VB40	Siemens										
			+U7CJA10	-2K01	/68.2:E	/91.1:B	/91.1:E					
			+U7CJA10	-2K02	/68.3:E	/91.2:B	/91.2:E					
			+U7CJA10	-2K03	/68.4:E	/91.3:B	/91.3:E					
			+U7CJA10	-2K04	/68.5:E	/91.4:B	/91.4:E					
1	Relej kontaktora Kontaktorski relej, 53E, DIN EN 50011, 5NO + 3NC, navojna stezaljka DC rad 24 V DC Narudžbeni br.: 3TH4253-0BB4 Prigušivač prenapona Tip: 3TX7402-3A Prigušivač prenapona Dioda za prigušivanje smetnji s odstojnikom kabela 24-250 V DC za montažu na stezaljku svitka za kontaktorske releje 3TH4	Siemens										
			+U7CJA10	-2K05	/68.6:E	/91.5:B	/91.5:E	/91.6:B	/91.6:B			

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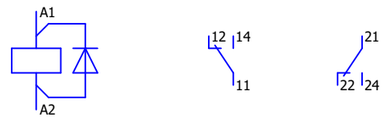
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.:	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 159 PAGE 162 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)		
2	Minijaturni plug-in relej - Zelio RXG 2 C/O 24VDC 5A Nazivni napon svitka: 24VDC Nazivna kontaktna struja: 5A 2 CO, baza: RGZE1S48M Zaštitna dioda: RXM040W Narudžbeni br.: RXG22BD Zaštitni modul Harmony, 6...230 V DC Tip: SE.RZM040W Harmony, zaštitni modul, s diodom, za RSZ, 6...230 V DC Utičnica Tip: RGZE1S48M Za RXG2.. releji	Schneider Electric					
			+U7CJA10	-2K101	/66.5:E	/90.6:B	/66.5:B
			+U7CJA10	-2K102	/66.4:E	/90.5:B	/66.4:B
4	RXG pomoćni relej, 1CO, 24VDC, 10A Nazivni napon svitka: 24VDC Nazivna kontaktna struja: 10A Baza: RGZE1S35M Zaštitna dioda: RXM040W Narudžbeni br.: RXG12BD Zaštitni modul Harmony, 6...230 V DC Tip: SE.RZM040W Harmony, zaštitni modul, s diodom, za RSZ, 6...230 V DC Utičnica RGZ - odvojeni kontakt - 10 A Tip: RGZE1S35M	Schneider Electric					
			+U7CJA10	-2K103	/66.7:E	/89.4:B	
			+U7CJA10	-2K104	/66.6:E	/89.3:B	
			+U7CJA10	-2K105	/66.3:E	/92.3:B	
			+U7CJA10	-2K106	/66.2:E	/92.2:B	

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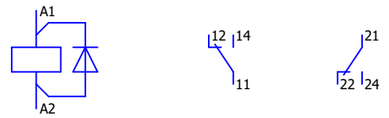
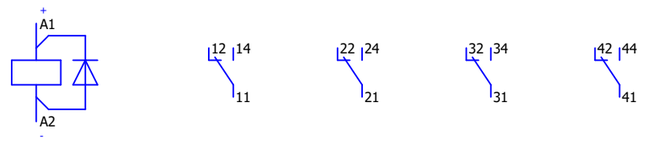
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.:	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 160 PAGE 163 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)	
	Minijaturni plug-in relej - Zelio RXG 2 C/O 24VDC 5A Nazivni napon svitka: 24VDC Nazivna kontaktna struja: 5A 2 CO, baza: RGZE1S48M Zaštitna dioda: RXM040W Narudžbeni br.: RXG22BD Zaštitni modul Harmony, 6...230 V DC Tip: SE.RZM040W Harmony, zaštitni modul, s diodom, za RSZ, 6...230 V DC Utičnica Tip: RGZE1S48M Za RXG2.. releji	Schneider Electric				
			+U7CJA10	-2K201	/50.2:C	/70.5:B
			+U7CJA10	-2K202	/50.3:C	/88.6:C
			+U7CJA10	-2K203	/50.4:C	
			+U7CJA10	-2K204	/50.5:C	/76.3:D
			+U7CJA10	-2K205	/50.6:C	/76.7:C
			+U7CJA10	-2K206	/51.3:C	/89.2:B
			+U7CJA10	-2K207	/52.3:C	/95.2:B
			+U7CJA10	-2K208	/52.4:C	/95.3:B
			+U7CJA10	-2K209	/52.5:C	/95.4:B
			+U7CJA10	-2K210	/52.6:C	/68.6:C
			+U7CJA10	-2K211	/53.1:C	/92.4:B
			+U7CJA10	-2K212	/53.2:C	/92.5:B
			+U7CJA10	-2K213	/53.3:C	/92.6:B
			+U7CJA10	-2K214	/53.4:C	/92.7:B
			+U7CJA10	-2K215	/53.5:C	/92.8:B
			+U7CJA10	-2K216	/53.6:C	/93.2:B
			+U7CJA10	-2K217	/53.7:C	/93.3:B
			+U7CJA10	-2K218	/53.8:C	/93.4:B
			+U7CJA10	-2K219	/54.1:C	
			+U7CJA10	-2K220	/54.2:C	
			+U7CJA10	-2K221	/54.3:C	/93.7:B
			+U7CJA10	-2K222	/54.4:C	/93.8:B
			+U7CJA10	-2K223	/54.5:C	/94.2:B
			+U7CJA10	-2K224	/54.6:C	/77.2:C
			+U7CJA10	-2K225	/54.7:C	/96.2:B
			+U7CJA10	-2K226	/54.8:C	/96.3:B
			+U7CJA10	-2K227	/55.1:C	/96.4:B
			+U7CJA10	-2K228	/55.2:C	/96.5:B

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.:	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 161 PAGE 164 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)			
	Minijaturni plug-in relej - Zelio RXG 2 C/O 24VDC 5A Nazivni napon svitka: 24VDC Nazivna kontaktna struja: 5A 2 CO, baza: RGZE1S48M Zaštitna dioda: RXM040W Narudžbeni br.: RXG22BD Zaštitni modul Harmony, 6...230 V DC Tip: SE.RZM040W Harmony, zaštitni modul, s diodom, za RSZ, 6...230 V DC Utičnica Tip: RGZE1S48M Za RXG2.. releji	Schneider Electric						
			+U7CJA10	-2K229	/55.3:C	/96.6:B		
			+U7CJA10	-2K230	/55.4:C			
			+U7CJA10	-2K231	/55.5:C	/96.7:B		
			+U7CJA10	-2K232	/55.6:C	/97.2:B		
			+U7CJA10	-2K233	/55.7:C	/97.3:B		
34			+U7CJA10	-2K234	/55.8:C	/97.4:B		
	Minijaturni utični relej - Zelio RXM 4C/O 24VDC 6A 6 A 4 CO Sa LED diodom S gumbom za testiranje koji se može zaključati 24 V DC Narudžbeni br.: RXM4AB2BD Utičnica Tip: RXZE2S114M Za RXM2/RXM4 releje Vijčani spojnici Odvojeni kontakt Zaštitni modul Harmony, 6...230 V DC Tip: SE.RZM040W Harmony, zaštitni modul, s diodom, za RSZ, 6...230 V DC	Schneider Electric						
1			+U7CJA10	-2K401	/69.5:E	/90.2:B	/69.6:B	

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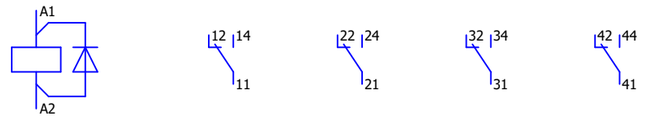
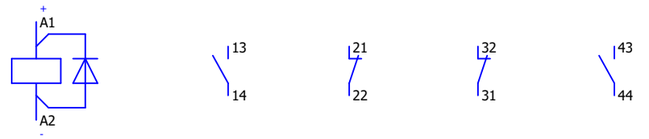
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 162
Checked by	Antonio Aleksić	01.04.2026.						PAGE 165 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			

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Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)								
1	SIRIUS kontaktorski relej Relej kontaktora, 4NO, 24VDC, s integriranom diodom, veličina S00, vijčani terminal Narudžbeni br.: 3RH2140-1FB40 POM. BLOK PREKIDAČA, 3NO+1NC USL. STAZE: 1NO, 1NC, Tip: 3RH2911-1GA31 Pomoćni prekidački blok, 3NO+1NC uvjetni putovi: 1NO, 1NC, 1NO, 1NO, za kontaktorske releje veličine S00, vijčani terminal DIN EN 50011	Siemens	+U7CJA10	-K01	/70.5:E	/79.4:C	/79.4:C	/79.5:C	/79.5:C	/81.5:C	/81.4:C	/70.4:C	/65.5:B
1	KONTAKT.RELEJ,2NO+2NC,DC24V,DIODA Kontaktorski relej, 2NO+2NC, DC 24V, s integriranom diodom, veličina S00, vijčana stezaljka Narudžbeni br.: 3RH2122-1FB40 Tip: 3RH2911-1GA04	Siemens	+U7CJA10	-K30.1	/76.a.4:E	/38.6:C	/94.6:B	/94.7:B		/94.3:B	/94.4:B	/76.b.5:B	
1	Relej kontaktora Kontaktorski relej, 44E, DIN EN 50011, 4NO + 4NC, navojna stezaljka DC rad 24 V DC Narudžbeni br.: 3TH4244-0BB4 Prigušivač prenapona Tip: 3TX7402-3A Prigušivač prenapona Dioda za prigušivanje smetnji s odstojnikom kabela 24-250 V DC za montažu na stezaljku svitka za kontaktorske releje 3TH4	Siemens	+U7CJA10	-K30.2	/76.a.4:E					/91.7:B	/91.7:B	/91.8:B	/76.b.5:C

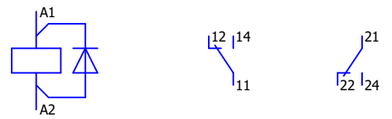
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.:	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 163
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 166 / 204

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Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)				
	Minijaturni utični relej - Zelio RXM 4C/O 24VDC 6A 6 A 4 CO Sa LED diodom S gumbom za testiranje koji se može zaključati 24 V DC Narudžbeni br.: RXM4AB2BD Utičnica Tip: RXZE2S114M Za RXM2/RXM4 releje Vijčani spojnici Odvojeni kontakt Zaštitni modul Harmony, 6...230 V DC Tip: SE.RZM040W Harmony, zaštitni modul, s diodom, za RSZ, 6...230 V DC	Schneider Electric							
			+U7CJA10	-K100	/65.4:E	/65.6:B	/70.4:B		
			+U7CJA10	-K101	/65.5:E	/78.2:B	/78.6:B	/77.3:C	
			+U7CJA10	-K102	/67.2:E	/68.2:B	/18.7:D	/81.6:C	
			+U7CJA10	-K103	/67.4:E	/68.4:B	/18.8:D	/81.7:C	
			+U7CJA10	-K104	/67.6:E	/68.6:B	/19.1:D	/81.8:C	/72.2:C
			+U7CJA10	-K105	/69.4:E	/69.3:C	/90.2:C	/90.4:C	
			+U7CJA10	-K106	/71.4:E	/18.5:D		/72.2:B	
			+U7CJA10	-K107	/71.4:E	/18.6:D	/65.5:C	/72.3:B	
9			+U7CJA10	-K108	/71.5:E	/65.6:C	/72.4:B	/72.2:E	/21.6:D
	KONTAKT.RELEJ,2NO+2NC,DC24V,DIODA Kontaktorski relej, 2NO+2NC, DC 24V, s integriranom diodom, veličina S00, vijčana stezaljka Narudžbeni br.: 3RH2122-1FB40	Siemens							
			+U7CJA10	-K150	/73.5:E	/101.4:D		/101.4:D	/83.3:C
2			+U7CJA10	-K151	/73.6:E	/101.4:D		/101.4:D	/83.5:C

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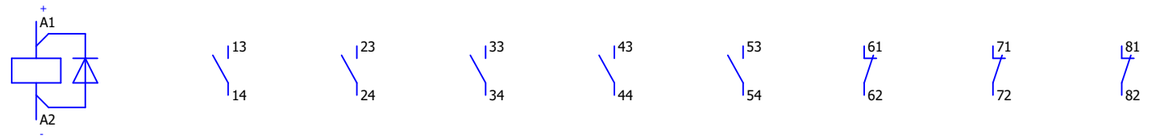
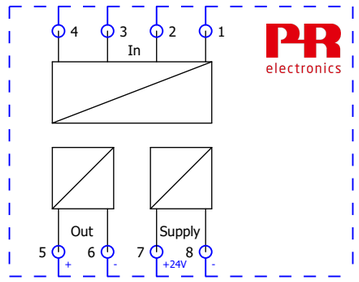
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 164
Checked by	Antonio Aleksić	01.04.2026.						PAGE 167 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)		
	Minijaturni plug-in relej - Zelio RXG 2 C/O 24VDC 5A Nazivni napon svitka: 24VDC Nazivna kontaktna struja: 5A 2 CO, baza: RGZE1S48M Zaštitna dioda: RXM040W Narudžbeni br.: RXG22BD Zaštitni modul Harmony, 6...230 V DC Tip: SE.RZM040W Harmony, zaštitni modul, s diodom, za RSZ, 6...230 V DC Utičnica Tip: RGZE1S48M Za RXG2.. releji	Schneider Electric					
			+U7CJA10	-K169	/98.6:B	/98.6:C	
			+U7CJA10	-K201	/17.2:E	/20.4:D	/86.4:C
			+U7CJA10	-K202	/17.3:E	/20.5:D	/86.4:C
			+U7CJA10	-K203	/69.2:E	/69.3:B	/90.2:C
			+U7CJA10	-K204	/69.3:E	/90.3:C	
			+U7CJA10	-K205	/73.2:E	/77.3:C	/83.6:C
			+U7CJA10	-K206	/73.2:E	/70.5:D	/83.7:C
			+U7CJA10	-K207	/74.2:E	/75.1:B	/75.2:B
			+U7CJA10	-K208	/73.3:E	/88.5:C	
			+U7CJA10	-K209	/74.2:E	/75.5:B	/75.6:B
			+U7CJA10	-K210	/74.3:E	/75.7:B	/75.8:B
			+U7CJA10	-K211	/74.3:E	/75.1:C	/75.2:C
			+U7CJA10	-K212	/74.4:E	/75.3:C	/75.4:C
			+U7CJA10	-K213	/74.4:E	/75.5:C	/75.6:C
			+U7CJA10	-K214	/74.5:E	/75.7:C	/75.8:C
			+U7CJA10	-K215	/74.5:E	/75.1:E	/75.2:E
			+U7CJA10	-K216	/74.6:E	/75.3:E	/75.4:E
			+U7CJA10	-K217	/74.7:E	/75.5:E	/75.6:E
			+U7CJA10	-K218	/74.7:E	/75.7:E	/75.8:E
			+U7CJA10	-K219	/74.8:E	/75.a.1:B	/75.a.2:B
21			+U7CJA10	-K220	/80.2:D	/80.2:C	/83.1:C

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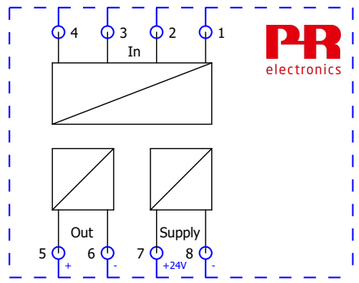
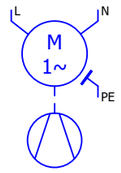
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.						
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 165
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			PAGE 168 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)
1	Relej kontaktora Kontaktorski relej, 53E, DIN EN 50011, 5NO + 3NC, navojna stezaljka DC rad 24 V DC Narudžbeni br.: 3TH4253-0BB4 Prigušivač prenapona Tip: 3TX7402-3A Prigušivač prenapona Dioda za prigušivanje smetnji s odstojnikom kabela 24-250 V DC za montažu na stezaljku svitka za kontaktorske releje 3TH4	Siemens	+U7CJA10	-K401	 /72.2:E /72.6:B /72.6:B /72.6:F /72.6:E
	Pt100 pretvarač Pt100 ulaz, mA/V izlaz, napajanje preko stezaljki Tip: 3102 Narudžbeni br.: 3102	PR electronics	+U7CJA10	-KF101	 /58.1:C
			+U7CJA10	-KF101	/58.1:C
			+U7CJA10	-KF102	/58.3:C
			+U7CJA10	-KF102	/58.3:C
			+U7CJA10	-KF103	/58.5:C
			+U7CJA10	-KF103	/58.5:C
			+U7CJA10	-KF104	/58.7:C
			+U7CJA10	-KF104	/58.7:C
			+U7CJA10	-KF105	/59.1:C
			+U7CJA10	-KF105	/59.1:C
			+U7CJA10	-KF106	/59.3:C
			+U7CJA10	-KF106	/59.3:C
			+U7CJA10	-KF107	/59.5:C
			+U7CJA10	-KF107	/59.5:C
			+U7CJA10	-KF108	/59.7:C
			+U7CJA10	-KF108	/59.7:C
			+U7CJA10	-KF109	/60.1:C
			+U7CJA10	-KF109	/60.1:C
			+U7CJA10	-KF110	/60.3:C
			+U7CJA10	-KF110	/60.3:C

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.						
Checked by	Antonio Aleksić	01.04.2026.				DEVICE TAG LIST	Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 166
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			PAGE 169 / 204

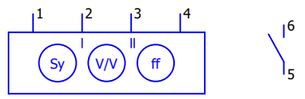
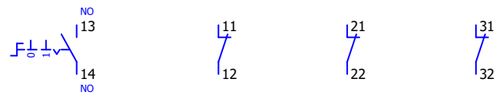
Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)
	Pt100 pretvarač Pt100 ulaz, mA/V izlaz, napajanje preko stezaljki Tip: 3102 Narudžbeni br.: 3102	PR electronics			
			+U7CJA10	-KF111	/60.5:C
			+U7CJA10	-KF111	/60.5:C
			+U7CJA10	-KF112	/60.7:C
			+U7CJA10	-KF112	/60.7:C
			+U7CJA10	-KF113	/61.1:C
			+U7CJA10	-KF113	/61.1:C
			+U7CJA10	-KF114	/61.3:C
			+U7CJA10	-KF114	/61.3:C
			+U7CJA10	-KF115	/61.5:C
			+U7CJA10	-KF115	/61.5:C
30			+U7CJA10	-KF115	/61.5:C
	Ventilator i rešetka Nazivni napon: 230VAC Nazivna frekvencija: 50Hz Nazivna struja: 0.21A Nazivna snaga: 35W Narudžbeni br.: SK3240.100+SK3240.200	Rittal			
			+U7CJA10	-M101	/6.7:F
2			+U7CJA10	-M102	/6.8:F
	analogni panel mjerač Narudžbeni br.: EA16N xxxxxxxxxx	LUMEL			
			+U7CJA10	-P110	/102.4:C
2			+U7CJA10	-P111	/102.6:C

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.				DEVICE TAG LIST	Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 167 PAGE 170 / 204
Checked by	Antonio Aleksić	01.04.2026.						
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			

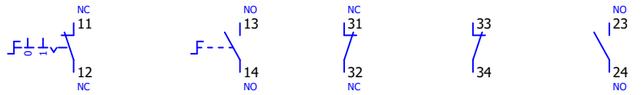
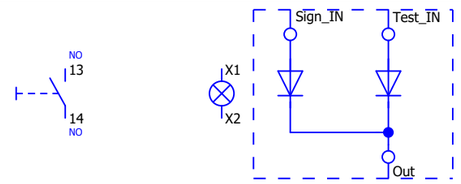
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Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)
	Sinhronoskop za ručnu sinhronizaciju s izlaznim kontaktom Skala: 360° X - prikazana vrijednost: 110 kV Prikazana vrijednost napona: 230 V Narudžbeni br.: SQ0214-N-57.7V-R3-N-1-S-A-X	ISKRA			
1			+U7CJA10	-P411	/79.4:D /72.2:D
	Grijač Nazivna snaga: 100W Ulazni napon: 110-250V Narudžbeni br.: HT100	Alfa Electric			
			+U7CJA10	-R101	/6.5:F
2			+U7CJA10	-R102	/6.6:F
	Krajnja sklopka Krajnji prekidač, granični prekidači XC Standard, XCKN, klip na metalnom kraju, 1NC+1 NO, uskočnik, Pg11 Narudžbeni br.: XCKN2110G11	Schneider Electric			
			+U7CJA10	-S101	/6.3:E
2			+U7CJA10	-S102	/6.4:E
	PREKLOPNI PREKIDAČ, O-I, CRNI, BIJELI Preklopni prekidač, s osvjetljenjem, 22 mm, okrugli, metalni, sjajni, bijeli, kratki, 2 položaja O-I, zadrživ (latching), kut zakretanja 90°, 10:30h/13:30h, s nosačem, 1 NO, na vijak Narudžbeni br.: 3SU1150-2BF60-1BA0 Modul s jednim kontaktom, 1 NC Tip: 3SU1400-1AA10-1CA0 Modul kontakata s 1 NC kontaktom, vijčane stezaljke KONTAKTNI MODUL 2NC Tip: 3SU1400-1AA10-1EA0 Kontaktni modul s 2 kontaktna elementa, 2 NC, navojna stezaljka, za montažu na prednju ploču	Siemens			
1			+U7CJA10	-S103	/18.1:D /78.2:C /78.6:C /65.4:B

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001		+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.				DEVICE TAG LIST	Designer No.:	SALES No.:	PAGE 168
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 171 / 204

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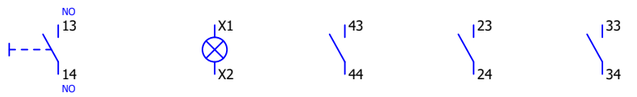
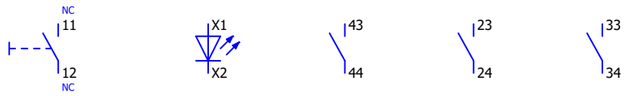
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Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)
1	PREKLOPNI PREKIDAČ, O-I, CRNI, BIJELI Izborna sklopka, osvijetljena, 22 mm, okrugla, metalna, sjajna, bijela, izborna sklopka, kratka, 2 položaja prekidača O-I, zasun, kut aktiviranja 90°, 10:30h/13:30h, s držačem, 1 NO, 1 NC, navojna stezaljka Narudžbeni br.: 3SU1150-2BF60-1MA0 Modul s jednim kontaktom, 1 NC Tip: 3SU1400-1AA10-1CA0 Modul kontakata s 1 NC kontaktom, vijčane stezaljke Nosač oznake Tip: 3SU1900-0AP10-0AA0 Držač etikete, 22mm, pravokutni okvir na dnu, crni, za etiketnu pločicu 17.5 mm x 27 mm, za lijepljenje	Siemens			
			+U7CJA10	-S104	/18.2:D /65.4:C /18.3:D
3	SVJETLEĆA TIPKA, BIJELA Osvijetljeno tipkalo, 22 mm, okruglo, metalno, sjajno, bijelo, tipkalo, ravno, trenutni kontakt, s držačem, 1 NO, LED modul s integriranom LED 24 V AC/DC, navojna stezaljka Narudžbeni br.: 3SU1152-0AB60-1BA0 Nosač oznake Tip: 3SU1900-0AP10-0AA0 Držač etikete, 22mm, pravokutni okvir na dnu, crni, za etiketnu pločicu 17.5 mm x 27 mm, za lijepljenje LED TESTNI MODUL Tip: 3SU1400-1CK10-1AA0 LED ispitni modul Radni napon 12-240 V AC/DC, vijčani terminal, za montažu na prednju ploču	Siemens			
			+U7CJA10	-S111	/67.2:B /81.6:D /81.6:C
			+U7CJA10	-S112	/67.4:B /81.7:D /81.7:C
			+U7CJA10	-S113	/67.6:B /81.8:D /81.8:C

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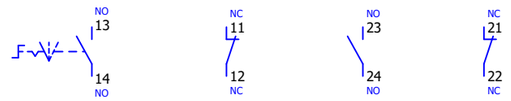
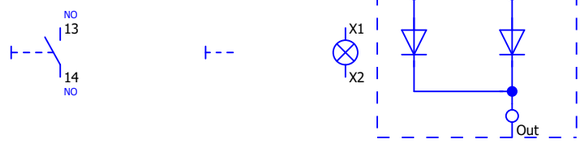
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.				DEVICE TAG LIST		
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 169
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			PAGE 172 / 204

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Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)
	SVJETLEĆA TIPKA, ZELENA Osvijetljena tipka, 22 mm, okrugla, metalna, sjajna, zelena, tipka, ravna, trenutni kontakt, s držačem, 1 NO, LED modul s integriranom LED 24 V AC/DC, navojna stezaljka Narudžbeni br.: 3SU1152-0AB40-1BA0 Nosač oznake Tip: 3SU1900-0AP10-0AA0 Držač etikete, 22mm, pravokutni okvir na dnu, crni, za etiketnu pločicu 17.5 mm x 27 mm, za lijepljenje Modul s jednim kontaktom, 1 NO Tip: 3SU1400-1AA10-1BA0 Modul kontakata s 1NO kontaktom, vijčane stezaljke Contact module with 1 contact element, 1 NO Tip: 3SU1400-1AA10-1BA0 Contact module with 1 contact element, 1 NO, screw terminal, for front plate mounting Contact module with 1 contact element, 1 NO Tip: 3SU1400-1AA10-1BA0 Contact module with 1 contact element, 1 NO, screw terminal, for front plate mounting	Siemens			
1			+U7CJA10	-S114	/19.2:D /68.6:C /68.2:C /68.4:C
	SVJETLEĆA TIPKA, CRVENA Narudžbeni br.: 3SU1152-0AB20-1CA0 Nosač oznake Tip: 3SU1900-0AP10-0AA0 Držač etikete, 22mm, pravokutni okvir na dnu, crni, za etiketnu pločicu 17.5 mm x 27 mm, za lijepljenje Modul s jednim kontaktom, 1 NO Tip: 3SU1400-1AA10-1BA0 Modul kontakata s 1NO kontaktom, vijčane stezaljke Contact module with 1 contact element, 1 NO Tip: 3SU1400-1AA10-1BA0 Contact module with 1 contact element, 1 NO, screw terminal, for front plate mounting Contact module with 1 contact element, 1 NO Tip: 3SU1400-1AA10-1BA0 Contact module with 1 contact element, 1 NO, screw terminal, for front plate mounting	Siemens			
1			+U7CJA10	-S115	/19.3:D /72.2:C /68.3:C /68.5:C

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.:	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 170 PAGE 173 / 204
Checked by	Antonio Aleksić	01.04.2026.							
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

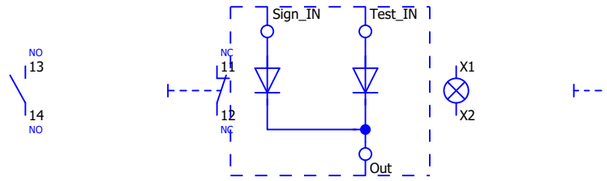
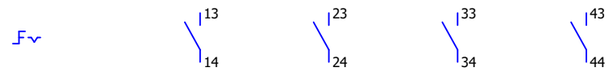
Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)
1	Preklopka %0-%1-%2 za izbor načina upravljanja Prekidač za odabir, osvijetljen, 22 mm, okrugli, metalni, sjajni, bijeli, prekidač za izbor, kratki, 3 položaja prekidača I-O-II, zasun, kut aktiviranja 2x45°, 10:30h/12h/13:30h, sa držačem, 2x1NO+1NC, vijčana stezaljka Narudžbeni br.: 3SU1150-2BL60-1LA0 Nosač oznake Tip: 3SU1900-0AP10-0AA0 Držač etikete, 22mm, pravokutni okvir na dnu, crni, za etiketnu pločicu 17.5 mm x 27 mm, za lijepljenje	Siemens	+U7CJA10	-S121	 /71.4:B /71.5:B /71.4:B /71.5:C
1	Modul s jednim kontaktom, 1 NO Modul kontakata s 1NO kontaktom, vijčane stezaljke Narudžbeni br.: 3SU1400-1AA10-1BA0 SVJETLEĆA TIPKA, CRVENA Tip: 3SU1051-0AB20-0AA0 Osvijetljena tipka, 22 mm, okrugla, metalna, sjajna, crvena, tipka, ravna, trenutni kontakt DRŽAČ Tip: 3SU1550-0AA10-0AA0 Držač za 3 modula, univerzalni, plastika/metal LED MODULE, RED Tip: 3SU1401-1BB20-1AA0 LED MODULE WITH INTEGRATED LED 24V AC/DC, RED, SCREW TERMINAL, FOR FRONT PLATE MOUNTING SIRIUS ACT Label holder Tip: 3SU1900-0AP10-0AA0 Label holder, 22mm, frame rectangular at the bottom, black, for labeling plate 17.5 mm x 27 mm, for gluing	Siemens	+U7CJA10	-S130	 /70.4:C /81.5:D /81.5:C

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.:	= U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.							
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.:	PAGE 171 PAGE 174 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER				

CONFIDENTIAL, ALL RIGHTS RESERVED DIN 34

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)
1	Modul s kontaktima 1NO+1NC Modul s 2 kontakta, 1NO+1NC, vijčane stezaljke, za montažu na vrata Narudžbeni br.: 3SU1400-1AA10-1FA0 DRŽAČ Tip: 3SU1550-0AA10-0AA0 Držač za 3 modula, univerzalni, plastika/metal LED TESTNI MODUL Tip: 3SU1400-1CK10-1AA0 LED ispitni modul Radni napon 12-240 V AC/DC, vijčani terminal, za montažu na prednju ploču LED MODULE, GREEN Tip: 3SU1401-1BB40-1AA0 LED module with integrated LED 24 V AC/DC, green, screw terminal, for front plate mounting ILLUMINATED PUSHBUTTON, GREEN Tip: 3SU1051-0AB40-0AA0 Illuminated pushbutton, 22 mm, round, metal, shiny, green, pushbutton, flat, momentary contact type	Siemens	+U7CJA10	-S131	 /70.4:C /81.4:C /81.4:D
1	PREKIDAČ I-O-II BIJELI Prekidač za izbor, osvijetljen, 22 mm, okrugli, metalni, sjajni, bijeli, prekidač za izbor, kratak, 3 položaja prekidača I-O-II, zasun, kut aktiviranja 2x45°, 10:30h/12h/13:30h Narudžbeni br.: 3SU1052-2BL60-0AA0 DRŽAČ Tip: 3SU1550-0AA10-0AA0 Držač za 3 modula, univerzalni, plastika/metal Modul s jednim kontaktom, 1 NO Tip: 3SU1400-1AA10-1BA0 Modul kontakata s 1NO kontaktom, vijčane stezaljke Contact module with 1 contact element, 1 NO Tip: 3SU1400-1AA10-1BA0 Contact module with 1 contact element, 1 NO, screw terminal, for front plate mounting Contact module with 1 contact element, 1 NO Tip: 3SU1400-1AA10-1BA0 Contact module with 1 contact element, 1 NO, screw terminal, for front plate mounting	Siemens	+U7CJA10	-S141	 /19.4:D /19.4:D /19.5:D /69.2:B /69.4:B

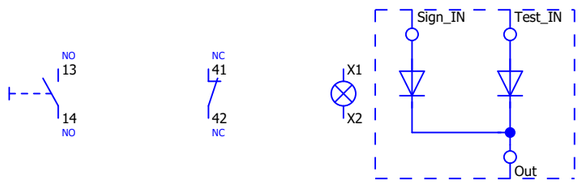
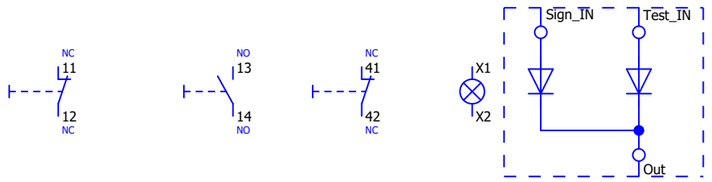
Format ISO/DIN A3

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.						
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 172
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			PAGE 175 / 204

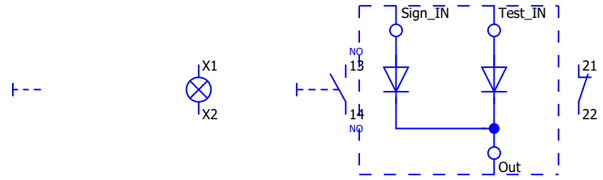
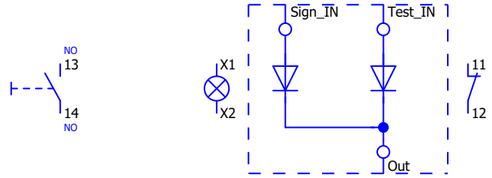
CONFIDENTIAL, ALL RIGHTS RESERVED DIN 34

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)
1	SVJETLEĆA TIPKA, CRVENA Osvijetljeno tipkalo, 22 mm, okruglo, metalno, sjajno, crveno, tipkalo, ravno, trenutni kontakt, s držačem, 1 NO+1 NC, LED modul s integriranom LED 24 V AC/DC, navojna stezaljka Narudžbeni br.: 3SU1152-0AB20-1FA0 Nosač oznake Tip: 3SU1900-0AP10-0AA0 Držač etikete, 22mm, pravokutni okvir na dnu, crni, za etiketnu pločicu 17.5 mm x 27 mm, za lijepljenje LED TESTNI MODUL Tip: 3SU1400-1CK10-1AA0 LED ispitni modul Radni napon 12-240 V AC/DC, vijčani terminal, za montažu na prednju ploču	Siemens	+U7CJA10	-S142	 /69.5:B /80.7:D /80.7:C
1	SVJETLEĆA TIPKA, ZELENA Osvijetljena tipka, 22 mm, okrugla, metalna, sjajna, zelena, tipka, ravna, trenutni kontakt, s držačem, 1 NO+1 NC, LED modul s integriranom LED 24 V AC/DC, navojna stezaljka Narudžbeni br.: 3SU1152-0AB40-1FA0 Nosač oznake Tip: 3SU1900-0AP10-0AA0 Držač etikete, 22mm, pravokutni okvir na dnu, crni, za etiketnu pločicu 17.5 mm x 27 mm, za lijepljenje LED TESTNI MODUL Tip: 3SU1400-1CK10-1AA0 LED ispitni modul Radni napon 12-240 V AC/DC, vijčani terminal, za montažu na prednju ploču	Siemens	+U7CJA10	-S143	 /69.5:C /80.8:D /80.8:C

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.						
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 173
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			PAGE 176 / 204

Format ISO/DIN A3

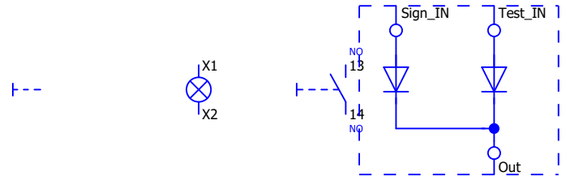
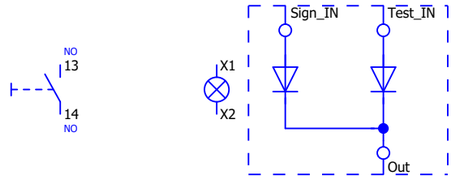
CONFIDENTIAL, ALL RIGHTS RESERVED DIN 34

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)
1	SVJETLEĆA TIPKA, CRVENA Osvijetljena tipka, 22 mm, okrugla, metalna, sjajna, crvena, tipka, ravna, trenutni kontakt Narudžbeni br.: 3SU1051-0AB20-0AA0 DRŽAČ Tip: 3SU1550-0AA10-0AA0 Držač za 3 modula, univerzalni, plastika/metal LED MODUL, CRVENI Tip: 3SU1401-1BB20-1AA0 LED MODUL S INTEGRIRANOM LED 24V AC/DC, CRVENI, NAVIJAČNI TERMINAL, ZA MONTAŽU NA PREDNJU PLOČU Contact module with 1 contact element, 1 NO Tip: 3SU1400-1AA10-1BA0 Contact module with 1 contact element, 1 NO, screw terminal, for front plate mounting SIRIUS ACT Label holder Tip: 3SU1900-0AP10-0AA0 Label holder, 22mm, frame rectangular at the bottom, black, for labeling plate 17.5 mm x 27 mm, for gluing	Siemens	+U7CJA10	-S151	 /81.2:D /66.4:B /81.2:C /66.5:B
1	SVJETLEĆA TIPKA, ZELENA Osvijetljena tipka, 22 mm, okrugla, metalna, sjajna, zelena, tipka, ravna, trenutni kontakt, s držačem, 1 NO, LED modul s integriranom LED 24 V AC/DC, navojna stezaljka Narudžbeni br.: 3SU1152-0AB40-1BA0 Nosač oznake Tip: 3SU1900-0AP10-0AA0 Držač etikete, 22mm, pravokutni okvir na dnu, crni, za etiketnu pločicu 17.5 mm x 27 mm, za lijepljenje LED TESTNI MODUL Tip: 3SU1400-1CK10-1AA0 LED ispitni modul Radni napon 12-240 V AC/DC, vijčani terminal, za montažu na prednju ploču	Siemens	+U7CJA10	-S152	 /66.5:B /81.3:D /81.3:C /66.4:B

Format ISO/DIN A3

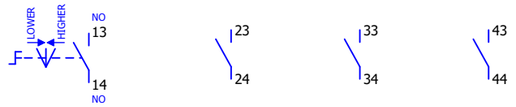
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 174 PAGE 177 / 204
Checked by	Antonio Aleksić	01.04.2026.						
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			

CONFIDENTIAL, ALL RIGHTS RESERVED DIN 34

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)
	SVJETLEĆA TIPKA, CRVENA Osvijetljena tipka, 22 mm, okrugla, metalna, sjajna, crvena, tipka, ravna, trenutni kontakt Narudžbeni br.: 3SU1051-0AB20-0AA0 DRŽAČ Tip: 3SU1550-0AA10-0AA0 Držač za 3 modula, univerzalni, plastika/metal LED MODUL, CRVENI Tip: 3SU1401-1BB20-1AA0 LED MODUL S INTEGRIRANOM LED 24V AC/DC, CRVENI, NAVIJAČNI TERMINAL, ZA MONTAŽU NA PREDNJU PLOČU Contact module with 1 contact element, 1 NO Tip: 3SU1400-1AA10-1BA0 Contact module with 1 contact element, 1 NO, screw terminal, for front plate mounting SIRIUS ACT Label holder Tip: 3SU1900-0AP10-0AA0 Label holder, 22mm, frame rectangular at the bottom, black, for labeling plate 17.5 mm x 27 mm, for gluing	Siemens			
1			+U7CJA10	-S161	/80.5:D /66.6:B /80.5:C
	SVJETLEĆA TIPKA, ZELENA Osvijetljena tipka, 22 mm, okrugla, metalna, sjajna, zelena, tipka, ravna, trenutni kontakt, s držačem, 1 NO, LED modul s integriranom LED 24 V AC/DC, navojna stezaljka Narudžbeni br.: 3SU1152-0AB40-1BA0 Nosač oznake Tip: 3SU1900-0AP10-0AA0 Držač etikete, 22mm, pravokutni okvir na dnu, crni, za etiketnu pločicu 17.5 mm x 27 mm, za lijepljenje LED TESTNI MODUL Tip: 3SU1400-1CK10-1AA0 LED ispitni modul Radni napon 12-240 V AC/DC, vijčani terminal, za montažu na prednju ploču	Siemens			
1			+U7CJA10	-S162	/66.7:B /80.6:D /80.6:C

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.				DEVICE TAG LIST	Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 175
Checked by	Antonio Aleksić	01.04.2026.						PAGE 178 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			

Format ISO/DIN A3

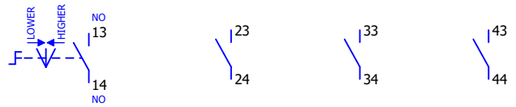
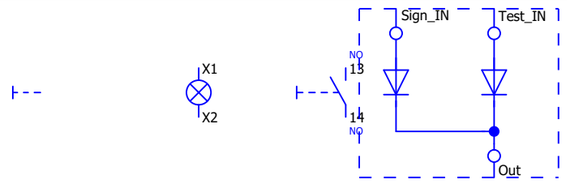
Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)
	TIPKA, CRNA Tipkalo, 22 mm, okruglo, metalno, sjajno, crno, tipkalo, ravno, trenutni kontakt, s držačem, 1 NO, navojna stezaljka Narudžbeni br.: 3SU1150-0AB10-1BA0 Nosač oznake Tip: 3SU1900-0AP10-0AA0 Držač etikete, 22mm, pravokutni okvir na dnu, crni, za etiketnu pločicu 17.5 mm x 27 mm, za lijepljenje	Siemens			
1			+U7CJA10	-S203	/80.2:C
	IZBORNI PREKIDAČ, I>O<II, CRVENI Prekidač za izbor, osvijetljen, 22 mm, okrugli, metalni, sjajni, crveni, prekidač za izbor, kratak, 3 položaja prekidača I>O<II, trenutni kontakt, kut aktiviranja 2x45°, 10:30h/12h/13:30h Narudžbeni br.: 3SU1052-2BM20-0AA0 Modul s jednim kontaktom, 1 NO Tip: 3SU1400-1AA10-1BA0 Modul kontakata s 1NO kontaktom, vijčane stezaljke Modul s jednim kontaktom, 1 NO Tip: 3SU1400-1AA10-1BA0 Modul kontakata s 1NO kontaktom, vijčane stezaljke Contact module with 1 contact element, 1 NO Tip: 3SU1400-1AA10-1BA0 Contact module with 1 contact element, 1 NO, screw terminal, for front plate mounting Contact module with 1 contact element, 1 NO Tip: 3SU1400-1AA10-1BA0 Contact module with 1 contact element, 1 NO, screw terminal, for front plate mounting	Siemens			
1			+U7CJA10	-S221	/19.8:D /78.6:D /78.7:D /20.1:D

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Format ISO/DIN A3

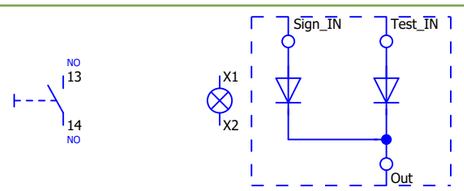
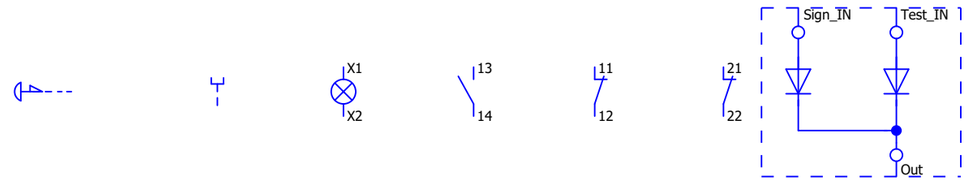
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company **Sintaksa - A Voith Company** Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.:	REV No.:	= U7CJA10GH100
Approved by Designer	Grga Kolić	01.04.2026.					PERU-01-17-170-31-001		+ U7CJA10
Checked by	Antonio Aleksić	01.04.2026.					Designer No.:	SALES No.:	PAGE 176
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER		22044-DD-EE-08-07-100		PAGE 179 / 204

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Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)
	IZBORNI PREKIDAČ, I>O<II, PLAVI Prekidač za odabir, osvijetljen, 22 mm, okrugli, metalni, sjajni, plavi, prekidač za odabir, kratak, 3 položaja prekidača I>O<II, trenutni kontakt, kut aktiviranja 2x45°, 10:30h/12h/13:30h Narudžbeni br.: 3SU1052-2BM50-0AA0 Modul s jednim kontaktom, 1 NO Tip: 3SU1400-1AA10-1BA0 Modul kontakata s 1NO kontaktom, vijčane stezaljke Modul s jednim kontaktom, 1 NO Tip: 3SU1400-1AA10-1BA0 Modul kontakata s 1NO kontaktom, vijčane stezaljke Contact module with 1 contact element, 1 NO Tip: 3SU1400-1AA10-1BA0 Contact module with 1 contact element, 1 NO, screw terminal, for front plate mounting Contact module with 1 contact element, 1 NO Tip: 3SU1400-1AA10-1BA0 Contact module with 1 contact element, 1 NO, screw terminal, for front plate mounting	Siemens			
1			+U7CJA10	-S222	/20.2:D /78.2:D /78.3:D /20.3:D
	SVJETLEĆA TIPKA, CRVENA Osvijetljena tipka, 22 mm, okrugla, metalna, sjajna, crvena, tipka, ravna, trenutni kontakt Narudžbeni br.: 3SU1051-0AB20-0AA0 DRŽAČ Tip: 3SU1550-0AA10-0AA0 Držač za 3 modula, univerzalni, plastika/metal LED MODUL, CRVENI Tip: 3SU1401-1BB20-1AA0 LED MODUL S INTEGRIRANOM LED 24V AC/DC, CRVENI, NAVIJAČNI TERMINAL, ZA MONTAŽU NA PREDNJU PLOČU Contact module with 1 contact element, 1 NO Tip: 3SU1400-1AA10-1BA0 Contact module with 1 contact element, 1 NO, screw terminal, for front plate mounting SIRIUS ACT Label holder Tip: 3SU1900-0AP10-0AA0 Label holder, 22mm, frame rectangular at the bottom, black, for labeling plate 17.5 mm x 27 mm, for gluing	Siemens			
1			+U7CJA10	-S223	/80.3:D /66.2:B /80.3:C

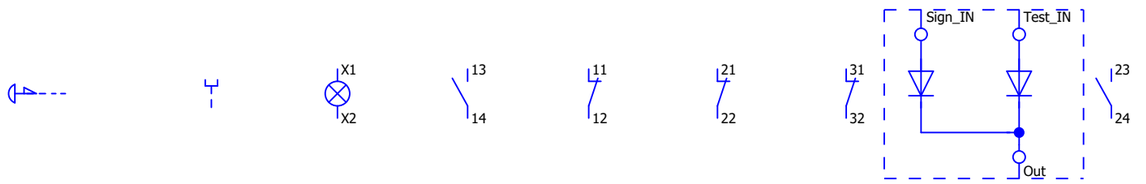
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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.				DEVICE TAG LIST	Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 177 PAGE 180 / 204
Checked by	Antonio Aleksić	01.04.2026.						
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)
1	SVJETLEĆA TIPKA, ZELENA Osvijetljena tipka, 22 mm, okrugla, metalna, sjajna, zelena, tipka, ravna, trenutni kontakt, s držačem, 1 NO, LED modul s integriranom LED 24 V AC/DC, navojna stezaljka Narudžbeni br.: 3SU1152-0AB40-1BA0 Nosač oznake Tip: 3SU1900-0AP10-0AA0 Držač etikete, 22mm, pravokutni okvir na dnu, crni, za etiketnu pločicu 17.5 mm x 27 mm, za lijepljenje LED TESTNI MODUL Tip: 3SU1400-1CK10-1AA0 LED ispitni modul Radni napon 12-240 V AC/DC, vijčani terminal, za montažu na prednju ploču	Siemens	+U7CJA10	-S224	 /66.3:B /80.4:D /80.4:C
1	Sklopka sa 12 pozicija - nazivna struja: 12 A, kut prekidača: 30°, broj polova: 1, - tip: GN12-106-U-P3094 Narudžbeni br.: GN12-106-U-P3094		+U7CJA10	-S251	 /74.2:B
1	ILLUM. TIPKA GLJIVICA, 40MM, CRVENA Osvijetljena gljivasta tipka, 22 mm, okrugla, metalna, sjajna, crvena, 40 mm, zatvaranje, mehanizam povuci i otključaj Narudžbeni br.: 3SU1051-1BA20-0AA0 DRŽAČ Tip: 3SU1550-0AA10-0AA0 Držač za 3 modula, univerzalni, plastika/metal LED MODUL, CRVENI Tip: 3SU1401-1BB20-1AA0 LED MODUL S INTEGRIRANOM LED 24V AC/DC, CRVENI, NAVIJAČNI TERMINAL, ZA MONTAŽU NA PREDNJU PLOČU PROTECTIVE COLLAR Tip: 3SU1900-0DY30-0AA0 PROTECTIVE COLLAR FOR EM. STOP MUSHROOM PUSHBUTTON, WITHOUT OR WITH RONIS LOCK, YELLOW, PLASTIC Contact module with 1 contact element, 1 NO Tip: 3SU1400-1AA10-1BA0 Contact module with 1 contact element, 1 NO, screw terminal, for front plate mounting	Siemens	+U7CJA10	-S301	 /19.6:D /82.2:D /82.2:C /76.3:B /19.6:D /82.2:C

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Format ISO/DIN A3

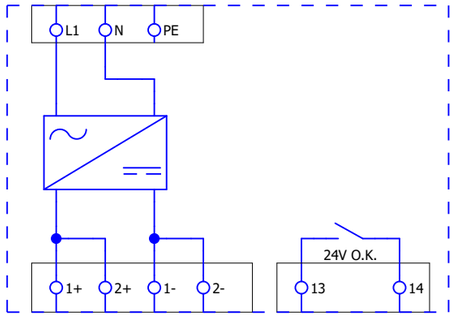
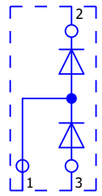
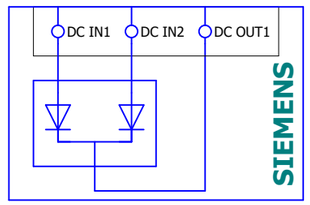
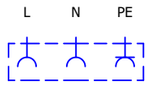
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 178
Checked by	Antonio Aleksić	01.04.2026.						PAGE 181 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)
1	ILLUM. TIPKA GLJIVICA, 40MM, CRVENA Osvijetljena gljivasta tipka, 22 mm, okrugla, metalna, sjajna, crvena, 40 mm, zatvaranje, mehanizam povuci i otključaj Narudžbeni br.: 3SU1051-1BA20-0AA0 DRŽAČ Tip: 3SU1550-0AA10-0AA0 Držač za 3 modula, univerzalni, plastika/metal LED MODUL, CRVENI Tip: 3SU1401-1BB20-1AA0 LED MODUL S INTEGRIRANOM LED 24V AC/DC, CRVENI, NAVIJAČNI TERMINAL, ZA MONTAŽU NA PREDNJU PLOČU PROTECTIVE COLLAR Tip: 3SU1900-0DY30-0AA0 PROTECTIVE COLLAR FOR EM. STOP MUSHROOM PUSHBUTTON, WITHOUT OR WITH RONIS LOCK, YELLOW, PLASTIC Contact module with 1 contact element, 1 NO Tip: 3SU1400-1AA10-1BA0 Contact module with 1 contact element, 1 NO, screw terminal, for front plate mounting	Siemens	+U7CJA10	-S302	 /19.7:D /82.3:D /82.3:C /76.3:C /19.7:D /82.3:C /76.7:C
2	2-polni DC rastavljač 2P istosmjerni rastavljač Nazivna struja: 50 A Izolacijski napon: 1000 VDC Narudžbeni br.: A9N61690	Schneider Electric	+U7CJA10	-SV1	/7.2:D
			+U7CJA10	-SV2	/7.3:D
3	Distribucijski blok Ovi Linergy DS razdjelni blokovi s vijcima mogu se ugraditi na standardnu din šinu ili na montažnu ploču. Namijenjen je 1P instalacijama. Narudžbeni br.: LGY112510	Schneider Electric	+U7CJA10	-U1	/7.1:A
			+U7CJA10	-U2	/7.1:B
			+U7CJA10	-U10	/8.1:F

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.						
Checked by	Antonio Aleksić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 179
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			PAGE 182 / 204

Qty. (pcs)	Equipment technical description	Manufacturer	Location (+)	Designation (-)	Symbol Position in shematics (Page, column)
	SITOP PSU8200 SITOP PSU8200M 20A STABILIZIRANO NAPAJANJE ULAZ: 120-230 V AC 110-220 V DC IZLAZ: 24 V/20 A DC Tip: 6EP1336-3BA10 Narudžbeni br.: 6EP1336-3BA10	Siemens			
			+U7CJA10	-U101	/8.2:C /8.3:D
2			+U7CJA10	-U102	/8.5:C /8.6:D
	Dioda Diskretni poluvodički moduli Diskretni poluvodički moduli 800V 100A Vf prednji napon: 1.35 V Vr -Reverzni napon: 800 V Ako: 100A Tip: DD100KB80 Narudžbeni br.: DD100KB80				
			+U7CJA10	-V1	/7.2:B
2			+U7CJA10	-V2	/7.3:B
	Redundancijski modul Ulaz/izlaz: DC 24-48V/40A Prikladno za odvajanje dva SITOP napajanja s max. 20 A izlazne struje svaki Tip: SITOP RED1200 Narudžbeni br.: 6EP4347-7RB00-0AX0	Siemens			
			+U7CJA10	-V101	/8.2:E
1					
	Napon: 250V, 50Hz Nazivna struja: 16A Narudžbeni br.: 5TE6800	Siemens			
			+U7CJA10	-X101	/6.2:F
1					

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Voith Company Sintaksa - A Voith Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit 7 process control circuit diagram DEVICE TAG LIST	DRAWING No.: PERU-01-17-170-31-001	REV No.: = U7CJA10GH100 + U7CJA10
Approved by Designer	Grga Kolić	01.04.2026.					Designer No.: 22044-DD-EE-08-07-100	SALES No.: PAGE 180
Checked by	Antonio Aleksić	01.04.2026.						PAGE 183 / 204
Drawn by	Luka Božičević	01.04.2026.	PROJECT NAME	EMPLOYER	DESIGNER			